Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 — Electrochemistry and corrosion · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_02 · Electrochemistry and corrosion

1 UE26CY151A · Unit_02: Electrochemistry and corrosion

This unit master organizes source material for Engineering Chemistry, Unit_02, officially titled Electrochemistry and corrosion. It includes 14 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26CY151A, Unit_02 (Electrochemistry and corrosion). Bucket folder present: yes. Videos: 14; slide decks: 12.

EV0000001

2 Origin of electrode potential, cell potential

This lesson is indexed from the source video titled Origin of electrode potential, cell potential. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

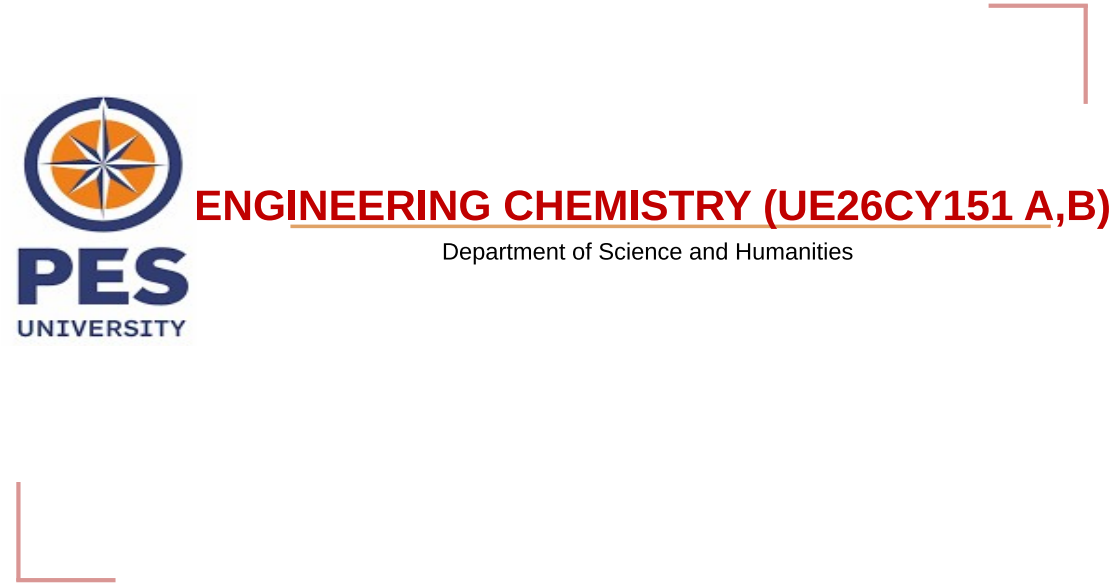

Figure G000001. Origin of electrode potential cell potential · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

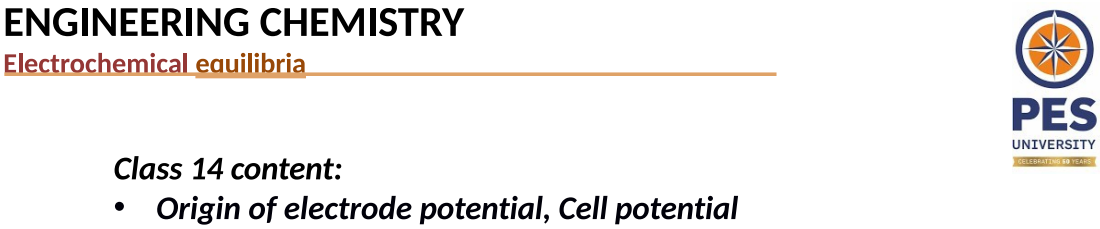

Figure G000002. Origin of electrode potential cell potential · source slide page 2.

EV0000270

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

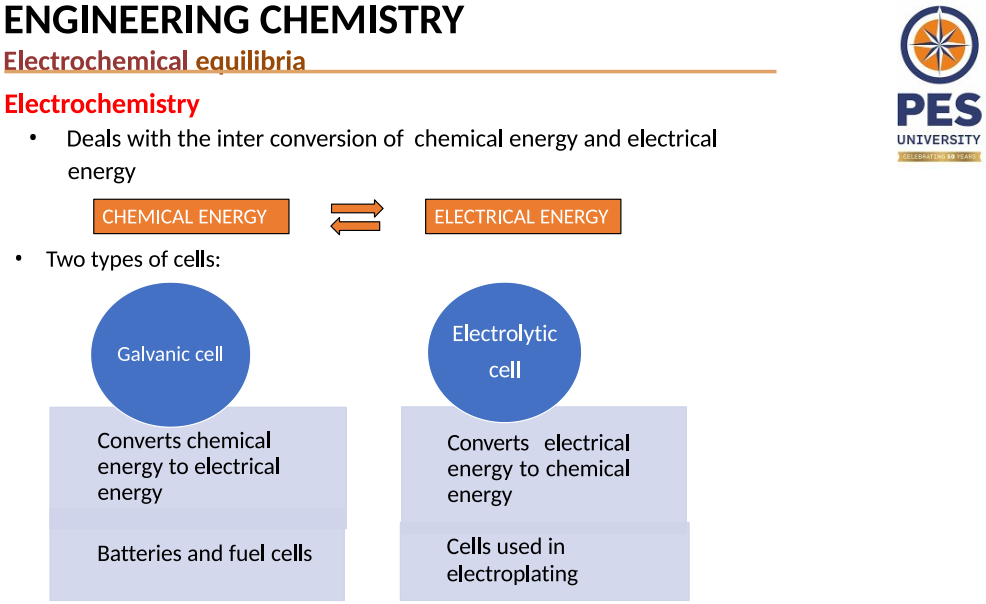

Figure G000003. Origin of electrode potential cell potential · source slide page 3.

EV0000273

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

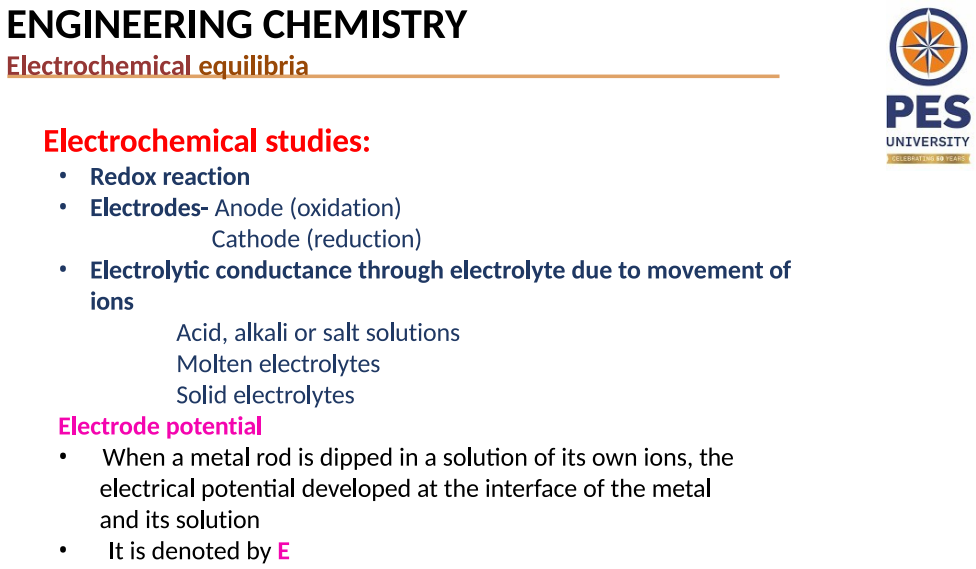

Figure G000004. Origin of electrode potential cell potential · source slide page 4.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

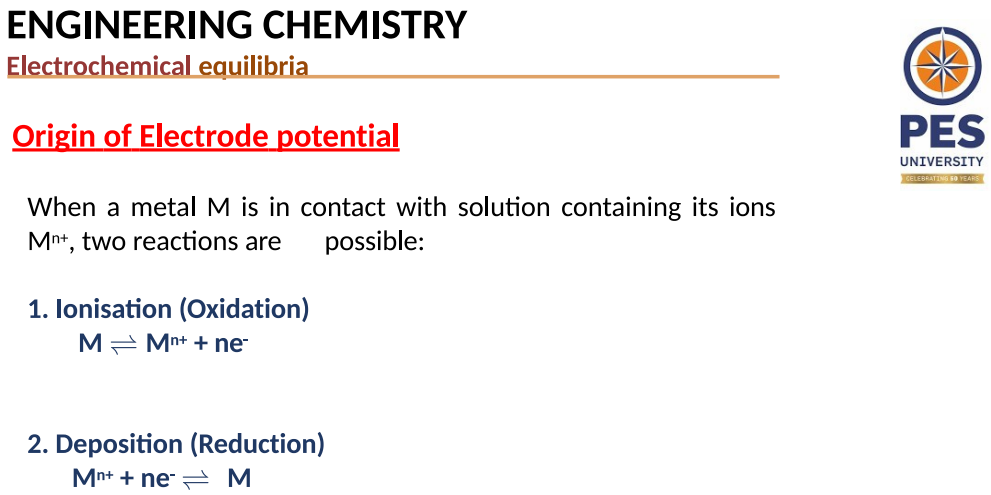

Figure G000005. Origin of electrode potential cell potential · source slide page 5.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

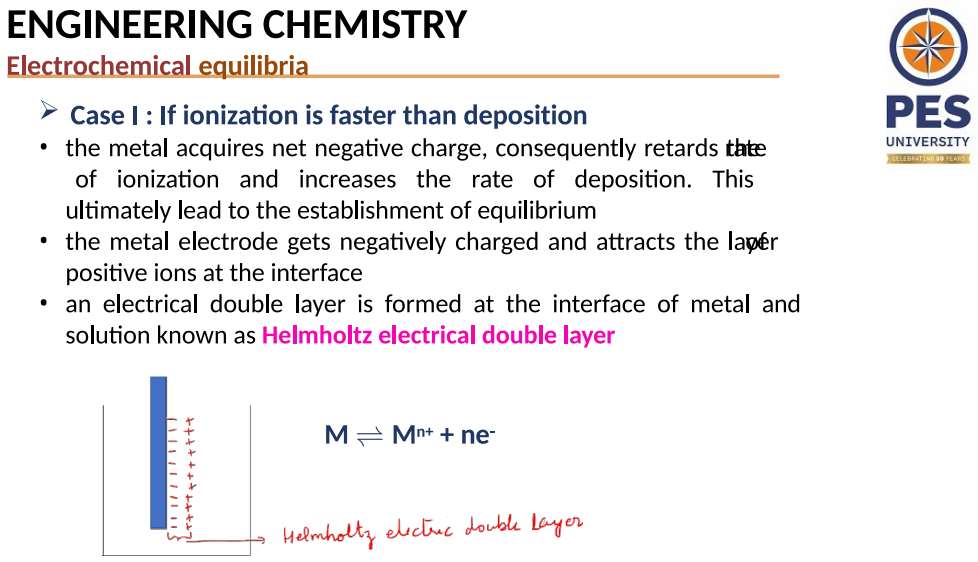

Figure G000006. Origin of electrode potential cell potential · source slide page 6.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

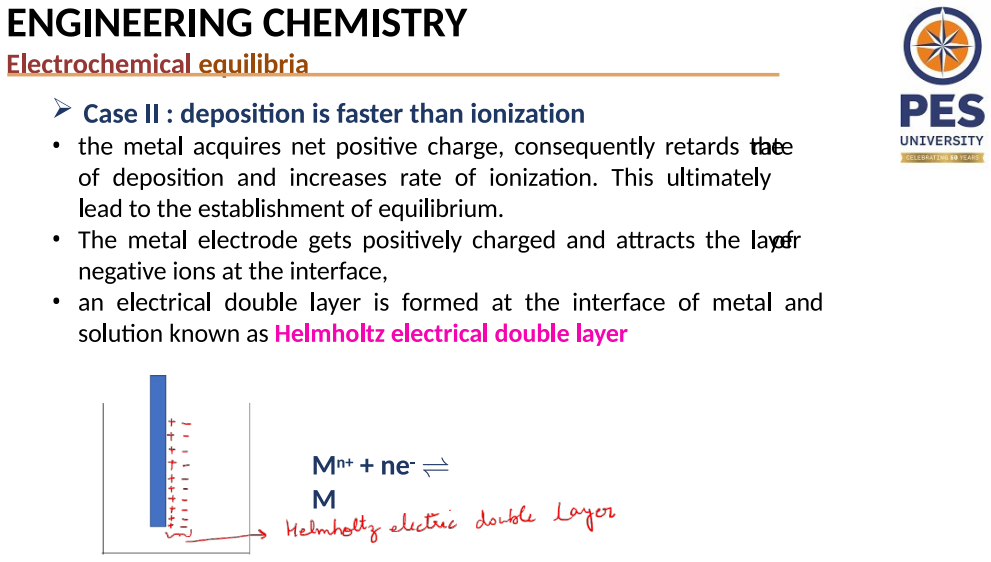

Figure G000007. Origin of electrode potential cell potential · source slide page 7.

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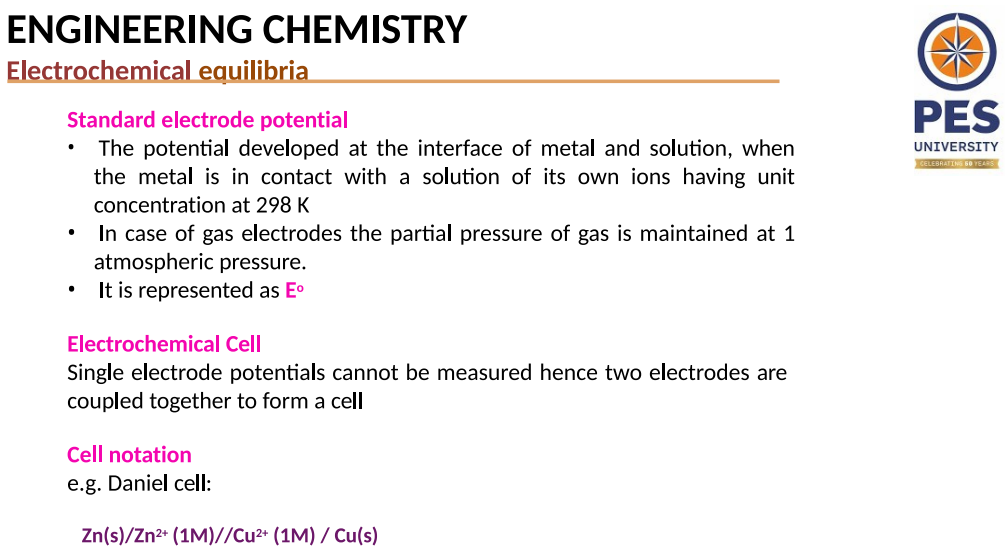

Figure G000008. Origin of electrode potential cell potential · source slide page 8.

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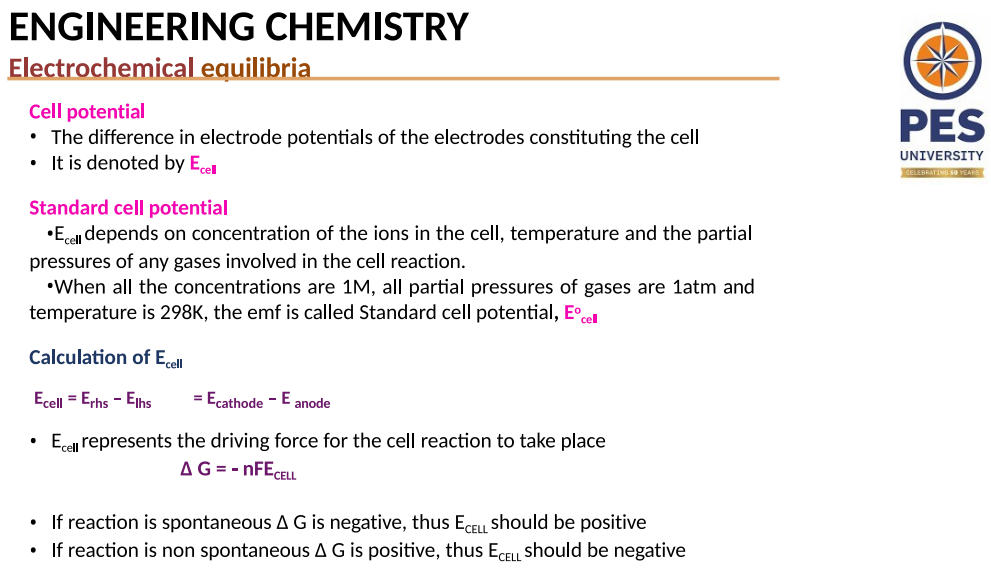

Figure G000009. Origin of electrode potential cell potential · source slide page 9.

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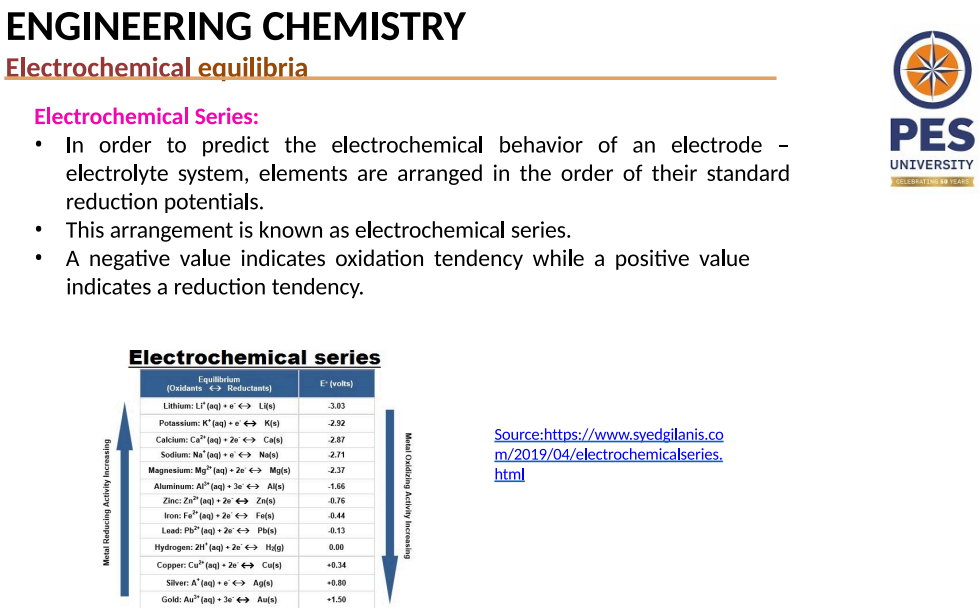

Figure G000010. Origin of electrode potential cell potential · source slide page 10.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000011. Origin of electrode potential cell potential · source slide page 11.

EV0000297

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000266 · EV0000267 · EV0000268
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Class 14 content: Origin of electrode potential, Cell potential EV0000269 · EV0000270 · EV0000271
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Electrochemistry Deals with the inter conversion of chemical energy and electrical energy CHEMICAL ENERGY ELECTRICAL ENERGY Two types of cells: Electrolytic cell Galvanic cell Converts chemical energy to electrical energy Converts electrical energy to chemical energy Cells used in electroplating Batteries and fuel cells EV0000272 · EV0000273 · EV0000274
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Electrochemical studies: Redox reaction Electrodes- Anode (oxidation) Cathode (reduction) Electrolytic conductance through electrolyte due to movement of ions Acid, alkali or salt solutions Molten electrolytes Solid electrolytes Electrode potential When a metal rod is dipped in a solution of its own ions, the electrical potential developed at the interface of the metal and its solution It is denoted by E EV0000275 · EV0000276 · EV0000277
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Origin of Electrode potential When a metal M is in contact with solution containing its ions Mn+, two reactions are possible: Ionisation (Oxidation) M ⇌ Mn+ + ne- Deposition (Reduction) Mn+ + ne- ⇌ M EV0000278 · EV0000279 · EV0000280
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Case I : If ionization is faster than deposition the metal acquires net negative charge, consequently retards the rate of ionization and increases the rate of deposition. This ultimately lead to the establishment of equilibrium the metal electrode gets negatively charged and attracts the layer of positive ions at the interface an electrical double layer is formed at the interface of metal and solution known as Helmholtz electrical double layer M ⇌ Mn+ + ne- EV0000281 · EV0000282 · EV0000283
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Case II : deposition is faster than ionization the metal acquires net positive charge, consequently retards the rate of deposition and increases rate of ionization. This ultimately lead to the establishment of equilibrium. The metal electrode gets positively charged and attracts the layer of negative ions at the interface, an electrical double layer is formed at the interface of metal and solution known as Helmholtz electrical double layer Mn+ + ne- ⇌ M EV0000284 · EV0000285 · EV0000286
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Standard electrode potential The potential developed at the interface of metal and solution, when the metal is in contact with a solution of its own ions having unit concentration at 298 K In case of gas electrodes the partial pressure of gas is maintained at 1 atmospheric pressure. It is represented as Eo Electrochemical Cell Single electrode potentials cannot be measured hence two electrodes are coupled together to form a cell Cell notation e.g. Daniel cell: Zn(s)/Zn2+ (1M)//Cu2+ (1M) / Cu(s) EV0000287 · EV0000288 · EV0000289

9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Cell potential The difference in electrode potentials of the electrodes constituting the cell It is denoted by Ecell Standard cell potential Ecell depends on concentration of the ions in the cell, temperature and the partial pressures of any gases involved in the cell reaction. When all the concentrations are 1M, all partial pressures of gases are 1atm and temperature is 298K, the emf is called Standard cell potential, Eocell Calculation of Ecell Ecell = Erhs – Elhs = Ecathode – E anode Ecell represents the driving force for the cell reaction to take place Δ G = - nFECELL If reaction is spontaneous Δ G is negative, thus ECELL should be positive If reaction is non spontaneous Δ G is positive, thus ECELL should be negative EV0000290 · EV0000291 · EV0000292
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Electrochemical Series: In order to predict the electrochemical behavior of an electrode – electrolyte system, elements are arranged in the order of their standard reduction potentials. This arrangement is known as electrochemical series. A negative value indicates oxidation tendency while a positive value indicates a reduction tendency. [Image text, OCR] Electrochemical series Equilibrium E(volts) (Oxidants(>Reductants) Lithium:Li(aq)+e<> Li(s) -3.03 Potassium:K+(aq)+e> K(s) -2.92 Calcium:Ca²+(aq) +2e> Ca(s) -2.87 Metal Sodium:Na (aq)+e<> Na(s) -2.71 Oxidizing Magnesium:Mg² (aq) +2e<> Mg(s) -2.37 Aluminum:Al² (aq) +3e<> Al(s) -1.66 Activity Zinc:Zn²+ (aq) +2e→ Zn(s) -0.76 lron:Fe² (aq) +2e<> Fe(s) -0.44 Lead:Pb² (aq) +2e> Increasing Pb(s) -0.13 Hydrogen:2H* (aq) +2e> H2(g) 0.00 Copper:Cu²(aq) +2e> Cu(s) +0.34 Silver:A (aq) +e> Ag(s) +0.80 Gold:Au²+ (aq) +3e> Au(s) +1.50 Source:https://www.syedgilanis.co m/2019/04/electrochemicalseries. html EV0000293 · EV0000294 · EV0000295
11. Source slide page 11 (extracted text): THANK YOU Department of Science and Humanities EV0000296 · EV0000297 · EV0000298

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.356] Hello everyone, welcome back to this course on engineering chemistry. Today we will start with a new module on electrochemical equilibrium. [00:00:10.788] In this module we are going to study about electrode potential and cell potential. We will derive the Nernst equation. Then we discuss the different types of electrodes. [00:00:22.212] We will look at reference electrodes and also talk about concentration cells. And finally, we will discuss about ion selective electrodes. [00:00:33.796] In today's class we will discuss the origin of electrode potential, we will talk about cell potential and we will derive Nernst equation. [00:00:45.348] Electrochemical equilibria deals with electrochemistry, which is a branch of chemistry which deals with the interconversion of chemical energy and electrical energy. [00:00:59.200] So we have chemical energy getting converted into electrical energy and vice versa. Now this can be done in two different types of cells galvanic cell and the electrolytic cell. [00:01:15.300] In the galvanic cell, we have chemical energy getting converted into electrical energy. Now where is this chemical energy coming from? It is the energy in a chemical reaction. [00:01:30.468] Now the reaction has to be a spontaneous reaction because we are not going to provide any energy from outside. The chemical energy has to produce electrical energy. So the reaction has to be a spontaneous reaction, meaning it should happen on its own. [00:01:49.220] And since we are going to produce electrical energy, that is power, there has to be exchange of electrons. Therefore, we have to have detox reaction. [00:02:00.480] So if we are talking of a galvanic cell, we

need a spontaneous redox reaction which is going to convert chemical energy to electrical energy. Examples of galvanic cell would be in batteries and fuel cells. [00:02:19.140] Now another type of cell is the electrolytic cell where the reverse is happening. We convert electrical energy to chemical energy. Electrical energy is provided externally and [00:02:34.224] We want to get chemical energy, meaning we want to make a chemical reaction happen. So obviously the reaction is not going to be spontaneous. We require a non-spontaneous redox reaction in an electrolytic cell. [00:02:50.692] We use such cells in electroplating, electro refining, and many other metallurgical processes. [00:02:58.500] Whenever we do electrochemical studies, the first thing we require is a redox reaction because we are talking of conversion of chemical energy into electrical energy or electrical energy into chemical energy, and both require an exchange of electrons. [00:03:16.272] So a redux reaction is one where we have oxidation and reduction taking place. And one thing we should always remember that oxidation and reduction have to happen simultaneously. [00:03:30.116] The second requirement is that of electrodes. These oxidation and reduction reactions have to happen at the anode and cathode respectively. So we need electrodes wherein oxidation and reduction can take place. [00:03:49.540] And the third requirement would be electrolytic conductance. So we need an electrolyte where [00:03:57.540] The conductance is through the movement of ions. We have to remember that electrons cannot move through the electrolyte. It's only the ions which are going to cause electrolytic conductance. [00:04:13.072] So the most common electrolytes are acids, alkali or salt solutions, which we are all familiar with and they are used in batteries and fuel cells. We can also have molten electrolytes. [00:04:28.612] Suppose we have sodium chloride, we know that when we melt it, then the ions become free to move, and therefore we can use molten electrolyte also in galvanic cells or electrolytic cells. Apart from that [00:04:45.728] There is a third category of electrolytes which are called solid electrolytes. Now solids acting as electrolytes is little strange, but there are certain [00:04:58.180] compounds which are going to allow the movement of ionside. That is the criteria for an electrolyte. For example, we have zirconium oxide when it is doped with Y_2O_3. What happens is

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.560] Oxide ion vacancies are created and this oxide ion can move through the solid electrolyte, therefore it acts as an electrolyte. Since it's allowing for the movement of the oxide ions, we can say that zirconium oxide dope by two or three [00:05:32.208] is also an electrolyte. Now let us define electrode potential. When a metal rod is dipped in a solution of its own ions [00:05:43.716] At the interface there is a potential which is developed and this is known as electrode potential and this happens at the interface of the metal and the solution and it is denoted by capital E. [00:06:02.916] Let us now discuss the origin of electrode potential. [00:06:07.556] So we know that when a metal rod is in contact with a solution of its own ions. So suppose we have a metal M which is in contact with M M plus ions, there are two reactions which are possible. The first one is ionization or [00:06:24.304] oxidation the metal rod can dissolve and get changed into Mn plus thereby giving out electrons. So when there is loss of electrons we know it is called oxidation and there is also a possibility [00:06:42.816] that deposition will happen and deposition will be a reduction reaction wherein the metal ions take up electrons and get deposited in the form of metal. [00:06:56.420] Let us take up a case where [00:07:00.032] The ionization reaction is faster than the deposition reaction. Just now I mentioned that whenever we have a metal rod dipped in a solution of its own ions, there are two possibilities. Either the metal gets ionized or the metal ions take up electrons and yet [00:07:17.712] deposit.

Now suppose the first reaction is faster where wherein the ionization reaction is faster. That is M is becoming Mn plus plus the N electrons. Now [00:07:29.968] Here we have this situation, say let us look at this diagram here. We have M and we have the solution which contains Mn plus. [00:07:39.172] When this reaction takes place we know that the metal is getting ionized. So mn plus ions are going to go into the solution. But what happens to the electrons? [00:07:50.976] They are left back on the surface of the electrode, and therefore the electrode acquires a negative charge. This at the surface of the electrode we have negative charge, and because there is a negative charge on the [00:08:06.848] uh electrode it is going to attract the positive ions from the solution. So what do we have here? We have a positive layer which is getting formed, and because of this we see that at the interface [00:08:22.672] There is an electric double layer which is getting formed. This electric double layer is responsible for the electrode potential and it is known as the hellholes electric double layer. [00:08:40.196] If we have the second case where the deposition is faster than the ionization, let us look at this. Now here we have. [00:08:48.548] metal rod and it is dipped in a solution of Mn plus. But this is case two where the deposition reaction is faster, meaning the reduction reaction is happening faster. So Mn plus is taking up electrons and getting deposited as M. So these metal ions are going to take up electrons. [00:09:06.320] And get deposited. But where are the electrons coming from? They are coming from the metal rod. So as a result, what happens is the metal rod is going to acquire a positive charge and now [00:09:25.956] the electrical the negative ions from the solution are going to get attracted to this positive [00:09:37.232] charge which is there on the surface and thereby giving rise to an electrical double layer. Again this electrical double layer is responsible for the electrode potential at the interface and this electrical double layer is known as the [00:09:54.992] And molds electrical double layer. So whether oxidation happens, that is ionization happens to be the faster reaction.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.792] or the reduction reaction that is deposition happens to be the faster reaction. In both cases we understand that there's going to be an electrical double layer which is formed at the interface between the metal rod and the solution and this is [00:10:20.496] known as the Helmor's electrical double layer and responsible for the electrode potential which we see in this system. [00:10:30.532] Whether the oxidation reaction will happen faster or the reduction reaction will happen faster is going to depend on the nature of the metal. If the metal has greater tendency to undergo oxidation, we find that ionization will be the faster reaction and if [00:10:48.576] The ions have a greater tendency to undergo reduction, then we find that the deposition reaction will be happening at a faster rate. So that means it clearly depends on the nature of the metal, and it also depends on the concentration of the ions which are being used, it depends on the temperature. [00:11:06.256] And if gases are involved on the pressure. Now with so many parameters changing, we will not be able to assign a particular value to any electrode. [00:11:18.416] If we want to compare electrodes, then we need to define a new quantity which is known as the standard electrode potential. [00:11:27.856] It is defined as the potential developed at the interface of the metal in the solution. When the metal is in contact with the solution of its own ions, but now we have placed a restriction that the concentration should be unity and the temperature is now fixed at two ninety eight Kelvin and in the [00:11:45.552] In case there are gases involved, then the pressure of the gas also has to be maintained at one atmosphere pressure. Now this quantity is called [00:11:56.416] as standard electrode potential denoted by E naught and now this is going to depend only on the nature of the metal. [00:12:06.660] But single

electrode potentials cannot be measured. What we can measure is the potential difference. Therefore, we need to form a cell where two electrodes are coupled together and such a cell would be called as an electrochemical cell. [00:12:25.092] Now if we want to represent this cell, we cannot draw the cell every time, so we need a notation by which we can represent this cell. And therefore we see [00:12:38.352] Suppose we take the example of Daniel Cell. We know that in a Daniel cell what is it? Zinc is dipped in a solution of zinc sulphate, copper is dipped in a solution of copper sulphate. So let us look at this representation and understand why it is written this way. Firstly [00:12:55.824] Zinc rod is dipped in a solution of zinc sulfate. We always write the anode first. Where oxidation happens, the anode is written first. Between zinc and copper we find that the zinc electrode will undergo oxidation. So zinc is dipped in a solution of Zn two plus. [00:13:13.456] The value in this parenthesis is the concentration of the zinc ions. So anode is written first and this single line over here represents the phase separation. So zinc [00:13:26.704] is solid and Zn two plus is in the liquid form, therefore there is a phase separation shown by the single line. Similarly when we talk of copper dipped in a solution of copper sulphate and here we can see again it's one molar concentration. So copper in copper sulphate [00:13:43.552] Again is separated by a single line, but and this is where this is the cathode. So we have the anode first and then we write the cathode, and in between we can see these double lines. This represents the salt bridge. What we have to remember is we cannot write [00:14:01.200] Zinc Zn2 plus and salt bridge copper and then CU2 plus. Why? Because the salt bridge is connecting the solutions. So we have the salt bridge, one end is dipped in zinc sulphate, the other end is zipped in dipped in copper sulphate. Therefore, this is the [00:14:18.768] representation and this is known as cell notation. [00:14:25.252] The difference in electrode potentials of the electrode is something which we can measure and that is known as the cell potential. So if we have two electrodes and we measure the difference in the electrode potential, that we will call it as the cell potential. [00:14:43.024] Potential and it is denoted by e cell. Again, cell potential is going to depend on the concentrations of the solutions used and also on the temperature and also on the pressure as well as the electrodes which have been used. So therefore, in order to

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.380] compare we need to have a standard value which is known as the standard electrode potential and e cell depends on the concentration temperature and partial pressure of the gases involved. So if we take all the concentrations as one molar [00:15:20.224] all partial pressure of the gases as one atmosphere and the temperature is maintained at 298 Kelvin, then this EMF which we measure will be known as the standard cell EMF or standard cell potential which is given by E naught cell. [00:15:37.424] Now how do we calculate the cell potential? Cell potential is [00:15:42.912] Because we know the cell is made of two electrodes, one electrode is the anode electrode, the other electrode is the cathode electrode. And also I mentioned that anode is written first and then that is on the left hand side and cathode is written on the right hand side of the salt bridge. So therefore [00:16:00.592] E cell is given by electrode potential of the right hand side electrode minus the electrode potential of the left hand side electrode or in other words we can say e cathode minus the e anode. [00:16:15.056] Now each cell is a very important quantity. Why? Because it represents the driving force for the cell reaction to take place. [00:16:25.840] because the change in free energy for any reaction any system tries to go to minimum free energy so the change in free energy is given by minus nf ec [00:16:39.748] What does it tell us that for any spontaneous reaction we know that

delta G has to be negative and therefore E cell should be positive. [00:16:51.940] And if the reaction is non-spontaneous, then delta G will be positive and therefore the E cell has to be negative. So E cell is the electromotive force, the difference between the electrode potentials [00:17:06.512] The uh of the two electrodes, and that will be the driving force for the reaction to take place, the difference in the [00:17:15.632] free uh the electrode potentials will decide the reaction the they decide the reaction and in other words thermodynamically that is expressed as delta g is equal to minus nfe c let us talk about the electrochemical series [00:17:34.084] The standard reduction potentials have been listed in a table in ascending order. This is to help us predict the electrochemical behavior of an electrode electrolyte system. This series is known as the electrochemical series. [00:17:53.636] Most often we find that it is the standard reduction potentials which have been used and such an arrangement is referred to as the electrochemical series. [00:18:05.860] This series is going to be very informative. A negative value of standard reduction potential indicates oxidation tendency while a positive value is going to indicate a reduction tendency. [00:18:21.008] For example, lithium is found to be having a standard reduction potential of minus three point zero three. So we know that lithium can easily undergo oxidation. Whereas copper we find that [00:18:36.000] CU2 plus taking up electrons giving you copper that is the reduction potential is plus 0.34. So the reduction tendency of C U two plus is very high. Hence the value is positive. [00:18:50.992] Let us derive the Nernst equation for a single electrode and the same concept will be extended to an electrochemical cell. [00:19:00.132] Electrochemical series lists the standard electrode potentials which are measured under standard conditions of unit concentration two ninety eight Kelvin temperature and one atmosphere pressure. But most of the times we do not deal with standard conditions. [00:19:17.584] So Nernst equation is going to give us the relation between the electrode potential and [00:19:25.360] and the concentration of the species with which the electrode is reversible, meaning how the electrode potential changes with concentration, temperature and pressure. [00:19:38.340] Now the reaction at the electrode is the reduction reaction, so we write it as Mn plus plus N electron is equal to M because we are talking of reduction potentials. The maximum work that can be obtained from any system is given by the decrease in free energy, so that can be reduced. [00:19:56.112] Written as minus delta G is equal to W max. The system we are talking about here is a metal rod dipped in a solution of its own ions. So let us find out what is the maximum work done for such a system.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:11.076] For any electrical chemical system, the maximum work done is given by the total charge available multiplied by the energy available per unit charge. [00:20:21.732] The total charge available here is equal to the number of moles of electrons exchanged in the redox reaction, that is n we can see here, multiplied by the charge carried per mole of electrons, and that is f. So the [00:20:37.376] Total charge is N into F and the energy available per unit charge is will be electrode potential because electrode potential is given by energy per unit charge. [00:20:50.352] That is if you uh if the unit is volts then that is given by joules per coulomb and that is the electrode potential which is given by E. Therefore, now the W max becomes equal to N F. [00:21:06.928] multiplied by E. That is the total charge available is Nf and the energy available per unit charge is E, therefore W max is equal to N F E and therefore delta G becomes equal to minus N FE by substituting in this equation. [00:21:24.260] Under standard conditions, delta G is delta G naught that is equal to N minus of Nfe. The reaction we are talking about is Mn plus plus N electron is giving you M. [00:21:42.084] A

thermodynamic equation which is going to react which relates the reaction quotient and the decrease in free energy is given by delta g is equal to delta g0 plus r t log q. So basically it is telling us how the delta g change [00:21:59.856] changes when q changes. What is q? It is the reaction quotient. [00:22:05.024] the extent of the reaction that can be given by the concentration of the products divided by the concentration of the reactants. For this particular reaction which we are talking about, this is given by Q is equal to M [00:22:20.384] concentration of M divided by concentration of Mn plus. Now we have an expression for delta G. We know what is delta G naught and we also know Q. So when we substitute we get [00:22:32.288] In this equation, we are going to substitute, we get minus of Nfe is equal to minus Nfe naught plus R T log M over Mn plus. [00:22:44.196] Well we know that E0 stands for the standard electrode potential, n is the number of electrons getting exchanged in the redox reaction, R is the gas constant, that is 8.314 joules per Kelvin per mole, P is the temperature in Kelvin and F is Faradin. [00:23:01.968] that is 96500 coulombs per mole. Now if we divide this equation throughout by minus n f then we get e is equal to e0 minus r t over n f log of m over mn plus. [00:23:17.380] For pure substances we take M concentration equal to one. So the equation now becomes E is equal to E0 plus R P over N F log of Mn plus. [00:23:29.824] Now if we calculate at 298 Kelvin the value of R T over F and we convert into common law then we find that [00:23:40.368] Substituting for R 8.314 joules per Kelvin per mole, temperature as two ninety-eight Kelvin, F is nine six five zero zero coulombs per mole. [00:23:50.480] multiplied by 2.303 then we get the value 0.0591. So at 298 Kelvin the learnst equation is e is equal to E naught plus 0.0591 over n log of mn plus. [00:24:07.424] This is for a single electrode. Now, if you want to extend the concept to an electrochemical cell. [00:24:15.428] The only thing which is going to change is the reaction taking place. Now we have AA plus BB is giving you C C plus D. So what changes is the Q value. Q value is concentration of the [00:24:30.432] Products raised to the respective stechometric coefficients divided by concentration of the reactants raised to the [00:24:39.524] respective stepometric coefficients. So what we are going to do is again use this equation delta G is equal to delta G0 plus R T over N F. [00:24:52.996] Ln Q. Now Q is different. Delta G remains the same. It is minus N F E and Delta G naught is going to be minus N F E naught. Now E and E naught are the

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:06.336] uh cell emf and the standard cell emf values so dividing by uh minus nf and of course we have to substitute for q the nonst equation at 298 Kelvin now is is equal to e naught minus 0.0591 by n log of [00:25:25.904] C raised to the power of small c d raised to the power of small b divided by a raised to the power of small a and b raised to the power of small b. [00:25:37.328] and n is the number of electrons transferred and e0 is the standard EMF of the cell. So this is the Nernst equation of course this is at 298 Kelvin for an electrochemical cell. [00:25:55.140] Now in the next class, we are going to discuss about the different types of electrodes. Thank you.

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Full transcript: 3578 normalized words, 111 timestamped segments, 1565.533333 seconds duration. Low-confidence segments flagged: 0.

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3 Types of electrodes

This lesson is indexed from the source video titled Types of electrodes. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000026 · EV0000027

Graphs and diagrams

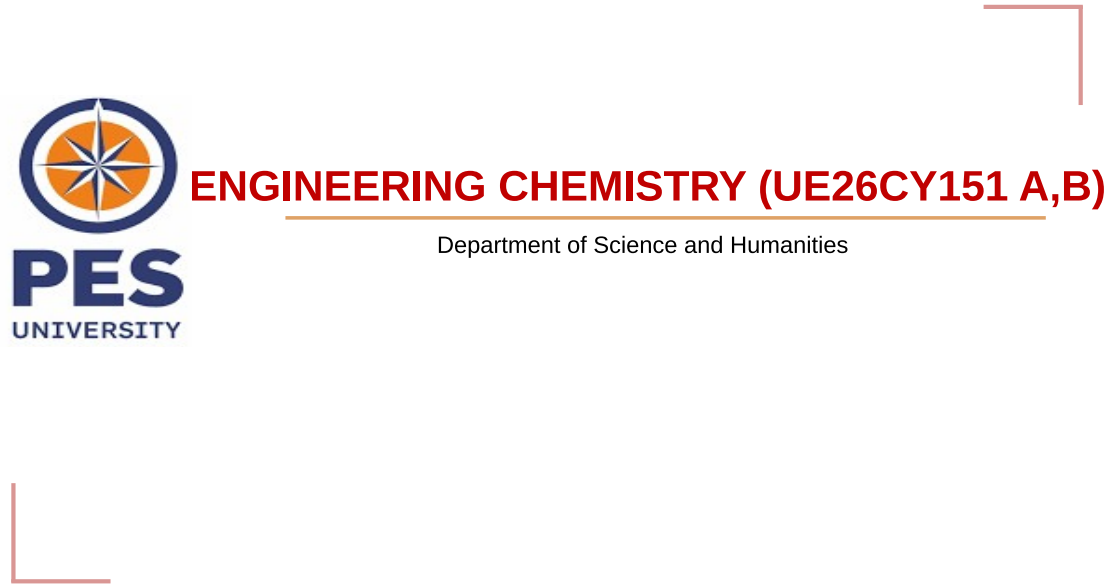

Figure G000018. Types of electrodes · source slide page 1.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

PES UNIVERSITY

Class 16 content:

- ***Types of electrodes***

Figure G000019. Types of electrodes · source slide page 2.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

PES UNIVERSITY

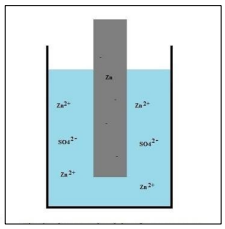
Types of electrodes

- In order to form a cell, 2 half cells or 2 electrodes are required
- Various types of electrodes are available which are constructed based on the application

1. Metal-metal ion electrode:

- Metal in contact with a solution of its own ions
  e.g., Zn/Zn^{2+}, Cu/Cu^{2+}, Ag/Ag^{+}
- $M^{n+} + ne^- \rightleftharpoons M$
- Nernst equation

$$E_{M/M^{n+}} = E^{o}_{M/M^{n+}} + \frac{2.303RT}{nF} \log\left[M^{n+}\right]$$



Source:http://www.valgetal.com/physics/Batteries/batteries.htm

Figure G000020. Types of electrodes · source slide page 3.

EV0000324

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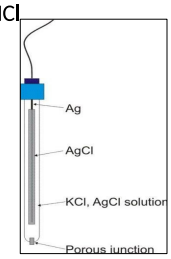
ENGINEERING CHEMISTRY

Electrochemical equilibria

2. Metal-Metal insoluble salt- ion electrode:

- These electrodes consist of a metal in contact with a sparingly soluble salt of the same metal dipped in a solution of soluble salt of the same anion
  e.g.,Calomel electrode $Hg/Hg_2Cl_2/KCl$, Ag/AgCl(s)/HCl
- For silver –silver chloride electrode
  $AgCl + e^- \rightleftharpoons Ag + Cl^-$
- Nernst equation:

$$E_{Ag/AgCl/Cl^-} = E^{o}_{Ag/AgCl/Cl^-} - \frac{0.0591}{1}\log[Cl^-]$$



Source:https://www.corrosion-doctors.org/Corrosion-Thermodynamics/Reference-Half-Cells-Silver.htm

Figure G000021. Types of electrodes · source slide page 4.

EV0000327

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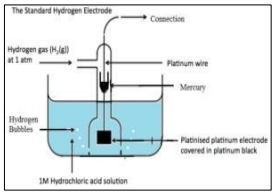
ENGINEERING CHEMISTRY

Electrochemical equilibria

3. Gas electrode:

- It consists of gas bubbling about an inert metal foil, immersed in solution containing ions to which the gas is reversible.
- The metal provides electrical contact and facilitates the establishment of equilibrium between the gas and its ions
  e.g.,Hydrogen electrode $Pt/H_2/H^+$,Chlorine electrode $Pt/Cl_2/Cl^-$
- For a hydrogen electrode
  $2H^+ + 2e^- \rightleftharpoons H_2$
- Nernst equation:

$$E_{Pt|H_2|H^+} = E^{0}_{Pt|H_2|H^+} - \frac{0.0591}{2}\log\left(\frac{p_H}{[H^+]^2}\right)$$



Source:https://thefactfactor.com/facts/pure_science/chemistry/physical-chemistry/reference-electrodes/5844/

Figure G000022. Types of electrodes · source slide page 5.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

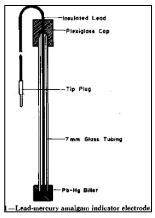
4. Amalgam electrode:

- It is similar to metal- metal ion electrode in which metal amalgam is in contact with a solution containing its own ions
  e.g., Lead amalgam electrode Pb-Hg/Pb^{2+}
- For lead amalgam electrode

Pb^{2+} + 2e⁻ ⇌ Pb-Hg

- Nernst equation:

$$E_{Pb^{2+}/Pb-Hg} = E^{0}_{Pb^{2+}/Pb-Hg} - \frac{0.059}{1\ 2}\log\left(\frac{[Pb-Hg]}{[Pb^{2+}]}\right)$$



Source:https://www.semanticscholar.org/paper/Potentiometric-Titration-of-Sulfate-in-Water-and-a-Robbins-Carter/c823ab0578481e876975ee707a5f8adca14c512f

Figure G000023. Types of electrodes · source slide page 6.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

5. Oxidation - reduction electrode :

- It consists of an inert metal such as platinum immersed in a solution containing an appropriate oxidized and reduced form of redox system.
- The metal merely acts as electrical contact.
- The potential arises due to the tendency of one form to change in to other form.
  e.g.,Pt/Fe^{2+},Fe^{3+}, Pt/Ce^{3+},Ce^{4+}, Pt/Sn^{2+},Sn^{4+}
- For stannous stannic electrode

Sn^{4+} + 2e⁻ ⇌ Sn^{2+}

- Nernst equation:

$$E_{Pt/Sn^{4+}/Sn^{2+}} = E^{0}_{Pt/Sn^{4+}/Sn^{2+}} - \frac{0.0591}{2}\log\left(\frac{[Sn^{2+}]}{[Sn^{4+}]}\right)$$

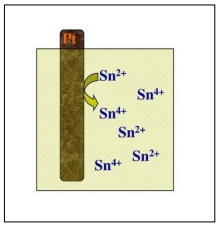

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Figure G000024. Types of electrodes · source slide page 7.

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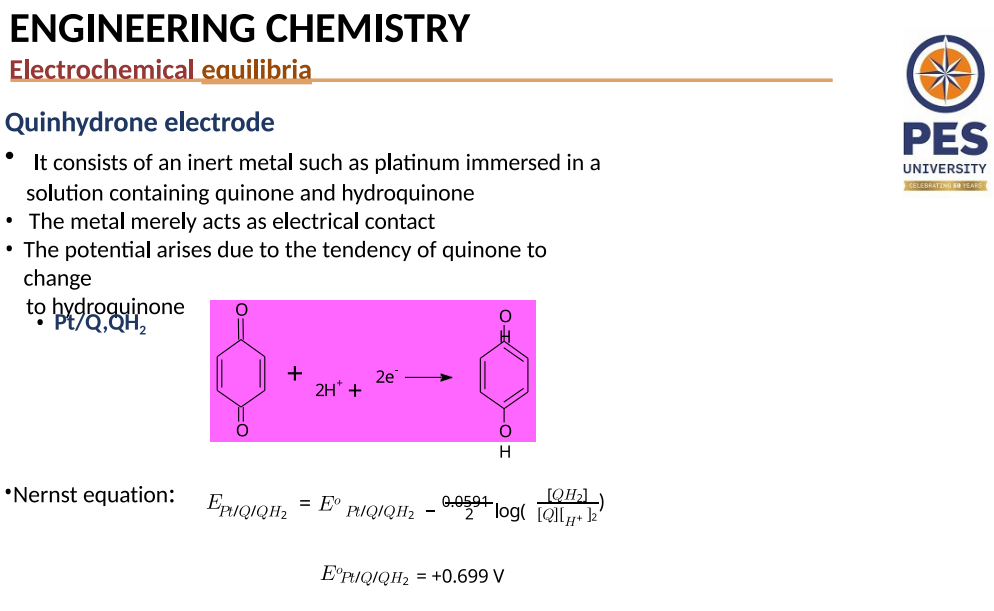

Figure G000025. Types of electrodes · source slide page 8.

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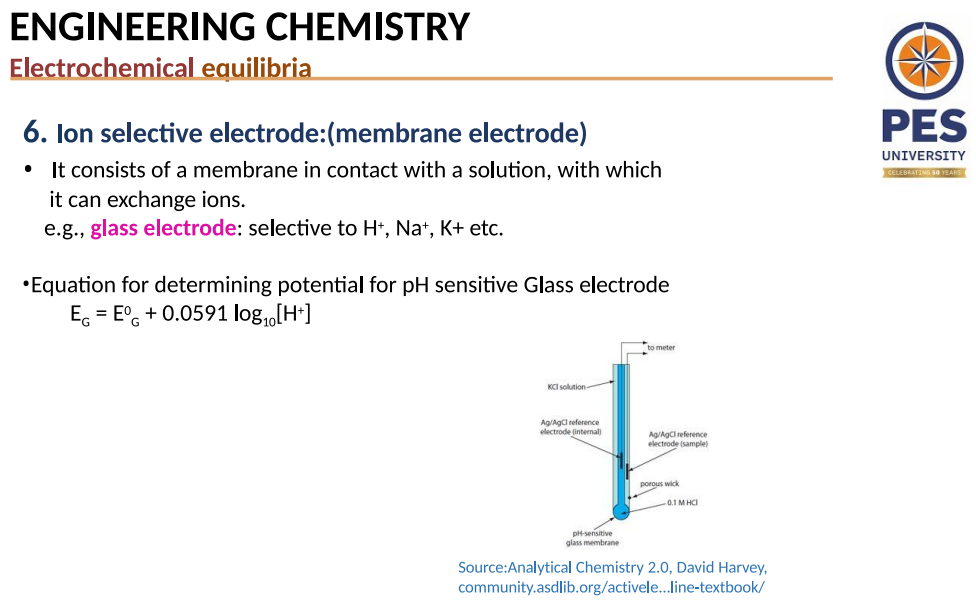

Figure G000026. Types of electrodes · source slide page 9.

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Figure G000027. Types of electrodes · source slide page 10.

EV0000345

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000317 · EV0000318 · EV0000319
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Class 16 content: Types of electrodes EV0000320 · EV0000321 · EV0000322
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Types of electrodes In order to form a cell, 2 half cells or 2 electrodes are required Various types of electrodes are available which are constructed based on the application Metal-metal ion electrode: Metal in contact with a solution of its own ions e.g., Zn/Zn2+, Cu/Cu2+, Ag/Ag+ Mn+ + ne- ⇌ M Nernst equation Source:http://www.valgetal.com/physics/ Batteries/batteries.htm EV0000323 · EV0000324 · EV0000325
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 2. Metal-Metal insoluble salt- ion electrode: These electrodes consist of a metal in contact with a sparingly soluble salt of the same metal dipped in a solution of soluble salt of the same anion e.g.,Calomel electrode Hg/Hg2Cl2/KCl, Ag/AgCl(s)/HCl For silver –silver chloride electrode AgCl + e- ⇌ Ag + Cl- Nernst equation: $EAg/AgCl/Cl- = EoAg/AgCl/Cl - 0.0591 \log[Cl-]$ 1 Source:https://www.corrosion-doctors.org/Corrosion- Thermodynamics/Reference-Half-Cells-Silver.htm EV0000326 · EV0000327 · EV0000328

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 3. Gas electrode: It consists of gas bubbling about an inert metal foil, immersed in solution containing ions to which the gas is reversible. The metal provides electrical contact and facilitates the establishment of equilibrium between the gas and its ions e.g.,Hydrogen electrode Pt/H2/H+,Chlorine electrode Pt/Cl2/Cl- For a hydrogen electrode 2H+ + 2e- ⇌ H2 Nernst equation: [Image text, OCR] The Standard Hydrogen Electrode Cormeetion Hydrogen gas (H,(g]) at1atm Platinum wire Mercury Hydrogen Bubbles Piatinisedplatinum electrode covered in platinum black IM Hydrochloric acid solution 0.0591 pH 0 − log(2) EPtTH TH+ = E PtTH TH+ H+ 2 2 2 2 Source:https://thefactfactor.com/facts/pure_science/chemistry/physical-chemistry/reference-electrodes/5844/ EV0000329 · EV0000330 · EV0000331
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 4. Amalgam electrode: It is similar to metal- metal ion electrode in which metal amalgam is in contact with a solution containing its own ions e.g., Lead amalgam electrode Pb-Hg/Pb2+ For lead amalgam electrode Pb2+ + 2e- ⇌ Pb-Hg Nernst equation: 0.0591 $Pb - Hg$ 0 E log( ) Pb2+/Pb−Hg = E Pb2+/Pb−Hg − 2 [Pb2+] Source:https://www.semanticscholar.org/paper/Potentiometric-Titration-of- Sulfate-in-Water-and-a-Robbins- Carter/c823ab0578481e876975ee707a5f8adca14c512f EV0000332 · EV0000333 · EV0000334
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 5. Oxidation - reduction electrode : It consists of an inert metal such as platinum immersed in a solution containing an appropriate oxidized and reduced form of redox system. The metal merely acts as electrical contact. The potential arises due to the tendency of one form to change in to other form. e.g.,Pt/Fe2+,Fe3+, Pt/Ce3+,Ce4+ , Pt/Sn2+,Sn4+ For stannous stannic electrode Sn4+ + 2e- ⇌ Sn2+ Nernst equation: 0.0591 [Sn2+] 0 log( ) EPt/Sn4+/Sn2+ = E Pt/Sn4+/Sn2+ − 4+ 2 [Sn] Source:https://slideplayer.com/slide /13860805/ EV0000335 · EV0000336 · EV0000337
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Quinhydrone electrode It consists of an inert metal such as platinum immersed in a solution containing quinone and hydroquinone The metal merely acts as electrical contact The potential arises due to the tendency of quinone to change to hydroquinone O OH Pt/Q,QH2 2H+ + + 2e- O OH Nernst equation: − 0.0591 log([QH2] E = Eo) H+ 2 $Pt/Q/QH$2 $Pt/Q/QH$2 2 Q Eo $Pt/Q/QH$2 = +0.699 V EV0000338 · EV0000339 · EV0000340
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 6. Ion selective electrode:(membrane electrode) It consists of a membrane in contact with a solution, with which it can exchange ions. e.g., glass electrode: selective to H+, Na+, K+ etc. Equation for determining potential for pH sensitive Glass electrode EG = E0G + 0.0591 log10[H+] Source:Analytical Chemistry 2.0, David Harvey, community.asdlib.org/activele...line-textbook/ EV0000341 · EV0000342 · EV0000343
10. Source slide page 10 (extracted text): THANK YOU Department of Science and Humanities EV0000344 · EV0000345 · EV0000346

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.156] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module four on electrochemical equilibria and in today's class we are going to discuss about the different types of electrodes.

[00:00:16.932] The metal metal ion electrode, the metal insoluble salt ion electrode, the gas electrode, amalgam electrode, redox electrode and iron selective electrode. [00:00:31.908] In order to form a cell, we need two half cells, and therefore two electrodes are required. [00:00:41.188] Now there are various types of electrodes which are available to us, and those are chosen based on the application where we are going to use the particular cell. So let us look at the different types of electrodes which are available to us. [00:00:59.972] The first one is a metal metal ion electrode. So what is this electrode? We have a metal rod which is in contact with a solution of its own ions. [00:01:13.252] So we could have zinc in contact with Zn2 plus, or we could have copper in contact with C U2 plus, and we have silver in contact with silver plus iron. [00:01:26.436] So whenever we have a metal rod dipped in a solution of its own ions, we know that an electrode potential is formed at the surface. [00:01:38.372] So what would be the reaction? Remember that we always write the reduction reaction because we all talk about reduction potentials. So we can write the equation Mn plus plus N electrons is giving you n. Since a potential is formed at the [00:01:58.532] Electrode and the solution interface. Let us see what would be the expression for the electrode potential, and that is given by the Nant equation. This is a familiar Nant equation which we have already learnt. So E is equal to E [00:02:15.808] M of MN plus is equal to E naught that is the standard electrode potential of M Mn plus plus two point three zero three R T over NF log of Mn plus. So that is the first type of electrode. [00:02:31.556] Let us come to the second type where we have metal, metal insoluble salt, iron electrode. How would we describe this electrode? As the name suggests, we have a metal [00:02:45.572] which is in contact with a sparingly soluble salt of the metal and that is again dipped in a solution of soluble salt of the same anion. [00:03:00.004] The most common example for this would be the calomel electrode. Let us write the electrode representation and see whether it satisfies the description. Now for a calomyl electrode we have [00:03:13.904] Mercury, which is a metal, and that is in contact with a sparingly soluble salt of mercury, that is [00:03:21.968] A G2 Cl2 and now this is dipped in a solution of KCl. So why did we choose Kl here? Because it should be the solution of a soluble salt of the same anion. Now here we have [00:03:41.348] silver silver chloride electrode also, which is the same type of electrode. [00:03:47.232] And here what do we have in silver silver chloride electrode? We have the silver metal which is in contact with silver chloride and that is in contact with HCl. So we have silver in contact with the sparingly soluble sort of silver, and then that is in contact with HCl, which is [00:04:04.976] solution of the sort. Now here if we want to write the equation. The equation is AG Cl plus electron is giving you AG plus Cl minus. What we have to [00:04:20.484] be careful about is that this is not A G plus and even if we write A G plus the equation would not be balanced. So it has to be only A G. So A G plus C M plus electron is giving you A G [00:04:34.064] Plus Cl minus and we can see this electrode over here in the diagram we have a silver wire which is coated with silver chloride that is the sparingly soluble salt of silver and that is dipped in a solution either of KCl or SCl. [00:04:50.640] It should contain it should be a solution containing the anion and the anion of the salt is Cl minus. [00:04:57.504] So uh this equation stands true for this type of electrode and the earnest equation is E of E aglcl minus is equal to E naught again the standard electrode potential of silver silver chloride C L minus minus.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.800] Zero point zero five nine one by one log of Cl minus. So the moment we say zero point zero five time one we know that we are talking of room temperature. [00:05:23.908] Let us come to the third type of electrode that is

the gas electrode. All of you are familiar with the hydrogen electrode. So hydrogen electrode is a good example for the gas electrode. [00:05:37.284] So it consists of a gas bubbling around an inert metal foil, usually platinum, that is immersed in a solution containing ions to which the gas is reversible. [00:05:49.616] Now, let us look at a picture here. We know that in a hydrogen electrode you have this kind of a bell-shaped jar which is being which is used here, and it has an inlet where we are going to let in hydrogen. So this hydrogen gas is now being bubbled around an inner [00:06:07.376] Metal. It is usually platinum or it could be a platinum wire or a platinum foil. So and then this is now this whole assembly is dipped into a solution of H plus ions because those are the ions to which the hydrogen gas is reversible. [00:06:25.284] Now what is the purpose of this metal contact here? The the platinum wire which we are using, what is the purpose of that? It provides the electrical contact and facilitates the establishment of equilibrium between the gas and its ions. [00:06:45.380] So, example would be hydrogen electrode, as I mentioned just now. We have platinum in contact with H2. So H2 gas is being bubbled around the platinum wire, and this is dipped in a solution of H plus ions. We also have another example of chlorine electrode. So what would be the difference here? We have to [00:07:03.152] Use chlorine gas which is going to be bubbled around the platinum wire and now this has to be dipped in a solution of Cl minus ions. [00:07:12.964] So for a hydrogen electrode, we will write the equation again reduction reaction. So 2H plus plus 2 electron is giving you H2. Now if we uh remember this sentence over here that the [00:07:28.528] Electrical contact provid facilitates the establishment of equilibrium. [00:07:34.948] between the gas and its ion. So that's exactly what it is doing here. So which equilibrium are we talking about? We are talking about this equilibrium. So your platinum foil or platinum wire is helping us to establish this equilibrium. [00:07:50.308] Now what would be the corresponding Nernst equation? Now in the Nernst equation we have E is equal to E naught minus 0.0591 by 2 because here we are using two electrons. So log of it's always products by the reactants. That's why we have to make sure we write the [00:08:07.456] uh reduction reaction only. So products by reactions here we have what is the product here? Hydrogen. But hydrogen is a gas, so we have to use the partial pressure of hydrogen here and [00:08:20.708] What is in the reactant side? We have H plus ions. So H plus raised to the power of 2. So we have to be careful when we write the Nernst equation for a hydrogen electrode. When we write the log term, it is partial pressure of hydrogen divided by H plus the concentration. [00:08:38.480] Raise to the power of 2. Now if it was a standard hydrogen electrode, obviously we use hydrogen at a partial pressure of 1 atmosphere and the hydrogen ion concentration is 1 molar. So this log term goes to 0, and therefore we get E is equal to E naught. [00:08:59.140] The fourth of electrode is an amalgam electrode. Now as the name suggested suggests, amalgam means any alloy of Merkle. So this electrode is exactly the same as the [00:09:14.272] Uh metal metaline electrode the only difference is that the metal is now taken in the form of an amalgam. So it is alloy with mercury and we are using that metal amalgam instead of the pure metal. [00:09:28.868] So here we have the metal amalgam and now when it just like we have in a metal metal and electrode, the metal amalgam is dipped in a solution containing [00:09:38.852] its own ions, the metal ions. So therefore we have say if we have a lead amalgam, then we use lead in the form of an amalgam and it has to be dipped in a solution of its own ions. So PB2 plus what would be the reaction taking place here? It is just like the [00:09:57.104] Uh metal metal ion electrode where we have metal ion taking up electron and becoming the metal. But here the only difference is we are taking the metal in the form of an amalgam. Now the question

arises why should we use the metal in the form of an amalgam? Now we know that sodium, for example

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.736] Is very active. Now, if we are going to dip sodium in a solution of its own ions that is sodium ions, we know that there's going to be a vigorous reaction and we will not be able to measure the potential. But [00:10:28.256] If we amalgamate the sodium, then we would have sufficiently reduced the activity of sodium. [00:10:36.580] And then we can make this kind of electrode. So we have sodium amalgam dipped in Na plus ion. So not only that we can actually control the activity of this electrode because we can decide the percentage of mercury which we are going to add into the metal to the [00:10:54.352] form the amalgam. So what would be the Nernst equation just like any other electrode E is equal to E naught minus 0.0591 by 2. 2 is the number of electrons log of [00:11:06.624] Let amalgam. Now if you remember when we talked about metal, metal and electrode, this here it was pure metal and hence it will it was taken as one. Whenever we have pure metal, the activity is taken as one. But here since we have amalgamated it. [00:11:24.400] Definitely the activity will not be one. In fact it will depend on the percentage of lead which is there in this um in this uh electron. [00:11:36.388] So we have to consider the activity of lead here. So in this case we are writing it as concentration. So concentration of lead amalgam divided by lead 2 plus. So that is coming from this equation over here. [00:11:52.900] Then we have redox electrodes, or they are also known as the oxidation reduction electrode. Now here we have an inert metal such as platinum, which is immersed in a solution containing [00:12:08.544] the appropriate oxidized and reduced form of the redox system. So you can see here what do we mean by that? We have say tin [00:12:19.104] Which is in the form of two species Sn2 plus and the Sn four plus. So Sn four plus is the oxidized and S N two plus is the reduced form. When we have both these species present in the solution and we place a platinum wire inside, then we know [00:12:36.816] There is a natural tendency of one of the oxidation states to get converted into other, and in that process electrons are getting exchanged. So if we can pick up those electrons, we will have a potential which is getting. [00:12:52.752] uh formed and that elect that potential is being picked up by these uh by the inert electrode. So the metal here merely is acting as an electrical contact. [00:13:05.840] The potential arises due to the tendency of one form to change into the other. So a good example of that would be your iron, which exists as Fe two plus and Fe three plus. And if we have both these species present in the same solution and we place a platinum [00:13:23.472] And wire there, we will get what we call as a redox electrode. Other examples would be platinum C3 plus C4 plus. [00:13:31.920] or platinum Sn two plus S N four plus as you can see in the picture over here. So what would be the corresponding reaction? It would be Sn four plus taking up electrons and forming S N two plus the Nernst equation will be based on this equation now. So it will be [00:13:49.632] ELE is equal to E naught minus 0.0591 by 2 log of S N2 plus that is the products divided by the reactants. [00:14:02.052] A special case of redox electrode is the quin hydron electrode. So what do we have here? It consists of an inert metal such as [00:14:12.432] platinum which is immersed in a solution containing quinone and hydroquinone. So what is the structure of quinone and hydroquinone? We can see here this is the reaction which is happening. So we have quinone here and this is hydroquinole [00:14:31.652] [ASR low confidence; check audio] Que no. [00:14:35.844] Now, what would be the reaction for conversion of quinone into hydroquinone? It involves an exchange of electrons in the presence of H plus ion. So this is the reaction where we have quinone changing into hydroquinone. So if we have

[00:14:53.632] Quinone and hydroquinone present together in the solution. And if we place a platinum electrode there, we will be able to get

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.312] a potential and this will form a redox reaction redox electrode and what would be the electrode platinum in contact how would you write the representation platinum in contact with q that is quinone and q h2 that is the hydroquinone let us write [00:15:20.944] the Nernst equation for this E is equal to E naught minus 0.0591 by number of electrons exchanged is 2 over here and this would be log of here [00:15:39.712] the concentration of hydroquinone. How do we represent Q H two and divided by that of quinone that is [00:15:50.148] Q concentration and we also have H plus. So H plus raised to the power of 2. So this will be the Nernst equation for a quin hydrone electrode. [00:16:05.700] Now we come to the last type of electrode that is the ion selective electrode. These are also referred to as membrane electrodes. An ion selective electrode, as the name suggests, it is going to select a particular ion in the presence of other ions. [00:16:25.860] and it is going to tell us the concentration of that particular ion or in other term in other words it is going to develop a potential based on the concentration of that one particular ion [00:16:43.008] So how is it being able to do it? It is because it has a membrane which is in contact with the solution, and this membrane is the most important constituent, and this membrane is being able to exchange the [00:17:00.032] ions and since it is being able to exchange the ions it can it develops a potential uh and that [00:17:07.684] That is why it can detect the concentration of one ion in the presence of other ions. So, an example for this would be glass electrode, and we know that glass electrode is used in pH meters. And what are pH meters which tell us the concentration of electricity. [00:17:25.456] H plus so the glass membrane is selective to H plus ion. Because it is selective to H plus ion, it can give us the [00:17:35.136] concentration of H plus N, or in other words, it can help us determine the pH of a solution. And it is also sensitive to not only H plus, sometimes to NE plus and K plus. [00:17:47.200] Now the equation for determining the potential for a pH sensitive glass electrode uh would be E is equal to E naught. [00:17:56.292] Pla E no E G would be equal to E naught G plus 0.0591 log of H plus Ion. So this is how based on the concentration of H plus a potential is being is uh getting developed and we say that these electrodes are ion selective electrodes [00:18:14.064] Extremely sensitive and they are also known as membrane electrodes because membrane is the most important constituent of a glass electrode. So these are the different types of electrodes which we came across. [00:18:34.660] So these are the different types of electrodes available to us. And in the next class we are going to talk about reference electrodes. Thank you.

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Full transcript: 2747 normalized words, 78 timestamped segments, 1126.336 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:14:31.652

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4 Reference electrodes, calomel electrode

This lesson is indexed from the source video titled Reference electrodes, calomel electrode. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json.

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Graphs and diagrams

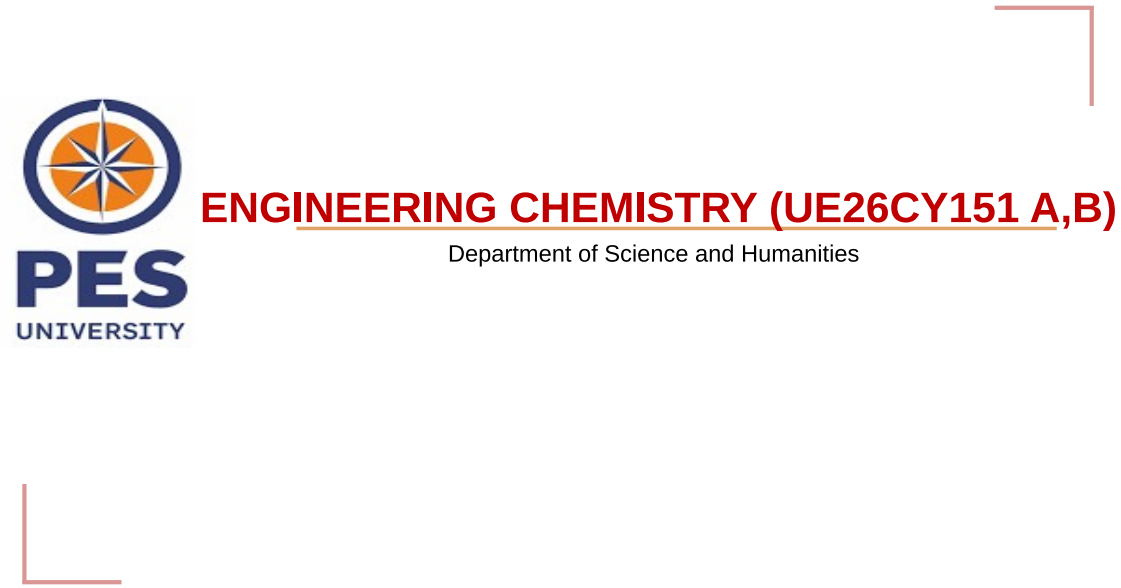

Figure G000028. Reference electrodes calomel electrode · source slide page 1.

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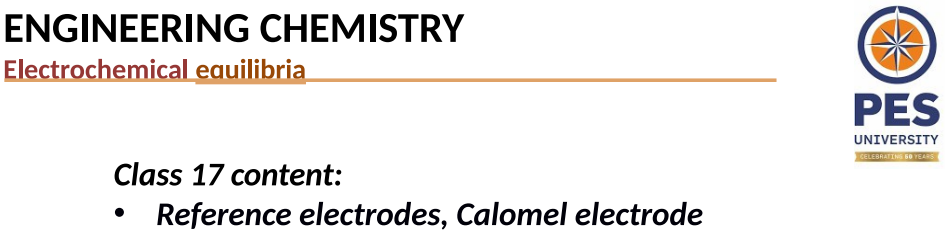

Figure G000029. Reference electrodes calomel electrode · source slide page 2.

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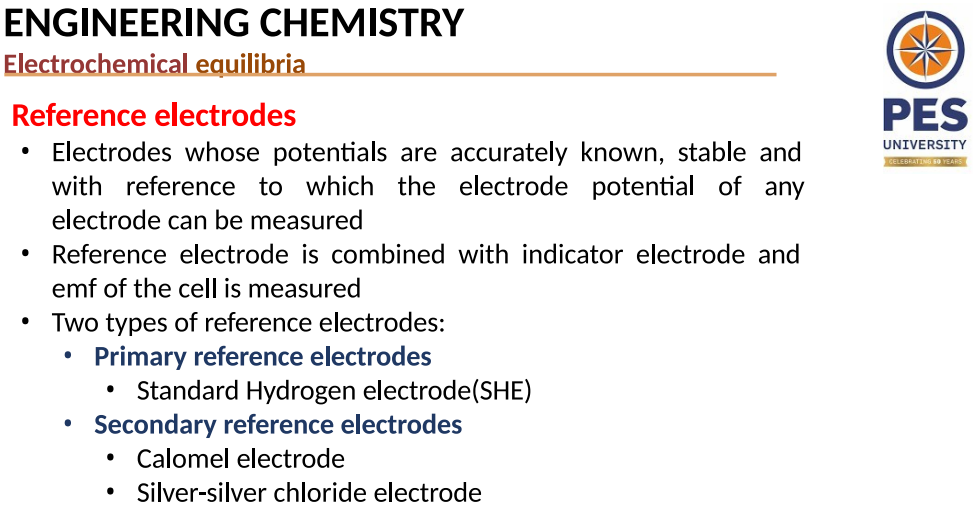

Figure G000030. Reference electrodes calomel electrode · source slide page 3.

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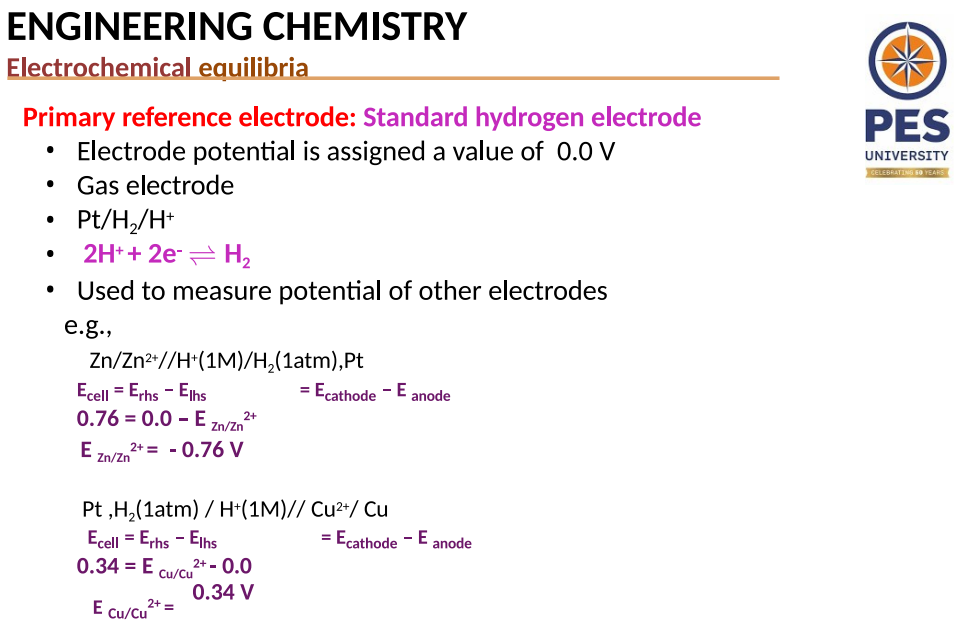

Figure G000031. Reference electrodes calomel electrode · source slide page 4.

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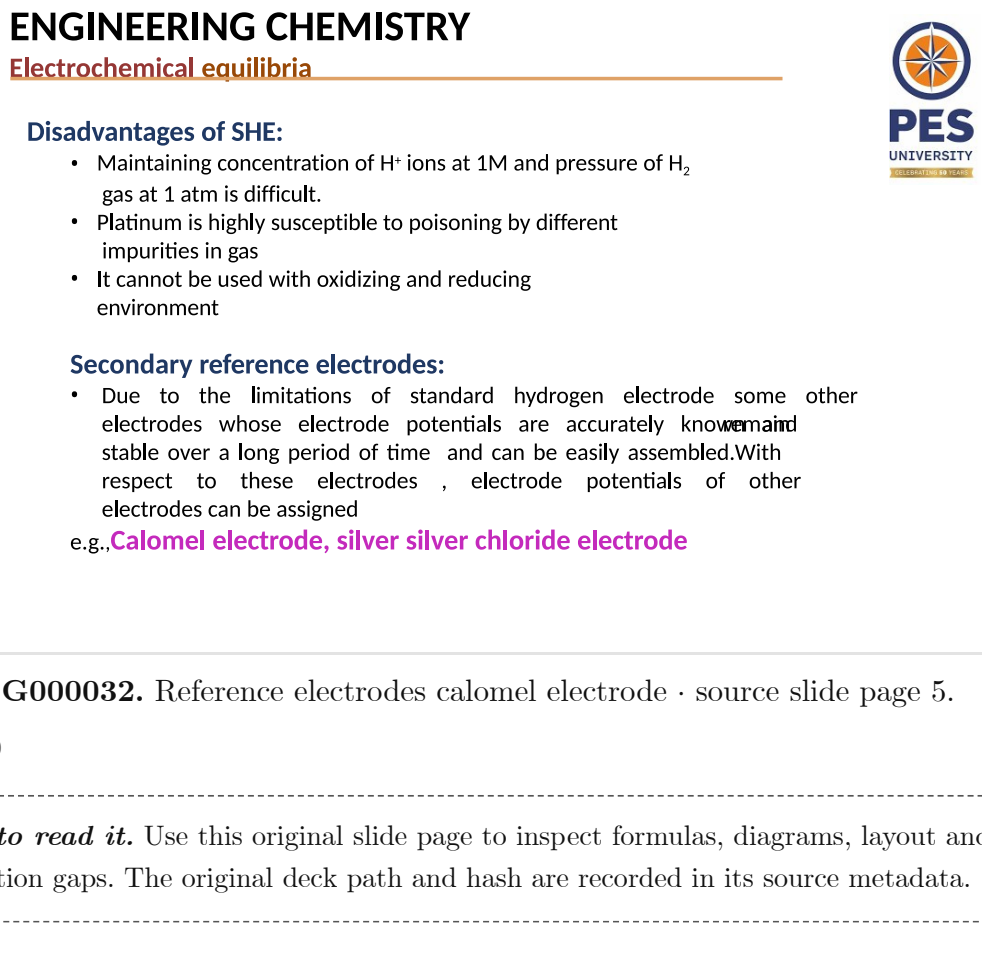

Figure G000032. Reference electrodes calomel electrode · source slide page 5.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

Secondary Reference electrodes: Calomel electrode

- Most widely used reference electrode
- Metal-insoluble salt –ion electrode

Construction:

- A glass tube containing a layer of mercury over which a paste of insoluble salt Hg_2Cl_2 (calomel) + Hg and the next layer is a solution of KCl
- A Pt wire dipped in Hg provides electrical contact
- Tube is fitted with a side tube to fill KCl solution of known concentration and another side tube which connects to the salt bridge

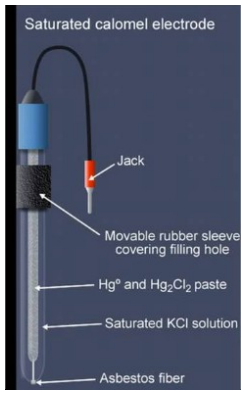

image of calomel electrode - Google Search

Figure G000033. Reference electrodes calomel electrode · source slide page 6.

EV0000363

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ENGINEERING CHEMISTRY

Electrochemical equilibriac

- $Hg/Hg_2Cl_2(s)/Cl^-$

Working :

- Can act as anode or cathode depending on the nature of the electrode with which it is coupled

As anode:

$$2Hg \rightleftharpoons Hg_2^{2+} + 2e^-$$
$$Hg_2^{2+} + 2Cl^- \rightleftharpoons Hg_2Cl_2$$
$$2Hg + 2Cl^- \rightleftharpoons Hg_2Cl_2 + 2e^-$$

As cathode:

$$Hg_2^{2+} + 2e^- \rightleftharpoons 2Hg$$
$$Hg_2Cl_2 \rightleftharpoons Hg_2^{2+} + 2Cl^-$$
$$Hg_2Cl_2 + 2e^- \rightleftharpoons 2Hg + 2Cl^-$$

Figure G000034. Reference electrodes calomel electrode · source slide page 7.

EV0000366

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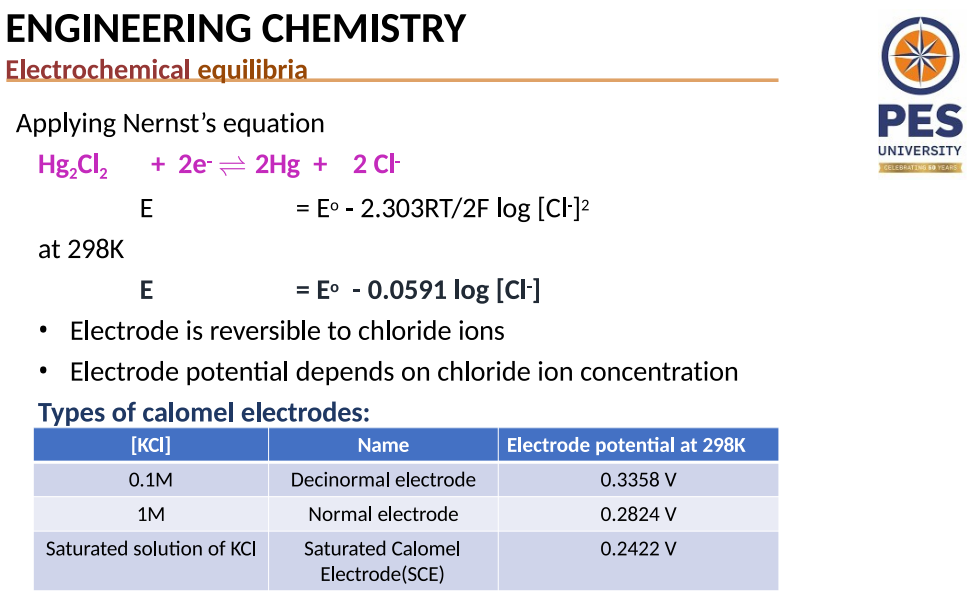

Figure G000035. Reference electrodes calomel electrode · source slide page 8.

EV0000369

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Figure G000036. Reference electrodes calomel electrode · source slide page 9.

EV0000372

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000347 · EV0000348 · EV0000349
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Class 17 content: Reference electrodes, Calomel electrode EV0000350 · EV0000351 · EV0000352
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Reference electrodes Electrodes whose potentials are accurately known, stable and with reference to which the electrode potential of any electrode can be measured Reference electrode is combined with indicator electrode and emf of the cell is measured Two types of reference electrodes: Primary reference electrodes Standard Hydrogen electrode(SHE) Secondary reference electrodes Calomel electrode Silver-silver chloride electrode EV0000353 · EV0000354 · EV0000355
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Primary reference electrode: Standard hydrogen electrode Electrode potential is assigned a value of 0.0 V Gas electrode Pt/H2/H+ 2H+ + 2e- ⇌ H2 Used to measure potential of other electrodes e.g., Zn/Zn2+//H+(1M)/H2(1atm),Pt Ecell = Erhs – Elhs = Ecathode – E anode 0.76 = 0.0 – E Zn/Zn2+ E Zn/Zn2+ = - 0.76 V Pt ,H2(1atm) / H+(1M)// Cu2+/ Cu Ecell = Erhs – Elhs = Ecathode – E anode 0.34 = E Cu/Cu2+ - 0.0 0.34 V E Cu/Cu2+ = EV0000356 · EV0000357 · EV0000358
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Disadvantages of SHE: Maintaining concentration of H+ ions at 1M and pressure of H2 gas at 1 atm is difficult. Platinum is highly susceptible to poisoning by different impurities in gas It cannot be used with oxidizing and reducing environment Secondary reference electrodes: Due to the limitations of standard hydrogen electrode some other electrodes whose electrode potentials are accurately known and remain stable over a long period of time and can be easily assembled. With respect to these electrodes , electrode potentials of other electrodes can be assigned e.g.,Calomel electrode, silver silver chloride electrode EV0000359 · EV0000360 · EV0000361
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Secondary Reference electrodes: Calomel electrode Most widely used reference electrode Metal-insoluble salt –ion electrode Construction: [Image text, OCR] Saturatedcalomelelectrode Jack Movablerubbersleeve coveringfillinghole Hg°andHg2Cl2paste Saturated KClsolution Asbestos fiber A glass tube containing a layer of mercury over which a paste of insoluble salt Hg2Cl2 (calomel) + Hg and the next layer is a solution of KCl A Pt wire dipped in Hg provides electrical contact Tube is fitted with a side tube to fill KCl solution of known concentration and another side tube which connects to the salt bridge image of calomel electrode - Google Search EV0000362 · EV0000363 · EV0000364
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibriac Hg/Hg2Cl2(s)/Cl- Working : Can act as anode or cathode depending on the nature of the electrode with which it is coupled As anode: 2Hg ⇌ Hg 2+ + 2e- 2 Hg 2+ + 2 Cl- ⇌ Hg Cl 2 2 2 2Hg + 2 Cl- ⇌ Hg2Cl2 + 2e- As cathode: Hg 2+ + 2e- ⇌ 2Hg 2 2+ - Hg2Cl2 ⇌ Hg2 + 2Cl Hg2Cl2 + 2e- ⇌ 2Hg + 2 Cl- EV0000365 · EV0000366 · EV0000367

8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Applying Nernst's equation Hg2Cl2 + 2e- ⇌ 2Hg + 2 Cl- E = Eo - 2.303RT/2F log [Cl-]2 at 298K E = Eo - 0.0591 log [Cl-] Electrode is reversible to chloride ions Electrode potential depends on chloride ion concentration Types of calomel electrodes: [KCl] | Name | Electrode potential at 298K 0.1M | Decinormal electrode | 0.3358 V 1M | Normal electrode | 0.2824 V Saturated solution of KCl | Saturated Calomel Electrode(SCE) | 0.2422 V EV0000368 · EV0000369 · EV0000370
9. Source slide page 9 (extracted text): THANK YOU Department of Science and Humanities EV0000371 · EV0000372 · EV0000373

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Hello everyone, welcome back to this course on engineering chemistry. We'll continue with module 4 on electrochemical equilibrium. [00:00:10.532] And in today's class we are going to look at reference electrodes. That would be the primary reference electrode which is your standard hydrogen electrode and then we will study about the secondary reference electrodes which are galomyl electrode and silver silver chloride electrode. [00:00:29.956] So what are reference electrodes? Let us first define reference electrodes. We know from the previous classes that whenever we form a cell, it will be two electrodes which are coupled together. What we can measure is just the potential difference. [00:00:50.084] Now if we want to assign an electrode potential value to any one of these electrodes, that is not possible unless we know the electrode potential of one of the electrodes. So that is where reference electrodes are used. [00:01:06.084] So we understand that reference electrode potential will be known to us. [00:01:10.576] and measuring the electrode cell potential we will be able to determine the potential of the other electrode. This other electrode is known as the indicator electrode. So whenever we do any electro potential electrode potential cell potential measurements [00:01:28.240] Then we combine the reference electrode with the indicator electrode. So what is the criteria for an electrode to be considered for a reference electrode? [00:01:43.940] So electrodes whose potentials are accurately known to us and they are stable and with reference to which the electrode potential of any electrode can be measured, they are called reference electrode. [00:02:00.772] So we have two types of reference electrodes. You're familiar with this, they are called [00:02:05.440] the primary reference electrode and the secondary reference electrode. An example for primary reference electrode in fact this is the only electrode which is used as a primary reference electrode is the standard hydrogen electrode. And for secondary reference electrodes we have [00:02:22.608] Calamile electrodes and the elect and the silver silver chloride electrode. [00:02:29.604] So let us look at first what is the primary reference electrode. As I mentioned just now, the standard hydrogen electrode, which is also referred to as the SHE, is used as a primary reference electrode. Why? Because [00:02:48.336] It is assigned a value of 0.0 volts. It is by convention and internationally we accept that electrode potential for a standard hydrogen electrode is assigned a value. [00:03:04.260] of 0.0 volts. Now we understand that standard hydrogen electrode is a gas electrode. We studied in the previous class it's a glass gas electrode and how do we represent this standard hydrogen electrode? We have [00:03:20.352] a gas being bubbled around an inert electrode which is dipped in a solution of ions to which the gas is reversible. [00:03:28.324] as we studied yesterday. So here the only difference is now hydrogen is at one atmosphere pressure and H plus concentration is one atmosphere one molar. [00:03:41.508] So this would be then become this would then become a standard hydrogen electrode. So the reaction taking place here

would be again we always consider reduction reaction. So 2H plus plus 2 electron is giving you hydrogen, and here we have to make sure that the pressure is all [00:03:59.280] Always one atmosphere and the H plus concentration is maintained at one molar concentration. Now using this, we are going to measure the potential of other electrodes. So how do we do that? As mentioned earlier, we have to form a cell. So [00:04:18.448] Suppose we want to determine for zinc electrode. So zinc electrode would be what? Zinc rod dipped in Z2 plus ions. Now this is coupled using a salt bridge with the standard hydrogen electrode. How do I know this is a standard hydrogen electrode? Because I've used one atmosphere pressure and one molar concentration of H plus ions. [00:04:38.352] So when this is coupled like this, you can measure the cell potential. So E cell can be measured for this particular system. And we know that E cell is always E right hand side minus the E left hand side, which is E cathode minus E anode. [00:04:56.048] Suppose when you do this arrangement, I get a value of 0.76

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.596] And this is equal to 0, as I said mentioned just now that standard hydrogen electrode potential is always given a value of 0 minus E of the zinc electrode. Now we measure this as 0.76. [00:05:18.704] Since it is a positive value, I know that the connections I have made here are right, meaning what that zinc electrode will act as the anode and the hydrogen electrode is going to act as a cathode. If there was any mistake in this, if I had put hydrogen electrode here and the [00:05:36.336] zinc electrode here then I would have measured E cell as a negative value but since I got a positive value I know that zinc will act as anode here and hydrogen electrode will act as the cathode. So now we have this expression 0.76 is equal to 0 minus E of [00:05:53.904] Zn Zn2 plus. So what would be the value of E Zn Zn2 plus is equal to minus 0.76. So let me tell you this is the standard reduction potential of a zinc electrode. So what does it tell me that the zinc ions used here. [00:06:11.492] Are also at one molar concentration. Nevertheless, from this arrangement, it can be any concentration of zinc, doesn't matter. I can get the value of the electrode potential if I can measure the cell potential. So that's how we assign the value to the electrode. Now let us take a [00:06:29.264] example where platinum uh hydrogen and h plus this electrode the hydrogen electrode is acting as the anode and in that case copper electrode is acting as the cathode again I do the same thing e cell is equal to e cathode minus e anode now I get a [00:06:46.832] value of 0.34 again this is a positive value indicating to me that the connections are right copper will act as the cathode and uh [00:06:56.868] standard hydrogen electrode will become the anode. So now we have copper electrode minus 0.0 that is for the anode electrode the electrode acting as the anode. So again [00:07:08.708] um sorry this should have been zinc uh this should have been copper cu2 plus is equal to 0.34 volts so the copper electrode now is is equal to so this is not right [00:07:27.972] so this is actually equal to e of obviously if I have used copper electrode it cannot be a zinc electrode it's just a typo sorry for that so e of copper cu2 plus is equal to 0.34 volts [00:07:47.620] Now, if it was so simple, like you just connect the standard hydrogen electrode with the [00:07:55.012] Any electrode you want to measure the potential, whatever cell potential you measure is the value which is to be assigned to that particular electrode. So that will become very simple. Why do we need any other type of reference electrode? The reason is that the standard hydrogen electrode has [00:08:12.784] certain disadvantages so what are the disadvantages let us have a look at it now maintaining H plus ions at one molar concentration and the pressure of hydrogen at one atmosphere is very difficult so [00:08:28.740] Continuously to maintain it at that particular value is very difficult. So that is the

first problem we face, and then we have platinum. [00:08:37.504] Which is highly susceptible to poisoning. We know that platinum gets poisoned easily by the different impurities which could be present in the gas, and the third point is that the high standard hydrogen electrode cannot be used wherever there is an oxidizing and reducing energy. [00:08:55.184] environment. Therefore, these are the problems associated or disadvantages or limitations of the standard hydrogen electrode. Hence we have to look for other type of electrodes which will act as [00:09:09.312] reference electrode. So there comes in the concept of secondary reference electrodes. Now what is it that we look for in the secondary reference electrode? [00:09:21.988] Now for a secondary reference electrode because we have these limitations with the standard hydrogen electrode, all these problems should be removed now. So we have to have an electrode which can be assembled easily, maintained easily. So we need such kind of electrodes. Apart from that. [00:09:39.760] Of course, this is very important, the electrode potential should be accurately known and it should remain stable. [00:09:47.904] for a long period of time. Over a long period of time we should have a stable potential so that when we measure the potential difference we know that this electrode has this potential and therefore the other part the difference of this

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.192] potential difference from the electrode potential of the reference electrode would give me the electrode potential of the [00:10:08.880] uh indicator electrode so one thing it should be accurately known second it should be stable over a long period of time and it can be easily assembled so these are the three things which we look for when we design or select a secondary reference electrode [00:10:26.576] And with this, of course, since it is a reference electrode, my definition gets complete when I say electrode potential of other electrodes can be assigned with the help of this electrode. So there are two electrodes which [00:10:39.664] Come under this category one is the caramel electrode, the other is the silver silver chloride electrode. So let us look at the secondary reference electrode that is the calomal electrode. Now the first thing if you are asked to [00:10:56.016] describe the Calomel electrode and explain the Calomar electrode first thing we need to write or we need to say is that it is used as a secondary reference electrode and it is a very widely used secondary reference electrode. [00:11:13.796] And then the next thing, what category does this electrode fall in? In the previous class I have taught you about types of electrodes. So the calomyl electrode comes into the second type of electrode, and what is that? Metal, insoluble salt. [00:11:29.600] ion electrode. So now I know how to construct a calomel electrode. Because in a calomyl electrode I know [00:11:39.360] I have to use a metal that is mercury is used, then I need to use H2Cl2, which is the insoluble salt or sparingly soluble salt of the metal, and then I need to use the KCl solution. So let us see the construction of this calomel electrode. So we take a glass tube. [00:11:58.916] And we place a layer of mercury in it. So you can let's go to this picture and we can see here there is this is your glass tube and we have a layer of mercury over here. Above this layer [00:12:12.912] We take a paste of mercury and HG2Cl2. Now H G2Cl2 HG2Cl2 is known as calomel. So that's why this electrode is known as a calomel electrode. So over this mercury we have a paste of H2 Cl2 plus A. [00:12:30.288] Hg and over that now we place KCl solution. [00:12:35.264] So you can see here everything now Hg is in contact with H2Cl2 which is in contact with KCl solution and inside this we dip a platinum wire. You can see a platinum wire is dipped here and it will go into the mercury. So what is the role of this platinum? [00:12:52.976] Platinum wire, it is providing the electrical contact. Mercury being a liquid, we don't have an electric contact electrical contact established

here. So platinum wire is used for that purpose, and you can see some more. [00:13:07.844] outlets or inlets over here. This inlet is for introducing KCl solution as in when we need to replace it and this is connecting to the salt bridge. This tube over here [00:13:24.612] uh outlet for the solvent this is connecting to the solder so this is the construction or this is what your calomel electrode looks like [00:13:36.772] Now the cell representation, sorry, the electrode representation, what would it be? Hg, H2, Cl2 solid that is and Cl minus. Okay. Now let us come to the working of this electrode. Now this electrode can act as anode as well as cathode. [00:13:56.736] Because that is true for any electrode, it always depends on the other electrode which is being combined. So one of them which has a greater oxidation tendency will be anode and the other electrode will be cathode. So depending on the other electrode which is connected to the [00:14:12.624] uh calomyl electrode we can have uh it acting as anode or it can act as cathode. Now let us look at the reactions taking place here. So we see that mercury [00:14:25.252] first forms a G22 giving out electron because it's an anode so electrons are being given out over here. Now this H22 plus the reaction doesn't stop here. It reacts with the chloride ions which we have provided there [00:14:39.760] in the form of KCl and this gives you HE2Cl2. So the overall reaction is going to look like this. What is it? We add this to and what we get 2 H G plus 2 Cl minus is equal 2 Cl minus is equal to H E two Cl2 plus 2 electrons. [00:14:59.684] This is the reaction when calomel acts as anode, and you can see electrons are being given out here. So definitely this is when it acts as anode. Now suppose it acts as cathode, exactly reverse of this is going to take place.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:17.456] is giving you 2 Hg and then H2Cl2 undergoes dissociation to compensate for the G2 plus which have been used up here. So H22 plus plus 2 electrons. So overall reaction is Eg2Cl2 plus 2 electron is giving you [00:15:33.264] 2 H G plus 2 Cl minus. So this is the reaction which is happening. [00:15:41.072] When the caromel electrode acts as cathode, again it is true we can understand because two electrons here, the electrons are being uh taken up, so it's a reduction reaction. [00:15:54.436] Now we will look at what is the potential of this electrode potential of this galomyl electrode. So we are remember again we always write the reduction reaction. So this would be the reaction NG2Cl2 plus 2 electron is giving you 2Hg plus 2Cl minus. So [00:16:14.244] Writing the Nernst equation. For the Nernst equation, what do we need? We need the Q value first. So what is Q here? We can see H G raised to the power of 2, Cl minus raised to the power of 2 divided by HE2 CL2. Now we know that H G as well as H2 Cl2. [00:16:32.016] Are pure form, so the value will be one for them. Therefore, the q value is just Cl minus squared. [00:16:40.912] Okay, so one now we write the Nest equation E is equal to E naught minus two point three zero three R T over two f two is because of these two electrons over here log of Cn minus square. Now if you want to remove the square we can cancel with this too and at two ninety eight [00:16:58.608] We get the expression that is E is equal to E naught minus 0.0591 log of Cl minus. [00:17:07.940] When we get this expression, what we say is the electrode is reversible to chloride ions. What does it mean? That the electrode potential [00:17:19.568] Is depending on the chloride ions, or in other words, if we know the electrode potential of this electrode, we will be able to determine the chloride ion concentration. So we see very easily from this uh [00:17:34.400] It's very obvious from this equation E is equal to E naught minus 0.059 1 log of Cl minus so electrode potential depends on the chloride ions. Now remember that we used KCl solution. [00:17:49.280] But we did not specify what is the concentration of KCl we are using. So since now we understand that the electrode potential is going to depend on the

chloride ion concentration, that means the KCl which is there in the electrode. [00:18:04.784] The potential is going to depend on the concentration of the KCL which we have used. So, based on that, we have different types of electrode. So [00:18:14.116] If we use 0.1 molar KCl or 0.1 normal, then it will be known as a decinormal electrode and the potential is 0.3358 volts. [00:18:24.832] And if we use a one molar concentration on one normal, then it is a called a normal electrode and it is 0.2824 volts, and whereas if it is [00:18:38.160] a saturated solution of KCl and then it is known as a saturated calom electrode and the electrode potential is 2244 volts. [00:18:50.864] So this is how we see that the electrode potential depends on the chloride ion concentration. [00:19:01.156] Now coming to the silver silver chloride electrode [00:19:06.212] Silver silver chloride electrode again is widely used as a reference electrode. So that's the first thing we need to note. So this is again a secondary reference electrode and what type of electrode it is again it is metal insoluble salt iron electrode. [00:19:24.100] Now the construction will be slightly different from that of the caloman electrode. Why? Because mercury being a liquid, we took a tube and we placed mercury [00:19:36.448] And then in that we place a platinum wire for electrical contact. But here we don't need to do that because silver is already in the solid form. So all we need to do is take a silver wire. Okay. So we are either going to take a silver wire. [00:19:55.972] Or we can have a silver clore coated platinum wire. Now on this silver wire, what do we need? It is a metal insoluble salt ion electrode. So the insoluble salt has to be coated over the over the silver wire. So we have

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:13.696] AGCl that is silver chloride which is electrolytically coated over the silver wire and now we need KCl solution. So we are going to dip this into a KCl solution. So the tube will contain KCl solution of or [00:20:33.024] H Cl is also allowed, we can use HCl also of known concentration. So we have AG in contact with AGCl which is in contact with Cl minus ions. So what would be the [00:20:46.032] cell electrode representation we have AG in contact with AGCl which is in contact with the Cl minus ion. So this is your electrode over here. So you can see the silver wire which is coated with silver chloride and that is dipped in the KCl solution. [00:21:04.868] So again when we look at working we know that it's exactly the same as the calomar electrode, that means it can act as anode as well as cathode, and if you see the reactions are also very [00:21:16.016] similar so we have silver which undergoes oxidation forming silver plus plus electron and this silver plus takes up chloride ions giving you agl so overall reaction would be ag plus cl minus is giving you a gcl plus electron and at the cat if it is acting [00:21:33.648] cathode then it is just the reverse what we saw. So Ag plus plus electron is giving you AG and AG Cl gives you Ag plus plus Cl minus and therefore overall reaction would be A G Cl plus electron is giving you AG plus Cl minus. [00:21:53.060] Now when we apply the Nernst equation then we see that AG plus plus electron is equal to AG plus Cl minus. If you remember when I taught second type of electrodes that is metal insoluble salt. [00:22:09.712] Ion electrode I gave this reaction over there and also mention it is not AG plus it is AG and it is balanced with the the negative charge is balanced in this way so now we know how we got this reaction so it is ag Cl plus electron is giving you [00:22:27.344] A G plus the Cl minus. So when we write the Nunst equation, now we are familiar with it, E is equal to E naught minus 2.303 R T over F log of Cl minus. So since it is one electron, this just [00:22:41.264] N is 1. So at 298 Kelvin the expression now becomes E is equal to E naught minus 0.0591 log of Cl minus and again we say that the electrode is reversible to chloride ions and the electrode potential now again we see very clearly the electrode. [00:22:58.896] potential is going to depend on the chloride ion concentration. So again

[00:23:05.408] Based on that we have three different types of silver silver chloride electrode because again why the chloride ion concentration is coming from KCl or if you have used HCl then HCl so based on the chloride ion concentration if it is 0.1 normal Kl [00:23:23.056] A Cl which you have used, then it is known as a decnormal electrode and the value is electrode potential at 298 Kelvin is 0.289 volts. If you use one normal, then it is known as a normal electrode and the electrode potential is 0.223 volts. [00:23:42.112] And if you use a saturated solution of Caseyl then it will it will be referred to as it will be called saturated silver silver chloride electrode. [00:23:54.404] And the potential is going to be 0.199 volts. So we see that the electrode potential of the [00:24:04.432] Silver silver chloride electrode also depends on the chloride ion concentration which we have used. So this is about the reference electrodes, galombol electrode and the silver silver chloride electrodes. So in the next class we are going to [00:24:21.360] take up concentration cells and also ion selected electrodes thank you

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Full transcript: 3601 normalized words, 101 timestamped segments, 1473.984 seconds duration. Low-confidence segments flagged: 0.

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5 Concentration cells, types of concentration cells

This lesson is indexed from the source video titled Concentration cells, types of concentration cells. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000062 · EV0000063

Graphs and diagrams

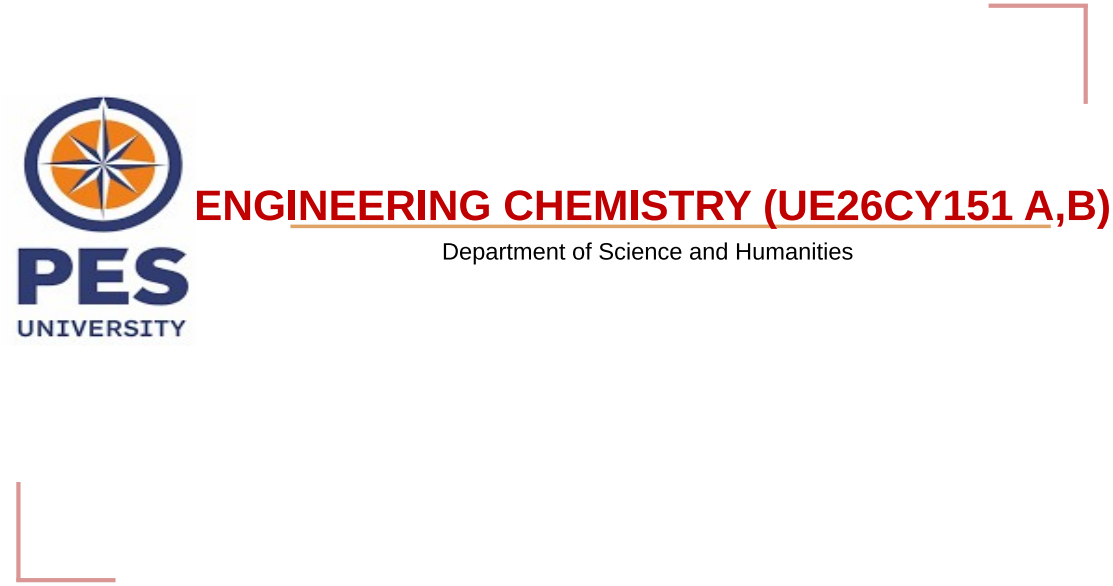

Figure G000037. Concentration cells types of concentration cells · source slide page 1.

EV0000375

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ENGINEERING CHEMISTRY

Electrochemical equilibria

Class content 18:

- ***Concentration cells and it's types***

Figure G000038. Concentration cells types of concentration cells · source slide page 2.

EV0000378

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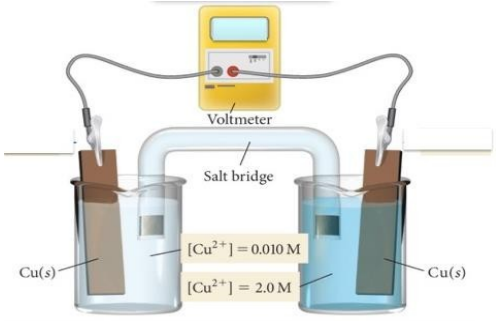
ENGINEERING CHEMISTRY

Electrochemical equilibria

Concentration cells:

- An electrochemical cell in which identical electrodes are in contact with a solution of identical species but of different activity/concentration of the electrode/electrolyte.



https://chemdemos.uoregon.edu/demos/Voltaic-Cell-CuCu-concentration-cell-Demonstration

Figure G000039. Concentration cells types of concentration cells · source slide page 3.

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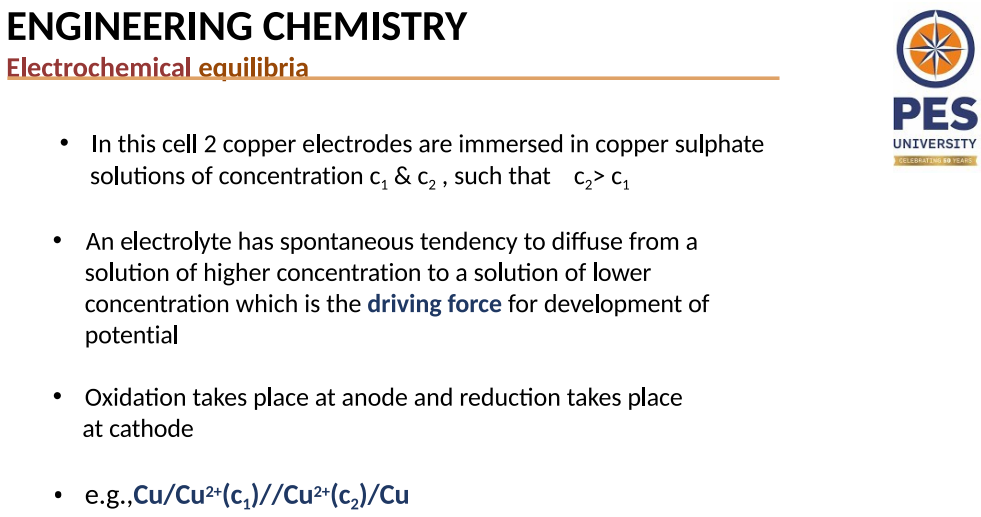

Figure G000040. Concentration cells types of concentration cells · source slide page 4.

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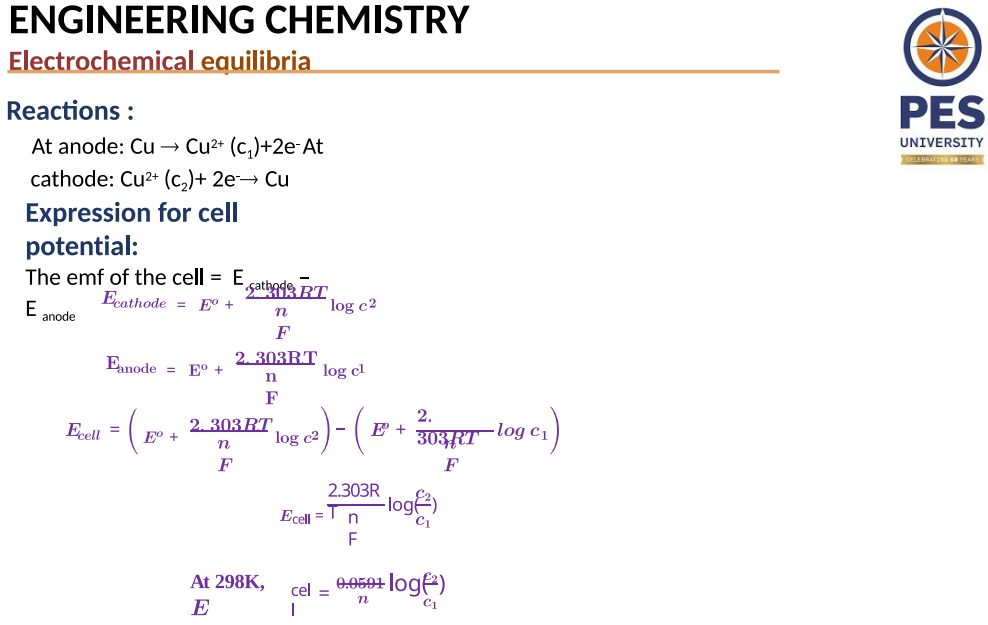

Figure G000041. Concentration cells types of concentration cells · source slide page 5.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

- The emf of the cell is positive only if $c_2 > c_1$

  i.e., conc of metal ion at cathode > conc. of metal ion at anode
- The emf of the cell depends upon the ratio c_2/c_1
- When $c_2 = c_1$, the emf of the cell becomes zero
- During working of the cell, concentration of ions increases at anode decreases at cathode
- When current is drawn from the cell c_1 increases and c_2 decreases
- The cell can operate only as long as the concentration terms are different

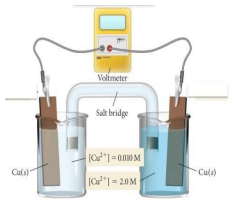

Figure G000042. Concentration cells types of concentration cells · source slide page 6.

EV0000390

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ENGINEERING CHEMISTRY

Electrochemical equilibria

Types of concentration cells:

- Electrolyte concentration cell
- Electrode concentration cell

Electrolyte concentration cell:

- Electrolyte concentration cell consists of two same electrodes that are dipped in the same electrolyte but with different concentrations of electrolytes
- $Cu/Cu^{2+}(c_1)//Cu^{2+}(c_2)/Cu$
- Cell potential is given by $E_{cell} = \frac{2.303RT}{nF}\log(\frac{c_2}{c_1})$

Electrode concentration cell

- Electrode concentration cell consists of two identical electrodes of different activity which are dipped in the same solution of electrolyte
- $Na\text{-}Hg(c_1)/Na^+/Na\text{-}Hg(c_2)$
- Cell potential is given by $E^{cell} = \frac{2.303RT}{nF}\log\frac{Na\text{-}Hg(c_1)}{Na\text{-}Hg(c_2)}$

Figure G000043. Concentration cells types of concentration cells · source slide page 7.

EV0000393

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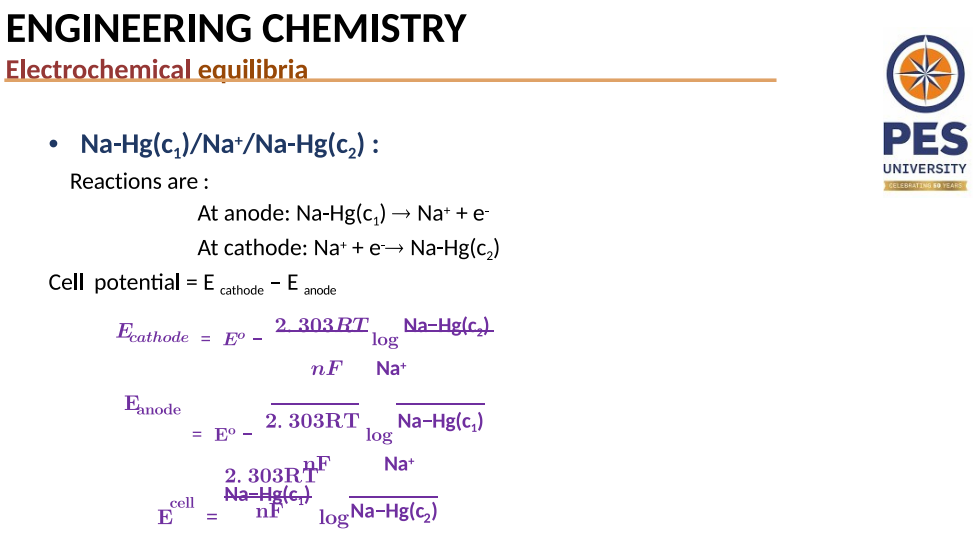

Figure G000044. Concentration cells types of concentration cells · source slide page 8.

EV0000396

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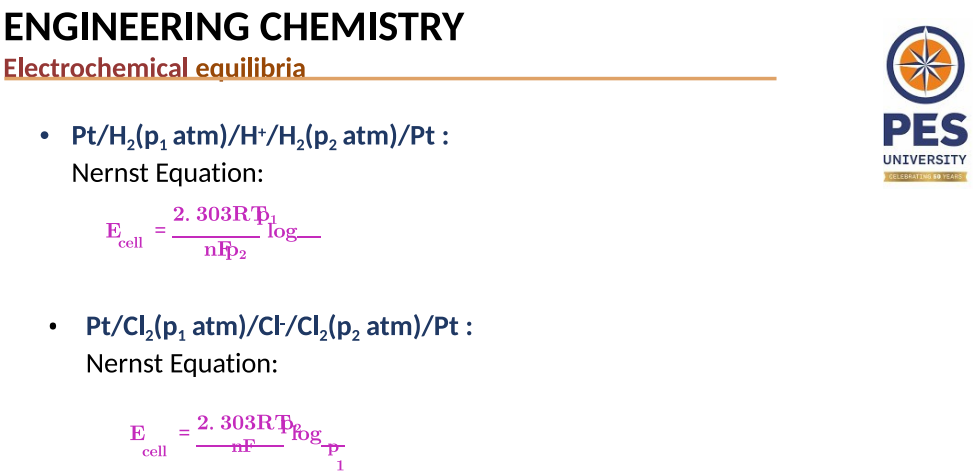

Figure G000045. Concentration cells types of concentration cells · source slide page 9.

EV0000399

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Figure G000046. Concentration cells types of concentration cells · source slide page 10.

EV0000402

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000374 · EV0000375 · EV0000376
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Class content 18: Concentration cells and it's types EV0000377 · EV0000378 · EV0000379
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Concentration cells: An electrochemical cell in which identical electrodes are in contact with a solution of identical species but of different activity/concentration of the electrode/electrolyte. [Image text, OCR] Voltmeter Saltbridge [$Cu^{2}+$]=0.010M Cu(s) Cu(s) [$Cu^{2}+$]=2.0M https://chemdemos.uoregon.edu/demos/Voltaic-Cell- CuCu-concentration-cell-Demonstration EV0000380 · EV0000381 · EV0000382
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria In this cell 2 copper electrodes are immersed in copper sulphate solutions of concentration c1 & c2 , such that c2> c1 An electrolyte has spontaneous tendency to diffuse from a solution of higher concentration to a solution of lower concentration which is the driving force for development of potential Oxidation takes place at anode and reduction takes place at cathode e.g.,Cu/Cu2+ (c1)//Cu2+(c2)/Cu EV0000383 · EV0000384 · EV0000385

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Reactions : At anode: Cu ☒ Cu2+ (c1)+2e- At cathode: Cu2+ (c2)+ 2e-☒ Cu Expression for cell potential: The emf of the cell = E cathode – E anode = ***Eo*** + **2. 303*RT* log *c E cathode 2 nF* = Eo** + **2. 303RT log c E anode 1 nF *Eo*** + **2. 303*RT* log *c* 2. 303*RT o E* = *– E* + *log c cell* 2 1 *nF nF*** 2.303RT ***c2*** log( ) ***E***cell = nF ***c1*** = **0.0591 *c2*** At 298K, ***E*** log( ) cell ***n c1*** EV0000386 · EV0000387 · EV0000388
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria The emf of the cell is positive only if c2 > c1 i.e., conc of metal ion at cathode > conc. of metal ion at anode The emf of the cell depends upon the ratio c2/c1 When c2 = c1, the emf of the cell becomes zero During working of the cell, concentration of ions increases at anode decreases at cathode When current is drawn from the cell c1 increases and c2 decreases The cell can operate only as long as the concentration terms are different EV0000389 · EV0000390 · EV0000391
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Types of concentration cells: Electrolyte concentration cell Electrode concentration cell Electrolyte concentration cell: Electrolyte concentration cell consists of two same electrodes that are dipped in the same electrolyte but with different concentrations of electrolytes Cu/Cu2+(c1)//Cu2+(c2)/Cu Cell potential is given by 2.303RT ***c2*** log( ) **E**cell = nF ***c1*** Electrode concentration cell Electrode concentration cell consists of two identical electrodes of different activity which are dipped in the same solution of electrolyte Na-Hg(c1)/Na+/Na-Hg(c2) **2. 303RT** Na−Hg(c1) **E** = **log** Cell potential is given by **cell nF** Na−Hg(c) 2 EV0000392 · EV0000393 · EV0000394
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Na-Hg(c1)/Na+/Na-Hg(c2) : Reactions are : At anode: Na-Hg(c1) ☒ Na+ + e- At cathode: Na+ + e-☒ Na-Hg(c2) Cell potential = E cathode – E anode = ***Eo*** **– 2. 303*RT* log** Na−Hg(c2) ***nF*** Na+ = **Eo – 2. 303RT log** Na−Hg(c1) **nF** Na+ ***E cathode*** **E anode 2. 303RT** Na−Hg(c1) **E** = **log cell nF** Na−Hg(c) 2 EV0000395 · EV0000396 · EV0000397
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Pt/H2(p1 atm)/H+/H2(p2 atm)/Pt : Nernst Equation: **2. 303RT p1 E** = **log nF p2 cell** Pt/Cl2(p1 atm)/Cl-/Cl2(p2 atm)/Pt : Nernst Equation: **2. 303RT p2 E** = **nF log p 1 cell** EV0000398 · EV0000399 · EV0000400
10. Source slide page 10 (extracted text): THANK YOU Department of Science and Humanities EV0000401 · EV0000402 · EV0000403

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.356] Hello, everyone. Welcome back to this course on engineering chemistry. We'll continue with module 4 on electrochemical equilibrium. [00:00:09.764] And in today's class we are going to talk about concentration cells, the different types of concentration cells, that is, electrolyte concentration cells and electrode concentration cells and then we study about ion selective electrodes. [00:00:25.040] The different types of ion selective electrodes, and then we derive an expression for the electrode potential of an ion selective electrode. So what are concentration cells? [00:00:38.404] An electrochemical cell in which identical electrodes are in contact with the solution of identical species but of different concentrations. What does this mean? Let us look at this picture. [00:00:51.616] Now suppose we have a copper plate which is dipped in a solution of copper sulfate of a particular concentration, we

know that it will have a potential say E1. [00:01:04.100] Now, if we have the same system like another copper plate, which is deep in the solution of copper sulphate, but this is of a higher concentration. [00:01:14.512] We know that the potential developed for this is going to be different from the one where the solution was of a dilute concentration. That we know from Nernst equation. So we have another potential which is found here, E2. Now if we combine these two. [00:01:33.636] electrodes and we have a salt bridge for electrical continuity, we find that the potential difference can be measured and this whole setup becomes of a galvanic cell. So that is how we define [00:01:51.332] a concentration cell when we have two different electrodes. [00:01:56.484] But each of these electrode is made of the same electrody material. The solutions are also of the same electrolyte, but the only difference is in the concentration of the [00:02:12.176] Solutions. So such a cell is known as a concentration cell. [00:02:18.628] Let us take an example just now you saw in the picture. We have taken a cell which has two copper electrodes and they are immersed in copper sulphate solutions such that C2 is greater than C1. One of the concentrations is more than the other. [00:02:38.180] Now we know that any electrolyte has a spontaneous tendency to diffuse from higher concentration to a lower concentration, and this will be the driving force for the potential to be formed between the cell, between the electrodes. [00:02:56.964] And as usual, like any galvanic cell, oxidation will take place at the anode and reduction will take place at the cathode. [00:03:06.288] So how will we represent the cell? The cell will be represented like this Cu, C U2 plus there's a soil bridge, and then we have again C U2 plus C U. So it doesn't make sense if you write like that. So what is important here? The only difference is in the concentration. So we have to mention the [00:03:23.952] The concentration. So let us say what are we assuming here that the concentration at the anode is C1 because remember cell representation, we always write the anode first and then the sol bridge and then the cathode. So the anode concentration is C1 and the cathode concentration is C2. [00:03:42.596] Let us write the reactions which are happening at the anode and the cathode. So at the anode we have copper going into solution, CU2 plus plus 2 electron, and since we are at the anode, the copper ion concentration is C1. And similarly, at the cathode, we have C U2 plus taking up [00:04:00.368] electrons and forming CO because there is a there has to be a reduction reaction again copper concentration is C2 at cathode is C2. So this is how we write the reaction at the cathode. [00:04:14.352] Now we know that the expression for cell potential is always the right hand side electrode minus the left hand side electrode, or in other words, e of the cathode minus the e of the animal. Now let us see what would be the expression for e cathode. If you [00:04:32.164] Look at this reaction, we have to know what is Q. So here we have the cathode reaction. So E0 plus 2.303 R T over NF log of C2. What type of electrode is this? I told in the previous classes that. [00:04:47.360] This electrode is metal rod dipped in a solution of its own ions. Therefore, the Nernst equation is given by this expression. So E0 plus 2.303 RT over NF log of C2. Similarly, the reduction potential.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.024] For the anode would also be E naught plus 2.303 RTORNF log of C1. And E cell. [00:05:11.296] is always E cathode minus E anode and all you have to remember here is that we always consider the reduction potential. So reduction potential at the cathode minus the reduction potential at the anode. When you write this expression we get E cell is equal to 2.303 R T over N F [00:05:28.944] log of C2 by C1. So at 298 Kelvin this expression now becomes 0.0591 over n log of C2 by C1. What is N? The number of electrons getting exchanged and this is at [00:05:46.512] Room temperature that is 298 Kelvin. So

when we substitute this is the expression we get. What we have to note here is that now E0 from both got [00:05:58.224] cancel the anode and cathode e naught is not there. So therefore the expression does not contain e naught. So it's a simple expression now 0.0591 over n log of C2 by C1 is the [00:06:14.084] expression for the cell potential of a concentration cell. [00:06:20.292] Now, there are certain important interpretations which we can make from that expression. The cell potential is going to be positive only when C2 is greater than C1, because [00:06:34.672] The expression was 0.0591 over n log of C2 by C1. So E cell will be positive only if C2 is greater than C1. Now what is C2 and what is C1? We know that we took [00:06:47.616] C2 as the cathode concentration and C1 as the anode concentration. So for this particular system, we see that the cathode concentration should always be higher than the anode concentration. You can see here, here the E cell will be positive. [00:07:05.264] Positive provided the cathode concentration is more than the anode concentration. Now the EMF of the cell also depends on the ratio C2 by C1. What does it mean? If the difference in the [00:07:22.084] Concentrations is very high, then the power which is generated, the EMF, is going to be high. If the concentration is not very high, we do not get a very high value of EMF because of the [00:07:39.588] Again, because of the expression where the ratio C2 by C1 is appearing. Therefore, if C2 if the ratio is high, then only we get the a good power. We can uh we have a high EMF being generated from this set. [00:07:57.284] Now, if C2 happens to be equal to C1, like if C2 is equal to C C1, then again using the same expression, what do we understand? That the cell potential will be 0. So you can have a potential only [00:08:14.880] When C1 is different from C2, and when will it be positive if C2 is greater than C1. This third interpretation, the third point which we discussed just now, when C2 is equal to C1, EMF of the cell becomes equal to 0. So what does it mean that [00:08:34.272] Finally, the cell wants to come to a condition where EMF of the cell becomes E is zero because that also means [00:08:43.920] Delta G is equal to 0, and what does that mean? That the cell has reached equilibrium. So what would be the equilibrium condition for a concentration cell? C2 becomes equal to C1. So that is where uh what we mean by there is a spontaneous tendency for C2 to change. [00:09:01.552] Into C1. And since we are saying C2 is a higher concentration, it's very obvious that higher concentration will try to lower the concentration, and that is possible only if a reduction reaction takes place. [00:09:17.284] The other case where at the anode we said the C1 concentration is low concentration. So it will try to increase its concentration. How will it increase only if the metal dissolves more and comes into the solution. Therefore, at the cathode, the concentration should be. [00:09:35.332] high as compared to the anode and once these concentrations are equalized then the cell PMF becomes zero. So in other words the cell has reach an equilibrium condition. Now concentration increases at the anode [00:09:52.000] and at the cathode the concentration decreases. So when the current is drawn from the cell we will find that C1 will go on increasing and C2 will go on decreasing. As long as there is a difference in the concentration the cell is going to function.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.068] Okay, now we come to the different types of concentration cell. Just now what we saw was an electrolyte concentration cell. There's another type of concentration cell which is known as [00:10:24.864] The electrode concentration cell. So electrolyte concentration cell is what we saw just now, and then we also have the electrode concentration cell. So what is an electrolyte concentration cell? We have [00:10:41.156] We have the electrolyte concentration cell where we have two same electrodes that are dipped in the same electrolyte but with different concentrations of the electrolyte. Just now we

saw this uh [00:10:53.520] uh example where we have two copper electrodes but of different concentrations of copper sulphate and when they are coupled together we see that EMF is generated and the cell potential is given by [00:11:06.096] 2.303 R T over N F log of C2 by C1. So that is an electrolyte concentration cell. So what is this electrode concentration cell? Let us concentrate on that. Now electrode concentration cell. [00:11:22.864] Would again, since it is a concentration cell, the electrodes are same, the electrolyte is same, but here. [00:11:31.268] The activity of the electrode itself is different. The concentration of the electrolyte remains the same, but the activity of the electrode itself is different. So when we have [00:11:44.496] Where what is an example for such an electrode? So such an electrode would be sodium amalgam. I taught you about sodium amalgam in types under types of electrode. This is another type of electrode where the metal is taken in the form of amalgam to reduce the activity of the metal. [00:12:02.160] So we have sodium amalgam here and also the other the electrode is also the sodium elect uh amalgam electrode. And therefore [00:12:12.048] These two electrodes now what is different? The electrolyte concentration, if you see there's only one electrolyte. So concentration of the electrolyte is not changing, but the concentration of [00:12:25.732] The activity of the electrode itself is different. So, in such a case, when we derive the potential, we will see how to do it. We will find that it will be given by 2.303 over R T over NF log of sodium amalgam electrode. [00:12:44.496] at the anode divided by the concentration of the sodium amalgam at anode divided by the concentration of the sodium amalgam at the cathode. Now let us derive this expression. How do we get to that expression? So what is our cell now? We have [00:13:02.596] Two sodium amalgam electrodes which are coupled together and both are dipped in a solution of Na plus ions. What would be the reaction at anode and cathode? So at the anode we have C1 concentration or activity, and at the cathode. [00:13:20.368] We have C2 activity as you can see from the cell representation. So the anode reaction is sodium amalgam with C1 activity or concentration of sodium is C1 gives you NA plus plus electron. That's an oxidation reaction. At the cathode we have NA plus plus electron is giving you sodium amalgam. [00:13:37.936] Of C2 concentration. Now again, E cell is always E cathode minus E anon. Let us write the expressions. So the expressions will be E0 minus 2.303 RT over NF log of the let us see this. We are writing the cathode. So q will be any. [00:13:57.380] H G C2 divided by NA plus. And similarly, the Nernst equation for the anode for the reduction potential will be E naught minus 2.303 R T over N F log of sodium amalgam with C1 concentration by NA plus. So E cell becomes equal to [00:14:15.200] E right minus e left, that is e cathode minus e anode, and finally you get this expression. Then now you see 2.303 R P R N F log of NAHG C1. [00:14:28.384] NAHG C2. So NA C1 divided by C2. So here what is C1? It's the anode concentration and C2 is the cathode concentration. So we understand that. [00:14:41.764] It is not that always we have the cathode concentration will should be higher than the anode concentration. Here you see is an example, E cell will be positive only if the activity of sodium amalgam at the anode is higher than. [00:14:59.200] at the one at the cat hole. So what does it tell us? That for every system we need to derive and see

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.752] What is the final expression? Only then can we suggest whether the anode concentration should be higher or cathode concentration should be higher. It is what do we understand? It is based on the reactions happening at the anode and cathode. So another example for [00:15:26.064] Such an electrode concentration cell would be if you have two hydrogen electrodes, you can see here. Now here the H

plus concentration again is maintained constant, so it cannot be. [00:15:39.492] Electrolyte concentration cell, it is an electrode concentration cell because the electrode here, the hydrogen itself has a different partial pressure as compared to the hydrogen at the [00:15:52.832] Cathode. So again we have used two hydrogen electrodes, but now what is different? They are at different partial pressures. So both of them are dipped in a common electrolyte that is H plus. So this is again another example for [00:16:08.708] Electrode concentration cell. The Nernst equation we can derive and find out that it will be given by 2.303 R T over N F log of P1 by P2. Again, here we see it is the and the cell potential will be positive provided the [00:16:26.160] P1. What is P1? The partial pressure of hydrogen at the anode is higher than that of the cathode. So we have to derive and find out what is the expression, and only then we can suggest. [00:16:39.936] That the anode concentration should be high or the cathod concentration should be high. So another exercise would be. [00:16:46.628] If we take a chlorine electrode, like two chlorine electrodes, now obviously the electrolyte is common, so there it is again an electrode concentration cell. So P1 and P2, when you write the reactions and then you find out the Nernst equation now, it becomes 2.303. [00:17:04.400] You are in a log of P2 by P1. It is different from what happened for the hydrogen electrode. So we cannot write just like that. We have to derive and find out what the expression. So here we see that in case of chlorine electrode, it is the [00:17:22.916] Cathode partial pressure should be higher of chlorine as compared to the partial pressure at the anode. So [00:17:31.620] We need to determine what is the E cell, what is the expression for E cell, in order to understand when the cell reaction is going to be spontaneous or when E cell will be. [00:17:46.720] positive and based on that we can decide the cathode and the anode concentration provided we want a positive emf then we have to see the anode and the con cathode concentration we will decide based on the expression which we get. [00:18:06.052] So now let us come to the next topic that is of ion selective electrodes. Now ion selective electrodes I gave a brief introduction when we talked about types of electrodes. So iron selective electrodes here they are [00:18:22.660] Very selective to a specific ion, and that is why they are so important. What does it mean? That in the presence of other ions, it can selectively detect one particular ion, not only selectively detect, it can actually tell us. [00:18:40.432] The concentration that means it can qualitatively detect the ion ignoring the other ions and it will also tell quantitatively tell us the concentration of that particular ion. So [00:18:53.840] Ion selected electrodes that is how it is named, and therefore the potential will be since it's an electrode, there will be a potential form, and that is going to be a function of the concentration of that particular ion. So since [00:19:11.120] It is determining concentration of one iron in the presence of other ions, so it can actually act as a chemical sensor. Now what is important? [00:19:22.624] They have a membrane. How do they achieve that? That finding only one ion when there are so many other ions present, it is because of the membrane which is capable of exchanging only that specific ion. So the membrane [00:19:37.168] The most important part is role is played by the membrane. It is capable of exchanging the specific ion with the solution with which it is in contact. So it is the membrane which is responsible for the ion selective electrodes. [00:19:53.636] Being able to recognize and detect and also quantitatively tell us the concentration of that particular ion, it is because of the membrane. So these electrodes are also known as membrane electrodes. So these electrodes, for example,

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:14.116] glass electrode. Glass electrode is an example for a membrane electrode. [00:20:20.996] So, what are the different types of

ion selective electrodes which are available to us? So, how are they classified? They are based on the membrane which is being used. As I said, membrane is the most important part of the ion selective electrode. So, based on the membrane material which is used. [00:20:38.768] Being used, there can be different types of ion selective electrodes. So the first type would be where we have crystalline. So crystalline or solid state membrane electrodes. [00:20:53.828] The best example for this is single crystal electrodes where we have which are selective to fluoride ions. We find that this [00:21:04.240] This is selective to fluoride ion. So what is the membrane being used over there? It is lanthanum fluoride. So these electrodes are going to have a membrane of lanthanum fluoride and they will be able to detect the fluoride ion concentration. [00:21:21.360] in the presence of other ions and where would this be uh used in water pollution you know there is a lot of fluoride contamination and that is very [00:21:32.416] uh dangerous. So if the fl uh we can determine the fluoride con concentration in water with the use of this mem this membrane that is Lanthanum fluoride. So such a [00:21:46.596] Such an ion selective electrode will be used to determine the fluoride ion concentration in the presence of other ions. Similarly, we can also have polycrystalline. [00:21:56.528] This was single crystal, this is polycrystalline, where you have AG2S which is being used as a membrane, and that is selective to sulfide ions. So if you want to determide in the presence of other ions, which ion selective electrode am I going to choose that has a membrane ag2s. [00:22:14.116] Okay, there can also be non-crystalline membrane electrodes. For example, glass. We know that glass electrode, what does it have? It has a glass membrane. So a glass membrane [00:22:26.560] And this glass membrane happens to be selective to H plus ions mostly, but it can also be selective to sodium ions also. So glass membrane is the most important constituent of the glass electrode, and since selective to H plus ions, that means [00:22:44.304] We can determine the pH of a solution using a glass electrode. Now there can also be liquid membrane electrodes where there is an ion exchanger which is dissolved in a viscous organic liquid membrane, and this is used [00:23:01.952] for ions like calcium, potassium, etc. And then we can also have immobilized liquid in a rigid polymer. So in a rigid polymer like PVC matrix, we have an immobilized iron exchanges. You know that iron exchangers as the name suggests are capable of exchanging ions to this [00:23:19.600] would be very very useful in ion selective electrodes so we have electrodes which are capable of detecting calcium and nitrate ions [00:23:32.580] Let us understand the principle behind how are how is the electrode potential getting formed in an ion selective electrode. Why is it a function of the concentration of the ion? [00:23:48.932] So let us have a schematic representation first, and then we'll understand how exactly we get [00:23:55.872] The electrode potential of an ion selective electrode. So let us derive an expression here for electrode potential of an ion selective electrode. Now as I pointed out, the membrane is the most important part of the [00:24:10.212] Ion selective electrode. So let us say this is the membrane of the ion selective electrode. Now one side of the membrane we have this analyte solution. [00:24:26.436] where the concentration is C1. So what is C1? Basically the [00:24:32.324] Concentration of the Mn plus ions which I want to determine using this ion selective electrode. Okay. Come to the other side, we have the we have a reference solution. Now this reference solution should be of the same metal ion which for which you want to [00:24:51.152] Determine the concentration. It will be the same solution on this side, but this concentration is known to us. Since we are calling it a reference solution, internal reference solution, we understand that this concentration is known to us. So C2 is known to me, and whereas C1 is

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:08.848] Is what I want to determine. And apart from this, there is also an inner reference electrode. So what is the role of this inner reference electrode we'll come to know in a [00:25:22.576] uh few minutes. Okay so now because there is a difference in concentration across the membrane, there is going to be a potential which is formed and that potential is known as the junction potential and sometimes it's also referred to as the membrane potential. [00:25:40.304] So, what is this junction potential? We know it is because of the difference in the concentration across the membrane. So, one side of the membrane we have concentration C1, the other side of the membrane we have concentration C2. Therefore, the juntial expression will be 2.303 RT over NF. [00:25:57.872] Log of C1 by C two. [00:26:01.604] Now, since C2 is known to us, we can remove it from this expression and get it in the form of a constant. So let us say the con we use a constant K. Now what happens to our expression now? The junction potential is given by 2.303 R T over NF. [00:26:19.376] Log of C1 and C2 has gone into the C2 has now become the constant. Now what is K? K will be [00:26:33.252] Minus, sorry, minus is missed over here, minus 2.303 R T over NF log of C2. [00:26:42.436] Now, we got an expression for the juntial as 0.0591, that is if you are talking at room temperature at 298 Kelvin by n log of C1 plus K. [00:26:58.628] Okay. But if you look at the membrane itself, the total membrane potential is going to be the sum of the junction potential plus the e reference electrode, because we have placed an [00:27:14.496] reference electrode also which is a part of the ion selective electrode. What is the role of this reference electrode, the internal reference electrode? Now the junction potential happens to be a very small value. [00:27:30.660] So in order to pick up that potential, we need to place an electrode inside. Moreover, the junction potential [00:27:39.888] It is there, but there is no external contact, no electrical contact by which we can measure this potential. Our aim is to find out what is this junction potential. Because why we know that junction potential will tell me what is the concentration C1. [00:27:57.344] So, I am going to place an internal reference electrode. The moment I say reference electrode, I know what is the potential of the internal reference electrode. So the total membrane potential is going to be the sum of this jun potential and the [00:28:15.492] E reference that is the internal reference electrode potential. So therefore the role of this reference potential is just to provide me an electrical contact so that I can [00:28:28.772] Find out what is this junction potential. So now the membrane potential becomes 2.303 R T over N F log of C1 plus K plus the E reference. So Ej, we already got an expression and that is what we have. [00:28:45.584] substituted here and plus the e reference. Now we know that k is a constant for the electrode e reference is also constant. So we can clump them together and get a new constant E0 M, which is going to be [00:29:04.164] Pertaining to that particular electrode which is being used. And what is E0 M? It is [00:29:11.296] K plus E reference. Now both are constants. So they are clubbed together as E0 M plus 2.303 R T over N F log of C1. So this is the expression for the membrane potential. [00:29:27.748] E m is equal to E naught M plus 2.303 R E over N F log of C1, where E0 M is equal to K plus E reference. [00:29:38.116] So we got an expression for the membrane potential. Now if we are working at 298 Kelvin, then this expression becomes E m is equal to E0 M plus 0.0591 by N log of C1. But [00:29:51.872] We still cannot determine C1. Why? Because this is a single electrode, and you know that single electrode potentials cannot be measured. So, what are we going to do? We are going to club this electrode with another reference electrode. So we are going to coup this with a

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Transcript block 7 (source interval 001800-002099 seconds): [00:30:09.712] That will be known as an external reference electrode. So why this word external has to be there? Because we have already used a reference electrode which is part of the in uh ion selective electrode. So that will be referred to as the internal reference electrode, and now to make a cell. [00:30:26.848] To make a cell using the membrane electrode, I need another reference electrode, and that will be referred to as the external reference electrode. [00:30:36.420] So now we made a cell where we have the external reference electrode which is in contact with the analyte, that is in contact with the membrane, which is now [00:30:47.280] in contact with reference solution and finally inside we have the internal reference electrode. So let's get the expression for the cell potential E cathode minus E anode. Now what is E cathode here? Here it is the [00:31:01.040] membrane electrode and then here on the anode we have the external reference electrode. Now we know that we can measure E cell [00:31:11.488] And this is constant or known to us. So therefore, from here I can determine what is E membrane. Okay. E membrane is the membrane potential, and once I know membrane potential, I can always find out what is [00:31:29.956] Concentration C1. So this is how I am going to determine the concentration C1. So therefore, when we are measuring using any potential meter, so basically it is all automated whenever you want to measure the [00:31:48.832] Concentration directly concentration is given to us. We can know what is the concentration, but what is happening inside is actually the cell potential is being measured and it is being subtracted, uh it is being [00:32:02.928] Uh subtract the membrane electrode is calculated using this expression. So we know what is the external reference electrode, and from there we determine membrane electrode potential and from there C1 is determined, but all the instruments are basically automated. [00:32:20.592] And we can get the concentration of the species directly when the electrode is dipped into the solution. Now, the only problem which we face here is that. [00:32:35.332] The membrane which is being used offers a very high resistance. Membrane offers a very high resistance and therefore we will we cannot use ordinary potential meters, therefore, special type of potential meters have to be used whenever. [00:32:53.104] We want to measure the cell potential where one of the electrode is an ion selective electrode. So, because the membrane which we are using offers very high resistance. Now there are a lot of applications for [00:33:08.420] These ion selective electrode they are used to determine concentration of number of cations and anions, such as we have hydrogen ion, we have lithium ion, sodium ion, potassium, etc., lead and arsenic also, and then we can also determine uh anions. [00:33:26.192] like cyanide, nitrate, fluorides or sulfide. So all these are possible if we have the particular [00:33:36.484] membran known to us and the ion selective electrode will be able to determine that particular ion in the presence of other ions. [00:33:46.820] So, in the next class, we are going to study a specific ion selective electrode that is the glass electrode, which is the most profusely used ion selective electrode. Thank you.

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Full transcript: 4866 normalized words, 139 timestamped segments, 2043.178667 seconds duration. Low-confidence segments flagged: 0.

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6 Electrochemical theory of corrosion

This lesson is indexed from the source video titled Electrochemical theory of corrosion. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000090 · EV0000091

Graphs and diagrams

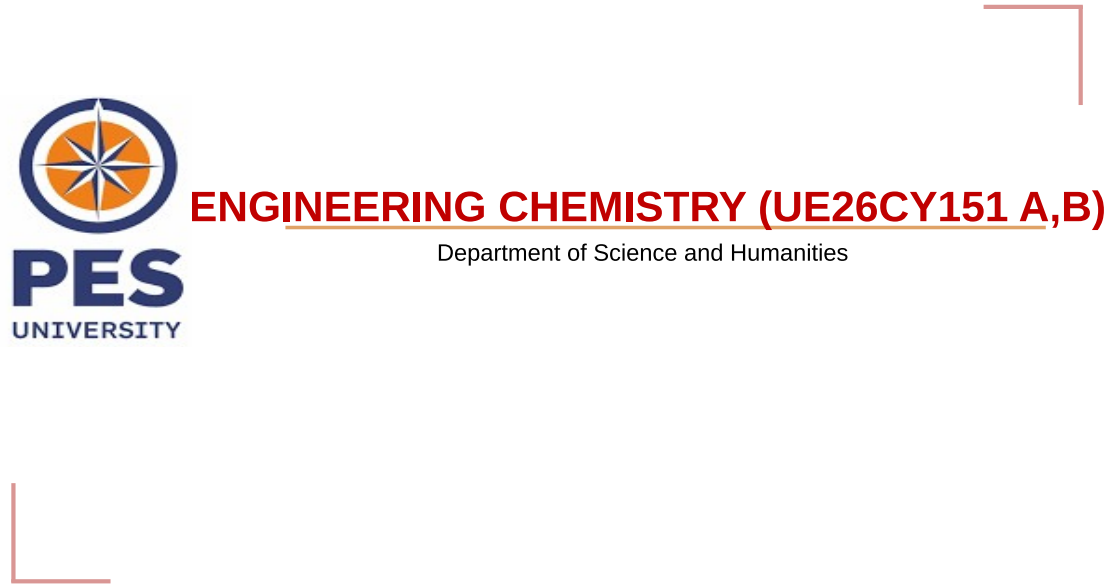

Figure G000047. Electrochemical theory of corrosion · source slide page 1.

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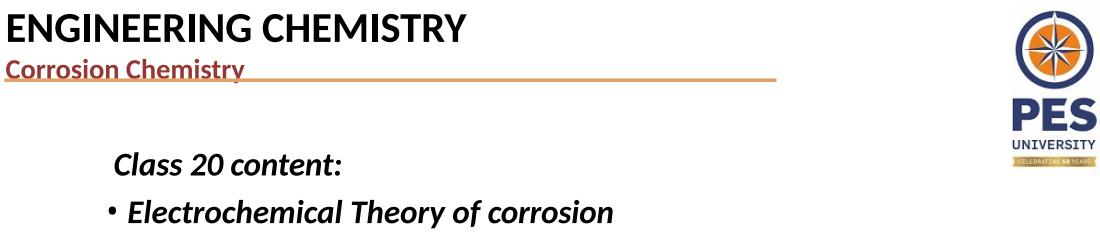

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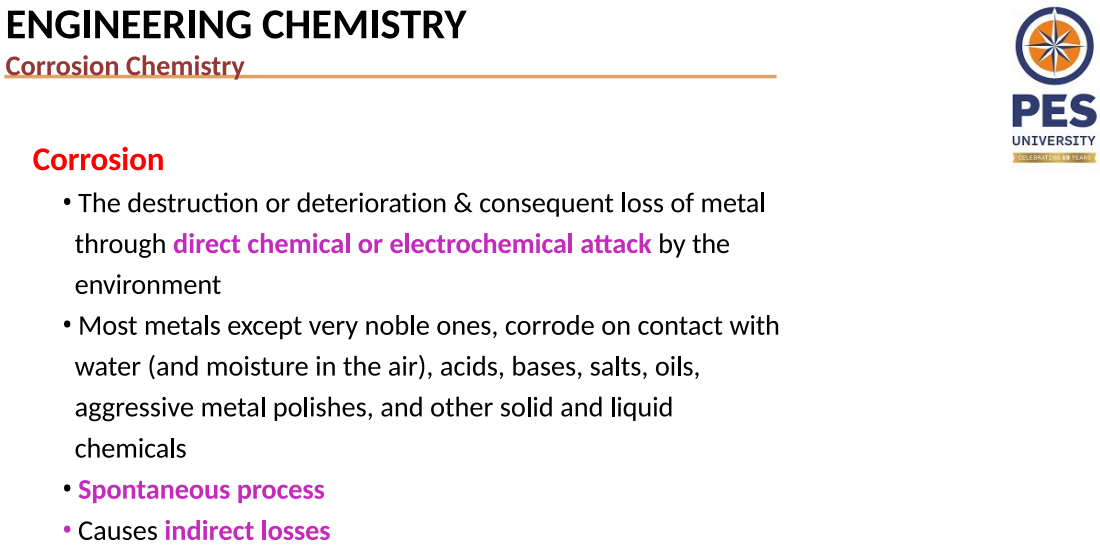

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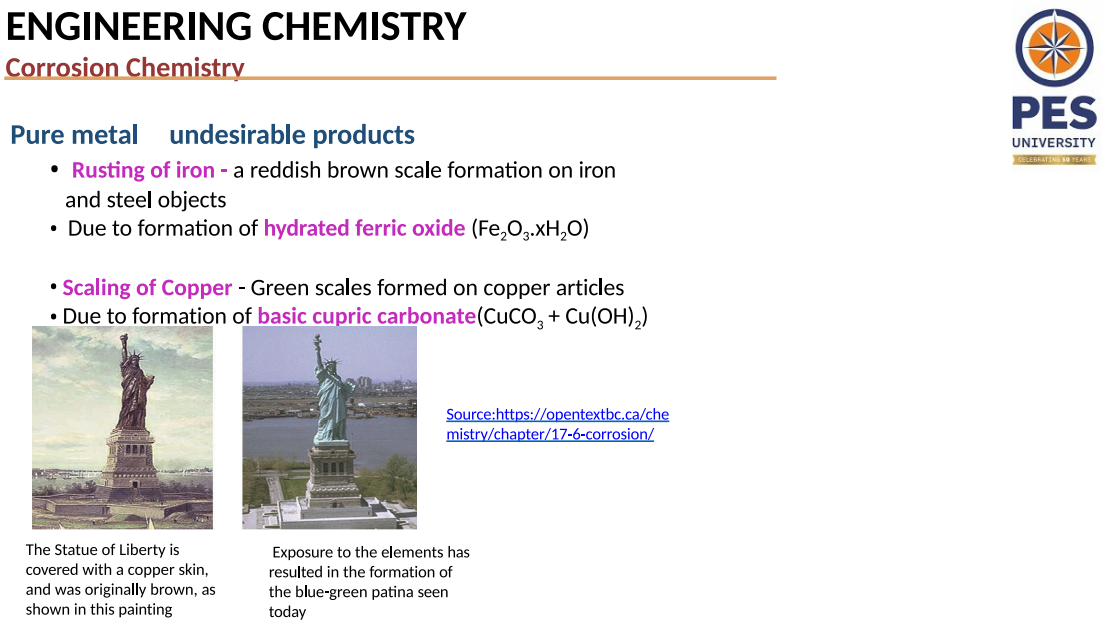

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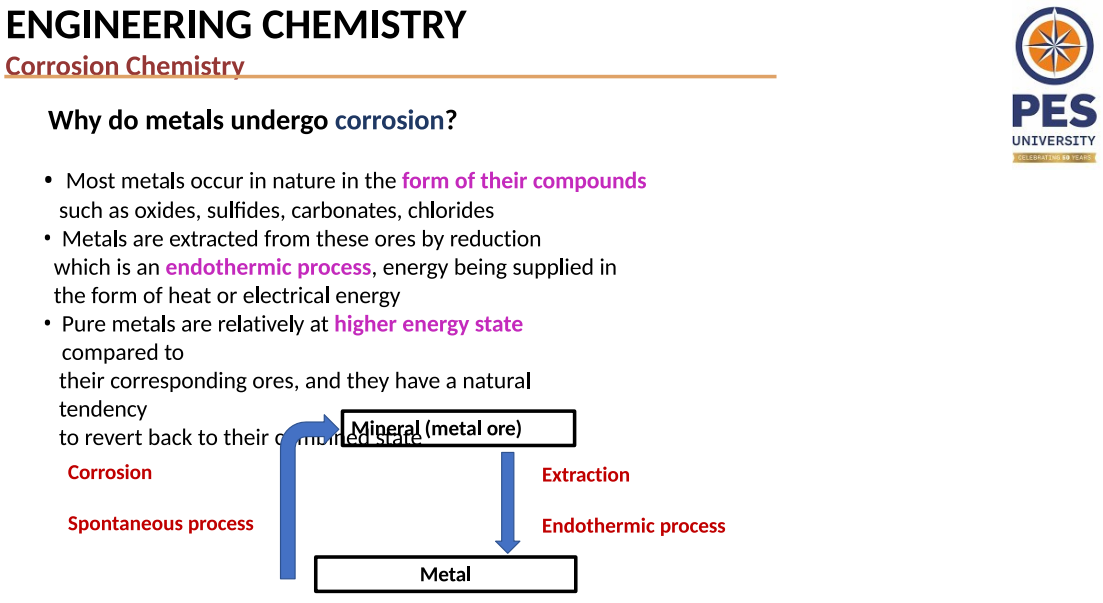

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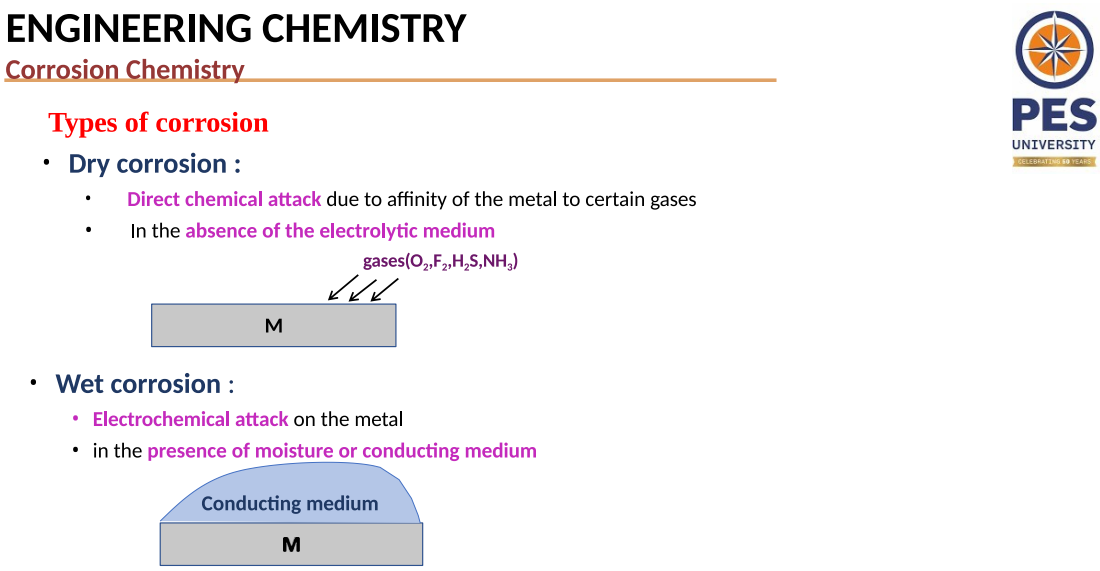

Figure G000052. Electrochemical theory of corrosion · source slide page 6.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

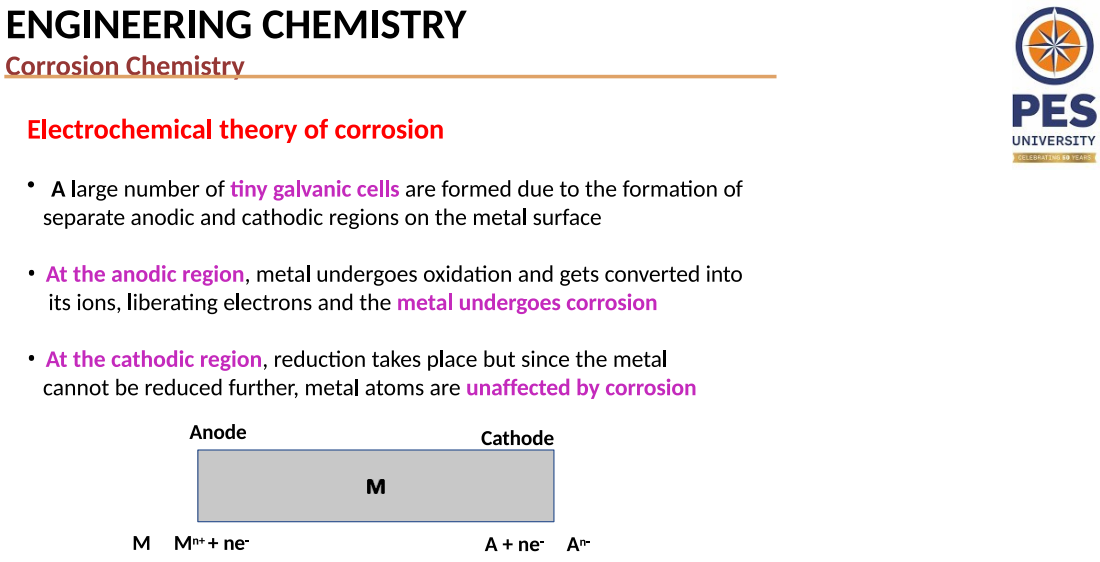

Figure G000053. Electrochemical theory of corrosion · source slide page 7.

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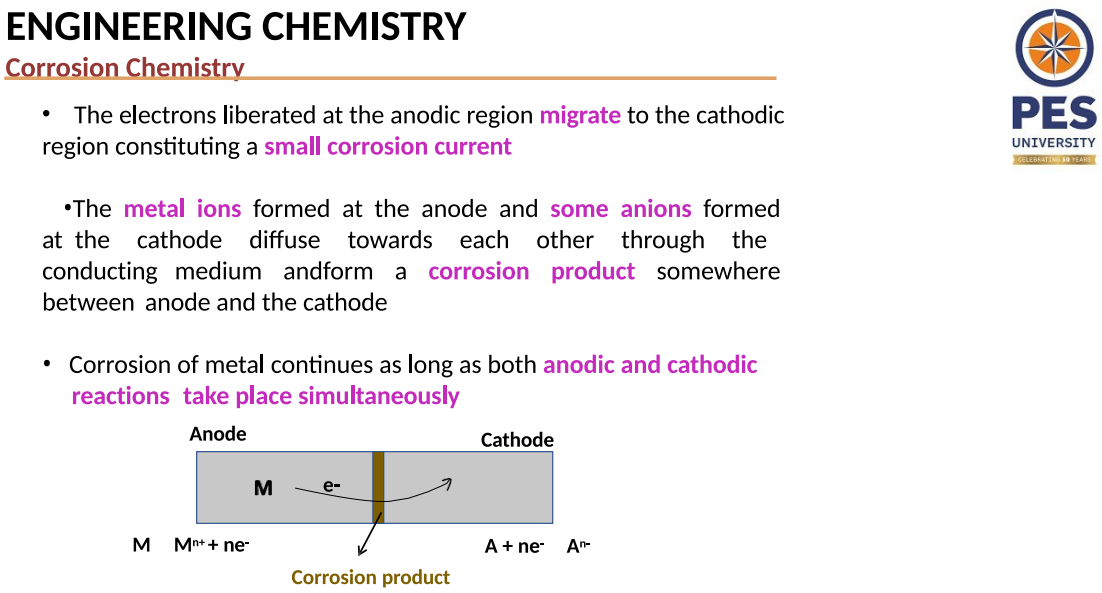

Figure G000054. Electrochemical theory of corrosion · source slide page 8.

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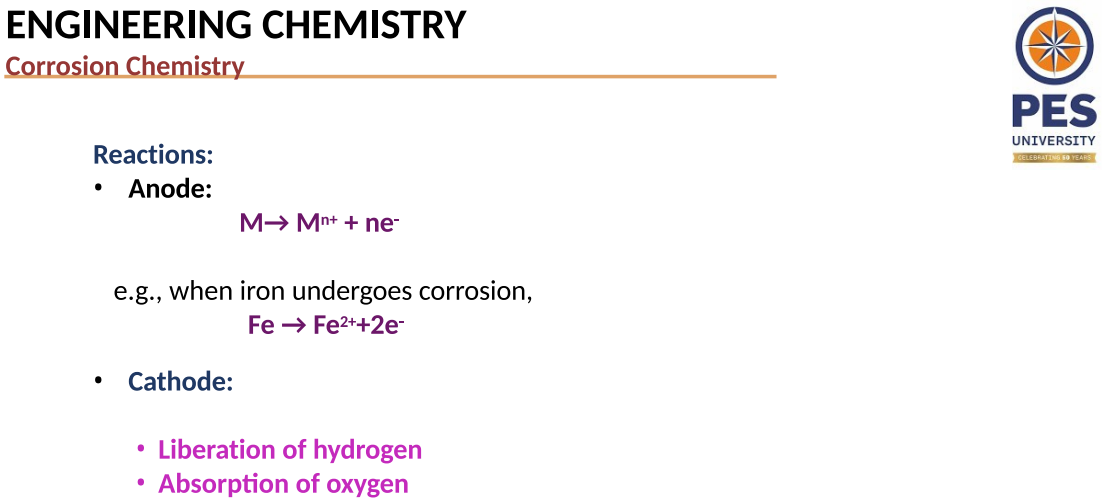

Figure G000055. Electrochemical theory of corrosion · source slide page 9.

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ENGINEERING CHEMISTRY

Corrosion Chemistry

- Liberation of hydrogen
  - In acidic medium
    $2H^+ + 2e^- \rightarrow H_2 \uparrow$
  - In neutral or alkaline medium
    $2H_2O + 2e^- \rightarrow 2OH^- + H_2 \uparrow$
- Absorption of oxygen
  - In acidic medium
    $4H^+ + O_2 + 4e^-$

Figure G000056. Electrochemical theory of corrosion · source slide page 10.

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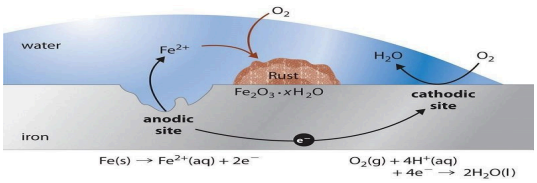
ENGINEERING CHEMISTRY

Corrosion Chemistry

Corrosion of Iron:



Source: https://chem.libretexts.org/Bookshelves/General_Chemistry/Book%3A_Chem1_(Lower)/16%3A_Electrochemistry/16.08%3A_Electrochemical_Corrosion

Anode : $Fe \rightarrow Fe^{2+} + 2e^-$

Cathode : $2H_2O + O_2 + 4e^- \rightarrow 4OH^-$

Fe^{2+} and OH^- ions move towards one another and $Fe(OH)_2$ is formed somewhere between anode and cathode regions

$$2Fe^{2+} + 4OH^- \rightarrow 2Fe(OH)_2$$

Figure G000057. Electrochemical theory of corrosion · source slide page 11.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

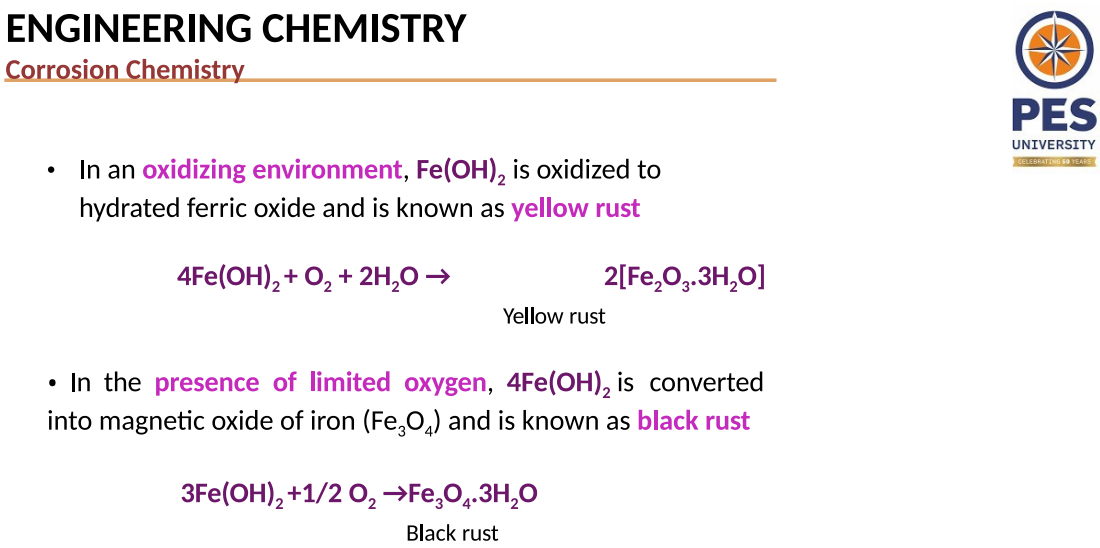

Figure G000058. Electrochemical theory of corrosion · source slide page 12.

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Figure G000059. Electrochemical theory of corrosion · source slide page 13.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000404 · EV0000405 · EV0000406
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Class 20 content: Electrochemical Theory of corrosion EV0000407 · EV0000408 · EV0000409
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Corrosion The destruction or deterioration & consequent loss of metal through direct chemical or electrochemical attack by the environment Most metals except very noble ones, corrode on contact with water (and moisture in the air), acids, bases, salts, oils, aggressive metal polishes, and other solid and liquid chemicals Spontaneous process Causes indirect losses EV0000410 · EV0000411 · EV0000412
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Pure metal ☒ undesirable products Rusting of iron - a reddish brown scale formation on iron and steel objects Due to formation of hydrated ferric oxide (Fe2O3.xH2O) Scaling of Copper - Green scales formed on copper articles Due to formation of basic cupric carbonate(CuCO3 + Cu(OH)2) Source:https://opentextbc.ca/che mistry/chapter/17-6-corrosion/ The Statue of Liberty is covered with a copper skin, and was originally brown, as shown in this painting Exposure to the elements has resulted in the formation of the blue-green patina seen today EV0000413 · EV0000414 · EV0000415
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Why do metals undergo corrosion? Most metals occur in nature in the form of their compounds such as oxides, sulfides, carbonates, chlorides Metals are extracted from these ores by reduction which is an endothermic process, energy being supplied in the form of heat or electrical energy Pure metals are relatively at higher energy state compared to their corresponding ores, and they have a natural tendency to revert back to their combined state Mineral (metal ore) Corrosion Extraction Spontaneous process Endothermic process Metal EV0000416 · EV0000417 · EV0000418
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Types of corrosion Dry corrosion : Direct chemical attack due to affinity of the metal to certain gases In the absence of the electrolytic medium gases(O2,F2,H2S,NH3) M Wet corrosion : Electrochemical attack on the metal in the presence of moisture or conducting medium Conducting medium EV0000419 · EV0000420 · EV0000421
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Electrochemical theory of corrosion A large number of tiny galvanic cells are formed due to the formation of separate anodic and cathodic regions on the metal surface At the anodic region, metal undergoes oxidation and gets converted into its ions, liberating electrons and the metal undergoes corrosion At the cathodic region, reduction takes place but since the metal cannot be reduced further, metal atoms are unaffected by corrosion Anode Cathode M ☒ Mn+ + ne- A + ne- ☒ An- EV0000422 · EV0000423 · EV0000424

8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry The electrons liberated at the anodic region migrate to the cathodic region constituting a small corrosion current The metal ions formed at the anode and some anions formed at the cathode diffuse towards each other through the conducting medium andform a corrosion product somewhere between anode and the cathode Corrosion of metal continues as long as both anodic and cathodic reactions take place simultaneously Anode Cathode e- M ☒ Mn+ + ne- A + ne- ☒ An- Corrosion product EV0000425 · EV0000426 · EV0000427
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Reactions: Anode: M→ Mn+ + ne- e.g., when iron undergoes corrosion, Fe → Fe2++2e- Cathode: Liberation of hydrogen Absorption of oxygen EV0000428 · EV0000429 · EV0000430
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Liberation of hydrogen In acidic medium 2H++2e-→H2↑ In neutral or alkaline medium 2H2O+2e- →2OH−+H2↑ Absorption of oxygen In acidic medium 4H+ + O2 + 4e- → 2H2O In neutral or alkaline medium 2H2O + O2 + 4e- →4OH- EV0000431 · EV0000432 · EV0000433
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Corrosion of Iron: [Image text, OCR] water Fe2+ H2O Rust Fe2O3·xH20 cathodic site anodic site iron Fe(s)→Fe2+ (aq)+2e O2(g) + 4H+(aq) +4e-→2H2O (1) Source: https://chem.libretexts.org/Bookshelves/General_Chemistry/Book%3A_Chem1_(Lower)/16%3A_Electrochemistry/16.08%3A_ Electrochemical_Corrosion Anode : Fe → Fe2++2e- Cathode : 2H2O + O2 + 4e- →4OH- Fe2+ and OH- ions move towards one another and Fe(OH)2 is formed somewhere between anode and cathode regions 2Fe2+ + 4OH- → 2Fe(OH)2 EV0000434 · EV0000435 · EV0000436
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry In an oxidizing environment, Fe(OH)2 is oxidized to hydrated ferric oxide and is known as yellow rust 4Fe(OH)2 + O2 + 2H2O → 2[Fe2O3.3H2O] Yellow rust In the presence of limited oxygen, 4Fe(OH)2 is converted into magnetic oxide of iron (Fe3O4) and is known as black rust 3Fe(OH)2 +1/2 O2 →Fe3O4.3H2O Black rust EV0000437 · EV0000438 · EV0000439
13. Source slide page 13 (extracted text): THANK YOU Department of Science and Humanities EV0000440 · EV0000441 · EV0000442

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.932] Hello everyone, welcome back to this course on engineering chemistry. We will start with a new module today on corrosion chemistry. [00:00:12.260] So in this module we are going to study about the electrochemical theory of corrosion, the types of corrosion, factors affecting rates of corrosion, and about corrosion control. [00:00:26.244] Today's class we will understand what is corrosion, the different types of corrosion, the electrochemical theory of corrosion, and finally we will look at corrosion of iron. Firstly, let us define what is corrosion. [00:00:45.668] Now corrosion is a phenomenon that is occurring all around us and we see that metals get corroded. Now how would we define this as a process? [00:01:03.588] So we'll say corrosion is the destruction and deterioration, and of course consequent loss of metal through direct chemical attack or an electrochemical attack by the environment. [00:01:21.892] So we understand that the metal as well as the environment are responsible for this phenomena of corrosion. Whenever the metal surface is exposed to the environment, the environment is going to

cause this destruction of the metal. [00:01:41.604] Now most of the metals except the very noble ones we know corrode on contact [00:01:49.892] with the environment. Now what is the environment? It could be water or it could be simply moisture in the air, acids, bases, salts, oils, aggressive metal polishes and other types of solid and liquid chemicals. [00:02:07.844] And what is important for us to know is that it is a spontaneous process. [00:02:13.424] means whenever a metal is exposed to the environment, the reaction is going to be spontaneous. We need not supply any kind of energy for the process to take place. And that is the reason why the moment the metal is exposed to any of these [00:02:31.088] uh chemicals it is undergoing corrosion. And most importantly, corrosion is a nuisance for us because it is not just the direct metal loss, it is the indirect losses which happen when corrosion takes place. [00:02:50.980] For example, it could result in a bridge collapsing, or it could result in air crashes. There could be oil pipelines which would get burst. [00:03:04.484] So all these are going to result in colossal losses, and therefore corrosion becomes very, very important study for us. What exactly happens during corrosion? Let us understand. We know that pure metal [00:03:24.448] Is getting converted into undesirable products. We made structures made of pure metal, but they are getting converted into undesirable products. So let us take the example. The most common example which we see happening all around us because most of the structures are made of iron. [00:03:42.160] or steel. Rusting of iron is something all of us have experienced. So what happens in rusting? There is a reddish brown scale which is getting formed on the iron and steel objects. And what is this? This is actually hydrated ferric oxide. [00:03:58.180] So we can say that during the rusting of iron we have Fe becoming Fe2O3XH2O. [00:04:07.940] Similarly, scaling of copper that is also a corrosion phenomena. And many times in the when you have these copper articles, we see that they get a greenish. [00:04:21.600] color on the surface. So what is that due to? It is due to basic pupric carbonate. So scaling of copper is [00:04:31.460] Another phenomenon which we obser uh observe and that is due to corrosion. So what is happening here? It is getting converted into CUCO3 plus C U O H twice. So we see that corrosion [00:04:46.256] basically results in the pure metal getting converted into a compound of the metal. Now very good example would be the Statue of Liberty. [00:04:57.840] which is actually covered with a thin copper skin and it was originally brown in colour as you can see in the picture. But with time due to exposure to the environment, this has resulted in the formation of the basic copper carbon

EV0000090 · EV0000091 · EV0000092 · EV0000093

Transcript block 2 (source interval 000300-000599 seconds): [00:05:15.472] Which we saw here, which is known as blue green patina, and as we see it today, it is this bluish green in color. [00:05:24.868] Apart from this, we can also have silver getting tarnished when it is exposed to an H2S atmosphere. We know that silver particles they get this black color. So basically we understand corrosion. [00:05:38.736] that the metal is getting converted into a compound which is n not uh desirable for us. [00:05:48.292] So let us answer the first question. Why do metals undergo corrosion? Or rather, why is corrosion a spontaneous process? [00:06:00.740] Let us understand this. What is happening is in nature, most of the metals are in the form of their compounds. [00:06:09.764] that is the mineral ores which are available to us, they are in the form of compounds like they could be oxides, sulphides, carbonates, chlorides, and and many many other compounds. Then when we do an extraction process [00:06:25.892] Extraction is always an endothermic process because we are either supplying heat or we are supplying some kind of electrical energy. So as a result when the metal when the metal ore is getting converted to metal, or rather when you are

extracting the metal [00:06:41.648] This metal has you have given this much energy, it's an endothermic process, so they end up at a higher energy state as compared to the ore from which they were formed. And because any system tries to go back to the uh lower uh lower energy state. [00:06:59.312] So there is a natural tendency to for the metal to go back into the combined state. That's why in nature we find that this process metal getting converted into the compound or uh any d undesirable product is [00:07:18.064] B is a spontaneous process because when you extracted the metal it you got it to a higher energy state by supplying energy, therefore there's a natural tendency for it to go back to the combined form. [00:07:33.508] Now, there are two types of corrosion. One is dry corrosion. The other is wet corrosion. In dry corrosion, the metal is being attacked [00:07:46.308] directly by the gases. So basically it is a chemical attack. These gases are attacking metal surface. Okay, we have to remember very clearly that corrosion is a surface phenomenon. [00:08:05.924] Throughout this chapter we have to [00:08:10.308] Understand that it only affects the surface of the metal, the bulk of the metal does not get affected by corrosion. So this is say the surface of the metal and there is direct chemical attack. Why? Because the metal has an affinity for these gases. Say it could be oxygen. [00:08:28.080] chlorine, H, ammonia, any any of the gases SO2 which are present in the environment around, they are directly attacking the metal surface and therefore it is giving rise to some kind of a product. [00:08:44.224] on the metal surface. Now this is happening in the absence of any electrolytic medium. [00:08:52.132] So, therefore, we say it's a direct chemical attack, and it is called dry corrosion. Now there's another type of corrosion which is called as wet corrosion, which happens only in the presence of moisture or a conducting medium. And when this happens, [00:09:08.832] we say that it is a electrochemical attack on the metal. So here we can see in this case the metal now is in the presence of a conducting medium and now the phenomena becomes an electrochemical phenomena. [00:09:26.064] So let us understand what is actually happening when the metal is exposed to the environment and in the presence of a conducting medium. So basically we'll study what we call as the electrochemical theory of corrosion. [00:09:45.732] Now the moment we say electrochemical, we understand that there is going to be anode and cathode formation. [00:09:56.420] Anode and cathode formation at as we studied in galvanic cells. So there is formation of anode and cathode. So basically the moment the metal is exposed to a conducting environment, then we understand that there is going to be formation of anode

EV0000094 · EV0000095 · EV0000096 · EV0000097

Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.288] and cathode on the surface of the metal. We have to remember always that it is happening only on the surface. So we find a t lot of uh tiny galvanic cells which are formed on the uh surface. So what are what do these gal galvanic cells constitute of? They are of course made of the anode and the cathode. [00:10:34.212] Now at the anode, we know that it's always an oxidation process which is going to take place. What do we have? We have this metal. Now the metal has to undergo oxidation. So the reaction would be metal is becoming Mn plus. So basically, metal is getting converted into a [00:10:53.764] ion that means into its compounds. So can we say this is corrosion? Yes. That's how we define corrosion. That metal gets converted into an undesired uh compound. So here at the anode [00:11:08.612] We see that the metal is getting corroded and it is going to get destroyed. Of course, there are n number of electrons which are getting generated in this process. Now coming to the cathode part of the metal. Remember, it is the same metal, and one part of it is acting as the cathode. [00:11:26.384] Now at the cathode we have to remember that metal can never undergo

reduction because at the cathode what is the process which is supposed to take place? It is reduction process. So but [00:11:42.000] The metal can never undergo reduction. Why? Because it's already in the lowest reduced form. Therefore, it is the environment. So you see we didn't write M here. It will never happen. So it is the environment because we know that again I would and cathode oxidation and reduction. [00:11:59.632] Should happen simultaneously. We cannot have just oxidation happening and no reduction. So what will take part in the reduction process? Some species in the environment is going to undergo reduction and that will result in the formation of AN minus. So what is the [00:12:17.056] Fact we have to remember that the anode part of the metal is getting corroded whereas the cathode part is completely unaffected by corrosion because the metal remains intact. It's only the environment which is taking part in the reduction process. [00:12:34.916] Now further, these electrons. Now we saw that electrons are getting generated here. These electrons are going to move towards the cathode and what would be the uh electronic uh conducting media? That would be the metal itself. [00:12:51.760] So on the metal surface now the electrons move towards the an from towards the cathode from the anode and at the cathode we have the species taking up these electrons and forming AN minus. Now we have Mn plus ions which are which got formed here. [00:13:09.792] And we have the AN minus which got formed at the cathode. Now due to simple diffusion, due to simple diffusion, we find that these Mn plus ions will move towards this side and the An minus are going to move towards this side, and somewhere in [00:13:27.536] Between there will be formation of corrosion product. These two are going to combine combine the Mn plus ions and the AN minus ions are going to move towards. [00:13:41.860] each other and somewhere in between we will find the formation of the corrosion product. And remember why do we need a conducting medium for such a phenomena to happen? Now we can understand that these ions need to move towards each other in order to come [00:13:59.632] So we have electrons which are moving through the [00:14:04.580] On the metal surface, whereas these ions need a medium to move, and therefore a conducting medium is what is required for corrosion to happen. And we should always remember that as long as the anode and the cathode reaction happens simultaneously. [00:14:22.352] Corrosion process is going to continue. Let us check out what are the reactions which are actually happening. At the anode, we don't have any other reaction which can take place, but that the metal is getting. [00:14:37.936] oxidized and therefore giving out n electrons and we have Mn plus ions which are getting formed. So suppose we take the case of iron then Fe becomes Fe two plus plus the two electrons. [00:14:50.224] But at the cathode, since it is not the metal which is involved in the reaction, you could you are having the environment which is taking part in the reaction. So there could be lot of species which are present in the environment. So

EV0000098 · EV0000099 · EV0000100 · EV0000101

Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.384] To simplify, we can understand there can be two types of reaction, either liberation of hydrogen or the absorption of oxygen. Let us see. [00:15:15.504] What happens when there is liberation of hydrogen? Now liberation of hydrogen obviously would happen when there is less amount of oxygen in the atmosphere. And again the medium could be acidic or it could be neutral or alkaline. If it is acidic medium, we observe [00:15:33.200] that obviously there will be lot of H plus ions and it's a cathodic reaction so electron will be taken up and we get liberation of hydrogen. In neutral or alkaline medium again reduction because we are at the cathode. So reduction of water takes place [00:15:49.888] and liberation of hydrogen happens. So both these will happen uh will be coming

under the category of liberation of hydrogen. And it is either a neutral or alkaline medium, then this is the reaction. If it's an acidic medium this will be the predominant reaction. Now coming to absorption of oxygen. [00:16:09.092] Again we could have uh in the acidic medium or in the neutral or alkaline medium. Absorption of oxygen will obviously happen when there is lot of oxygen present in the atmosphere. [00:16:21.392] So in an acidic medium we can understand there will be a lot of H plus ions oxygen will be absorbed along with the uh electrons taking part in the reaction and we get water molecules getting formed. If it is a neutral or alkyne medium, then again we have water, but now oxygen is getting. [00:16:39.088] absorbed with four electrons taking part in the reaction and H minus. So these are the four reactions which can happen when at the cathode [00:16:52.868] during the corrosion process. So which reaction will take place is going to depend on what kind of environment the metal is exposed to. Let us now look at a specific case and that is corrosion of iron. [00:17:08.804] Now if you see here there is again this metal bar and it is undergoing corrosion. So one part acts as the anodic side, the other part is acting as the cathodic side. [00:17:21.568] Add the anode we know that Fe2 plus ions are going to get formed and add the cathode [00:17:29.536] Most common reaction would be in a neutral medium oxygen is getting absorbed. So we will see oxygen taking part in the reaction here, and we get the water and uh water. So what happens here now? We have [00:17:47.108] Fe becoming Fe two plus plus two electron and uh most common would be of course if it's an acidic medium as in this case then we will find that water will be liberated, oxygen is being used and water is liberated. But most commonly we find neutral mediums. So let us assume that it's a neutral medium. [00:18:04.880] medium and we get 2 H2O plus O2 plus 4 electron is giving you 4 O H minus. Now what happens next? [00:18:15.120] The Fe2 plus ions which are got which were formed here, and this is the conducting medium as can be seen here. This Fe2 plus ions and the OH minus ions move towards one another and we get the formation of FeOH twice. [00:18:32.848] somewhere between the anode and the cathode regions. You can see here that the corrosion product is getting formed, of course it will be on the surface because corrosion is a surface phenomenon. [00:18:44.848] So what would be the reaction now? Uh Fe2 plus and OH minus are diffusing towards one another and we get FeOH twice, but we know that rust is not FeOH twice. So what happens next? Now if the environment is very oxidizing. [00:19:03.744] Fe2 plus is going to get converted or oxidized into Fe three plus. We know that Fe three plus is a more stable state as compared to Fe two plus. So ferrous is going to get oxidized to ferric. So ferrous hydroxide. [00:19:21.424] Now is becoming hydrated ferric oxide and this is known as yellow rust. So in the presence of plenty of oxygen and water we get what we call as the yellow rust. So basically what we have to understand here your F [00:19:37.536] Two plus has got converted to F three plus and that is yellow dust. [00:19:44.804] That is because you provided an oxidizing atmosphere. Now suppose there was a limited amount of oxygen, then FeH twice will get converted to Fe3O4. [00:19:58.820] And this is known as black rust. So F E O H twice plus half of H to O

EV0000102 · EV0000103 · EV0000104 · EV0000105

Transcript block 5 (source interval 001200-001499 seconds): [00:20:06.916] So ferric uh ferrous hydroxide is converted into magnetic oxide of iron that is Fe three O four and this is known as black crust now. So you can see the reaction there is limited amount of oxygen and we get Fe three O four. So why this Fe three O four when there is less oxygen. [00:20:24.688] of oxygen FE3O4 if you know is actually Fe O [00:20:30.208] Fe two O three. So basically iron plus two and plus three. So it has not got completely converted to plus three. It is known as ferrosopheric oxide, Fe three

O4, and therefore in limited oxygen we will see the formation of [00:20:47.920] black rust. So this is how iron is going to get corroded and the electrochemical theory of corrosion explains this phenomenon uh very uh [00:21:01.124] very clear. So now uh in the next class we are going to talk about the different types of corrosion. Thank you.

EV0000106 · EV0000107 · EV0000108 · EV0000109

Full transcript: 2966 normalized words, 86 timestamped segments, 1274.176 seconds duration. Low-confidence segments flagged: 0.

EV0000090 · EV0000091 · EV0000092 · EV0000093

7 Numerical problems based on electrode and cell potential determination

This lesson is indexed from the source video titled Numerical problems based on electrode and cell potential determination. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000110 · EV0000111

Graphs and diagrams

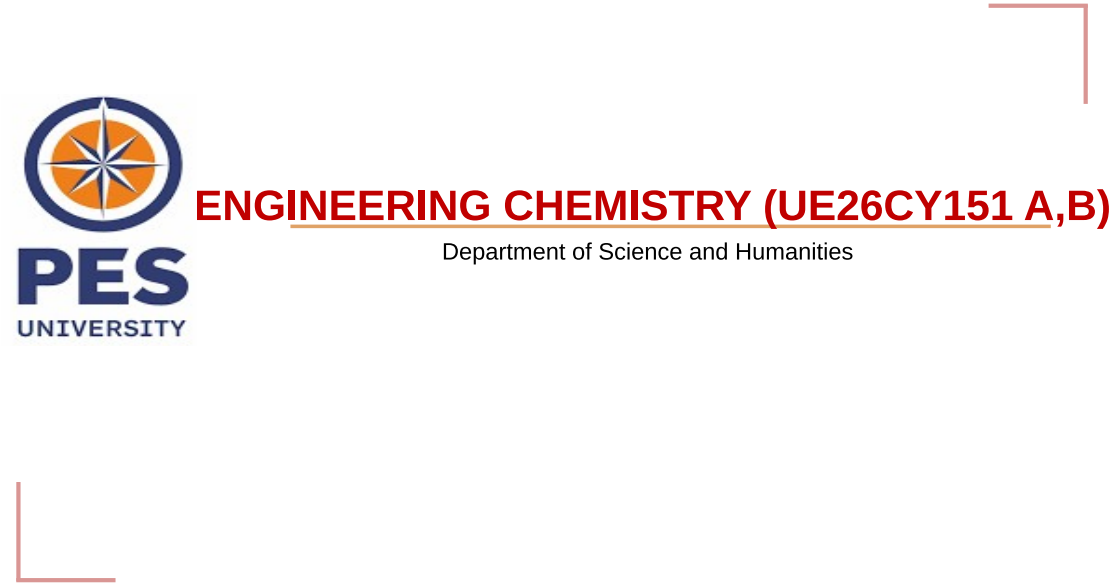

Figure G000060. Numerical problems based on electrode and cell potential determination · source slide page 1.

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Figure G000061. Numerical problems based on electrode and cell potential determination · source slide page 2.

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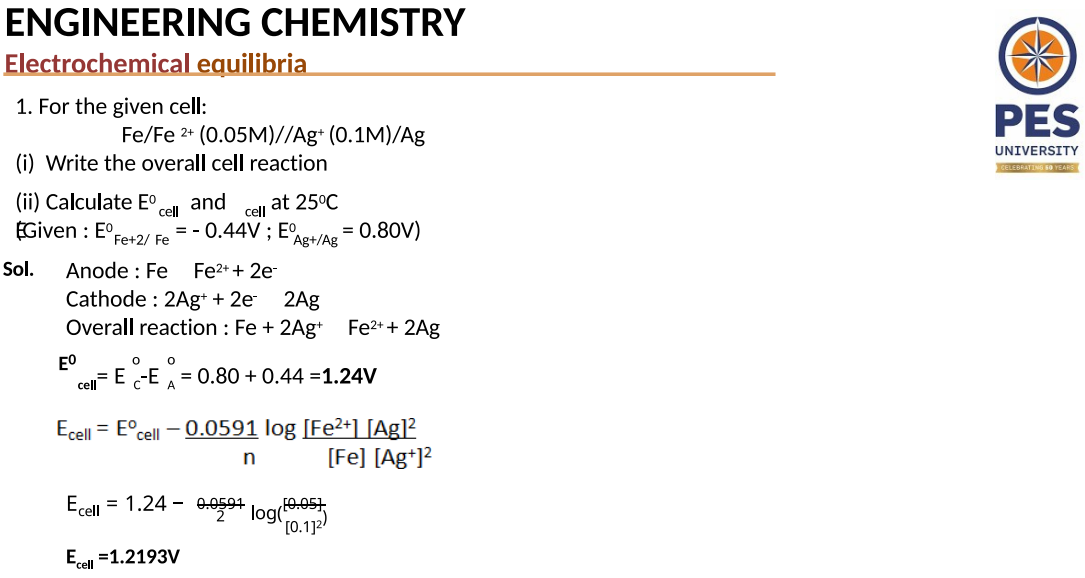

ENGINEERING CHEMISTRY

Electrochemical equilibria

1. For the given cell:

Fe/Fe $^{2+}$ (0.05M)//Ag^{+} (0.1M)/Ag

(i) Write the overall cell reaction

(ii) Calculate E^0_{cell} and $_{cell}$ at 25^{0}C

(Given : $E^0_{Fe+2/\ Fe}$ = - 0.44V ; $E^0_{Ag+/Ag}$ = 0.80V)

Sol. Anode : Fe $Fe^{2+} + 2e^-$

Cathode : $2Ag^+ + 2e^-$ 2Ag

Overall reaction : $Fe + 2Ag^+$ $Fe^{2+} + 2Ag$

$E^0_{cell} = E^o_C - E^o_A = 0.80 + 0.44 =$ **1.24V**

$$E_{cell} = E^o_{cell} - \frac{0.0591}{n} \log \frac{[Fe^{2+}]\,[Ag]^2}{[Fe]\,[Ag^+]^2}$$

$$E_{cell} = 1.24 - \frac{0.0591}{2} \log\left(\frac{[0.05]}{[0.1]^2}\right)$$

E_{cell} **=1.2193V**

Figure G000062. Numerical problems based on electrode and cell potential determination · source slide page 3.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

2. For the following concentration cell:
Pt/H_2 (8atm)/HCl(0.3M)/H_2(2atm)/Pt
Calculate potential of the cell at 25°C.

Sol.

$$E_{cell} = \frac{0.0591}{n} \log \frac{p_{H\,2(anode)}}{H\,2(cathode)}$$

$$E_{cell} = \frac{0.0591}{n} \log \frac{8}{2}$$

$$E_{cell} = 0.01779V$$

Figure G000063. Numerical problems based on electrode and cell potential determination · source slide page 4.

EV0000453

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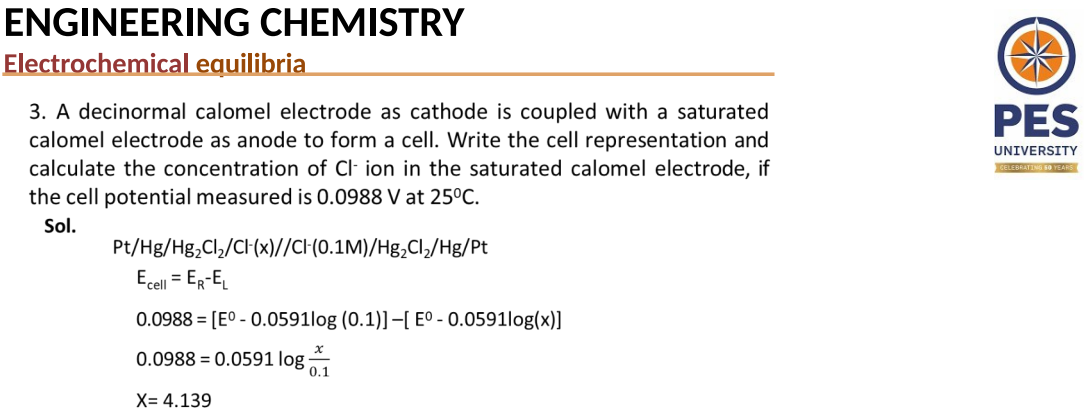

ENGINEERING CHEMISTRY

Electrochemical equilibria

PES UNIVERSITY

3. A decinormal calomel electrode as cathode is coupled with a saturated calomel electrode as anode to form a cell. Write the cell representation and calculate the concentration of Cl^- ion in the saturated calomel electrode, if the cell potential measured is 0.0988 V at 25^{0}C.

Sol.

$Pt/Hg/Hg_2Cl_2/Cl^-(x)//Cl^-(0.1M)/Hg_2Cl_2/Hg/Pt$

$E_{cell} = E_R - E_L$

$0.0988 = [E^0 - 0.0591\log(0.1)] - [E^0 - 0.0591\log(x)]$

$0.0988 = 0.0591 \log \frac{x}{0.1}$

$X = 4.139$

Figure G000064. Numerical problems based on electrode and cell potential determination · source slide page 5.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

4. For the following cell:

Ag/AgCl/Cl⁻(0.1M)// Fe^{2+} (0.29M), Fe^{3+} (0.18M)/Pt

(i) Write the half cell reactions and overall cell reaction.

(ii) Calculate E^{o}_{Cell} and E_{Cell} at 298 K

(Given: $E^{0}_{Fe^{3+}/Fe^{2+}}$ =0.77 V , $E^{0}_{Calomel}$ =0.222 V, R = 8.314 J/K/mol, F =96500 C/mol)

Sol. (i) Anode: $Ag + Cl^- \rightarrow AgCl + e^-$

Cathode : $Fe^{3+} + e^- \rightarrow Fe^{2+}$

Overall : $Ag + Cl^- + Fe^{3+} \rightarrow AgCl + Fe^{2+}$

(ii) $E^{0}_{cell} = E^{o}_{C} - E^{o}_{A}$ = 0.77-0.222 = **0.548V**

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{n} \log \left[\frac{([Fe^{2+}])}{([Fe^{3+}][Cl^-])} \right] \qquad E_{cell} = 0.548 - \frac{0.0591}{1} \log \left[\frac{[0.29]}{[0.18]x[0.1]} \right]$$

$E_{Cell} = 0.4767V$

Figure G000065. Numerical problems based on electrode and cell potential determination · source slide page 6.

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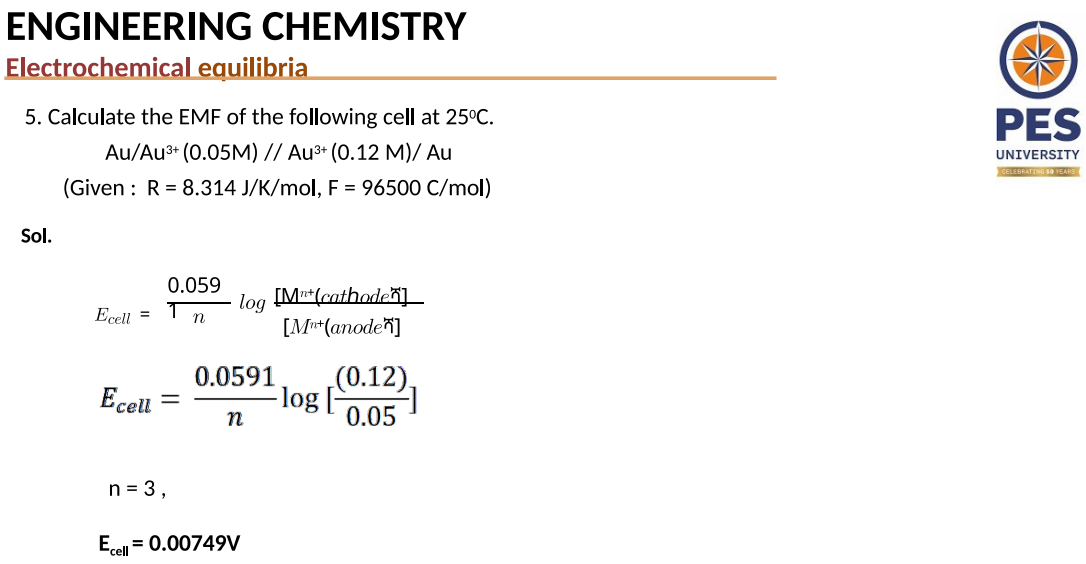

Figure G000066. Numerical problems based on electrode and cell potential determination · source slide page 7.

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ENGINEERING CHEMISTRY

Electrochemical equilibria

6. A decinormal calomel electrode is used to determine the potential of the following redox electrode : Pt/Cu^{2+}(0.58 M),Cu^{+}(0.08M)
(i) Write cell representation.
(ii) Write the reactions at the electrodes
(iii) Calculate E^{o}_{cell} and E_{cell} at 298 K.
(Given : $E^{o}_{Hg/Hg2Cl2/Cl^-}$ = 0.3358V , $E_{Cu2+/Cu}$ = 0.153 V)

Sol. (i) Pt/Cu^{2+}(0.58 M),Cu^{+}(0.08M)//Cl^{-}(0.1 M)/Hg_2Cl_2/Hg

(ii) $2Cu^{+} \longrightarrow 2Cu^{+2} + 2e^{-}$ At anode

$Hg_2Cl_2 + 2e^{-} \longrightarrow 2Hg + 2Cl^{-}$ At cathode

$2Cu^{+} + Hg_2Cl_2 \longrightarrow 2Cu^{+2} + 2Hg + 2Cl^{-}$ Net cell reaction

(iii) $E^{o}_{cell} = E^{o}_{C} - E^{o}_{A}$ = 0.3358 - 0.153 = **0.1828 V**

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{n} \log \frac{([Cl^-]^2[Cu^{2+}]^2}{([Cu\ ]^2}]$$

$$E = E^{0}_{CELL} - \frac{0.0591}{2} \log \frac{0.58^2 \times 0.1^2}{0.08^2}$$

E_{cell} = 0.1745 V

Figure G000067. Numerical problems based on electrode and cell potential determination · source slide page 8.

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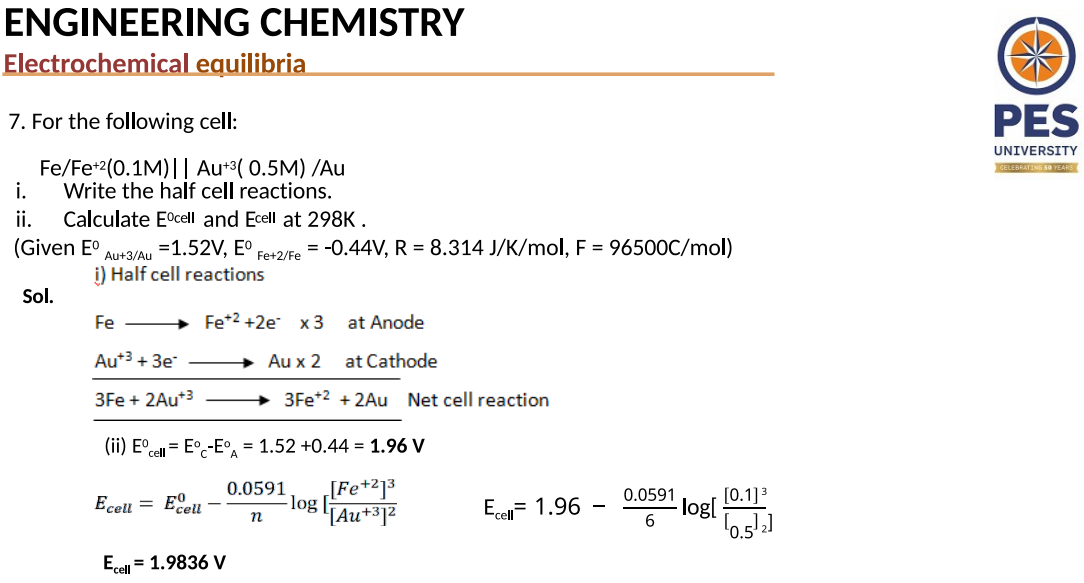

Figure G000068. Numerical problems based on electrode and cell potential determination · source slide page 9.

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Figure G000069. Numerical problems based on electrode and cell potential determination · source slide page 10.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000443 · EV0000444 · EV0000445
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Class 19 content: Numericals on electrochemistry (Nernst equation) EV0000446 · EV0000447 · EV0000448
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 1. For the given cell: Fe/Fe 2+ (0.05M)//Ag+ (0.1M)/Ag Write the overall cell reaction (ii) Calculate E0 and E at 250C cell cell (Given : E0 = - 0.44V ; E0 = 0.80V) Fe+2/ Fe Ag+/Ag Sol. Anode : Fe ☒ Fe2+ + 2e- Cathode : 2Ag+ + 2e- ☒ 2Ag Overall reaction : Fe + 2Ag+ ☒ Fe2+ + 2Ag E0 o o cell= E C-E A = 0.80 + 0.44 =1.24V 0.0591 log([0.05] E = 1.24 − [0.1]2) cell 2 Ecell =1.2193V EV0000449 · EV0000450 · EV0000451
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 2. For the following concentration cell: Pt/H2 (8atm)/HCl(0.3M)/H2(2atm)/Pt Calculate potential of the cell at 250C. Sol. ☒ 0.0591 log pH 2(anode) E cell n p ☒ 0.0591 log 8 n 2 H 2(cathode) E cell Ecell = 0.01779V EV0000452 · EV0000453 · EV0000454
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria EV0000455 · EV0000456 · EV0000457

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 4. For the following cell: Ag/AgCl/Cl-(0.1M)// Fe2+ (0.29M), Fe3+ (0.18M)/Pt Write the half cell reactions and overall cell reaction. Calculate EoCell and ECell at 298 K (Given:E0 3+/ 2+=0.77 V , E0 =0.222 V, R = 8.314 J/K/mol, F =96500 C/mol) Fe Fe Calomel Anode: Ag + Cl- → AgCl + e- Cathode : Fe3+ + e- → Fe2+ Overall : Ag + Cl- + Fe3+→ AgCl + Fe2+ E0cell = EoC-EoA = 0.77-0.222 = 0.548V Sol. ([Fe2+] ന 0.0591 o log[ ] $Ecell = E\ cell - -$ 3+ n ([Fe][Cl]ന ECell= 0.4767V EV0000458 · EV0000459 · EV0000460
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 5. Calculate the EMF of the following cell at 250C. Au/Au3+ (0.05M) // Au3+ (0.12 M)/ Au (Given : R = 8.314 J/K/mol, F = 96500 C/mol) Sol. 0.0591 [Mn+($cathode$ന] $log\ Ecell = n$ [Mn+ ($anode$ന] n = 3 , Ecell = 0.00749V EV0000461 · EV0000462 · EV0000463
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 6. A decinormal calomel electrode is used to determine the potential of the following redox electrode : Pt/Cu2+(0.58 M),Cu+(0.08M) Write cell representation. Write the reactions at the electrodes (iii) Calculate E0 and E at 298 K. cell cell (Given : EoHg/Hg2Cl2/Cl- = 0.3358V , E Cu2+/Cu = 0.153 V) (i) Pt/Cu2+(0.58 M),Cu+(0.08M)//Cl-(0.1 M)/Hg2Cl2/Hg Sol. (ii) (iii) E0cell = EoC-EoA = 0.3358 - 0.153 = 0.1828 V 0.0591 ([$Cl-$]2[Cu2+]2 log[ ] $Ecell = Eocell - n$ ([Cu]2 Ecell = 0.1745 V EV0000464 · EV0000465 · EV0000466
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria 7. For the following cell: Fe/Fe+2(0.1M)|| Au+3(0.5M) /Au Write the half cell reactions. Calculate E0 and E at 298K . cell cell (Given E0 Au+3/Au =1.52V, E0 Fe+2/Fe = -0.44V, R = 8.314 J/K/mol, F = 96500C/mol) Sol. [Image text, OCR] Halfcellreactions Fe Fe+2 +2e x3 atAnode Au+3 + 3e Aux2 at Cathode 3Fe + 2Au+3 3Fe+2 +2Au Netcellreaction (ii) E0cell = EoC-EoA = 1.52 +0.44 = 1.96 V 0.0591 0.1 3 Ecell= 1.96 − log[0.5 2] 6 Ecell = 1.9836 V EV0000467 · EV0000468 · EV0000469
10. Source slide page 10 (extracted text): THANK YOU Department of Science and Humanities EV0000470 · EV0000471 · EV0000472

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello everyone, welcome back to this course on engineering chemistry. [00:00:05.124] We'll continue with module four on electrochemical equilibria and in today's class we are going to discuss some numericals on electrochemistry. First we'll look at some problems on Nernst equation and then we look at a problem on ion selective electrode that is the glass electrode. [00:00:22.436] So the first question is we have been given a cell where we have coupled the iron electrode with the silver electrode. [00:00:30.020] And the values in the parenthesis stand for the concentrations of Fe2 plus ions and AG plus ions. The question says write the overall cell reaction and determine E naught cell and E cell at 25 degrees centigrade. What is given to us is the [00:00:46.512] Standard reduction potential of the Fe two plus Fe electrode that is minus 0.44 volts and the standard reduction potential of AG plus AG that is 0.80 volts. So let us see how to solve this problem. [00:01:01.764] First and foremost we need to write the overall cell reaction, iron being at the anode. [00:01:07.556] We have to write the reaction as Fe giving you Fe2 plus plus 2 electron and this being silver being at the cathode, the reaction will be reduction reaction. So 2 AG plus plus 2 electron is equal to 2 ag. We have multiplied this reaction with 2. [00:01:24.548] In order to balance the number of electrons,

overall reaction is Fe plus 2A G plus is giving you Fe2 plus plus 2 ag. [00:01:34.276] Let us find out the E naught cell, which is given by E naught of the cathode minus E0 of the anode. So the values are mentioned here given to you. So we can substitute. So 0.80 plus 0.44 is equal to 1.24 volts. [00:01:49.988] Now we need to find out E cell at 25 degrees centigrade. We know the Nernst equation. [00:01:57.924] So from the Nernst equation we will be able to determine E cell because we know what is E naught cell. But first and foremost, whenever you have a reaction, when you know the overall reaction, it's always a good practice to find out what is Q. [00:02:13.792] Before you go ahead and substitute in the Narnst equation as done here. Now what would be Q here? We have Fe2 plus and AG the whole square that is concentration of the reactants by Fe [00:02:31.876] and AG plus square. So once we know q and we know e0 cell, we can substitute we also know the number of electrons n that is 2 in this case. So once we substitute all these values we will be able to determine E cell. [00:02:51.460] So it will be 1.24 0.0591 by 2 log of 0.05 by 0.1 square. Now we have not taken AG and FE into consideration because we know that for pure substances. [00:03:07.568] The concentration is taken as 1. So that is that leaves your Q as Fe2 plus by AG plus square, and that's exactly what we have substituted here. And the answer we get is 1.2193 volts. So this is how you're going to determine E cell. [00:03:27.780] Let us come to the next problem where we have to find [00:03:33.284] This potential, cell potential at 25 degrees for a concentration cell. Now when we learnt concentration cells we learnt that there were two types of there are two types of concentration cell. One is the electrode type, the other one is the electrolyte type. [00:03:49.104] Now, first and foremost, let us decide what is the type of concentration cell here. We see that the electrolyte concentration is just one, but the activity of the electrode itself is different. So this will classify as electrode type of. [00:04:09.380] Concentration set. [00:04:12.068] So now we can uh determine what is the cell potential for electrode type of concentration cell. We are calling it electro type electrode type because here the activity of hide the partial pressure of hydrogen is 8 atmosphere. [00:04:29.760] And here the partial pressure of hydrogen at the cathode is 2 atmosphere. Let us find out the solution. [00:04:38.372] We know from the expression E cell is equal to 0.0591 by n log of partial pressure of hydrogen at the anode by partial pressure of hydrogen at the cathode. So what is the reaction actually taking place here? [00:04:55.616] We know that we always consider the reduction reaction H plus plus electron is giving you hydrogen. Now to balance we need two H plus plus two electrons is giving you hydrogen. Now

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.424] At the cathode, this is the reaction. At the anode, the reverse reaction would be taking place. H2 is giving you 2H plus plus 2 electron that is at the anode. So anode partial pressure of hydrogen is 8 atmosphere as is given. [00:05:27.632] And the cathode partial pressure would be 2 atmosphere. When we find out the cell potential expression, it comes out as 0.0591 by n log of partial pressure of hydrogen at anode by partial pressure of hydrogen at the cathode. [00:05:43.328] And understand here E0 is not there because E0 is same for the anode as well as the cathode and hence cancels out. So then we have E cell but just by substitution 8 by 2 for the partial pressures of hydrogen at the anode and cathode. [00:06:00.288] And the answer now is 0.1779 volts. Let us look at a third type of problem where we have a decinormal calomyl electrode. What do you understand by decinormal calomel electrode? We know that the [00:06:19.556] Potential of the calomar electrode depends on the chloride ion concentration or the KCl concentration which is used. So, in if we say a decenol calomar electrode, it is obvious that the concentration of KCl is 0.1 normal, and this is coupled

with [00:06:37.232] A saturated calomel electrode. So saturated calomyl electrode would be what saturated calomyl electrode would be where the Cl solution is a saturated solution. That forms the anode. So this is at the cathode, this is at the anode. [00:06:52.608] We are asked to write the cell representation and calculate the concentration of Cl minus IN in the saturated calomel electrode. Please understand that we already know that the cathode concentration of KCl is 0.1 molar or normal. [00:07:08.832] And we need to determine the chloride ion concentration for the saturated calomel electrode. So if the cell potential measured is 0.0988 volts at 25 degrees centigrade. Now how will we determine this? [00:07:26.500] First, let us write the cell representation. Now we know that at the cathode we have the decinormal electrodes, so calomel electrodes. So this will be the representation for the cathode and at the [00:07:41.328] Anode, it is a saturated calamar electrode for which the concentration is not known. So the aim is now to so this will be at the anode and the aim is to determine what is X. [00:07:56.164] So again we have E cell is given by E right minus E left. So what is E right? The decinormal calomel electrode, and for any calomal electrode, we know the expression is E naught. [00:08:10.020] Minus 0.0591 log of the concentration of Cl minus. Now here looking at the cathode, it's a decinormal calomyl electrode, therefore Cl minus is 0.1, and here it is [00:08:24.848] X because we want we don't know what is the concentration of Cl minus. So we have used the expression for the [00:08:33.124] Calomyl electrode when it is a decenormal calomal electrode, the Nernst equation for the decenormal calomal electrode minus the Nernst equation for the saturated calomel electrode. [00:08:45.092] Therefore, when we substitute we get 0.0988 is the e cell divided by 0.0591 log of x by point one. And therefore, when we substitute the values. [00:08:59.936] We get we need to take antilog over here. When we do the antilog, we find that x is equal to 4.69 molar. So what is it that you have determined here that a saturated solution [00:09:15.076] of KCl which has been used and in the saturated calomin electrode has a concentration 4.69 molar. [00:09:26.756] Come to the next problem now. Here we have silver silver chloride chloride 0.1 molar concentration of chloride ion. So what is this electrode? Can we identify this is a silver silver chloride electrode. [00:09:42.592] Which is a DC normal or having a concentration of 0.1 normal or 0.1 molar chloride concentration. KCl concentration or HCl concentration is 0.1 molar. And the other side we have Fe two plus [00:09:56.688] Which has a concentration 0.29 molar, Fe 3 plus, which has a concentration 0.18 molar, and this is in contact with an inert electrode platinum. If you recognize what is this type of electrode, where we have

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.396] Both the oxidized and the reduced species present together, this is a redox electrode. So we have combined a silver silver chloride electrode with a redox electrode. So what is the question? It says write the half cell reactions and overall cell reaction. [00:10:28.004] And calculate E naught cell and E cell at 298 Kelvin. Given the standard reduction potential values of both the electrodes are given to us, R is [00:10:39.684] 8.314 joules per Kelvin per mole per mole and f is equal to nine six five zero zero faraday is nine six five zero zero coulombs per mole. What is the solution? Let us see. So first we need to write the half cell reactions. [00:10:57.540] So at the anode there should be oxidation. What would be the oxidation reaction for silver silver chloride electrode? We have already learned it. So it will be AG plus Cl minus is giving you AGCl plus electron, and at the cathode [00:11:14.032] We will have the reaction Fe3 plus taking up electron and becoming Fe2 plus. That is the reduction reaction. So overall reaction stands like this AG plus Cl minus plus Fe 3 plus is giving you AG Cl plus Fe2 plus. Whenever we have such a question. [00:11:32.868] It's always a good practice to

write the half cell reactions and then the overall cell reaction, even though it is the question just says write the overall cell reaction. And next, the first thing we should do now is to find out what is Q because this has to be done very carefully. [00:11:52.388] F E two plus by Cl minus. [00:12:01.860] Fe three plus. So we have Fe two plus. [00:12:09.412] Cl minus and Fe 3 plus. That is the value of Q. Now let us see if we can find out E0 cell. Yes, that would be E naught cathode minus E0 anode. The values are given to us. So it is [00:12:25.124] 0.77 here and minus of 0.222. These values are given to us. So it comes out to be 0.548. We got E naught cell. Now let us determine E cell. [00:12:39.520] So we know that again from nonst equation this will be the expression. So E naught cell I have found out here. [00:12:46.336] And N I know is 1 because there's only one electron getting exchanged in the overall reaction also. And now I have an expression here that is the Q expression. How did I get this? This I got from the overall cell reaction. We are the product concentration divided by the [00:13:03.488] Reactant concentration and therefore we find if we substitute 0.548 is the E naught cell and for Q we substitute the values 0.29, 0.18 and 0.1. And we find that [00:13:19.920] The overall cell E cell is 0.4767 volts. Next, let us calculate EMF of the following cell at 25 degrees centigrade. Now looking at the cell, there are two [00:13:38.468] gold electrode au three plus 0.05 molar AU3 plus 0.12 molar AU. So that means we are straight away looking at a concentration cell. So again let us find out what type of concentration cell is this. [00:13:55.456] This is now we can make out the electrolyte concentrations are different. Therefore, this is an electrolyte type of concentration cell. Let us find the solution to this. [00:14:13.796] It will be given by E cell is equal to 0.0591 by N log of Mn plus at the cathode by Mn plus at the anode. So let us see what is N in this case. Now since you can see here the reaction will be AU. [00:14:32.964] Three plus plus three electrons is giving you [00:14:38.372] AU and this will happen at the cathode and at the anode we have the reverse that is AU is giving you AU3 plus plus three electrons. So what is the number of electrons being exchanged? It is three. Therefore, if we substitute three here [00:14:53.488] And we know the concentrations at the cathode and the anode, we find that since n is equal to 3, E cell comes out to be 0.00749 volts. What you have to understand here is that whenever we do concentration cells as explained.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:13.508] During the when I was talking about concentration cells teaching you, e cell will be a very small value. You can see here E cell is a very small value, that's because the main contribution to e cell comes from E naught cell and E0 cell. [00:15:32.288] From the cathode and anode being same as uh E naught of the anode and cathode being same, so E0 cell comes out to be zero. E0 for anode and cathode, they've got cancelled, therefore there is no E0 cell at all, and hence the concentration is [00:15:48.848] The cell potential is 0.00749 volts. That means it's a very small value. [00:15:56.324] Let us look at the next problem where we have a decommal calomel electrode and it is used to determine the potential of the following redox electrode. So we have Cu2 plus and C U plus with this 0.58 and 0.08 Muller concentration respectively. [00:16:14.000] And since both the oxidized and the reduced species are present together in the same solution and an inert wire is placed to collect the potential, this is a redox electrode. We have to write the cell representation, write the reactions at the electrodes, and we have to calculate E naught cell and E cell at 298 Kelvin. [00:16:31.332] So given the standard reduction potential for the calomar electrode that means that the chloride concentration is 1 molar and therefore this is 0.281 volts and C U2 plus C U, the standard reduction potential is given as 0.153. [00:16:51.492] Let us look at the solution. So we have [00:16:55.652] Here uh we have to write

the cell representation. So since decenormal electrode uh calomyl electrode potential is 0.281 and this is point copper is 0.153. Therefore this we place at the anode and this will be the cathode. [00:17:18.980] So now let us look at the reactions. The reactions at anode would be C U plus is giving you CU2 plus plus 2 electron and here at the [00:17:31.140] Cathode we have this reaction, the calom electrode is getting reduced. So this is the reaction we are familiar with. Hg2Cl2 plus 2 electron is giving you 2 H G plus 2 Cl minus. Overall reaction would be 2CU plus plus H2Cl2 is giving you 2 C U2 plus plus 2 Hg plus 2 Cl minus. [00:17:51.524] Now, let us calculate the E0 cell. E0 cell will be E cathode, E0 of the cathode minus E0 of the anode, that is 0.281 minus 0.153, and the value for E0 cell is 0.128 volts. Now let us calculate the [00:18:11.460] E no E cell. For E cell I need Q. So it's always advisable to write the Q here. So first itself, so C U2 plus using the overall reaction. C U2 plus squared. And here it is Cl minus. [00:18:28.848] squared and then that is divided by [00:18:34.724] C u plus square. And so we substitute that expression over here. And n in this case we can see is two electron. So we are going to substitute those values which are given to us for the concentrations. [00:18:49.488] of C U2 plus Cl minus and C U plus and these are substituted here and N is equal to 2. So overall we find that the E cell comes out to be 0.1362 volts. Now let us look at another uh [00:19:09.796] problem where we have Fe, Fe two plus point one molar and AU3 plus concentration is 0.5 molar. [00:19:18.724] AU. Now we have to write the half cell reactions. We have to write the E naught cell and E cell at 298 Kelvin. E naught for the gold electrode is 1.52 volts and E naught [00:19:33.472] For an iron electrode is minus 0.44 volts. R is given as eight point three one four joules per kelvin per mole. F is nine six five zero zero coulombs per mole. Now the solution let us see. [00:19:48.804] We know the half cell reactions that is Fe iron is undergoing oxidation, so Fe 2 plus plus 2 electron, and here we have AU3 plus plus 3 electron at the cathode undergoing reduction A giving you AU.

EV0000122 · EV0000123 · EV0000124 · EV0000125

Transcript block 5 (source interval 001200-001499 seconds): [00:20:04.516] Now, since the number of electrons is not the same, we need to balance it. So this reaction has to be multiplied by 3, and this reaction has to be multiplied by 2. That's exactly what we have done here, and the overall reaction now we get 3 FE plus 2AU3 plus, giving you 3 Fe plus. [00:20:21.520] 2 plus plus 2 AU. That is the net cell reaction. And let us write the Q again. So Q has to be written very, very carefully and [00:20:31.204] As I mentioned before, always good to write the Q first before proceeding with the calculation. So AU3 plus square. [00:20:44.420] So we know what is Q, now we can find out what is E0 cell, E naught cathode minus E0 anode. So here E cathode is the gold electrode, so that is 1.52 minus of minus 0.44 that is for the iron electrode, and we get the answer as 1.96 volts. [00:21:03.876] Now let us find out the E cell as that is what we is to be determined E cell at 298 Kelvin, therefore 0.0591. If it was not at 298 Kelvin, then we would require these values. [00:21:19.536] Because in that case, what is the expression you would use? 2.303 R T over NF. So in that case, we need to substitute the actual temperature, but after substitute 25 degrees centigrade or 298 Kelvin, we get the value of 0.0591 by N log of Fe2 plus cubed. [00:21:39.312] By AU plus 3 squared. That's exactly the Q value here. And N. What is N now here? Because this. [00:21:48.228] has this will become 6, this will also become 6, therefore the number of electrons are going to cancel each other. And then we get n as 6. And when you substitute the substitute the values, this is the expression we get and the final answer we get it as [00:22:04.976] One point nine eight three six votes. [00:22:09.828] Let us come to the last problem. Now this is involving the ion selective electrode that is the glass electrode. So a glass electrode is coupled with a saturated

calomel electrode to measure the unknown pH. So we want to measure the pH of a solution. [00:22:25.888] which is not known to us. So how would you do that? The cell potentials are 0.215 volts and 0.385 volts, which are when the electrode is in contact with the solution of pH is equal to 7 and with the solution of unknown pH. [00:22:43.620] uh respectively. So when we have a solution of 0.215, the pH is 7, and when we have a potential of 0.385, we don't know the pH, and that's exactly what we want to find out. And given [00:22:58.320] is E of saturated calamite electrode is 0.244 volts. So if you remember the glass electrode expression, what do we have? E naught g is given by [00:23:09.888] E cell plus 0.0591 PH plus E saturated calomol electrode. So this expression first will help us to determine E naught G because E naught G value is not known to us unless we use [00:23:26.256] A solution of known pH. So we have to use above a solution of known pH. What is the pH here? 7. [00:23:33.504] So we know the pH here, we can find out what is E0 G. So first and foremost we find out what is E0 G, and that is 0.8727 volts, and then we use this expression. [00:23:47.248] P H is equal to E naught G minus E SE minus E cell divided by 0.0591. So now we know that we know E0 G. [00:23:57.856] And E saturated calomin electrode is known to us, and E cell measured. Now we have to use this value because now the pH is unknown. Here, what is the E cell we use? 0.215. Why? Because this is the [00:24:13.296] pH seven, we know the pH seven. So this was E cell with pH seven. Now here we have point three eight five that is the [00:24:21.348] T cell we are going to use now. When we do that, we will substitute these values and what we get is ph is equal to 4.12. So this is how we are going to solve numericals on this. [00:24:38.128] Module of electrochemical equilibria. Now in the next class we will start with unit three, that is on energy storage devices. Thank you.

EV0000126 · EV0000127 · EV0000128 · EV0000129

Full transcript: 3605 normalized words, 103 timestamped segments, 1492.778667 seconds duration. Low-confidence segments flagged: 0.

EV0000110 · EV0000111 · EV0000112 · EV0000113

8 Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1

This lesson is indexed from the source video titled Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000130 · EV0000131

Graphs and diagrams

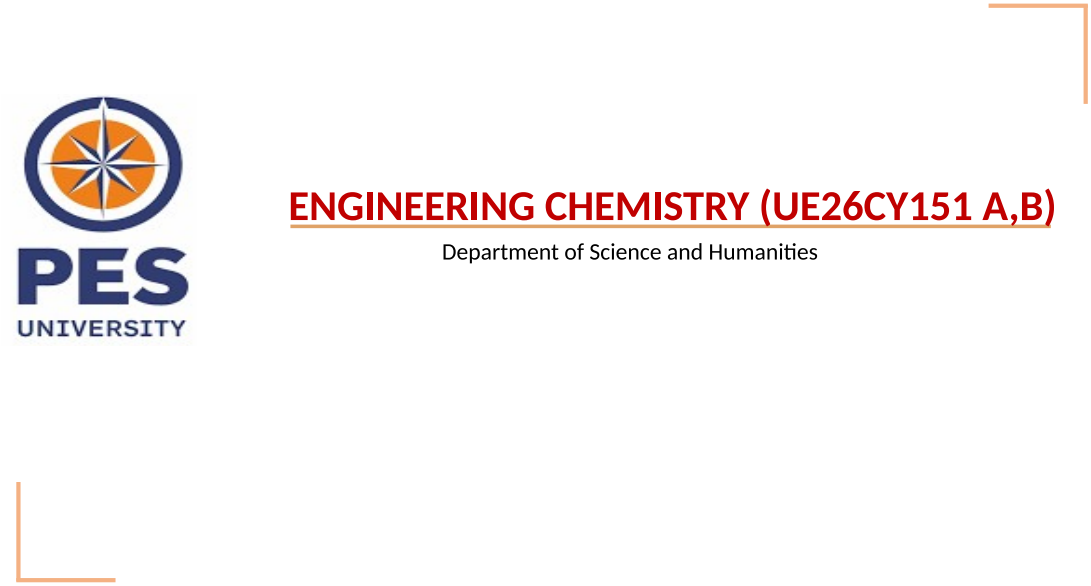

Figure G000070. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 1.

EV0000474

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000071. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 2.

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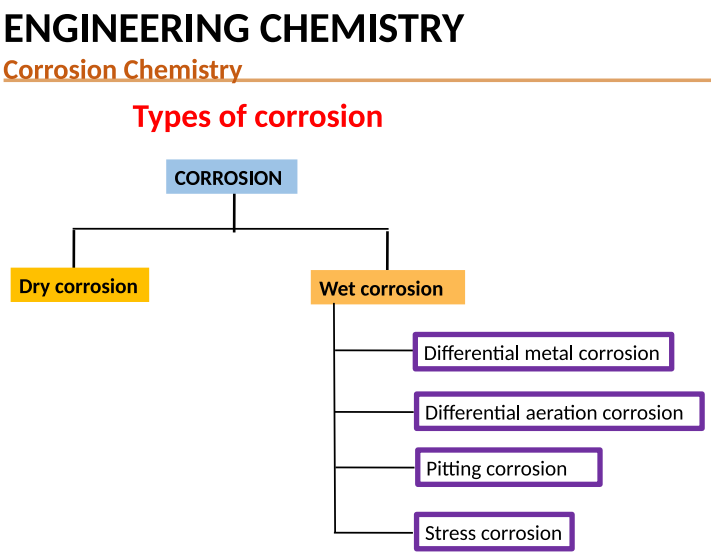

Figure G000072. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 3.

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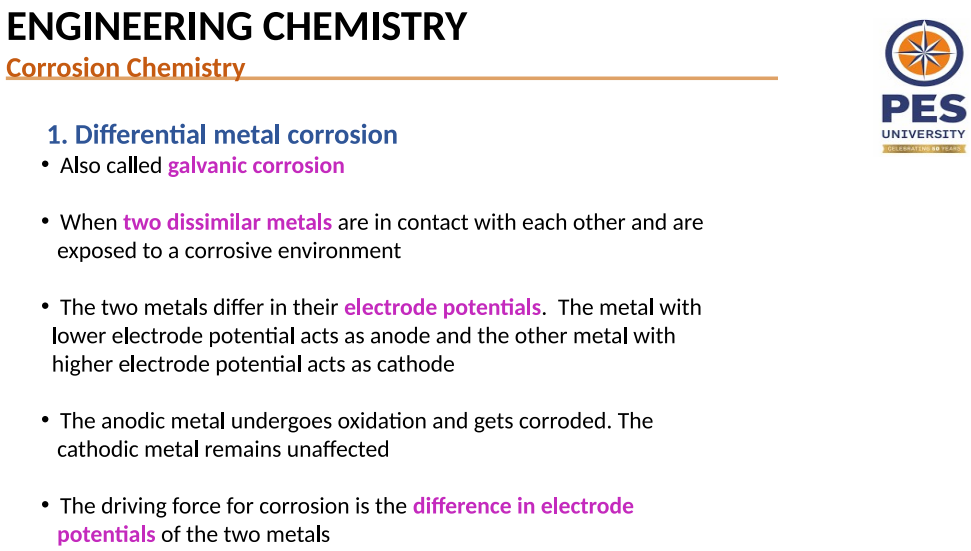

Figure G000073. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 4.

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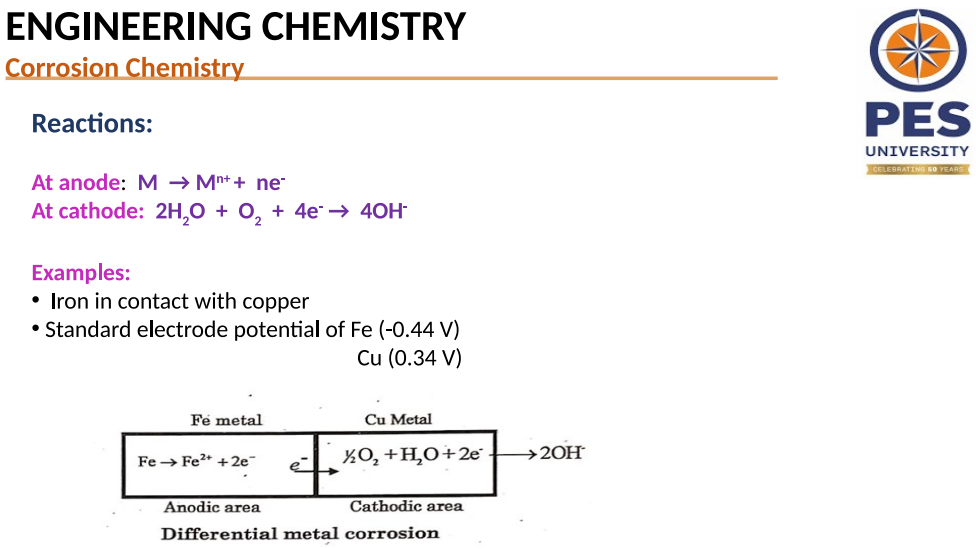

Figure G000074. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 5.

EV0000486

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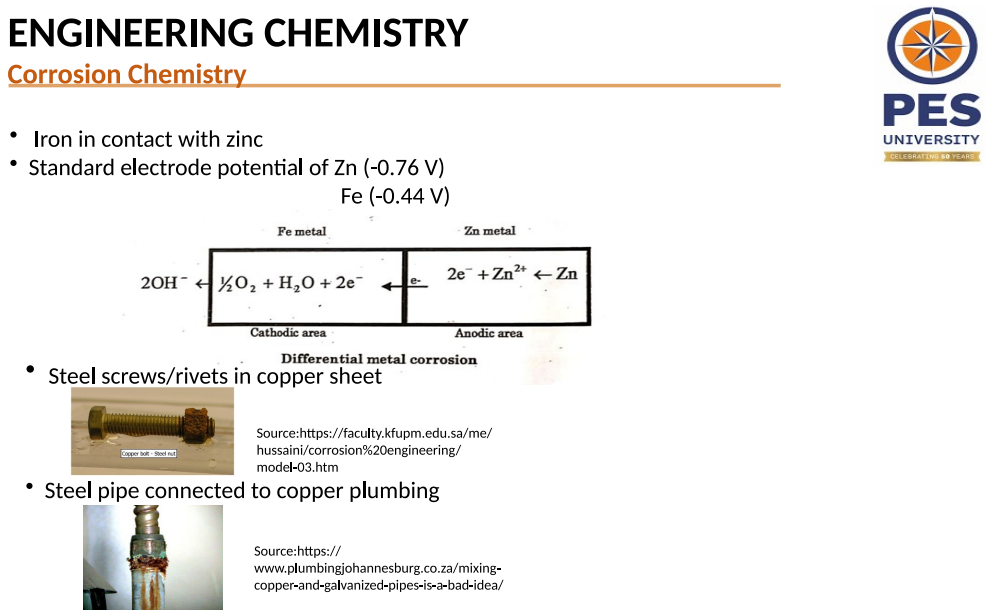

Figure G000075. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 6.

EV0000489

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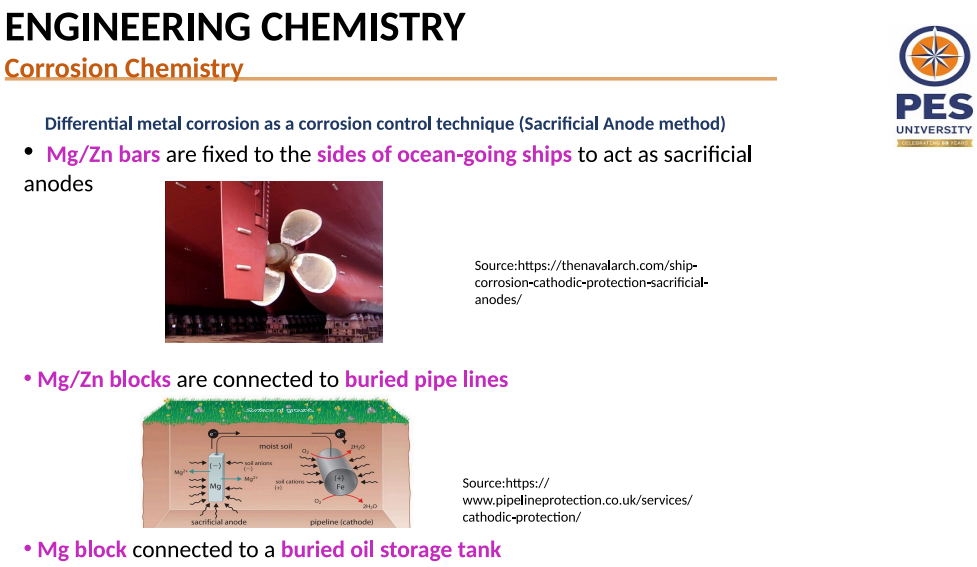

Figure G000076. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 7.

EV0000492

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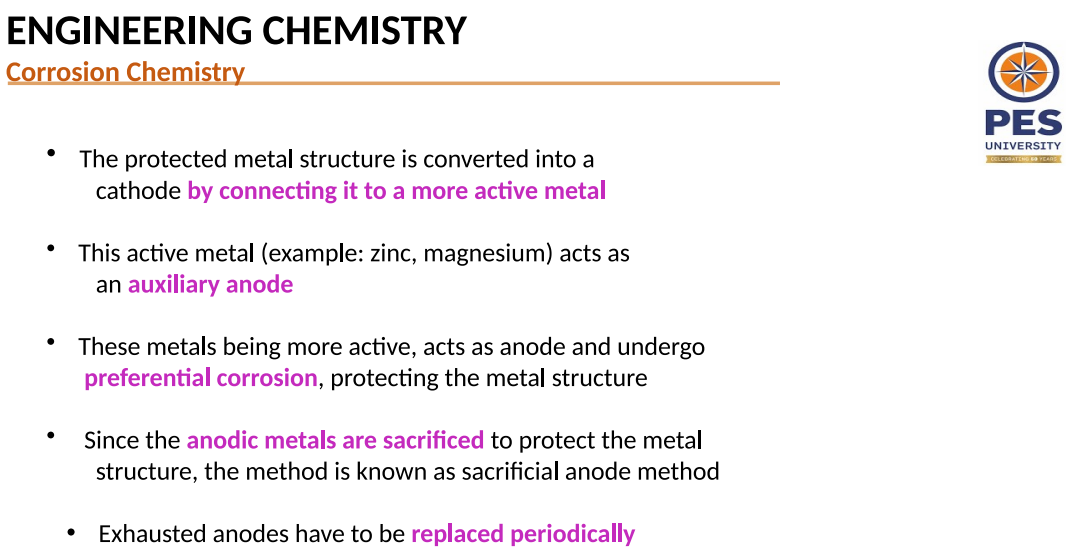

Figure G000077. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 8.

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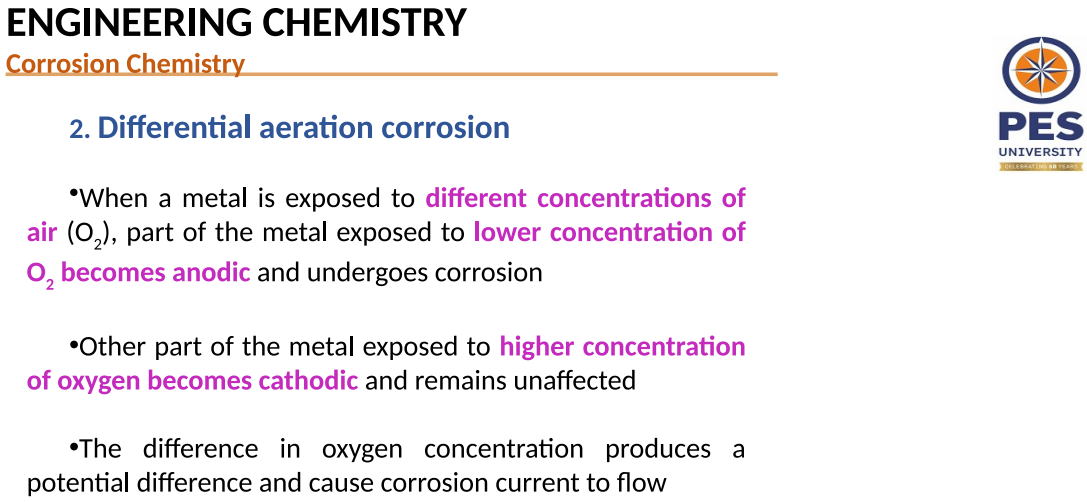

Figure G000078. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 9.

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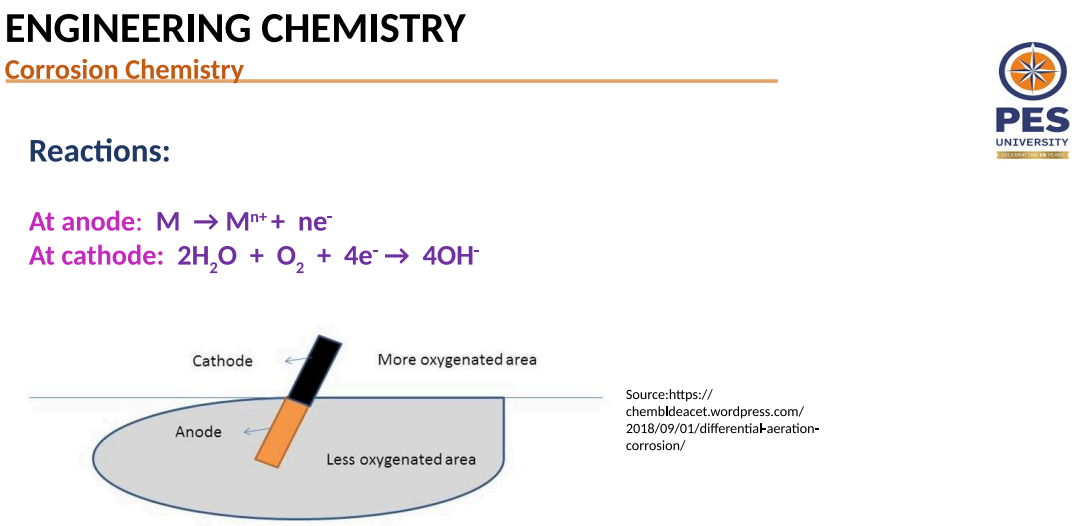

Figure G000079. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 10.

EV0000501

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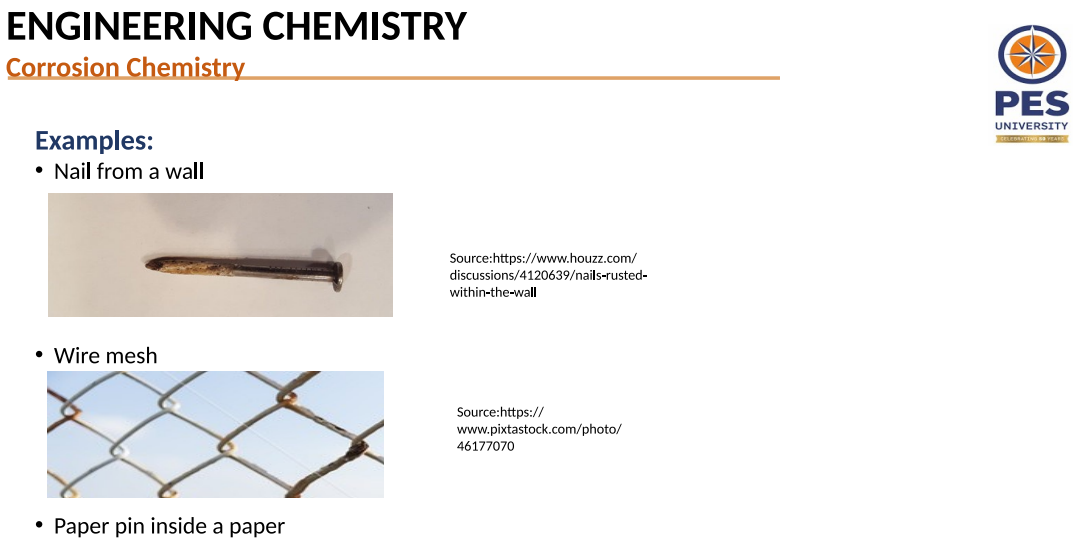

Figure G000080. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 11.

EV0000504

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000081. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 12.

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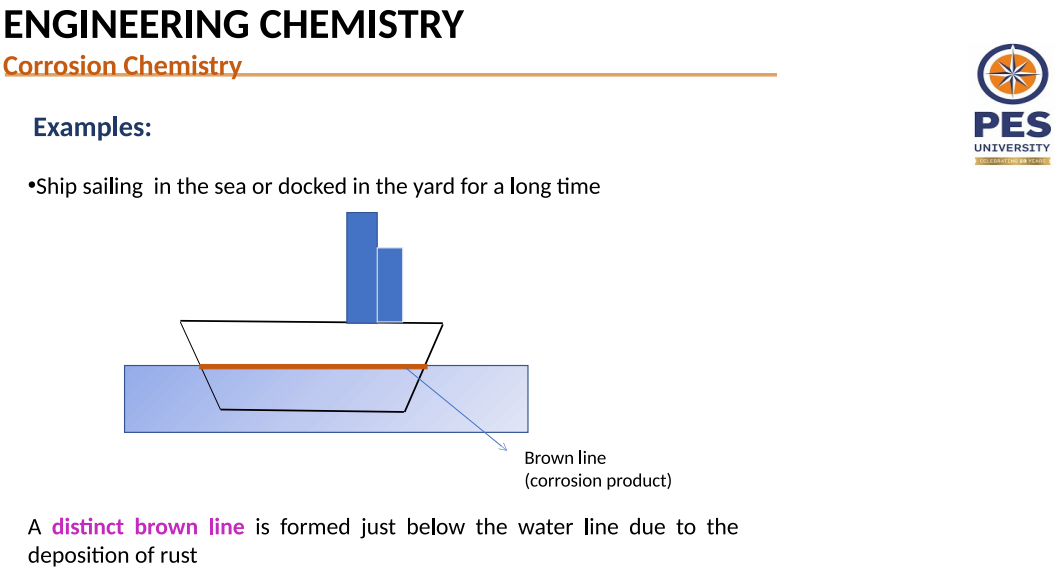

Figure G000082. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 13.

EV0000510

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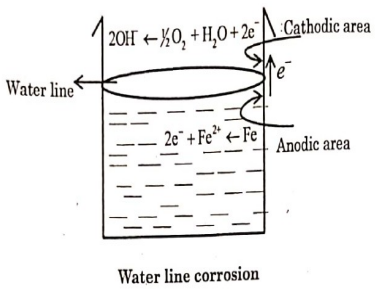
ENGINEERING CHEMISTRY

Corrosion Chemistry

• A steel tank is partially filled with water for a long time



Water line corrosion

Source: Basuchandra's Engineering chemistry, Banbayalu (2014)

• The portion below the water line acts as anode and undergoes corrosion. The upper portion acts as cathode and remains unaffected

Figure G000083. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 14.

EV0000513

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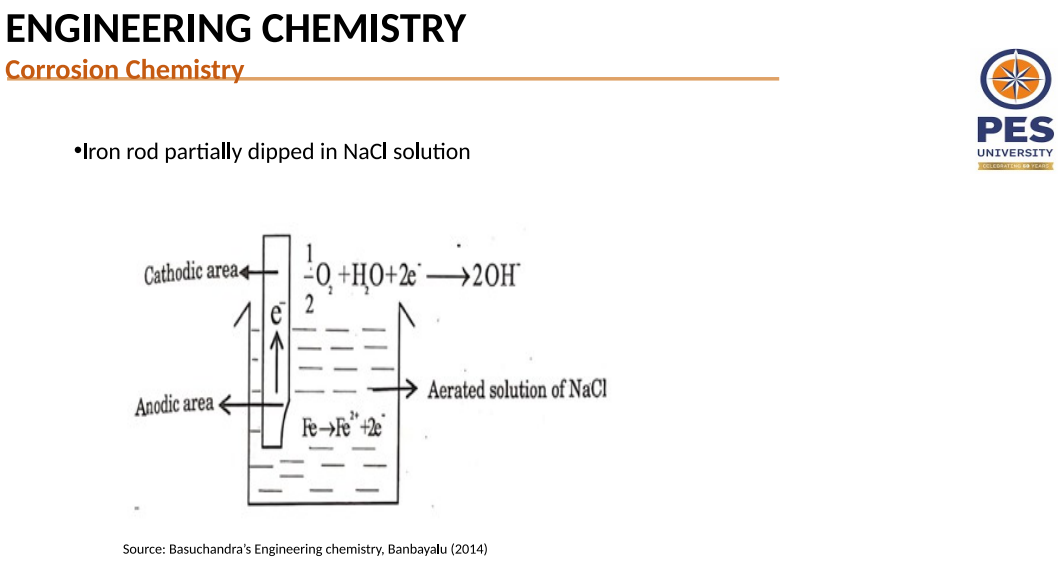

Figure G000084. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 15.

EV0000516

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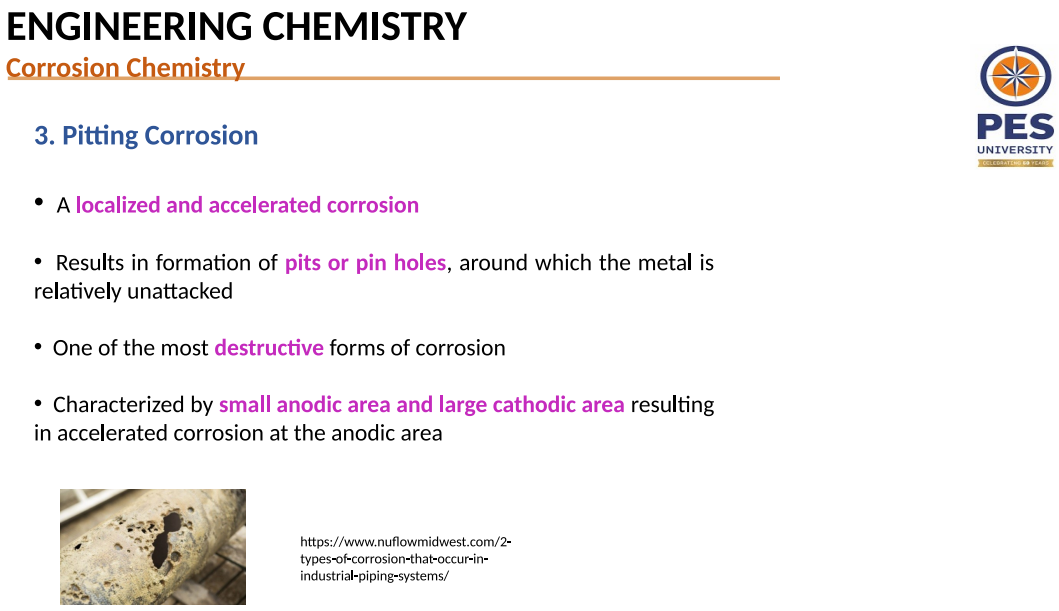

Figure G000085. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 16.

EV0000519

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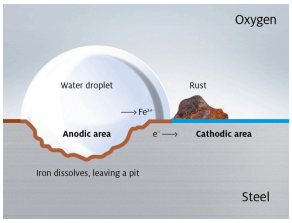
ENGINEERING CHEMISTRY

Corrosion Chemistry

- **Case 1**

- When **dust particles or oil drops** get deposited over the metal surface
- The portion of the metal covered by dust which is less aerated becomes anodic
- The adjacent area of the metal which is exposed to higher concentration of O_2 becomes cathodic



https://www.bonderite.com/en/technologies/pretreatment/corrosion-protection.html

Figure G000086. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 17.

EV0000522

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

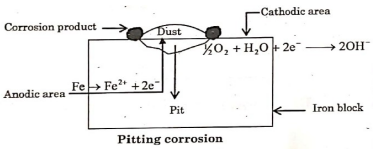
Corrosion Chemistry

- A small anodic region surrounded by a large cathodic region is formed
- The demand for electrons is high from the cathode
- Metal underneath the surface of dust particle being anode undergoes accelerated corrosion forming a deep and narrow pit

•Reactions:

At anode: $M \rightarrow M^{n+} + ne^-$

At cathode: $2H_2O + O_2 + 4e^- \rightarrow 4OH^-$



Source: Basuchandra's Engineering chemistry, Banbayalu (2014)

Figure G000087. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 18.

EV0000525

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

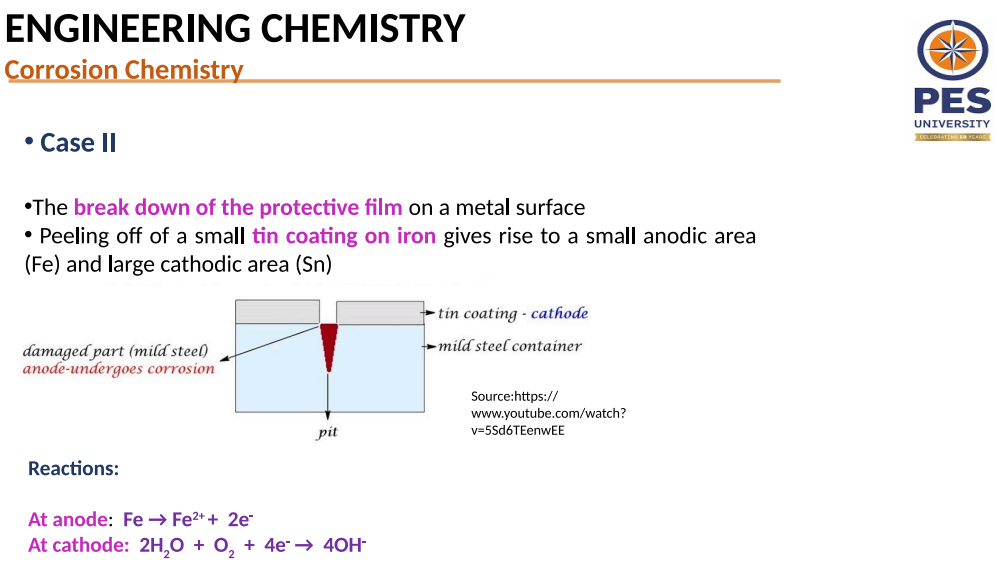


Figure G000088. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 19.

EV0000528

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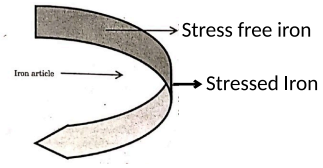
ENGINEERING CHEMISTRY

Corrosion Chemistry

4. Stress Corrosion:

- Also called **Stress Corrosion Cracking**(SCC)
- A type of corrosion that occurs when some part of the metallic material is **under stress** and exposed to **specific corrosive environment**
- During the manufacture or fabrication of the articles, when the metals are subjected to **mechanical operations** such as pressing, hammering, rolling, bending, quenching, welding and riveting
- The stress can be **external stress** acting on the metal during service conditions **or residual stress or both**



Source: Basuchandra's Engineering chemistry, Banbayalu (2014)

Figure G000089. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 20.

EV0000531

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

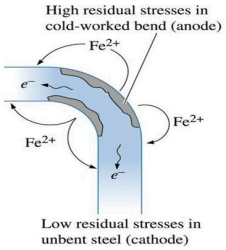
ENGINEERING CHEMISTRY

Corrosion Chemistry

•Due to stress a **crack is initiated**

•The metal atoms under stress are always at **higher energy levels** therefore becomes **more reactive** than the stress free part

•As a result, a corrosion cell is formed with the **stressed part acting as anode** and the **stress free part acting as cathode**



Source:https://slideplayer.com/slide/8594552/

Figure G000090. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 21.

EV0000534

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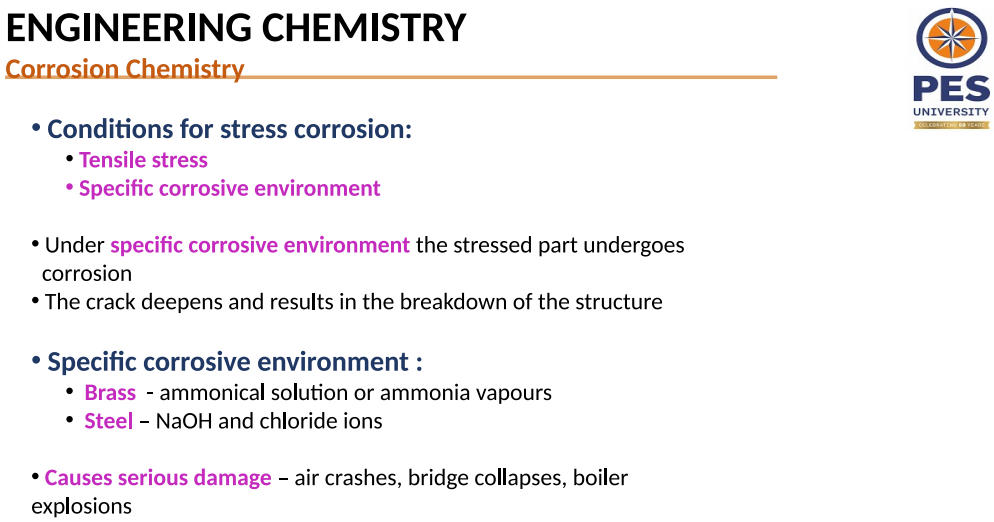

Figure G000091. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 22.

EV0000537

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000092. Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio · source slide page 23.

EV0000540

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000473 · EV0000474 · EV0000475
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Class content: Types of electrochemical corrosion EV0000476 · EV0000477 · EV0000478
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Types of corrosion CORROSION Dry corrosion Wet corrosion Differential metal corrosion Differential aeration corrosion Pitting corrosion Stress corrosion EV0000479 · EV0000480 · EV0000481
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 1. Differential metal corrosion Also called galvanic corrosion When two dissimilar metals are in contact with each other and are exposed to a corrosive environment The two metals differ in their electrode potentials. The metal with lower electrode potential acts as anode and the other metal with higher electrode potential acts as cathode The anodic metal undergoes oxidation and gets corroded. The cathodic metal remains unaffected The driving force for corrosion is the difference in electrode potentials of the two metals EV0000482 · EV0000483 · EV0000484
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Reactions: At anode: M → Mn+ + ne- At cathode: 2H2O + O2 + 4e- → 4OH- Examples: Iron in contact with copper Standard electrode potential of Fe (-0.44 V) Cu (0.34 V) [Image text, OCR] Femetal CuMetal Fe→Fe²++2e O+HO+2e >2OH Anodicarea Cathodicarea Differentialmetalcorrosion EV0000485 · EV0000486 · EV0000487

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Iron in contact with zinc Standard electrode potential of Zn (-0.76 V) Fe (-0.44 V) [Image text, OCR] Femetal Zn metal 2e+Zn²+←Zn 20H4 %O2+H2O+2e Cathodic area Anodic area Differentialmetalcorrosion Steel screws/rivets in copper sheet Steel pipe connected to copper plumbing Source:https://faculty.kfupm.edu.sa/me/hussaini/corrosion%20engineering/model-03.htm Source:https://www.plumbingjohannesburg.co.za/mixing-copper-and-galvanized-pipes-is-a-bad-idea/ EV0000488 · EV0000489 · EV0000490
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Differential metal corrosion as a corrosion control technique (Sacrificial Anode method) Mg/Zn bars are fixed to the sides of ocean-going ships to act as sacrificial anodes Mg/Zn blocks are connected to buried pipe lines Mg block connected to a buried oil storage tank Source:https://thenavalarch.com/ship-corrosion-cathodic-protection-sacrificial-anodes/ Source:https://www.pipelineprotection.co.uk/services/cathodic-protection/ EV0000491 · EV0000492 · EV0000493
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry The protected metal structure is converted into a cathode by connecting it to a more active metal This active metal (example: zinc, magnesium) acts as an auxiliary anode These metals being more active, acts as anode and undergo preferential corrosion, protecting the metal structure Since the anodic metals are sacrificed to protect the metal structure, the method is known as sacrificial anode method Exhausted anodes have to be replaced periodically EV0000494 · EV0000495 · EV0000496
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 2. Differential aeration corrosion When a metal is exposed to different concentrations of air (O2), part of the metal exposed to lower concentration of O2 becomes anodic and undergoes corrosion Other part of the metal exposed to higher concentration of oxygen becomes cathodic and remains unaffected The difference in oxygen concentration produces a potential difference and cause corrosion current to flow EV0000497 · EV0000498 · EV0000499
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Reactions: At anode: M → Mn+ + ne- At cathode: 2H2O + O2 + 4e- → 4OH- [Image text, OCR] Cathode Moreoxygenatedarea Anode Less oxygenated area At the anode,Less oxygenatedarea M >Mn++ne At the cathode,more oxygenatedarea O2+H2O+2e- 2OH- Source:https://chembldeacet.wordpress.com/2018/09/01/differential-aeration-corrosion/ EV0000500 · EV0000501 · EV0000502
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Examples: Nail from a wall Wire mesh Paper pin inside a paper Source:https://www.houzz.com/discussions/4120639/nails-rusted-within-the-wall Source:https://www.pixtastock.com/photo/46177070 EV0000503 · EV0000504 · EV0000505
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Waterline corrosion: A special case of differential aeration corrosion Observed when metal is half immersed in water The part immersed in water is exposed to less O2, hence acts as anode and gets corroded while the part not in water is exposed to more O2 and acts as cathode; remains unaffected by corrosion EV0000506 · EV0000507 · EV0000508

13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Examples: Ship sailing in the sea or docked in the yard for a long time A distinct brown line is formed just below the water line due to the deposition of rust Brown line (corrosion product) EV0000509 · EV0000510 · EV0000511

14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry A steel tank is partially filled with water for a long time The portion below the water line acts as anode and undergoes corrosion. The upper portion acts as cathode and remains unaffected [Image text, OCR] 2OH ←O, +HO + 2e Cathodicarea Waterline 2e+Fe²+←Fe Anodicarea Waterlinecorrosion Source: Basuchandra's Engineering chemistry, Banbayalu (2014) EV0000512 · EV0000513 · EV0000514

15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Iron rod partially dipped in NaCl solution [Image text, OCR] Cathodicarea 0 +HO+2e—→2OH 2 丰 AeratedsolutionofNaCl Anodic area Fe-→Fe"+2e =三= Source: Basuchandra's Engineering chemistry, Banbayalu (2014) EV0000515 · EV0000516 · EV0000517

16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 3. Pitting Corrosion A localized and accelerated corrosion Results in formation of pits or pin holes, around which the metal is relatively unattacked One of the most destructive forms of corrosion Characterized by small anodic area and large cathodic area resulting in accelerated corrosion at the anodic area https://www.nuflowmidwest.com/2-types-of-corrosion-that-occur-in-industrial-piping-systems/ EV0000518 · EV0000519 · EV0000520

17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Case 1 When dust particles or oil drops get deposited over the metal surface The portion of the metal covered by dust which is less aerated becomes anodic The adjacent area of the metal which is exposed to higher concentration of O2 becomes cathodic [Image text, OCR] Oxygen Waterdroplet Rust >Fe2+ Anodic area Cathodicarea Iron dissolves,leaving a pit Steel https://www.bonderite.com/en/technologies/pretreatment/corrosion-protection.html EV0000521 · EV0000522 · EV0000523

18. Source slide page 18 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry A small anodic region surrounded by a large cathodic region is formed The demand for electrons is high from the cathode Metal underneath the surface of dust particle being anode undergoes accelerated corrosion forming a deep and narrow pit Reactions: At anode: M → Mn+ + ne- At cathode: 2H2O + O2 + 4e- → 4OH- [Image text, OCR] Cathodicarea Corrosionproduct Diust %O+H2O +2e →2OH Fe→Fe²++2e Anodicarea Pit Ironblock Pitting corrosion Source: Basuchandra's Engineering chemistry, Banbayalu (2014) EV0000524 · EV0000525 · EV0000526

19. Source slide page 19 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Case II The break down of the protective film on a metal surface Peeling off of a small tin coating on iron gives rise to a small anodic area (Fe) and large cathodic area (Sn) [Image text, OCR] RP PITTINGCORROSION/ GALVANICCORROSION tin coating-cathode mildsteelcontainer damagedpart(mildsteel) anode-undergoescorrosion pit Damagedtincoatedmildsteelcontainerleftoutinrain Note:Thesameeffectoccurswhentheprotectivemetaloxide layeroverthemetalgetsdamaged. Source:https://www.youtube.com/watch?v=5Sd6TEenwEE Reactions: At anode: Fe → Fe2+ + 2e- At cathode: 2H2O + O2 + 4e- → 4OH- EV0000527 · EV0000528 · EV0000529

20. Source slide page 20 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 4. Stress Corrosion: Also called Stress Corrosion Cracking(SCC) A type of corrosion that occurs when some part of the metallic material is under stress and exposed to specific corrosive environment During the manufacture or fabrication of the articles, when the metals are subjected to mechanical operations such as pressing, hammering, rolling, bending, quenching, welding and riveting The stress can be external stress acting on the metal during service conditions or residual stress or both Stress free iron Source: Basuchandra's Engineering chemistry, Banbayalu (2014) Stressed Iron EV0000530 · EV0000531 · EV0000532
21. Source slide page 21 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Due to stress a crack is initiated The metal atoms under stress are always at higher energy levels therefore becomes more reactive than the stress free part As a result, a corrosion cell is formed with the stressed part acting as anode and the stress free part acting as cathode [Image text, OCR] StressCorrosionCracking Examples of stress cells Coldworkrequired tobend a steel barintroduceshighresidual stresses at thebend,whichthenisanodicandcorrodes. Highresidualstressesin cold-workedbend(anode) Fe2+ Fe2+ Lowresidualstressesin unbentsteel(cathode) Source:https://slideplayer.com/slide/8594552/ EV0000533 · EV0000534 · EV0000535
22. Source slide page 22 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Conditions for stress corrosion: Tensile stress Specific corrosive environment Under specific corrosive environment the stressed part undergoes corrosion The crack deepens and results in the breakdown of the structure Specific corrosive environment : Brass - ammonical solution or ammonia vapours Steel – NaOH and chloride ions Causes serious damage – air crashes, bridge collapses, boiler explosions EV0000536 · EV0000537 · EV0000538
23. Source slide page 23 (extracted text): THANK YOU Department of Science and Humanities EV0000539 · EV0000540 · EV0000541

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.092] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module six on corrosion chemistry and today we will discuss the types of electrochemical corrosion, that is differential metal corrosion and differential aeration corrosion. [00:00:21.636] When we say types of corrosion, we have studied that there are two types. One is dry corrosion and the other one is wet corrosion. Dry corrosion, as we understand, is due to [00:00:33.584] the chemical attack of the environment on the metal surface and it happens in the absence of any conducting medium. But for wet corrosion it has to take place only in conducting medium because what we have [00:00:51.824] is an electrochemical attack. And since most of the times metals are exposed to conducting medium because there's always moisture in the air, therefore wet corrosion is of more importance to us. So we need to understand what are the different mechanisms [00:01:09.552] by which this electrochemical attack can take place. So we have different types of corrosion which can happen, that is differential metal corrosion, differential aeration corrosion, piting corrosion and stress corrosion. [00:01:27.108] Let us understand what is differential metal corrosion. Now, this is also referred to as galvanic corrosion. When we have two [00:01:36.048] Dissimilar metals in contact with each other and they are exposed to the corrosive environment, that is when we say galvanic corrosion will take place or differential metal corrosion. Now this is a very common situation which we come across where we have two. [00:01:53.744] Two different metals in contact with each other.

Therefore, galvanic corrosion can happen very frequently in metals. Now why does this happen? When we have two different metals [00:02:08.080] They will have different electrode potentials. The moment they are in contact with each other, one of them which has a greater oxidation tendency becomes the anode, and the one which has lesser oxidation tendency will remain as the cathode. [00:02:25.536] Now, because there will always be in a change in the electrode potentials, this situation [00:02:32.496] or formation of anode and cathode can take place very easily. Now because we are we are talking of electrochemical attack, the most important thing which we should see under any situation is how are this anode and cathode getting formed on the surface and in the presence of the [00:02:48.688] uh medium conducting medium we will have corrosion happening. Now here because one of the metal has a lower electrode potential or rather [00:03:00.580] Negative electrode potential that becomes the anode, the other automatically becomes a cathode. And we know that it is only the anode which undergoes oxidation and hence gets corroded, whereas the cathode, the metal which has become cathode, will remain unaffected by carrot. [00:03:18.352] This is what we learned in the previous class when we talked about electrochemical theory of corrosion. Now the driving force for corrosion to happen here is the difference in the electrode potential. More the difference in the electrode potentials of the two metals. [00:03:35.536] More is going to be the corrosion which will happen, of course, it is only the anode which will corrode. So the difference in the electrode potential will be the driving force for this type of corrosion to take place. [00:03:52.260] Let us see what will be the reactions. Now the metal part which acted as the anode obviously undergoes oxidation. So M is giving you Mn plus plus N electrons, and at the cathode we have water and [00:04:05.828] Uh in the presence of oxygen and four electrons giving you four OH minus. Now why did we take these reactions in electrochemical theory of corrosion? We learned that at the cathode there can be different reactions based on the medium. Now we'll consider this most commonly happening because there's always plenty of oxygen. [00:04:23.600] In the atmosphere, and there is moisture. So, therefore, this is the most common reaction which can happen at the cathode. Now, let us take up some examples. Like we have iron in contact with copper. [00:04:37.504] Now let us look at what is the electrode potential of iron. It is minus 0.44, and copper is 0.34. So if you look at this situation where we have iron and copper in contact with each other, we will understand from the electrode potentials that [00:04:53.904] immediately iron is going to become the anode and [00:04:58.960] Copper will become the cathode. So at the anode we know that there will be corrosion. The metal will get converted into uh its ions, therefore the metal is getting destroyed. Whereas at the cathode we know that copper cannot undergo reduction. So copper remains completely

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.592] Unaffected by this type of corrosion when it is in contact with iron. And what will be the reaction which is happening? Just now we said that oxygen with the H2O plus two electron is giving you 2 OH minus. So [00:05:33.828] Another example, let us take where we have now iron in contact with zinc. Earlier it was iron in contact with copper. Now when you check the electrode potential, standard electrode potential of zinc is minus 0.76, whereas iron is minus 0.44. Therefore, what will happen? [00:05:51.600] Zinc is a more active metal as compared to iron. Therefore, zinc will now act as the anode and the cathode will now be the iron. So in the case where iron was in contact with copper, [00:06:06.704] Iron was undergoing corrosion, whereas when it is in contact with zinc, it is zinc which will undergo corrosion. So here zinc is getting [00:06:15.696] converted to z in two plus plus

two electrons and these electrons are moving towards the cathode and at the cathode we have these electrons being taken up with water and oxygen giving you hydroxyl ions. So the product here in this case will be zinc hydroxide whereas [00:06:33.328] In the first case, where we had iron in contact with copper, we will get copper hydroxide iron hydroxide. Now there could where do we commonly find this kind of corrosion happening? Very common is when we have steel screws or rivets in copper sheets. [00:06:50.896] Now when we have this copper sheet and we have these steel screws, then that will what will happen we notice that it is the steel screws which are getting [00:07:03.312] uh corroded whereas the copper remains intact. Now here we have a case where we have a copper uh bolt and iron it or a steel nut. So we have the copper bolt and so bolts and nuts [00:07:20.644] we see many times that they are made of different metals. Now since we have copper here and this is steel, what will we observe? That it we observe here in this case, in this picture, that it is the steel [00:07:32.480] uh nut which is getting uh corroded whereas the bolt remains absolutely intact. And similarly if we have a copper sheet with steel screws, there also we will find that the copper sheet remains in sack intact and it's only the [00:07:46.320] uh the steel screws which will get corroded. And very common again you can see this nice picture which is showing the steel pipe connected to copper plumbing. So [00:07:57.476] Now we have of course shifted to plastic pipes, but earlier steel pipes used to be used and we have this copper plumbing holding the steel pipe. So we will observe here you see that the edges get completely corroded after some time. But what is getting corroded? It is only the [00:08:15.248] the steel pipe and not the copper plumbing. So this is a very clear example of galvanic corrosion or differential metal corrosion. [00:08:26.500] Now let us talk about another type of corrosion that is differential aeration corrosion. Now as the name suggests, now we have the same metal getting exposed to different concentrations of air or oxygen. [00:08:42.432] Uh this is very commonly seen if you have an iron rod which is lying, you can have a part of it which is exposed to different uh different types of uh uh [00:08:54.208] concentration of the oxygen could be different in one area and it could be different in the other area. So there will be difference in the concentration of oxygen, and we find that the part of the metal [00:09:09.392] Which is exposed to lower concentration of oxygen becomes anodic in nature and undergoes corrosion, whereas the other part of the metal which has a higher oxygen concentration becomes cathodic and is unaffected by the [00:09:24.208] Corrosion. Now, the because of the difference in this oxygen concentration, we are getting something like a oxygen concentration cell on the surface of the metal. And this potential difference will [00:09:39.012] uh cause the uh current to a small amount of current corrosion current to flow. Let us see what are the reactions and you can see here [00:09:51.940] We have anode the same metal will undergo corrosion. So M is giving you Mn plus plus N electrons, and at the cathode we have oxygen taking part in the reaction. And because we are talking of less oxygenated and more oxygenated, so this is the reaction we are going to consider here.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.712] Oxygen absorption is the reaction we are going to look at. And let us assume that it's a neutral medium. So we chose this reaction. Now let us see how exactly this anode and cathode are getting formed. We have one part which is [00:10:24.224] Immersed in water or is less oxygenated, and the other part is more oxygenated. It could be water, it could be any other area which is less oxygenated. One part is exposed to less oxygenated, another part is exposed to uh more [00:10:41.872] oxygenated region so that is exposed to more oxygen. Now between the two, which should act as anode and which

should act as cathode. [00:10:51.332] Now because the oxygen absorption takes place at the cathole, so higher concentration of oxygen wherever we have. [00:11:02.192] That area becomes the cathode, whereas the other part where there is less oxygen now becomes the anode, and because it's the anode which undergoes corrosion, that is the metal will take part in this reaction here, whereas here there's a lot of oxygen, so oxygen will be involved. [00:11:19.824] the is always involved at the cathodic uh region so this area exposed to more oxygen is going to become the cathode whereas the part exposed to less oxygen is going to become the anode. [00:11:36.964] Now, let us look at some examples of differential aeration corrosion. This is a wonderful example. When you pull out a nail from the wall, what do you observe? The part of the nail which is inside the wall, you can see here the part which was inside the wall. [00:11:54.736] The wall has got rusted, whereas the part which was outside did not get rusted. The reason that because this part was exposed to less oxygen, that this acted as the cathode and the part exposed to uh [00:12:11.492] Sorry, this part which was exposed to less oxygen became the anode and the part exposed to more oxygen now has become the cathode. So we see [00:12:27.812] We see in this case that the part exposed to [00:12:32.192] Exposed to less oxygen becomes the anode and the part exposed to more oxygen becomes the cathode. And therefore we see the formation of anode and cathode on the surface of the nail and the part inside the wall which was exposed to less oxygen is getting the same. [00:12:49.904] getting corroded whereas the part exposed to more oxygen is not getting corroded. Another good example would be a wire mesh. As you can see in the picture, the part where they are crossing each other, the two [00:13:06.464] wires are crossing crossing each other, you can see that it has got corroded, whereas the part exposed [00:13:14.212] outside it is not getting corroded. The reason again the part which is crossing each other there the amount of oxygen is less and therefore that acts as the anode whereas the part exposed to more oxygen now this other part so [00:13:31.984] These parts you can see are more corroded. You can see the yellowish orangish color over here, the rust getting formed, whereas the other part look intact. That is acting as the cathode. So again, this is a very good example where you can see that the lower oxygen, wherever there is less oxygen, [00:13:49.552] That part is getting corroded, whereas the part exposed to more oxygen is not getting corroded. And another example would be paper pen inside a paper left for a long time. [00:14:02.624] What do we see there? The part which is inside the paper has got corroded. All of you would have experienced this. The part which is inside the paper has got corroded, whereas the part which was outside of the paper has not got corroded. Again, this is a very clear [00:14:20.272] example of differential aeration corrosion the part which was inside the it was exposed to less oxygen and hence acted as the anode and got corroded the part which was outside is exposed to more oxygen and therefore it acted as [00:14:37.840] the cathode and did not get corroded. [00:14:42.084] So this is about differential aeration corrosion. Now in tomorrow's class, in the next class, we are going to talk about a special type of differential aeration corrosion. [00:14:58.848] And that is waterline corrosion. And we also go to another type of corrosion that is pitting corrosion. Thank you.

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Full transcript: 2231 normalized words, 62 timestamped segments, 910.186667 seconds duration. Low-confidence segments flagged: 0.

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9 Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II

This lesson is indexed from the source video titled Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000142 · EV0000143

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:02.500] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module six on corrosion chemistry and we will talk about different types of electrochemical corrosion and today's discussion will be on waterline corrosion and petting corrosion. [00:00:24.420] Last class we discussed about differential aeration corrosion where an oxygen concentration cell was getting formed due to difference in the concentration of oxygen to which the metal surface is exposed. [00:00:40.368] Now waterline corrosion is a special case of differential aeration corrosion. [00:00:46.016] This happens when a metal is half immersed in water. Now when we have a metal half immersed in water, we know that the part which is inside the water is exposed to only the dissolved oxygen, and therefore the oxygen concentration is [00:01:03.696] very less whereas the part exposed to more oxygen is now [00:01:09.872] uh in the air I mean is exposed to air therefore that part has it has been exposed to more oxygen and as we studied in differential radiation corrosion the part exposed to less oxygen is going to act as the anode and therefore the part inside the water we find [00:01:27.504] gets corroded whereas the part outside is not getting corroded because that is acting as cathode that remains intact whereas the part inside the water is getting corroded. So this is [00:01:40.352] due to the difference in the concentration of oxygen on the surface of the metal. And when this happens we call it as waterline corrosion. [00:01:51.460] Let us see, take up some examples and understand this. Now here, suppose we have a ship sailing in the sea or docked in the yard for a long time. Now when there is a ship sailing in the sea, then we find that partly it is immersed in water, whereas the other part is exposed to the [00:02:10.192] Uh atmospheric oxygen. Now the part immerse to water is exposed only to the dissolved oxygen. So as we understood in differential airish corrosion, the part exposed to less oxygen is going to act as the anode, whereas the part exposed to more [00:02:27.824] oxygen becomes the cathode and hence what do we know that the it is only the anode which undergoes corrosion so the metal of the ship this is the surface we are talking about so on this surface we find formation of anode and cathode so the anode undergoes uh [00:02:45.392] corrosion the metal gets destroyed and we find corrosion happening at the anode. Whereas at the cathode because there is more oxygen. So in this case we have [00:02:55.780] more oxygen and here we have less oxygen. So this part will get corroded and

whereas the other part which is exposed to more oxygen does not get corroded. And if you remember electrochemical theory of corrosion, I said that the [00:03:13.220] Products which are formed at the anode and cathode, the ions which are formed at the anode and cathode move towards each other and in between between the anode and cathode we get the formation of the corrosion product. The corrosion product will deposit in between the anode and cathode. [00:03:30.992] And that's exactly what happens here. A brown line on the surface of the ship. Many of you would have noticed this, or there will be a brown line on the surface of the ship, and where do you see it? Wherever the water line. So exactly where the water level is, we see a brown line. [00:03:48.560] Brown line and what is that brown line on the surface? It is due to the corrosion product. So we can see here this brown line which is getting formed at the waterline is due to the deposition of the corrosion product. So let us say Fe2 plus ions from here and the OH minus ions from here. [00:04:06.128] they move towards each other and somewhere in between they will get deposited and that is happening at the water level. So we see a distinct brown line. [00:04:16.356] Which is formed just below the waterline, and because of this, this type of corrosion is referred to as waterline corrosion. So basically it is differential aeration, but because it is happening in the presence of water, especially, it is called waterline correction. [00:04:34.128] Corrosion and the brown line appears at the waterline. Now let us take up another example where we have a steel tank which is partially filled with water and left for a long time. What happens here? The part which is [00:04:47.728] inside the inside the water now here again exposed to less oxygen that becomes the anode and the part which is above exposed to more oxygen is going to become the cathodic area and in between we will see a brown line getting

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.360] formed on the surface of this steel tank. So this has become the anode undergoes corrosion, this does not undergo corrosion, this part doesn't undergo corrosion, but the corrosion product the ions the Fe2 plus here and the OH minus are going [00:05:22.928] to move towards each other and a brown line can be seen here due to the formation of the corrosion product that is rust. [00:05:32.580] Another example would be when an iron rod is partially dipped in a sodium chloride solution. Now here we have this iron rod which is dipped partially. So half of it is inside the water, again the same reason we have less oxygen, therefore iodic area is this oxy this part. [00:05:50.352] And the more oxygen part would be the cathodic area, and somewhere at the waterline we will see the formation of the corrosion product. And now, since there is sodium chloride, we know that. [00:06:04.996] The conductivity of the solution will be very high, and therefore we find corrosion also will be more in this case when we have a solution. So sodium chloride solution instead of just water. Of course, it is aerated, so obviously there will be oxygen in it, but never. [00:06:22.768] the less less oxygen. So these are examples for waterline corrosion. So [00:06:28.480] We have to understand that waterline corrosion is also differential aeration corrosion, but only a special case because we consider these cases wherever the metal is partially immersed into the water. [00:06:44.996] Now let's talk about another type of corrosion that is pitting corrosion, and this is one of the most destructive types of corrosion. What happens here? It is a localized and accelerated corrosion, and because of this, there is formation of pits and [00:07:02.768] and pin holes around which the metal is relatively unattacked. You can see in this picture how pits have got formed on the metal surface, and it's only this area which has got corroded, whereas the rest of it looks good. [00:07:16.004] Looks intact. Now what could be the reason for this? And therefore we find that formation of pits and

pinholes. Therefore it is called pitting corrosion and the rest of the area of the metal looks intact. [00:07:28.944] Now this is one of the most destructive forms of corrosion. Why? Because if there are very small pinholes getting formed, the corrosion product itself is going to cover the pinholes and we will not even know till the destruction has happened. For example, if this is a pipeline. [00:07:46.576] Which is carrying oil or water. So the moment the corrosion product will be covering the surface, then there could be a pit getting formed there. And the pipe will suddenly start till it is completely damaged, we will not come to know and unless it starts leaking, it will just give out. [00:08:04.144] the water or the oil and that is where that is when we come to know but the damage is already done. So pitting corrosion we say is the most destructive one of the most destructive forms of corrosion and it is a localized and accelerated corrosion. So what is the characteristic very important [00:08:21.712] here for us to understand it is characterized by small anode area and a large cathode area because the anode area is very small it will result in accelerated corrosion. [00:08:35.876] Now let us take up some cases to understand what exactly is happening here. Now suppose we have a dust or a water drop or an oil drop which has got deposited on the surface of the metal. Now this is a very common phenomenon where we can have some rust or a oil drop or a water drop. [00:08:53.648] Let's take the most common thing water droplet on the surface as you can see in the picture. Now, because the water droplets is covering a small part of the metal surface, the entire surface is now not exposed to the atmosphere atmosphere, a small part [00:09:11.216] That is covered by this water droplet. Now, inside the water droplet, the amount of oxygen is very less. So this area now is exposed to less oxygen, whereas the whole of this area is now exposed to [00:09:24.544] More oxygen. So let us say this is a steel bar which has been exposed to the environment and there is a droplet of water sitting on the surface of the metal bar, the steel bar. So this part now basically which is exposed to less oxygen is going to become the [00:09:42.192] anodic area whereas this large part now which is exposed to atmospheric oxygen is going to become the cathodic area. So we know that wherever there is less oxygen is anodic area will get corroded and the large [00:09:56.512] area is not going to get corroded. As a result, what is going to happen? You can see there is a small anode and a large cathode, as I said just now that a small anode and large cathode is characteristic of fitting corrosion, and we will find that the anode

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.160] Anode being small undergoes corrosion at a very fast rate. Let us understand this better. [00:10:21.604] Now we have a small anodic area which is surrounded by a large cathode region. You can see again here there's a small anode and there's a large cathode which is there. Now because the area the cathode is so large, the demand for electrons is very high. [00:10:39.684] from the cathode. So it is going to demand a lot of electrons. Now what will the anode do? The anode will undergo corrosion at a faster rate. And because the anode is very small, we will find that it undergoes corrosion at a very fast rate being small there is like a [00:10:57.456] Which is getting formed in this anode area. So we can understand very clearly pitting will happen only when there is a small anode and a large cathode. So under the dust particle or the water droplet, that being anode, there is accelerated corrosion. [00:11:15.024] Why accelerated corrosion? We can understand because the demand for electrons is very high, the cathode being large. So it will undergo accelerated corrosion and forms a deep and narrow pit. Now the reactions would be M is giving you Mn plus plus N electron, and at the cathode we have the oxygen. [00:11:32.592] Taking part in the reaction, of course, here because it is exposed to more oxygen, we will say this is the reaction which is happening under the dust

particle or the water particle, it is anode, so it is the metal. Say we were talking of iron or steel, so iron will become Fe2 plus. So the under the anode. [00:11:50.160] There is destruction, and this pit will go on deepening slowly till it reaches the end. And till that time we will not be able to even discover that there is a pit, and that will lead to uh a damage of the structure. Another case. [00:12:08.228] could be of fitting corrosion where we have this protective film which got or formed on the surface which is be been made on the surface has got damaged. For example, this is a very common thing which we see iron being covered by tin. This process is calling. [00:12:26.000] Now when we have tin covering the entire surface of the iron, it is forming a nice protective film. But the moment there is a damage or there's a small hole or it got scraped, then this is going to expose the [00:12:42.144] iron metal to the surface. So the surface now what we have here this small part is iron which got exposed to the surface. Now if we look at the uh values for tin it is minus 0 point one eight [00:13:01.828] And this is minus zero point four four. [00:13:09.316] These are the electrode potential values. Now what do we understand? Iron is going to act as the uh anode and tin is going to act as the cathode. So as a result, what will happen? This [00:13:23.936] Now we have a case where we have a small anode and a very large cathode. So this now will result in [00:13:34.692] the anode undergoing corrosion at a very fast rate and therefore since the anode is very small it will be like digging a pit in this area. So this is a similar situation of [00:13:49.120] the one where we had a droplet on the surface, but there the type of corrosion was differential aeration corrosion, whereas here it is differential metal corrosion. [00:14:03.440] So therefore, we find that in this case tin is acting as the cathode. The value is minus 0.18 or maybe 0.14. And here it is 0.44, and therefore we find that this metal. [00:14:21.168] The steel is going to act as anode, whereas the tin is going to act as cathode, and now anode being very small, the demand for electrons from the cathode, the tin coating is the cathode large cathode, demand for electrons is very high, and therefore it forces the anode. [00:14:38.736] To undergo corrosion at a very fast rate, and because the anode is small, it will be uh a pit will be dug here, and then this is how the damage is going to happen. So here we say it is due to differential medical corrosion or differential. [00:14:54.160] Or we say galvanic corrosion, whereas in the case one we saw it was differential aeration corrosion. But what is common to go? That they have a small anode and a large cathode, and that is what is characteristic of pitting corrosion.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.568] So what are the reactions here we are seeing iron at which is covered by tin. Now iron acting as anode, so iron becomes Fe2 plus plus two electrons, whereas the cathode which is the tin. [00:15:21.680] Tin is not going to be involved in the reduction reaction, it is only the environment, and here we have oxygen. So therefore this is the reaction which will take place. Hence the anode will get corroded and a deep pit, narrow pit will be formed at this position. So this is [00:15:39.376] is known as pitting corrosion. What we have to remember small anode and large cathode. Whenever we have this situation, it is going to result in pitting corrosion. [00:15:53.476] Thank you. So before that let me tell you that in the next class we are going to discuss the last type of corrosion that is stress corrosion. Thank you once again.

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Full transcript: 2530 normalized words, 63 timestamped segments, 964.970667 seconds duration. Low-confidence segments flagged: 0.

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10 Types of corrosion - stress corrosion - Part III

This lesson is indexed from the source video titled Types of corrosion - stress corrosion - Part III. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000158 · EV0000159

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.676] Hello, everyone. Welcome back to this course on engineering chemistry. We will continue with module six on corrosion chemistry. [00:00:09.860] So in today's class we are going to study about types of electrochemical corrosion and in that we will look at stress corrosion and we'll also understand what is caustic embittlement. Then we move on to studying some of the factors which affect the rate of corrosion. [00:00:29.796] First, let us look at what is stress corrosion. Stress corrosion is also referred to as stress corrosion cracking, and this is a most common type of corrosion which happens with metals when they are under stress. [00:00:46.692] and at the same time they are exposed to a specific corrosive environment. Now how does a metal come under stress? We know that whenever [00:00:58.176] We are doing any kind of manufacturing or fabrication of articles, structures, etc., the metal is subjected to lot of mechanical operations such as pressing, hammering, rolling, bending, etc. Now, because of these operations on the [00:01:15.856] metal to get the right structure there will be some kind of a residual stress left inside the metal. [00:01:23.748] Apart from that, that there could also be some kind of external stress due to the service conditions. Now, either external stress or the residual stress, or both when they are present inside the metal, then this will give rise to [00:01:41.520] what we call as the stress corrosion cracking provided we [00:01:45.648] Give the right kind of corrosive environment. Now let us look at this picture. We have an iron article. Now this part will be the stress-free iron, and this bent part is going to be the stressed iron. Now, how does a metal relieve the stress? Now, since we are bent [00:02:03.344] Is so much here, then a small crack is going to be initiated for inside the metal in order to relieve the stress. Now, when such a case happens, like here you can see the vent. [00:02:18.064] uh the bend uh uh which is here, the bent article over here. So the metal atoms now under stress like in the bent area will be at a higher energy level compared to the metal part which is stress free. [00:02:33.872] Now this being at higher energy level, they will be more reactive as compared to this part, and therefore the reactive part will become the anode and the stress free part becomes the cathode. So in any electrochemical corrosion [00:02:49.776] We have to see how the anode and cathode are getting generated, and in this stress corrosion, this is how it is being formed wherever there is stress in the uh metal. So the stress part now became [00:03:06.608] uh anode and the stress free part becomes the cathode. So we got the generation of anode and cathode, and now if this part is [00:03:17.840] Exposed to a specific corrosive environment, we can see that stress corrosion cracking is going to take place. So what are the conditions for stress corrosion? We just now saw that there could be tensile stress in the [00:03:35.056] uh metal but apart from that the specific corrosive environment is what is required. Now both have to be present simultaneously. Just having the specific corrosive environment or the tensile strength is tr stress is not

[00:03:51.312] Yeah sufficient both these factors the tensile stress as well as the specific corrosive environment when they act upon this metal, then the metal undergoes stress corrosion. Now what happens is we know understood that one part is acting as the amp. [00:04:09.008] The stressed part and the other part is acting as the cathode. So if we provide the right kind of environment, and that is very very specific, what happens is the stressed part undergoes corrosion. So as we saw. [00:04:23.520] Suppose we have a metal which is bent like this and a small craitiated in it. So what happens now? This crack when we subject it to a [00:04:35.312] uh corrosive environment we will find that due to corrosion this part being anode undergoes corrosion and we know what happens in corrosion the metal is going to get converted into iron. So basically we are losing metal and slowly this c crack is going to propagate. It's going to [00:04:52.944] to deepen and by the time it reaches here it has caused damage or breakdown of the structure. So such a type of corrosion is known as stress corrosion cracking.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.976] But what we have to remember is both the conditions should be satisfied. The tensile stress as well as specific corrosive environment needs to be present for such a corrosion to take place. [00:05:20.676] Now, specific corrosive environment. Let us talk about it. Now this is highly specific. So [00:05:28.260] If the metal is break uh brass or the uh structure is made of brass, which is an alloy, this will be corroded if there is stress in the structure. This will get corroded. [00:05:42.096] only in the presence of ammoniacal solution or ammoniacal vapor. So it is very very specific. Now if the structure is made of steel, mild steel, uh then it is found that sodium hydroxide is going to affect or provide the [00:05:59.728] corrosive environment and if such a structure is in the in the presence of sodium hydroxide, we find that there will be stress corrosion because there will be some parts which are stressed and if they are exposed to sodium hydroxide or even chloride ions [00:06:16.176] we find that for steel this is the specific corrosive environment and for brass uh ammonia acid solution or ammonia vapors are the uh specific corrosive environment required for stress corrosion. [00:06:29.472] uh cracking to take place. So therefore in the presence of both these uh conditions we will find that stress corrosion can happen and one more [00:06:40.452] Fact about stress corrosion is that it causes serious damage as we saw this crack once it starts deepening due to the corrosive environment, it can cause breakdown of structures and it has in the past causes caused air crashes, bridge collapses, and even border. [00:06:58.224] boiler explosion. So it is again a very dangerous type of corrosion and has to be taken care of. Let us talk of how distress corrosion takes place in boilers. [00:07:14.464] Just now I mentioned that boiler explosions could happen due to this stress corrosion. So in boilers, in water boilers, when we have the stress corrosion, it is referred to as caustic embrittlement. Let us see what exactly is happening. [00:07:29.988] Now so what is caustic embittlement? Basically it's a form of stress corrosion and it takes place in boilers and we know that in what boilers what are we doing? We are heating water. So we it has a high temperature as well as high pressure and there will be certain regions in the boiling. [00:07:47.760] boiler which have which are under stress for example the rivets or the joints etc they are under stress now and this these boilers suppose they are made of mild steel just now I mentioned very specific environment is required so what will happen now [00:08:05.424] If this comes in constant contact with concentrated alkali, then there is stress in the boiler and also you provided the specific corrosive environment, stress corrosion is going to happen. So suppose this is the surface. [00:08:19.328] So there is a small crack which will be formed due to the uh residual stress when the

in certain areas where there is stress. So in order to relieve the stress, there will be a crack which will be uh generated. So let us see how exactly this will lead to [00:08:37.508] uh boy uh boiler explosion. Now what is the mechanism? Let us understand [00:08:44.576] So, whenever we are boiling water, we know that we are going to use soft water. Now, due to softening, there could be some amount of sodium carbonate which is present in the boiler water, and this sodium carbonate gets hydrolysed at high temperature. [00:09:02.256] we get sodium hydroxide. And if we uh remember that for mild steel, what is the [00:09:10.240] uh corrosive environment we require sodium hydroxide. So inside the water we have generated the sodium hydroxide. Now as I mentioned just now there is a small crack which is formed on the [00:09:23.680] uh the on the on the stressed areas now this sodium hydroxide is going to enter this crack. So sodium hydroxide goes into this crack and is [00:09:36.336] Uh there now what happens is because we are heating the boiler, it is at a uh high temperature, this sodium hydroxide which went inside the crack. [00:09:48.928] is slowly getting more and more concentrated. When it gets concentrated to about 10%, then this sodium hydroxide starts dissolving all the iron with which it is in contact.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.536] And when it dissolves it forms a compound called sodium ferroate. After this, the sodium ferroate undergoes further hydrolysis, and when it undergoes further hydrolysis, it is going to [00:10:18.848] look at this reaction sodium fydrohy ferro weight is undergoing hydrolysis giving you Fe three O four plus H two plus NOH. So what happened now? Again you have regenerated the sodium hydroxide. So that means throughout the you while you are using the boiler [00:10:36.528] continuously you have sodium hydroxide in the water and apart from that we also got Fe three O four and Fe three O four is known to be brittle in nature. So what happened to the iron now? The iron which was around this [00:10:54.116] uh crack which was initially uh gener initiated due to the due to the stress in the uh structure. Now around this iron [00:11:03.888] Got dissolved and now it is forming something like Fe 3O4, which is brittle in nature. So once this has got converted into uh brittle FE3O4, what is going to happen? The crack now starts deepening. So it goes further, further, and when it reach the other [00:11:23.760] the side the other surface of the boiler we know that now suddenly it will be exposed to low temperature and different pressure and this is going to result in an explosion. [00:11:39.588] So these are the various reactions which take place. So following the pro the crack the crack starts to propagate and this could sometimes result in explosion of the boiler. Now why are we calling this caustic embrittlement? Because it is happening in the presence of sodium hydrogen. [00:11:57.360] hydroxide and it is in a very specific environment. Uh the mild steel is uh the water boiler made of mild steel is undergoing corrosion. Plus we are making the [00:12:09.604] iron into something brittle like FE3O4 I mentioned just now. Therefore, this is called as this is known as caustic embrittlement. So because of this causing embrittlement, we get a corrosion cell. [00:12:24.448] Whenever we talk of electrochemical corrosion, we know there should be a formation of anode and cathode. So what is the anode and cathode here? Now we are writing a cell over here. We'll always follow the same rule. We'll write the anode first and then the cathode. So the iron which is under stress is going to act as the anode. [00:12:42.756] And it is in contact with concentrated sodium hydrox, whereas the stress free iron is going to act as the cathode, and that is in contact with the dilute sodium hydroxide. So you get a corrosion cell of uh this kind. And that is the reason [00:12:59.872] why electrochemical uh corrosion takes place, the stress as well as the specific environment in this case we have provided the sodium hydroxide. Now

of course we have to think of ways by which this can be prevented because it could be very dangerous. So it can be prevented by adding [00:13:19.136] what are known as corrosion inhibitors. So some examples would be sodium sulphate, tannin, lignin, phosphates, etc. So these are added into the boilers and they can help in preventing [00:13:34.800] this kind of corrosion that is the stress corrosion or caustic caustic embridment in the borders. Now let us study some factors which affect the rate of corrosion or we can say the extent of corrosion. [00:13:50.724] So these this is a complete list of the factors. Now we could the most important is nature of the metal. Then we have the difference in electrode potentials of anode and cathode, nature of corrosion product, ratio of the anode to cathode area, then we have overvoltage [00:14:08.112] Polarization, pH, temperature, conductivity of the medium and humidity. So all these factors are going to affect the amount of corrosion which will take place or the rate at which the corrosion is going to happen for the metal. [00:14:27.780] Let us take up some of them. The first one, nature of the metal. Of course. [00:14:35.104] How much corrosion is going to happen is going to depend on how reactive the metal is. If it is a noble metal, we don't expect much reactivity. Therefore corrosion is also very very limited. If it's very noble, then you hardly see any corrosion. But [00:14:52.784] Those metals which are very reactive we find that they will be undergoing corrosion. [00:14:59.044] In fact, they are more susceptible to corrosion. So we can say that the ones with lower electrode potential, that is where the reduction potentials are negative, that means they want to undergo oxidation, and we know that corrosion is an oxidation process. So we expect that those with

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:17.840] Negative reduction potentials will be more reactive than those which have positive reduction potential values. So these reactive metals obviously will be more susceptible to corrosion. So we can see that very clearly: sodium, potassium, magnesium, zinc, zinc, etc. [00:15:35.472] etc. They all have very negative values, and we know that they are highly susceptible to corrosion. On the other hand, we have noble metals like silver, gold, and platinum which have positive values and we find that they are less susceptible to [00:15:52.784] Uh corrosion. So can we say that the electrochemical series which lists the [00:15:59.556] metals uh uh electrode potentials or the standard reduction potentials of these metals should give us an idea about the order in which the metals will undergo corrosion. And that's exactly what we found the ones which are lying high above in the series will undergo corrosion. The ones [00:16:17.328] which are very low in the series with positive values do not undergo corrosion. But there are exceptions like chromium, aluminium, titanium. Now chromium aluminum they have very negative values of electron potentials but they do not undergo corrosion. [00:16:34.784] Why? Because they become passive. [00:16:38.596] Now that means electrochemical series is not sufficient for us to give us the order in which corrosion will happen. Therefore, a new series is required, which is known as the galvanic series. And what does this galvanic series do? It is going to arrange the metal. [00:16:56.368] And now we have an advantage, it also can arrange the alloys in the order of their [00:17:02.736] uh tendency to undergo corrosion and in a particular environment because we know that corrosion depends not only on the metal or the alloy, it also depends on the environment to which the metal is subjected. [00:17:18.596] So we can see here the electrochemical series and we know that lithium, potassium, calcium, etcetera, all these yes they do, you can see the values are uh negative and then they become more and more positive when we come to gold, silver, it's copper, etcetera, which are below hydrogen, they have a positive value. [00:17:36.368] Yes, to a good extent we can say that the electrochemical series gives us an order in

which the metals will show tendency to undergo corrosion, but nevertheless, electrochemical series are standard values. That means we are using standard solutions and standard conditions, and moreover. [00:17:55.492] Aluminium, you see here it has a very negative value. Similarly, magnesium uh of course undergoes corrosion. Aluminium, chromium also you'll see they have very negative values, but we know that aluminium is uh passive in nature. So that means electrochemical series is not sufficient. [00:18:13.264] For us to decide the corrosion tendency. So we need to have another series which is known as the galvanic series. Now here you can see the galvanic series where the metals now are arranged according to the [00:18:30.852] Tend and see not only metals, you can see some alloys also. All these are uh arranged in an order in which [00:18:38.624] They will undergo corrosion. So what is a galvanic series where we have a list of metals as well as the uh alloys in the order of the tendency to undergo corrosion, but here most importantly in a particular electrolyte solution. [00:18:55.200] So sometimes you will find galvanic CDs in seawater. So basically all this study is done experimentally and in seawater it is tested. [00:19:07.280] and listed here which metal or alloy will be more reactive or undergo corrosion more as compared to the uh others. So we we can find that if the solution is different the order could also change. So whenever we list a galvanic series [00:19:24.944] Always the environment is also listed over there. Coming to the next. [00:19:33.924] reason or the factor which is going to affect the rate of corrosion and here the difference between the potential of the anode and the cathode region. We know that many times two metals are in contact with each other and [00:19:50.960] This is say M1 and this is M2. Now when these two metals are in contact with each other, we know that one of them will become the anode, the other becomes the cathode based on their electrode potentials. Now

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:04.580] We know also that the driving force here is the difference in the electrode potential. So what kind of uh corrosion are we talking about? We are talking about galvanic corrosion. So larger the difference between the potential difference between the anode and cathode area. [00:20:22.144] Higher is going to be this kind of differential metal corrosion or galvanic corrosion. Why? Because more is the potential difference, diff uh more is the potential difference, the free energy decrease accompanying the process is higher, and therefore we find that [00:20:39.792] extent of corrosion also increases. But what we have to remember here, more the difference between the electrode potentials of the two metals, more is the anode which is going to undergo [00:20:54.992] corrosion. It is a more reactive metal which undergoes corrosion. So if we have two metals like this, say if this is anode and this is cathode, of course this is based on the electrode potential values. If we understand that this is the anode and this is the cathode, we will find more the difference between this anode and cathode [00:21:12.624] More is the corrosion of M1 metal in this case. Let us take up a case one where we have iron which is minus 0.44 volts and copper 0.34 volts. So what is the first of all which metal will undergo corrosion? Iron, we can see very clearly. [00:21:32.292] Because this will become the anode. And the potential difference here comes out to be 0.78 volts. Let us take another case where we have iron at minus 0.44 and tin is [00:21:43.712] the metal which is in close contact with it and that uh tin has an electrode potential standard electrode potential of minus 0.14. What is the metal which is undergoing corrosion? Again we can see that it is iron which will undergo corrosion. So in this case also we have iron acting as iron [00:22:01.392] Here also I am acting as anode, but now the potential difference has come down to 0.3 that is e cathode e0 cathode minus the e0

anode. Here it is 0.78, here it is 0.3. Therefore, though in both the cases it is iron which [00:22:18.960] We find that iron will undergo corrosion faster when it is in contact with copper. So [00:22:29.600] The use of this similar metal should be avoided wherever it is possible. For example, nuts and bolts. Should be always made of the same metal, otherwise, we'll find that the metal which is more reactive. [00:22:47.280] Will undergo corrosion. Yesterday I gave an example of a copper bolt with a nut which was made of steel, and we saw that the copper remains intact, but the steel screw has got completely the steel nut has got completely uh corroded. So in order to avoid that. [00:23:04.848] Because nuts and bolts are always in close contact, they should be made of the same metal. Now coming to the next very important factor which affects the rate of corrosion, nature of corrosion product. [00:23:20.292] Now in some cases, just now we saw in the first factor that cr uh when we talked about nature of metal, chromium, aluminium, etc., they do not undergo corrosion. [00:23:31.936] They become passive. So, what is the reason for that? Because once the corrosion product is formed on the surface of the metal, it acts as a protective film. So let us say, and we know that corrosion is a surface phenomena. This we should always remember. So say this is [00:23:49.648] The metal, and now the corrosion product has got formed on the surface. Now this corrosion product is forming a protective film on the surface, where and it is actually protecting the metal underneath, and this is the corrosion. [00:24:06.912] product which is formed on the surface. So it's forming a nice film over the surface and protecting the metal how it is creating a barrier between the surface and the environment and therefore we will find that the metal is now cut off from the environment and hence there can be no further corrosion. [00:24:24.624] So, if the corrosion product is insoluble, stable, uniform, and non-porous, we find that it forms a protective film and therefore prevents further corrosion. So [00:24:38.416] It acts as a barrier, a thin, invisible, impervious sometimes continuous film is formed, and this acts as a barrier between the fresh metal surface and the corrosion environment. So such metals are known to become [00:24:54.256] Passive, and that's exactly what we see in the case of chromium and aluminium. Now if we have

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:01.472] metals like iron, zinc and magnesium. Now all of us know what rust looks like. Now when we have a metal like this metal now the corrosion is happening here also and the corrosion product will be on the surface but we find that it will be porous and [00:25:19.120] A non-continuous film will be formed. Now in this case, what happens? Even though the corrosion product is formed, it is not protecting the uh surface underneath, and every time there is a new surface after the corrosion product is formed, there's a new surface which is getting exposed. [00:25:36.688] and therefore the corrosion will continue to happen continuously. [00:25:42.852] So metals like iron, zinc, and magnesium, the corrosion product is soluble, unstable, non-uniform, and porous, and therefore there is no stopping corrosion. It will keep on getting uh corroded as a new metal surface, a fresh metal surface is getting extra. [00:26:00.624] So if especially this term soluble is very important. Now if you have a corrosion product [00:26:10.116] Which is covering the surface, but it is soluble in the surrounding environment. Now you have say water over here or any kind of conducting medium here. Now if this corrosion product gets dissolved, so what is happening the moment it is getting formed, it is dissolving. So a new surface is getting formed. [00:26:27.888] getting exposed every time. So now what happens? Uh that this [00:26:35.216] will result in more and more corrosion of the metal. So we can understand now why steel gets corroded but stainless steel does not. We all use uh stainless

steel utensils for cooking and uh uh other work in the [00:26:52.292] uh at homes. But we know that it does not get corroded. Why? Because it has [00:26:58.672] It is an alloy which contains chromium. Now chromium we know can form a passive layer, so impervious passive layer is getting formed on the surface and that prevents the stainless steel utensils or any article from getting further corroded, whereas steel we know that. [00:27:16.368] uh steel or that is iron it iron containing some amount of carbon so that steel uh articles we find that they get corroded easily like your nails or any kind of steel article will get corroded but stainless steel due to the uh [00:27:33.504] Passive nature of chromium which has been added there. This chromium oxide forms a nice layer on the surface and forms a protective film and prevents further corrosion of the particle. So nature of corrosion product decides to a great extent whether corrosion is [00:27:51.216] going to uh happen or not. So these were some of the factors we discussed. Now in tomorrow's uh class in the next class we are going to talk about take up some more factors like overvoltage, polarization, pH temperature, etc, which were listed [00:28:08.784] It just now, and then we will learn how those factors are going to affect corrosion rate. Thank you.

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Full transcript: 4164 normalized words, 115 timestamped segments, 1698.005333 seconds duration. Low-confidence segments flagged: 0.

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11 Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage)

This lesson is indexed from the source video titled Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000182 · EV0000183

Graphs and diagrams

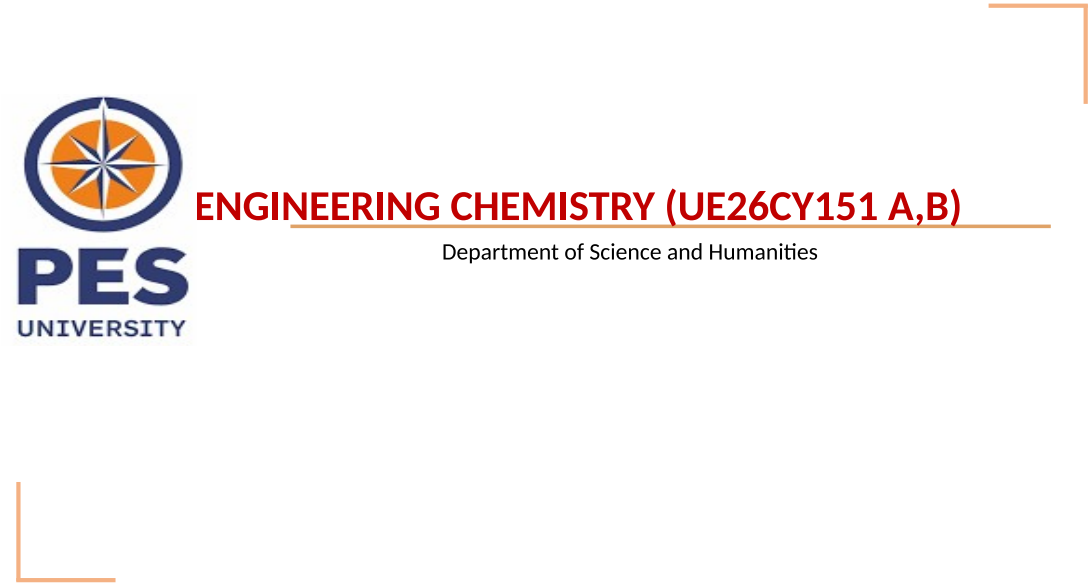

Figure G000093. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 1.

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Figure G000094. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 2.

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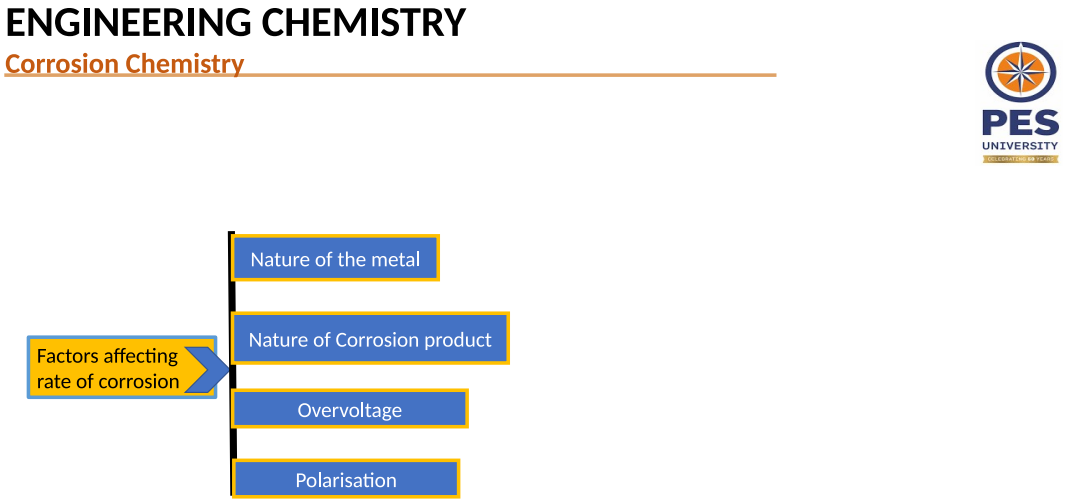

Figure G000095. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 3.

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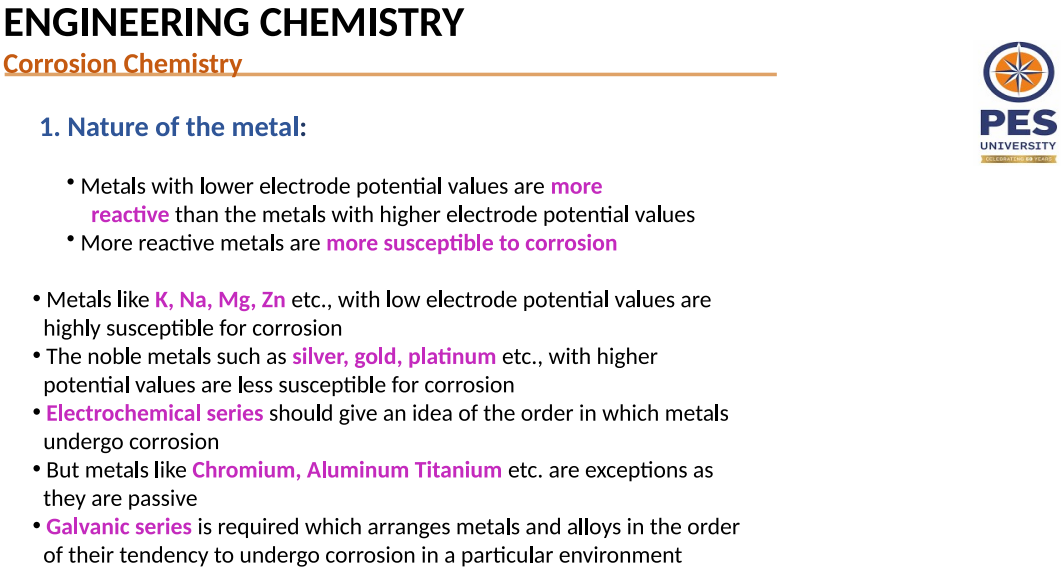

Figure G000096. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 4.

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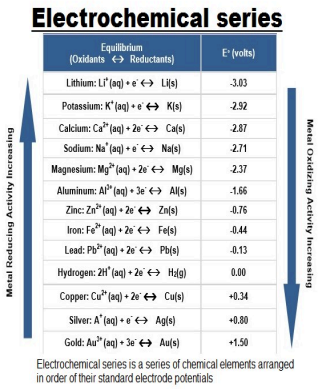
ENGINEERING CHEMISTRY

Corrosion Chemistry



The electrochemical series is built up by arranging various redox equilibria in the order of their standard electrode potentials (redox potentials). The most negative E° values are placed at the top of the electrochemical series, and the most positive at the bottom

Source:https://www.syedgilanis.com/2019/04/electrochemicalseries.html

Figure G000097. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 5.

EV0000555

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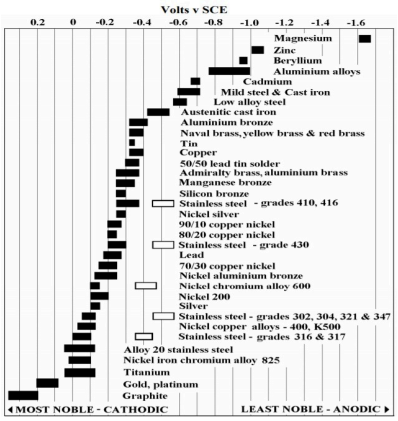
ENGINEERING CHEMISTRY

Corrosion Chemistry

Galvanic Series



Sorce: https://www.corrosionpedia.com/an-introduction-to-the-galvanic-series-galvanic-compatibility-and-corrosion/2/1403

A "galvanic series" lists metal and alloys in order of their tendency to undergo corrosion in a particular electrolyte solution, hence for each specific solution which is expected to be encountered for actual use, a different order will ensue

Figure G000098. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 6.

EV0000558

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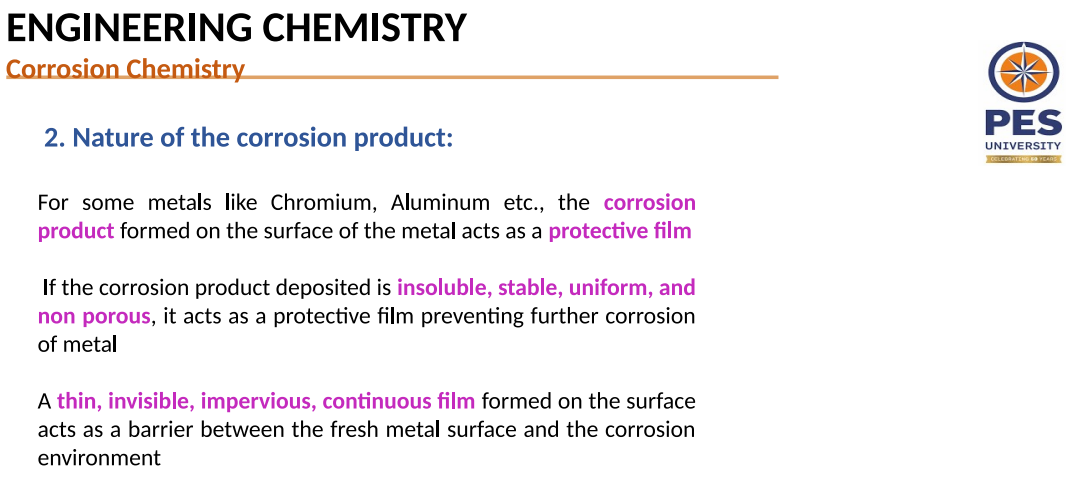

Figure G000099. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 7.

EV0000561

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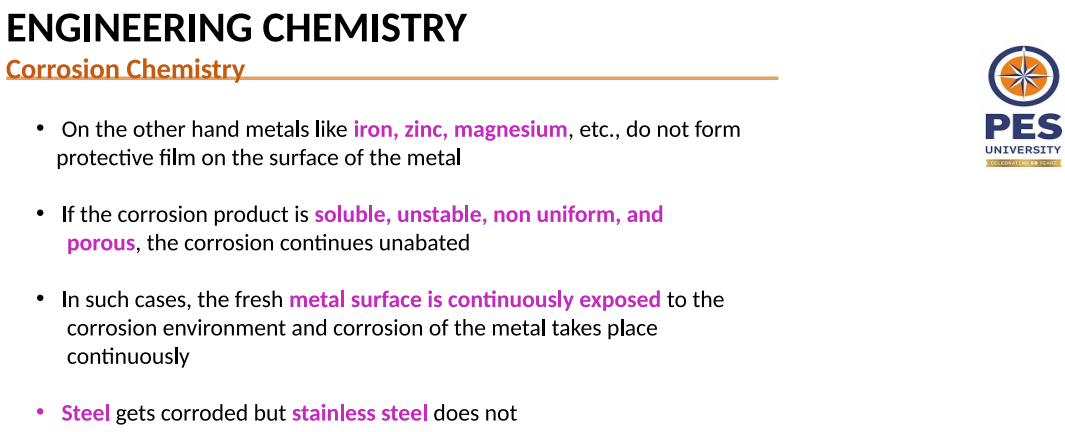

Figure G000100. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 8.

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ENGINEERING CHEMISTRY

Corrosion Chemistry

3. Hydrogen overvoltage:

Overvoltage is defined as the **extra voltage over the theoretical voltage** required for an electrode reaction to take place

Applicable especially for evolution of **gases**

Depends on the **surface**; if overvoltage of a gas is high on a surface, liberation of the gas on that surface will be less

TABLE 19–3
Overvoltage for Hydrogen and Oxygen Formation at Various Electrodes at 25°C*

Electrode Composition	Overvoltage (V) (Current Density 0.001 A/cm²)		Overvoltage (V) (Current Density 0.01 A/cm²)		Overvoltage (V) (Current Density 1 A/cm²)	
	H_2	O_2	H_2	O_2	H_2	O_2
Smooth Pt	−0.024	−0.721	−0.068	−0.85	−0.676	−1.49
Platinized Pt	−0.015	−0.348	−0.030	−0.521	−0.048	−0.76
Au	−0.241	−0.673	−0.391	−0.963	−0.798	−1.63
Cu	−0.479	−0.422	−0.584	−0.580	−1.269	−0.793
Ni	−0.563	−0.353	−0.747	−0.519	−1.241	−0.853
Hg	−0.9[a]		−1.1[c]		−1.1[e]	
Zn	−0.716		−0.746		−1.229	
Sn	−0.856		−1.077		−1.231	
Pb	−0.52		−1.090		−1.262	
Bi	−0.78		−1.05		−1.23	

* National Academy of Sciences, *International Critical Tables*, Vol. 6, pp. 339–340, McGraw-Hill, New York, 1929.
[a] −0.556 V at 0.000077 A/cm²; −0.929 V at 0.00154 A/cm².
[c] −1.065 V at 0.00760 A/cm².
[e] −1.126 V at 1.153 A/cm².

Source:https://chemistry.mdma.ch/hiveboard/serious/000490034.html

Figure G000101. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 9.

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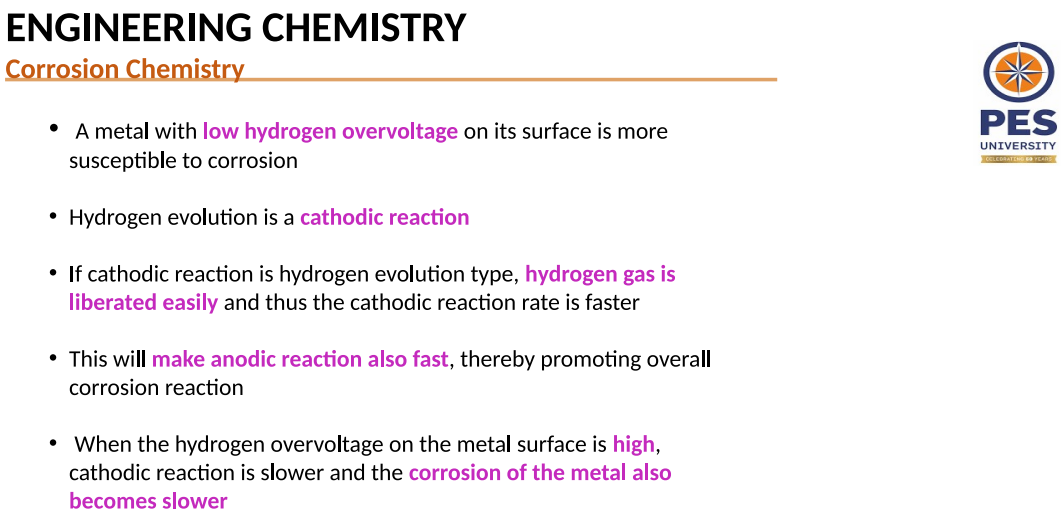

Figure G000102. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 10.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

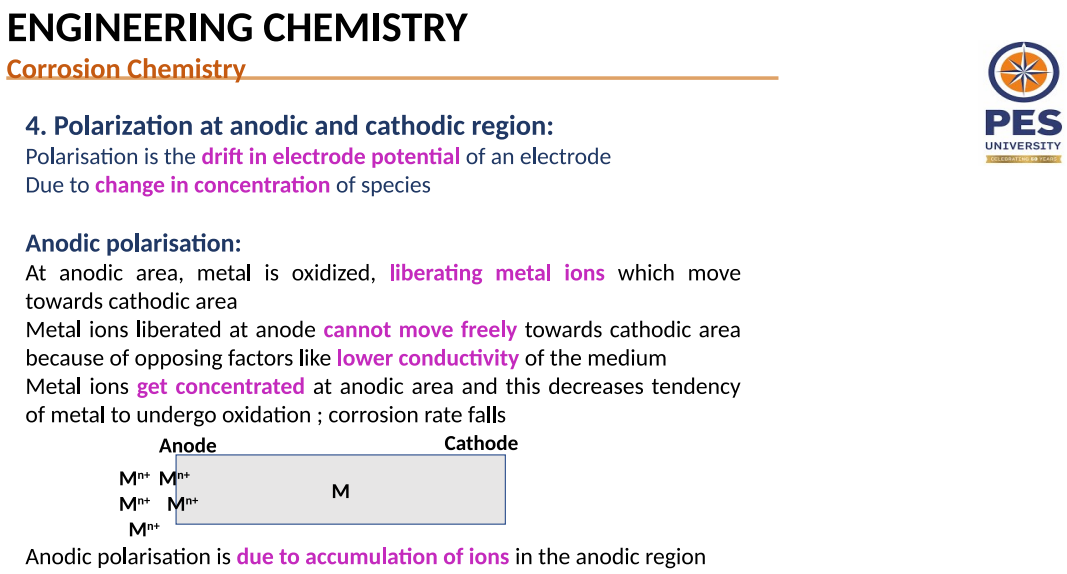

Figure G000103. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 11.

EV0000573

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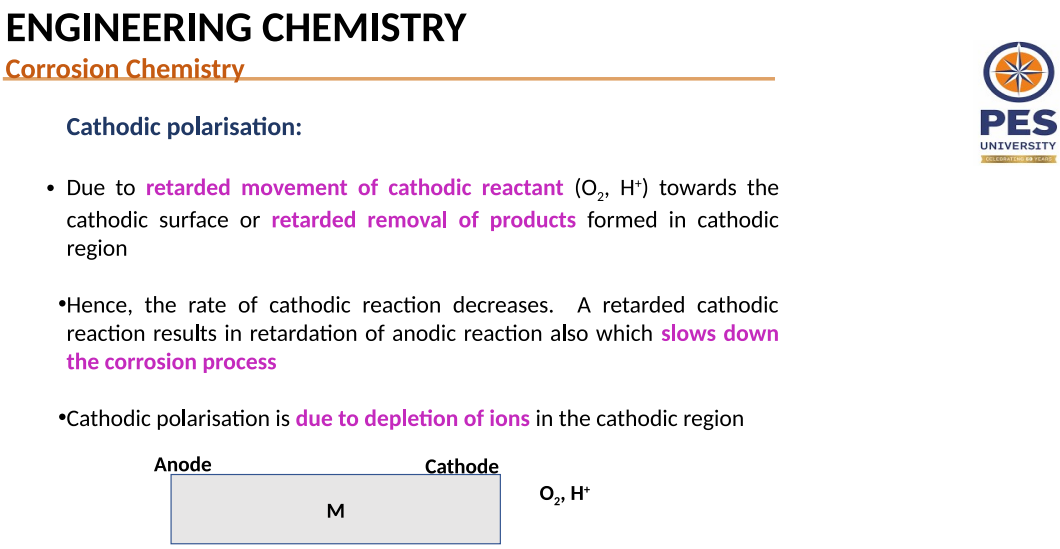

Figure G000104. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 12.

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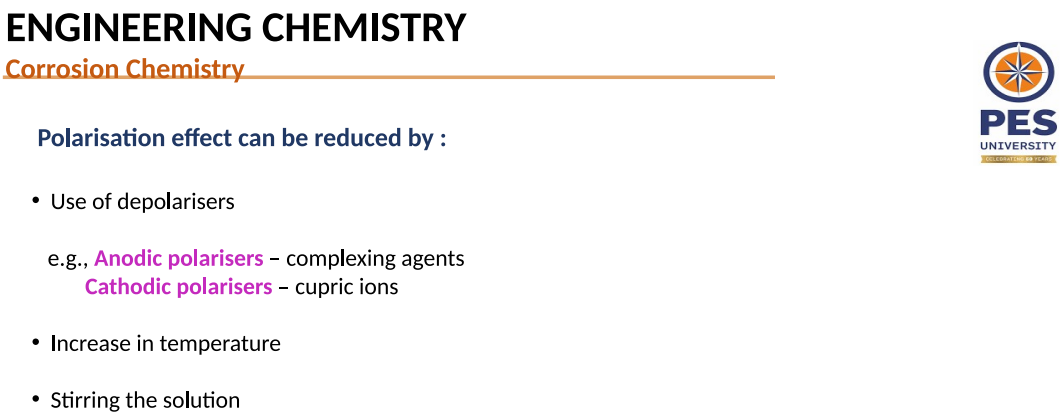

Figure G000105. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 13.

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Figure G000106. Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage · source slide page 14.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000542 · EV0000543 · EV0000544
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Class 22 content: Factors affecting rate of corrosion EV0000545 · EV0000546 · EV0000547
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Nature of the metal Nature of Corrosion product Factors affecting rate of corrosion Overvoltage Polarisation EV0000548 · EV0000549 · EV0000550
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 1. Nature of the metal: Metals with lower electrode potential values are more reactive than the metals with higher electrode potential values More reactive metals are more susceptible to corrosion Metals like K, Na, Mg, Zn etc., with low electrode potential values are highly susceptible for corrosion The noble metals such as silver, gold, platinum etc., with higher potential values are less susceptible for corrosion Electrochemical series should give an idea of the order in which metals undergo corrosion But metals like Chromium, Aluminum Titanium etc. are exceptions as they are passive Galvanic series is required which arranges metals and alloys in the order of their tendency to undergo corrosion in a particular environment EV0000551 · EV0000552 · EV0000553

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry [Image text, OCR] Electrochemical series Equilibrium E"(volts) (Oxidants>Reductants) Lithium:Li*(aq) +e<>Li(s) -3.03 Potassium:K(aq)+e'<>K(s) -2.92 Calcium:Ca²(aq)+2e<→>Ca(s) -2.87 Metal Oxidizing Activity Increasing Sodium:Na*(aq)+e'<>Na(s) -2.71 Magnesium:Mg²*(aq)+2e>Mg(s) -2.37 Aluminum:Al²+(aq)+3e<>Al(s) -1.66 Zinc:Zn²(aq)+2e<→>Zn(s) -0.76 Iron:Fe²(aq) +2e'<>Fe(s) -0.44 Lead:Pb²*(aq)+2e<>Pb(s) -0.13 Hydrogen:2H*(aq)+2e'<>H2(g) 0.00 Copper:Cu2(aq)+2e'(→Cu(s) +0.34 Silver:A*(aq)+e'<>Ag(s) +0.80 Gold:Au²(aq)+3e>Au(s) +1.50 Electrochemical series is a series of chemical elements arranged in order of their standard electrode potentials The electrochemical series is built up by arranging various redox equilibria in the order of their standard electrode potentials (redox potentials). The most negative E° values are placed at the top of the electrochemical series, and the most positive at the bottom Source:https://www.syedgilanis.com/2019/04/electrochemicalseries.html EV0000554 · EV0000555 · EV0000556
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Galvanic Series [Image text, OCR] VoltsvSCE 0.2 0 -0.2 -0.4 -0.6 -0.8 -1.0 -1.2 -1.4 -1.6 Magnesium Zinc Beryllium Aluminiumalloys Cadmium Mildsteel&Castiron Low alloysteel Austenitic castiron Aluminiumbronze Navalbrass,yellowbrass&redbrass Tin Copper 50/50leadtinsolder Admiraltybrass,aluminiumbrass Manganesebronze Siliconbronze Stainless steel -grades 410,416 Nickelsilver 90/10copper nickel 80/20 copper nickel Stainless steel-grade 430 Lead 70/30coppernickel Nickelaluminiumbronze Nickelchromiumalloy600 Nickel200 Silver Stainless steel-grades302,304,321&347 Nickel copper alloys-400,K500 Stainlesssteel-grades316&317 Alloy20stainlesssteel Nickelironchromium alloy825 Titanium Gold,platinum Graphite MOSTNOBLE-CATHODIC LEASTNOBLE-ANODIC A "galvanic series" lists metal and alloys in order of their tendency to undergo corrosion in a particular electrolyte solution, hence for each specific solution which is expected to be encountered for actual use, a different order will ensue Sorce: https://www.corrosionpedia.com/an-introduction-to-the-galvanic-series-galvanic-compatibility-and-corrosion/2/1403 EV0000557 · EV0000558 · EV0000559
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 2. Nature of the corrosion product: For some metals like Chromium, Aluminum etc., the corrosion product formed on the surface of the metal acts as a protective film If the corrosion product deposited is insoluble, stable, uniform, and non porous, it acts as a protective film preventing further corrosion of metal A thin, invisible, impervious, continuous film formed on the surface acts as a barrier between the fresh metal surface and the corrosion environment EV0000560 · EV0000561 · EV0000562
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry On the other hand metals like iron, zinc, magnesium, etc., do not form protective film on the surface of the metal If the corrosion product is soluble, unstable, non uniform, and porous, the corrosion continues unabated In such cases, the fresh metal surface is continuously exposed to the corrosion environment and corrosion of the metal takes place continuously Steel gets corroded but stainless steel does not EV0000563 · EV0000564 · EV0000565

9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 3. Hydrogen overvoltage: Overvoltage is defined as the extra voltage over the theoretical voltage required for an electrode reaction to take place Applicable especially for evolution of gases Depends on the surface; if overvoltage of a gas is high on a surface, liberation of the gas on that surface will be less [Image text, OCR] TABLE19-3 OvervoltageforHydrogen andOxygenFormation atVariousElectrodesat25°C Overvoltage(V) Overvoltage (V) Overvoltage (V) (Current Density (Current Density (CurrentDensity 0.001A/em²) 0.01A/cm) 1A/cm) Electrode Composition H: 0: H 0. H 0: SmoothPt -0.024 -0.721 -0.068 -0.85 -0.676 -1.49 Platinized Pt -0.015 -0.348 -0.030 -0.521 -0.048 -0.76 Au -0.241 -0.673 -0.391 -0.963 -0.798 -1.63 Cu -0.479 -0.422 -0.584 -0.580 -1.269 -0.793 Ni -0.563 -0.353 -0.747 -0.519 -1.241 -0.853 Hg -0.96 -1.1 -1.14 Zn -0.716 -0.746 -1.229 Sn -0.856 -1.077 -1.231 Pb -0.52 -1.090 -1.262 Bi 0.78 -1.05 -1.23 Natonal Academy ofSeiences.IrernatonalCritical Tables,Vol.6.pp.339-340,McGraw-Hil:NewYork,1929. -0.556Vat0.000077A/cm²:0.929Vut0.00154A/cm² 1.063Vat0.00769A/cm 1.126Vat1.153A/cm Source:https://chemistry.mdma.ch/hiveboard/serious/000490034.html EV0000566 · EV0000567 · EV0000568
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry A metal with low hydrogen overvoltage on its surface is more susceptible to corrosion Hydrogen evolution is a cathodic reaction If cathodic reaction is hydrogen evolution type, hydrogen gas is liberated easily and thus the cathodic reaction rate is faster This will make anodic reaction also fast, thereby promoting overall corrosion reaction When the hydrogen overvoltage on the metal surface is high, cathodic reaction is slower and the corrosion of the metal also becomes slower EV0000569 · EV0000570 · EV0000571
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 4. Polarization at anodic and cathodic region: Polarisation is the drift in electrode potential of an electrode Due to change in concentration of species Anodic polarisation: At anodic area, metal is oxidized, liberating metal ions which move towards cathodic area Metal ions liberated at anode cannot move freely towards cathodic area because of opposing factors like lower conductivity of the medium Metal ions get concentrated at anodic area and this decreases tendency of metal to undergo oxidation ; corrosion rate falls Anodic polarisation is due to accumulation of ions in the anodic region Cathode Anode M Mn+ Mn+ Mn+ Mn+ Mn+ EV0000572 · EV0000573 · EV0000574
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Cathodic polarisation: Due to retarded movement of cathodic reactant (O2, H+) towards the cathodic surface or retarded removal of products formed in cathodic region Hence, the rate of cathodic reaction decreases. A retarded cathodic reaction results in retardation of anodic reaction also which slows down the corrosion process Cathodic polarisation is due to depletion of ions in the cathodic region Anode Cathode O2, H+ M EV0000575 · EV0000576 · EV0000577
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Polarisation effect can be reduced by : Use of depolarisers e.g., Anodic polarisers – complexing agents Cathodic polarisers – cupric ions Increase in temperature Stirring the solution EV0000578 · EV0000579 · EV0000580
14. Source slide page 14 (extracted text): THANK YOU Department of Science and Humanities EV0000581 · EV0000582 · EV0000583

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.060] Hello everyone, welcome back to this course on engineering chemistry. We continue with module six on corrosion chemistry, and today we are going to study about some more factors affecting the rate of corrosion. [00:00:19.588] Let us study the ratio of anodic to cathodic area. The size of the anode and the cathode plays a major role in deciding the rate at which [00:00:34.496] the anode undergoes corrosion. We remember from electrochemical theory that it's only the anode which undergoes corrosion and the cathode remains unaffected. Now what happens if this ratio where the anode is small and the cathode is large is that the [00:00:52.452] Anode area being very small and a cathode area being very large, corrosion is very [00:00:59.472] intensive and the anode will undergo corrosion at a very fast rate and as this ratio becomes smaller and smaller the corrosion rate is going to increase rapidly and this is what we studied under pitting corrosion. So what [00:01:17.104] Is the reason for that. We know that when the anode is small, as in this case, and the cathode is large, the electrons are getting liberated at the anode and they are getting [00:01:31.088] consumed at the cathode and since the cathode is so large there is rapid consumption of these electrons and the cathode and like and because of that uh demand for electrons is very high from the cathode. Therefore the anode has to undergo reaction [00:01:48.720] at a very fast rate. That means it will get corroded at a very fast rate. Now if the ratio of the anode to cathode area is large then intense corrosion is not going to take place. [00:02:05.412] For example, let us say we have an iron article which is covered with tin. [00:02:11.312] Now we know that the standard electrode potential of iron is minus 0.44 volts and that of tin is minus 0.14 volts. Now between the two we know that iron is acting as the anode more active than tin. So if the tin covering is in [00:02:29.040] that dentin is going to undergo corrosion. But [00:02:32.784] If the tin coating is broken due to some reason, then we find that iron surface is getting exposed to the environment, and therefore, because iron is anode, it will be iron which will undergo corrosion. And since we have a case of small anode and large case. [00:02:50.448] So the ratio of anode to cathode is very small and we will find that the iron is going to undergo corrosion at a very fast rate. [00:03:01.188] Whereas if we have covered iron with zinc. Now for zinc, we know that the electrode potential value for zinc is minus 0.76, whereas that of iron just now we mentioned it is minus 0.44 volts. [00:03:21.284] Now here between the two we know that it is zinc which will act as the anode. So even if there is a gap on the surface, we find that it's only zinc which gets corroded and the iron is intact and at the same time there is no intense corrosion which is happening. [00:03:40.432] So if the ratio of anode to cathode area is very small, in that case we find that anode undergoes uh corrosion at a very fast rate. The next [00:03:53.028] factor is about hydrogen over voltage. So let us first understand what is overvoltage. Overvoltage is defined as the extra voltage which we need to provide over the theoretical voltage for a particular electrode reaction to take place. [00:04:12.400] Now we know that for a particular reaction, for any electrode reaction, there is an standard electrode potential or uh electrode potential is a fixed value. [00:04:22.832] But we find that the reaction does not take place specially if there is evolution of gases. It is noticed that [00:04:33.008] In the case like say hydrogen gas or oxygen gas is getting evolved at the electrode due to the reaction, it is found that we need to provide this extra voltage, and that voltage is referred to as the overvoltage. And this overvoltage value depends on the [00:04:50.704] The surface. So if the overvoltage value on a particular surface will be high, then the liberation of that gas is going to be that much more difficult. For example, you can see the list of

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.932] Hydrogen over voltage values and also ox oxygen over voltage values. Now hydrogen over voltage, just by looking at platinum we can understand. When we have just uh platinum, we have a value [00:05:23.136] uh which is close to uh zero point uh one. But if you take the case of platinized platinum it's almost going to zero. That means the same hydrogen evolution on a platinum surface [00:05:38.672] would not be uh would be more facilitated when it's platinized platinum why because the surface area as has become high so in the case of platinum we find that platinized platinum has lower hydrogen over voltage so what would be the result the hydrogen [00:05:56.304] gas evolution which we will be much higher. Similarly on copper we find that it is close to 0.34, whereas for zinc it is 0.76 minus values. So we find that the liberation of hydrogen [00:06:12.272] This is the hydrogen voltage values. So it the liberation of hydrogen is going to be much more difficult on zinc as compared to copper. And then coming to platinum, it is even more easier. [00:06:26.704] to uh for the hydrogen liberation and when it comes to platinized platinum it's even more uh the hydrogen over voltage is very less. [00:06:36.260] This shows that hydrogen over voltage is dependent on the surface of the metal on which the hydrogen evolution is taking place. So [00:06:51.888] If a metal has low hydrogen overvoltage on its surface, then it is more susceptible to corrosion. Why would we say that? Because hydrogen liberation H plus plus electron is giving you half H2 is a cathodic reaction. [00:07:10.820] Now, if in a particular during a particular corrosion, if cathodic reaction is hydrogen evolution type, then we will find that hydrogen gas gets liberated easily, and that means [00:07:26.432] The cathodic reaction is very fast. Now if the cathodic reaction is very fast, it is going to make the anodic reaction also fast because we know that anodic and cathodic reactions will take place simultaneously, and if the anode is undergoing reaction at a fast rate, it means that the high [00:07:44.112] The over uh overall corrosion is going to happen at a fast rate. So if low hydrogen over voltage over a metal surface is there, then the corrosion of that metal will be faster. And if the [00:08:02.212] Hydrogen overvoltage on the metal surface is high, that means the cathodic reaction has become slow and therefore the oxidation also becomes less and therefore the reac the corrosion of that metal will also become slower. [00:08:21.316] Now, let us talk about temperature. We know that for any chemical reaction, as temperature increases, there is going to be an increase in the rate of reaction. And similarly, that is what we observe in the case of [00:08:37.828] metals also or in the case of corrosion process. So when we have a corrosion process which is actually a redox reaction which is taking place, we find that the rate of corrosion increases as the temperature increases. So [00:08:54.724] What are the various reasons? So we see, of course, we know any reaction increase in the rate of the reaction happens as temperatures increase. [00:09:04.112] Apart from this, there are several other reasons which are going to promote corrosion due to high temperature. Now there will be an increase in conductance of the medium. You know conductivity of the ionic conductance will increase with temperature. So ions can move faster. So as [00:09:21.808] I have uh mentioned before, whether it's in batteries or in corrosion, it's not just the anode and cathode reaction which will decide the fate of the reaction. It's also the movement of ions in the electrolyte. Now since the movement is faster, we expect that. [00:09:39.044] the reaction will also be faster. Now we can decrease the uh polarization effect. [00:09:46.436] We will study about this polarization soon and we will uh understand what is this polarization at the anode and cathode sites. So by decreasing the uh polarization the also the

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.648] the the because of high temperature polarization effect is less and therefore we find that corrosion will be faster. Also the s corrosion product solubility of the corrosion product [00:10:15.076] generally increases at with high temperature. Solubility of any substance we know with increase in temperature it is going to increase. So here because this for the product is on the surface and if it is getting more dissolved [00:10:32.848] into the solution uh into the medium we can understand that it's going to expose new surface every time and therefore we find that the rate of corrosion is also going to increase and apart from that another reason could be [00:10:45.648] breakdown of the protective film. Now we know that many times an ice protective film gets formed over the surface but at a higher temperature it is possible that this uh the the covering is going to the protective film is going to break down and therefore that could be another [00:11:03.280] Reason for uh height as the rate of reaction being higher or rate of corrosion being higher at a higher temperature. But there is an exception which we should know that oxygens like uh sorry, gases like oxygen carbon dioxide, if they [00:11:21.572] They dissolve, they are and are causing corrosion, then the rate of corrosion decreases with increase in temperature as the solubility of these gases in the medium decreases. So if the temperature is more than oxygen carbon dioxide, [00:11:39.344] solubility is less and therefore we find that the rate of corrosion actually uh decreases. Now pH. [00:11:49.956] We know in general that when the pH is low, we expect that the rate of corrosion will be high, meaning that in acidic medium we find that the rate of corrosion is higher. Why is it so? Because when we have low pH. [00:12:06.016] we find that the concentration of H plus ions is very high and we know that H plus ions [00:12:12.740] are responsible for one of the cathodic reaction. If this H plus ion concentration is more, obviously we expect that this reaction will take place at a very fast rate. Now if the if this reaction happens fast, this being a cathodic reaction, it is also going to make the anodic [00:12:30.512] oxidation uh faster and therefore we expect that the rate of corrosion will increase. Apart from that another reason in acidic medium is that corrosion product becomes more uh soluble in acetic medium then [00:12:44.324] Every time again a new surface is getting exposed and therefore we find that the rate of corrosion increases. But there is an exception here which we have to note that metals like aluminium actually undergo corrosion faster in highly alkaline solution. [00:13:02.096] Now what could be the reason? Because we know that the corrosion product, for example, in case of aluminium A L2O3 is more soluble in alkaline medium. So even if L2O3 is getting formed, because it dissolves in the alkaline medium, we find that [00:13:19.664] for uh for aluminium in alkaline medium the rate of corrosion is going to be more let us look at another important factor that is called polarization of the anode and the cathode [00:13:34.944] First, let us understand what is polarization. Whenever there is a drift in the electrode potential of the electrode, it is known as polarized. We say that the electrode got polarized. So we know that every electrode has a certain uh electrode potential, but this electrode potential if [00:13:52.656] it changes, if it drips, then we say the electrode has got polarized and it will be referred to as [00:13:59.200] two different electrodes each one will have a different polarization. So if it happens at the anode, they are independent of each other. If it happens at anode we say it's anode anode has got polarized. If it happens at cathode we say cathode has got polarized. So do why would there be a change [00:14:16.848] Electrode potential one of the important reasons could be a change in concentration. We know that when concentration changes, the electrode potential is also going to change. So there are two types of polarization as I mentioned just now, the anode could get polarized or the cathode. So then let us look at anodic polarization. [00:14:36.164] Now what

happens? This here, look at the picture here, we have a metal surface, and at the metal, at the anode, we have anode and cathode getting formed on the surface. Now at the anode, what is the reaction which is taking place? Metal is forming. [00:14:52.608] Mn plus plus n electrons. Now the mn plus as corrosion progresses we find that the mn plus ions are getting formed around the anode.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.800] Now as a rule what would happen due to diffusion these metal ions have to move towards the cathode. But many times the rate at which the uh anode is undergoing oxidation and the rate at which the metal ions are moving towards the cathode is different. So if that rate of formation of [00:15:24.464] Metal ions is very high. In that case, we find there is an accumulation of metalions around the anode, and when this happens, obviously the tendency for this reaction to go in the forward direction will be reduced, and that means the tendency of the metal to undergo. [00:15:42.032] Oxidation is reduced, or in other words, corrosion will be less now. So the rate of corrosion is less, and the anode is said to be polarized because there is a drift in the electrode potential due to accumulation of ions. [00:16:00.672] So this is known as anodic polarization and there could be several factors like lower conductivity of the medium etc which are is causing the anode to be polarized nevertheless the [00:16:12.448] What we have to understand that whenever anodic polarization happens, it is going to reduce the rate of corrosion because the tendency of the metal to uh undergo oxidation is actually decreasing, the reason there is accumulation of ions around the anodic region. [00:16:30.308] Let us talk of cathodic polarization. Now here when we have a metal surface, the cathode reaction involves the uh [00:16:40.928] I reduction of oxygen or reduction of H plus ions, H plus ions taking up electrons, oxygen taking up electrons. Now when would we say that the cathode is polarized? When we find that there is not enough oxygen and H plus which are [00:16:57.312] close to the cat hole. So there's a depletion of oxygen and H plus as corrosion progresses, proceeds. [00:17:04.784] We find that there is a depletion of oxygen and H plus around the cathode, and therefore there is no uh reactant to pick up these electrons and that would result in cathodic polarization. So what happens due to that? [00:17:21.072] Because there are less number of oxygen or H plus ions around the cathode, the rate of um cathodic reaction also decreases. [00:17:29.728] And therefore, we say that the cathode has got polarized and again this is resulting in uh slowing down of the corrosion product. So it could be either due to retarded movement of the cathodic reactant. [00:17:43.904] And also it could be due to retarded removal of the product. So we can say that anodic polarization was due to accumulation of the metal ions around another and cathodic polarization is due to the depletion of either ions or the [00:18:01.540] material which has to undergo or the substances which have to undergo reduction at the cathode. Now polarization effect actually is helping us here. When the anode and cathode get polarized we see that the rate of corrosion comes down. [00:18:21.316] But in certain cases, like in batteries, etc., we don't want polarization. So in that case, depolarizers are used in order to remove polarization and [00:18:36.016] One uh the types of depolarizers are anodic depolarizers, depolarizers, sorry, not polarizers, and cathodic depolarizers. So anodic depolarizers would be complexing agent which are going to react with the MN plus and remove the MN plus from the system so that the reaction goes in the [00:18:55.504] Forward direction. Similarly, cathodic depulverizers like cuplicons, which are going to take up the electrons themselves, and therefore this results in [00:19:06.276] So basically we are removing the depol uh polarization effect and that would again increase the rate of corrosion. So that's why if duplicants are present in the atmosphere or complexing agents, then we

find that the [00:19:24.048] rate of reaction will be more. Another way of in removing the polarization effect is to increase the temperature. Just now while studying temperature we understood that depolarization, we talked about the polarization effect if it is there, then [00:19:44.048] Then it is with increase in temperature, the polarization effect is decreased. We can understood understand this because with increase in temperature, there will be a decrease in polarization. Why? [00:19:54.960] Because now the ions are more temperature, their ions can move easily, the migration of ions can happen and therefore polarization will be uh removed. So increase in temperature can remove polarization. Also simple things like stirring the solution. Basically you are

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:12.624] removing the ions from the metal ions from the anode or you are causing the material which has to undergo reduction, the species which has to undergo reduction to come closer to the capsule. [00:20:27.940] Next, electrical conductivity of the corrosion medium. Now, conductivity of the corrosion medium as it increases, we know corrosion rate also increases. Just now we have discussed it several times that [00:20:44.740] More is the conductivity, faster will the ions move between the anode and the cathode region of the corrosion cell. Now when this happens, [00:20:56.528] we know that the exchange of electrons at the electrode surfaces is going to happen very quickly, very fast, and therefore the corrosion rate is going to increase increase. And there what we can uh one very good [00:21:13.840] example would be that in saline water corrosion happens at a very fast rate. Saline water what does it mean? It contains salt. And s we know are [00:21:24.416] uh good ionic conductors and therefore when there is lot of uh ions in this uh in the water then we find that they will migrate very fast and therefore the exchange of electrons will also happen at the ion and cathode very fast and therefore the rate of corrosion will always [00:21:42.128] is be high in saline water as compared to normal water or distilled water. [00:21:50.244] The last factor we need to talk about is humidity. This is again understood that the very fact that electrochemical corrosion is happening, we know that we need a conducting medium. Now, corrosion, of course, will have to increase with increase in the humidity of the medium. [00:22:08.016] So what happens when the uh the it is humid, the atmosphere is humid, then there is lot of moisture which [00:22:17.024] acts as the conducting medium and it is a and we can get a nice uh galvanic cell over there and therefore this will facilitate corrosion of the metal. In arid areas we find therefore that corrosion will not happen to that great extent but whenever the the the atmosphere [00:22:34.672] Is moisture laden, then we find that corrosion will be fast because it is providing the conducting medium required for the migration of ions. So these are the various factors responsible for corrosion. And in the next class, now we are [00:22:52.272] going to start talking about corrosion control. Thank you.

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Full transcript: 3267 normalized words, 92 timestamped segments, 1378.474667 seconds duration. Low-confidence segments flagged: 0.

EV0000182 · EV0000183 · EV0000184 · EV0000185

12 Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II

This lesson is indexed from the source video titled Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000202 · EV0000203

Graphs and diagrams

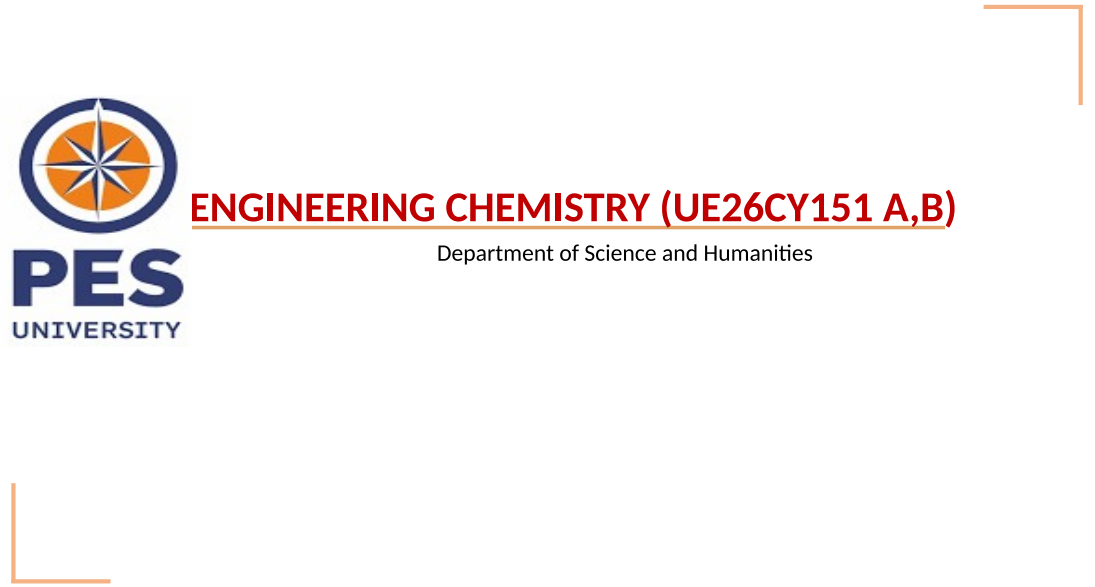

Figure G000107. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

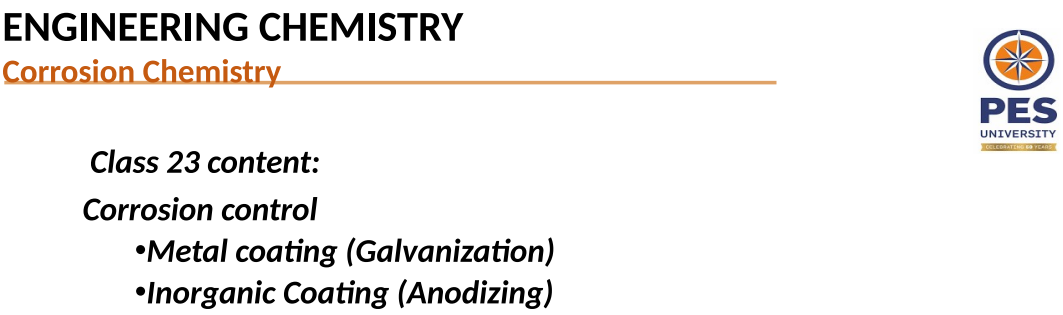

Figure G000108. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 2.

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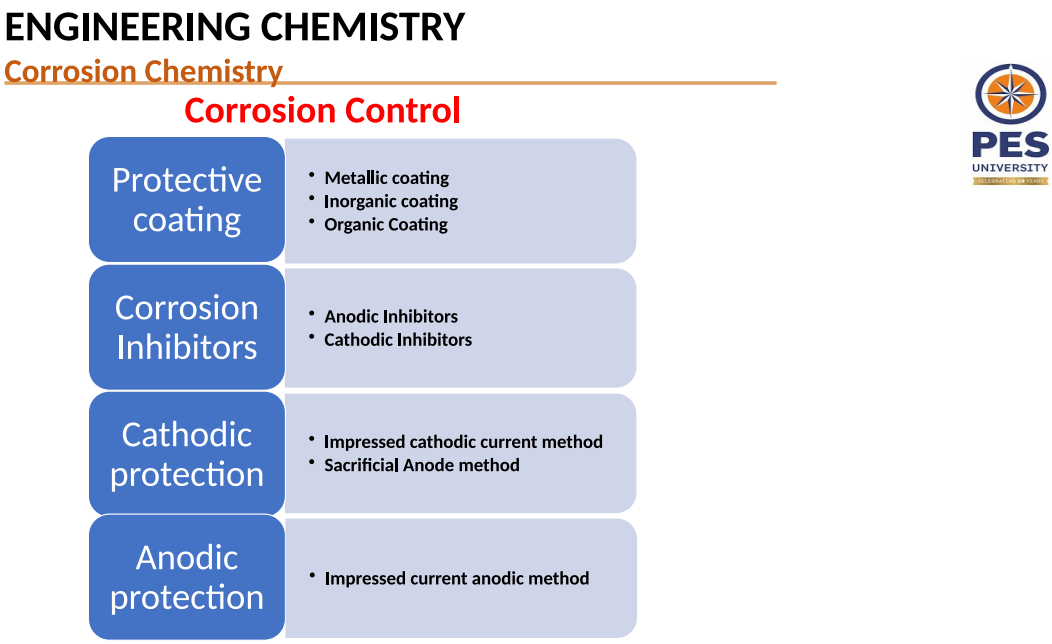

Figure G000109. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 3.

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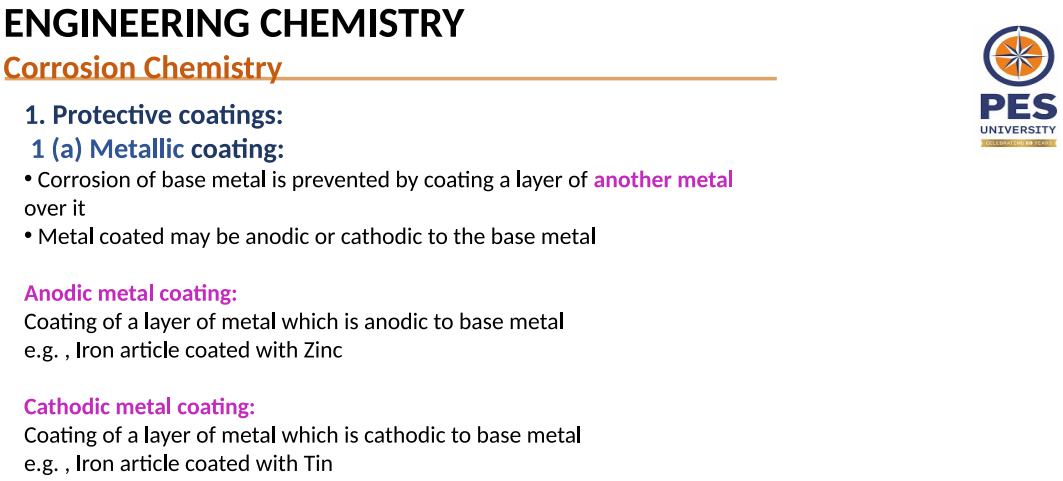

Figure G000110. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 4.

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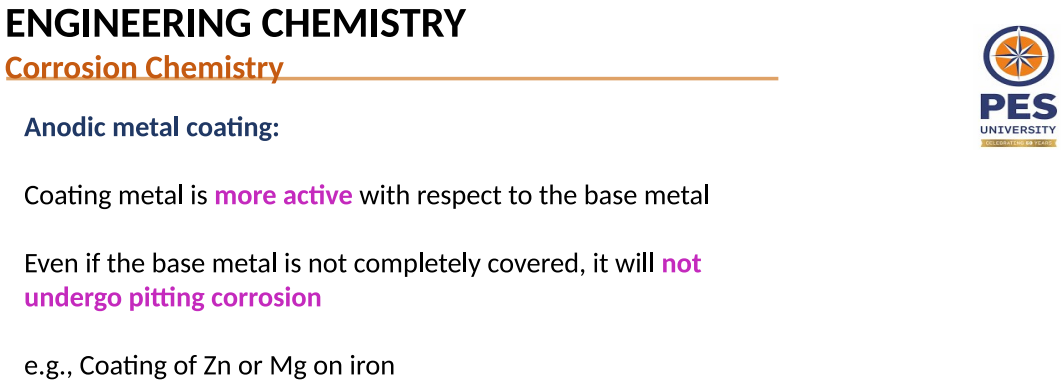

Figure G000111. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 5.

EV0000597

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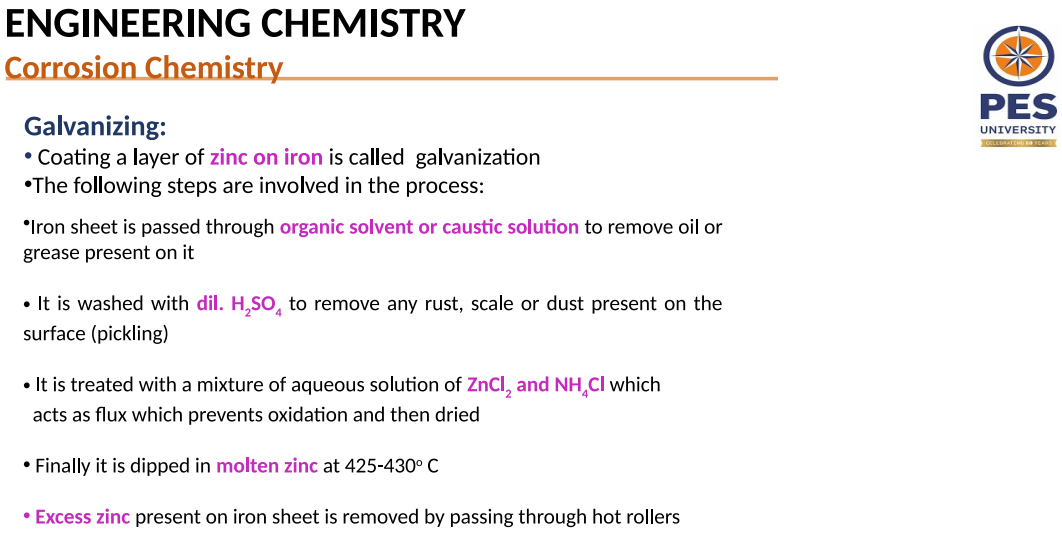

Figure G000112. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 6.

EV0000600

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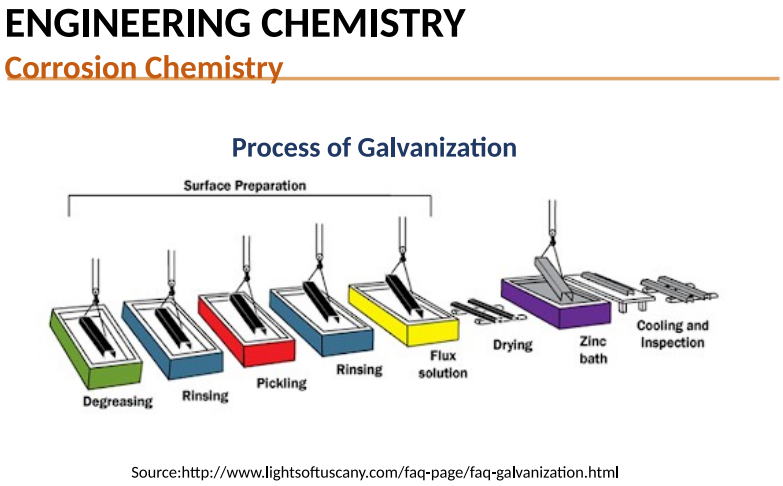

Figure G000113. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 7.

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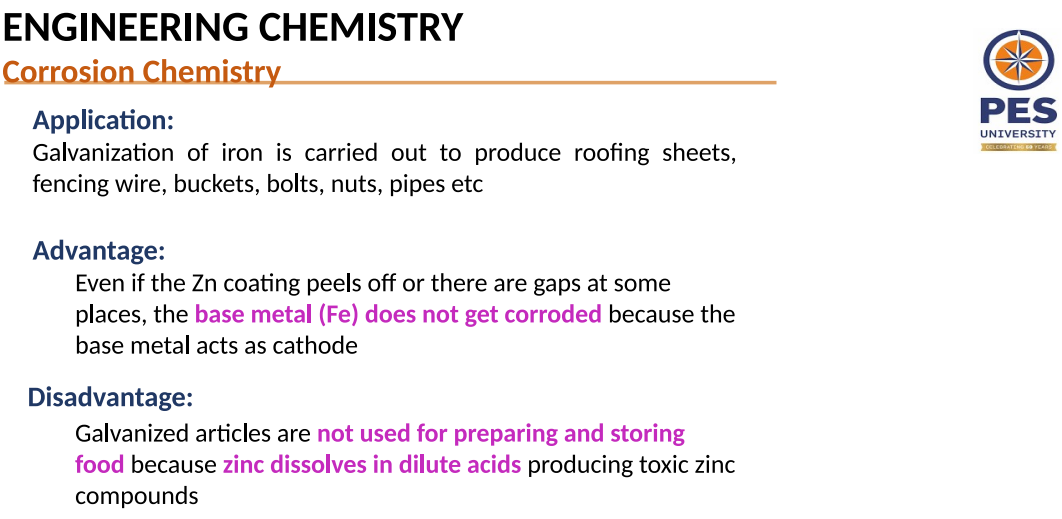

Figure G000114. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 8.

EV0000606

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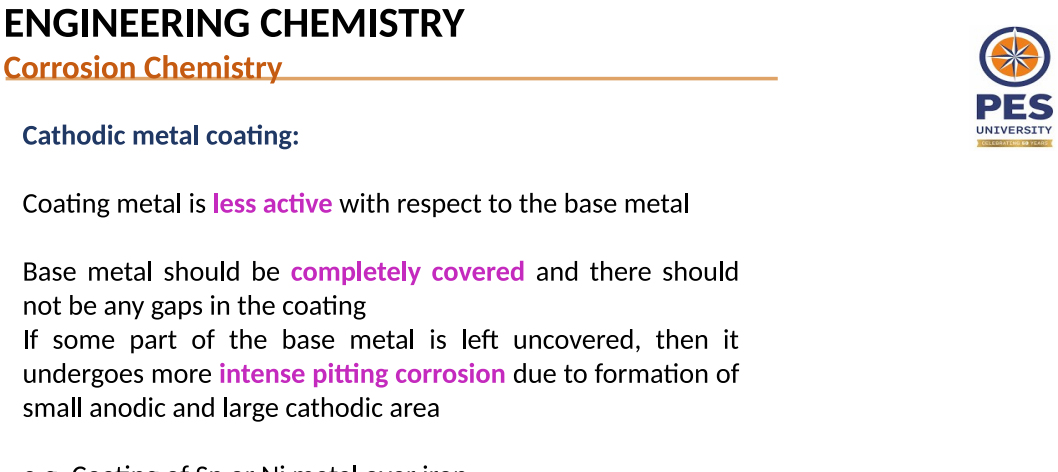

Figure G000115. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 9.

EV0000609

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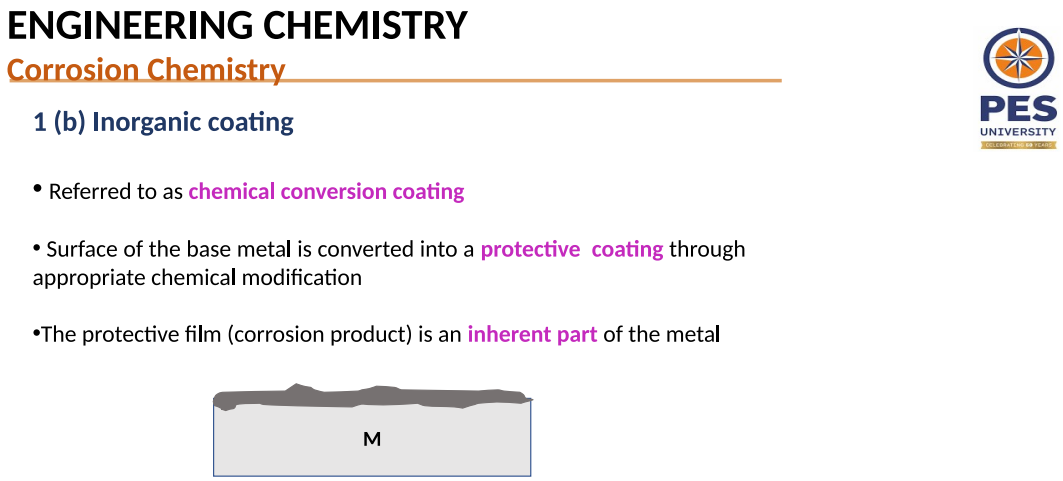

Figure G000116. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 10.

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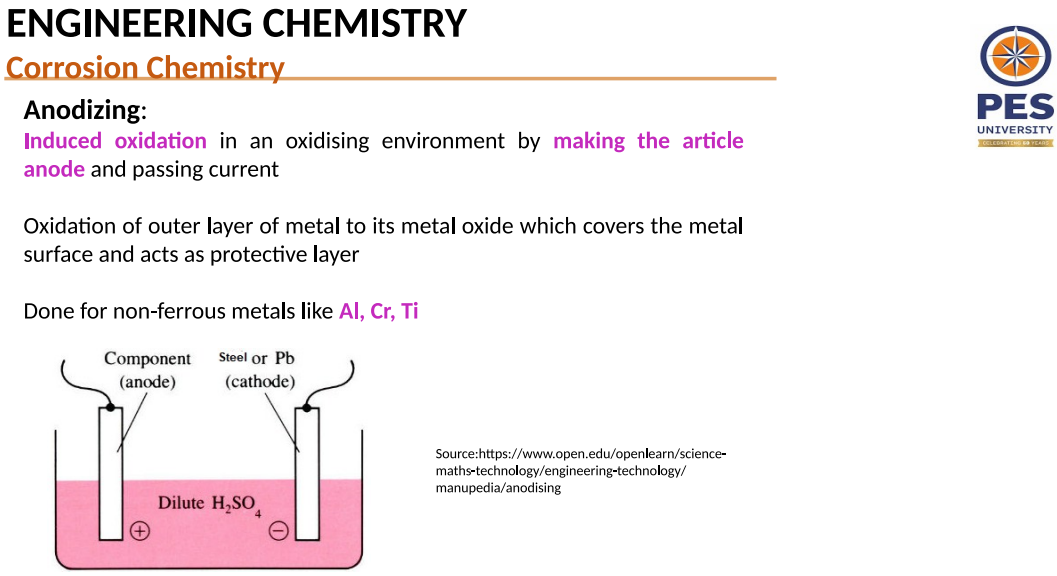

Figure G000117. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 11.

EV0000615

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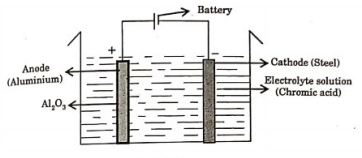
ENGINEERING CHEMISTRY

Corrosion Chemistry

Anodizing of Aluminum:

The process:

- Article is cleaned, degreased, and polished and taken as anode
- It is immersed in an electrolyte consisting of 5-10% chromic acid, sulfuric acid, phosphoric acid, oxalic acid or their mixtures maintained at around 40° C
- Lead or steel is taken as cathode
- Voltage above 40V is applied
- Outer layer of Al is oxidised to Al_2O_3



$$2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2$$

Source: Basuchandra's Engineering chemistry, Banbayalu (2014)

Figure G000118. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 12.

EV0000618

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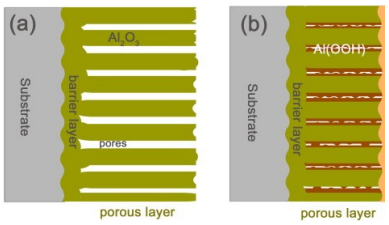
ENGINEERING CHEMISTRY

Corrosion Chemistry

Al_2O_3 formed on the surface is **slightly porous** in nature

The porous layer may be made compact by **sealing** which is done by immersing the anodised article in boiling water or steam

Al_2O_3 is converted into $Al_2O_3.H_2O$ which occupies **higher volume**; the pores are sealed



Source:https://www.sciencedirect.com/science/article/abs/pii/S0257897215304540

Figure G000119. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 13.

EV0000621

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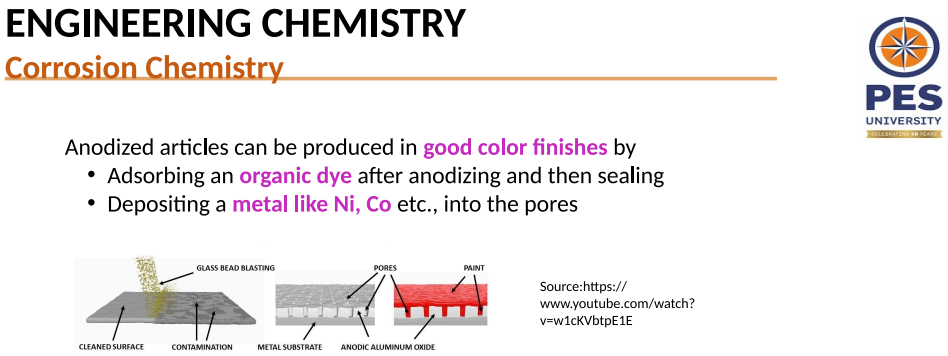

Figure G000120. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 14.

EV0000624

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Figure G000121. Corrosion control metal coating galvanizing inorganic coating anodizing · source slide page 15.

EV0000627

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000584 · EV0000585 · EV0000586
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Class 23 content: Corrosion control Metal coating (Galvanization) Inorganic Coating (Anodizing) EV0000587 · EV0000588 · EV0000589
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Corrosion Control Metallic coating Inorganic coating Organic Coating Protective coating Anodic Inhibitors Cathodic Inhibitors Corrosion Inhibitors Impressed cathodic current method Sacrificial Anode method Cathodic protection Anodic protection Impressed current anodic method EV0000590 · EV0000591 · EV0000592
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 1. Protective coatings: 1 (a) Metallic coating: Corrosion of base metal is prevented by coating a layer of another metal over it Metal coated may be anodic or cathodic to the base metal Anodic metal coating: Coating of a layer of metal which is anodic to base metal e.g. , Iron article coated with Zinc Cathodic metal coating: Coating of a layer of metal which is cathodic to base metal e.g. , Iron article coated with Tin EV0000593 · EV0000594 · EV0000595

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Anodic metal coating: Coating metal is more active with respect to the base metal Even if the base metal is not completely covered, it will not undergo pitting corrosion e.g., Coating of Zn or Mg on iron EV0000596 · EV0000597 · EV0000598
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Galvanizing: Coating a layer of zinc on iron is called galvanization The following steps are involved in the process: Iron sheet is passed through organic solvent or caustic solution to remove oil or grease present on it It is washed with dil. H2SO4 to remove any rust, scale or dust present on the surface (pickling) It is treated with a mixture of aqueous solution of ZnCl2 and NH4Cl which acts as flux which prevents oxidation and then dried Finally it is dipped in molten zinc at 425-430o C Excess zinc present on iron sheet is removed by passing through hot rollers EV0000599 · EV0000600 · EV0000601
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Process of Galvanization [Image text, OCR] Surface Preparation Coolingand Drying Zinc Flux Inspection Rinsing bath Pickling solution Degreasing Rinsing Source:http://www.lightsoftuscany.com/faq-page/faq-galvanization.html EV0000602 · EV0000603 · EV0000604
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Application: Galvanization of iron is carried out to produce roofing sheets, fencing wire, buckets, bolts, nuts, pipes etc Advantage: Even if the Zn coating peels off or there are gaps at some places, the base metal (Fe) does not get corroded because the base metal acts as cathode Galvanized articles are not used for preparing and storing food because zinc dissolves in dilute acids producing toxic zinc compounds Disadvantage: EV0000605 · EV0000606 · EV0000607
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Cathodic metal coating: Coating metal is less active with respect to the base metal Base metal should be completely covered and there should not be any gaps in the coating If some part of the base metal is left uncovered, then it undergoes more intense pitting corrosion due to formation of small anodic and large cathodic area e.g.,Coating of Sn or Ni metal over iron EV0000608 · EV0000609 · EV0000610
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 1 (b) Inorganic coating Referred to as chemical conversion coating Surface of the base metal is converted into a protective coating through appropriate chemical modification The protective film (corrosion product) is an inherent part of the metal Two methods: Anodising and Phosphating M EV0000611 · EV0000612 · EV0000613
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Anodizing: Induced oxidation in an oxidising environment by making the article anode and passing current Oxidation of outer layer of metal to its metal oxide which covers the metal surface and acts as protective layer Done for non-ferrous metals like Al, Cr, Ti [Image text, OCR] Component SteelorPb (anode) (cathode) Dilute HSO 4 Source:https://www.open.edu/openlearn/science-maths-technology/engineering-technology/manupedia/anodising EV0000614 · EV0000615 · EV0000616

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Anodizing of Aluminum: The process: Article is cleaned, degreased, and polished and taken as anode It is immersed in an electrolyte consisting of 5-10% chromic acid, sulfuric acid, phosphoric acid, oxalic acid or their mixtures maintained at around 40o C Lead or steel is taken as cathode Voltage above 40V is applied Outer layer of Al is oxidised to Al2O3 2Al + 3H2O → Al2O3 +3H2 [Image text, OCR] Battery Anode >Cathode(Steel) (Aluminium) Electrolyte solution (Chromicacid) A1O 幸 Anodizing Source: Basuchandra's Engineering chemistry, Banbayalu (2014) EV0000617 · EV0000618 · EV0000619
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Al2O3 formed on the surface is slightly porous in nature The porous layer may be made compact by sealing which is done by immersing the anodised article in boiling water or steam Al2O3 is converted into Al2O3.H2O which occupies higher volume; the pores are sealed [Image text, OCR] (a) (b) AI,O AI(OOH) Substrate barrierlayer Substrate barrierlayer pores porous layer porouslayer PAfilm Source:https://www.sciencedirect.com/science/article/abs/pii/S0257897215304540 EV0000620 · EV0000621 · EV0000622
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Anodized articles can be produced in good color finishes by Adsorbing an organic dye after anodizing and then sealing Depositing a metal like Ni, Co etc., into the pores Applications: Anodized Al is used as an attractive, highly durable, corrosion resistant material in exteriors for roofs, walls, buildings and also in window frames, office partitions, tiffin carriers, etc. [Image text, OCR] BASIC STEPS IN ANODIZING CAPTAIN CORROSION PRE-TREATMENT ANODIZING SEALING GLASSBEADBLASTING PORES PAINT CLEANEDSURFACE CONTAMINATION METALSUBSTRATE ANODICALUMINUMOXIDE Source:https://www.youtube.com/watch?v=w1cKVbtpE1E EV0000623 · EV0000624 · EV0000625
15. Source slide page 15 (extracted text): THANK YOU Department of Science and Humanities EV0000626 · EV0000627 · EV0000628

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.348] Hello everyone, welcome back to this course on engineering chemistry. [00:00:05.860] We continue with module 6 on corrosion chemistry, and today's class we are going to talk about corrosion control and we will take up the next type of coating that is inorganic coating, wherein we will talk about anodysing and phosphating, and then we will look at what is organic coating. [00:00:27.364] Inorganic coating. So in the previous class we studied about metal coating where [00:00:34.560] Uh one of one metal was taken, the metal which has to be protected, and it was covered with another metal. Now here we are going to, as the name suggests, we are going to cover the surface with an inorganic compound. [00:00:48.288] But there's a small difference, and let us understand what that is. Now this method is also referred to as chemical conversion coating. [00:00:57.860] Now, let us see what exactly we do here. The surface of the base metal is converted into a protective coating through appropriate chemical modification. So, what does that mean? Basically, [00:01:11.840] We have this metal, we can look at the picture here. Now the top surface of the metal is allowed to undergo corrosion, but what happens after that is this inorganic compound, the compound which is formed due to corrosion. [00:01:29.456] Is now forming a protective film over the surface of the metal and that creates a barrier between the metal and the environment. So this film is actually act as a protective

coating for the metal. [00:01:47.524] So, by chemical modification, we have created a protective coating which is going to protect the metal from further corrosion. [00:01:56.164] The difference from metallic coating is when we talked about metallic coating, we have a metal M1, and we coated it with another metal, say M2, and in this case we know that M1 is the base metal, M2 is an example. [00:02:13.936] External another metal which is being coated onto the surface. But what is the what are we doing here in inorganic coating? It is like we are the metal itself is undergoing corrosion and creating this protective film over the surface. So we can say that the [00:02:31.504] Protective film which is formed here is an inherent part of the metal, it's an integral part of the metal itself. So it is rightly known as chemical conversion coating. So by using a chemical reaction, we are converting the metal surface. [00:02:49.072] surface into an inorganic compound and hence the name inorganic coating. Now there are two methods in this one is anodizing and one is phosphating. Let us study what is anodizing. [00:03:00.228] The definition would be we are going to induce oxidation by providing an oxidizing environment, and we are going to make the article into anode and then pass current. Now making the article into anode sounds a little strange. Why? Because [00:03:20.128] We know that it is the anode which undergoes corrosion and cathode remains unaffected. But we are making the metal into cat anode and we are passing current also. So we are making sure that the metal [00:03:33.092] Will undergo oxidation, but there's a purpose here. The oxide which is getting formed, the metal oxide which is getting formed on the surface is going to cover the metal surface and act as a protective layer. [00:03:50.608] So it's very obvious we cannot do it for all types of metals, so only certain category of metals. [00:03:58.368] can be anodized and they are told that they are usually the non-ferrous metals like aluminium, chromium, titanium. So we understand why aluminium particles can be anodized because aluminium when it undergoes corrosion it becomes [00:04:15.872] Passive and further corrosion is prevented because of this protective layer which has got formed over the surface. So on the surface we have Al2O3, which is going to form a nice protective film over the surface, and therefore will create a barrier and hence further corrosion is prevented. So [00:04:35.876] What are we doing here as the definition goes, we have taken the article as the anode and of course we are using a power supply here, there has to be a power supply. [00:04:50.208] And this article is going to be connected like this. And here we have to when we have an anode, of course, we have to have a cathode. That will be some kind of uh inert metal or some conducting metal has to be used. And this is uh usually aluminium or lead that is used as a cathode.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:07.888] And we are providing the uh atmosphere which is oxidizing in nature. So dilute H2SO4. So what are let us check out the definition. So this we can see is a DC supply here. So what let us check out definition now. [00:05:24.736] We are saying that we are going to induce oxidation, yes, by passing the by passing current. So we are going to induce oxidation on the anode. [00:05:34.544] and then it will uh it will be done in an oxidizing environment and only such metals which can form a protective film obviously can be anodized. Let's understand the full process here. So anodizing of aluminium let us take aluminium. [00:05:51.492] Because we know, and it's a big industry where anodizing of aluminium is done, and that passive layer is created onto the surface of aluminium by inducing the oxidation. So that means [00:06:05.968] Why are we doing it? We know that aluminum, anyway, when it's exposed to the when it's exposed to the environment, it is going to undergo uh it's going to become passive. Now, why are we doing it here? Because we can control the thickness of the

layer by [00:06:21.200] Inducing the oxidation ourselves. Now what is done? Of course, any coating when you do, we have to clean the article. That's exactly what is done. And the first step, the article is clean, decreased, and polish, and then it is taken as the anode. [00:06:37.008] So the first step is you have chosen your anode that is the article which has to be coated. Now it is immersed into an electrolyte which is and should be an oxidizing environment. So it could be chromic acid. [00:06:51.876] Sulfuric acid, phosphoric acid, oxanic acid, or it could be a mixture, and this is maintained at around 40 degrees. Now we need a cathode. So lead or steel is usually taken as cathode, anything which is conducting. And the voltage is about [00:07:09.200] 40 volts which is applied here, and what happens because of this now when you pass current, there will be an oxide layer getting formed over the surface. So around this aluminium, you will find that aluminium oxide has got form. So outer layer. [00:07:29.028] Of aluminium is oxidized to Al2O3. So 2AL plus 3H2O is going to form Al2O3 plus 3H2. [00:07:42.148] Aluminium oxide now formed on the surface is slightly porous in nature. [00:07:49.892] Now, how if it is porous in nature, then it is not going to be very useful. So we need to make this layer compact, and that is done by a process called sealing. [00:08:05.860] So, what is sealing? Let us see. The article is dipped in boiling water or steam. So, we are going to immerse the article in boiling water or steam so that now the porous layer which is formed here of AL2O3. [00:08:25.552] It gets converted to Al2O3HOT, but hydrated. And since this is a larger molecule, it is going to occupy a higher volume and all the pores are going to get sealed. You can see in this nice picture over here, this is the substrate. [00:08:43.492] And aluminium oxide has got formed over the surface. Now the initial lay are compact layer, it is forming a nice barrier, but the layer later you will find that these pores exist in the aluminium oxide layers now. What is done for that? [00:09:00.896] If we dip this anodized article into hot water, boiling water, or steam, it is found that [00:09:10.192] It forms ALOOH or it gets hydrated, and you can see that the volume of this hydrated ALOOH is much larger, and therefore it is going to seal all the pores. So this process of sealing has to be done after. [00:09:26.368] The process of anonymizing is done for any aluminium article. [00:09:32.516] Now, if sometimes we want to give a good color finish, most of these uh coated articles are generally dull in color. So we want to give it a good color finish. That is also possible in the case of anodised articles. Why? Because we know again that the layers are poor. [00:09:50.288] In nature. So if we use an organic dye, it can it gets absorbed into the pores. After anodising, we are going to use the dip the article in

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.324] An organic dye that color seeps in through the pores and then it is sealed, so you get a nice colored finish. You can see here that first the [00:10:12.976] uh surface is being cleaned and it has got anodized but there are these pores you can see the pores here now when you dip it in any paint any organic dye this color is going to get in and you will get a very good colored finish for this anodized article. [00:10:30.692] Or another method is by depositing a metal like nickel cobalt. You know that nickel cobalt salts are very uh colourful, so we can deposit these metals, nickel, cobalt, etc., into the pores, and again this will give us a nice color finish. So where are the applications for uh [00:10:48.496] Such anodizing or anodized aluminium. We know that it is used to as an attractive, highly durable, and corrosion resistant material in exteriors for roofs, walls, building buildings, and also in window frames, office partitions, stiffen carriers, mostly. [00:11:08.400] you will find that anodized articles are used for hookware. They are very very popular in cookware. The aluminum articles are anodized and they are used for hookware. Now let's talk of the next method [00:11:26.832] Which

is phosphate. Again, this is also a chemical conversion coating or an inorganic coating matter. So from the name we can make out that we are going to uh make a phosphate layer over the surface. So how are we going to do that? So let is a definition [00:11:44.496] Converting the surface metal atoms into their phosphates by either by chemical or electrochemical reactions. So we are trying to create a phosphate layer. Now originally phosphates were done only for ferris a lot. [00:12:01.924] like iron or steel, but now it has been extended to zinc, aluminum, and alloy alloys. Nevertheless for iron and steel, this is very commonly done getting a phosphate layer over the surface. So how do we do this? [00:12:20.192] If you want to do chemical phosphating, then it is just done by dipping the particle into a phosphate bar. But if it is electrolychical phosphate, then the process of electrolysis is used and current is used to get the phosphate layer onto the surface. So we will see [00:12:37.936] study the simple method of chemical phosphating. What is the process? So as I mentioned, we need a phosphating mark in which the article is going to be there. So what would be the components of the [00:12:52.228] phosphoric so first and foremost of course we need phosphoric acid because we want the iron [00:12:59.908] to react with the to form the phosphate layer. So obviously it has to react with phosphoric acid. Now next we require another component that is the metal phosphates. Now metal phosphates of zinc [00:13:17.120] Or manganese. What is their role? The zinc and manganese phosphates are [00:13:24.100] Going to increase the adherents of this coating which is getting formed. The phosphate layer, which is getting formed, at the same time, it is going to decrease the porosity of the coating because we are basically trying to create a [00:13:39.600] uh film on the surface so it does film appears and also there is it is less porous so metal phosphates of zinc and manganese are used for this purpose and of course here we don't we are not using any um power [00:13:56.580] So there has to be an accelerator which is going to speed up the reaction and those could be nitrites, chlorates, peroxides, etc., which will help [00:14:06.224] The reaction to happen at a fast rate. So these are the three important components of a phosphate bar, and the pH is maintained around 3 and the temperature is about 35 to 40 degrees centigrade. [00:14:23.460] What exactly is happening when you dip the article in this phosphing part? It's very clear the reaction will include what? The dissolution of the metal, and then there is a formation of [00:14:34.928] phosphate layer which is due to the reaction with the bark solution and this phosphate layer is going to get deposited onto the surface. [00:14:46.304] So there are three steps, the metal gets dissolved and then it forms a phosphate with the bar solution, and then subsequently it is going to get deposit onto the surface of the metal. So this would be uh what would be the reaction taking place here? Iron plus

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.696] Phosphoric acid is going to give you Fe3 BO4 twice, which is going to get deposit on the surface and 3 hydrogen. Now this layer which is getting formed on the surface is porous in nature. [00:15:20.836] So it is not forming a compact layer. In fact, it is known for its porosity. Any item compound usually will be porous. So uh [00:15:32.464] You will find that it is porous in nature. Now, when it is porous in nature, how is it forming uh protective film or how is it going to help us in preventing corrosion? Well, on its own, it is does not have. [00:15:47.300] Very high corrosion resistance, but it is used as a base for painting. So before painting the article. [00:15:58.724] The article is first subjected to phosphating. So when it is subjected to phosphating, it creates this nice porous phosphate layer upon the surface. And now when we paint on this [00:16:13.712] porous layer the paint the paint will get interlocked onto the surface very strongly. So it is going to adhere to the surface very strongly. So

this is the uh reason why phosphating is done. Though it is forming a porous layer it [00:16:31.376] It is because this porous layer, once on this porous layer paint is uh applied, then it is going to uh give it a very good addition quality and because it is going to interlock the paint due to the pores. So phosphate coatings are usually very thin. [00:16:48.384] So it will be about 3 micrometers and then because it is porous, as I mentioned just now, it is going to interlock the paint very well. [00:16:57.348] So, what are the applications? As you can understand, it has a very big industry phosphating because it is the pre-treatment which is done for all automobile bodies. [00:17:11.584] refrigerators, washing machines, all steel sheet fabrications. So they have to undergo phosphating before they are subjected to painting. [00:17:26.244] Now coming to organic coating, we are not going to study in detail, but we should know what is organic coating. So in this year, as the name suggests, we are going to use paints and lacquers. We know this is a very common method of corrosion control. So paints and lacquers are [00:17:44.016] Most widely used as anti-corrosion coatings, and it's understood why because they keep out the air and the [00:17:52.544] Moisture from coming in contact with the metal surface, they're basically creating a barrier between the metal surface and the corrosion environment. So these are the different methods. [00:18:06.320] by which we created a protective coating on the surface. The last class we talked about metal coating and then we talked about protect uh inorganic coating and then the organic coating. [00:18:18.340] In the next class, we are going to talk about another method of corrosion control and that is about corrosion inhibitors. Thank you.

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Full transcript: 2662 normalized words, 76 timestamped segments, 1108.501333 seconds duration. Low-confidence segments flagged: 0.

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13 Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I

This lesson is indexed from the source video titled Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000218 · EV0000219

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.676] Hello, everyone, welcome back to this course on engineering chemistry. We will continue with module 6 on corrosion chemistry, and today's class we are going to talk about corrosion control. In that we will look at metallic coating and specifically galvanization and tinning. [00:00:21.700] When we talk of corrosion control, the first thing which comes to our mind is to create a barrier between the surface and the environment. Now, except for the very noble metals, we know that all metals will undergo corrosion when exposed to the right kind of environment. [00:00:39.748] So protective coating is a very good

method by which we can control corrosion to a great extent. In protective coating, we have metallic coating, inorganic coating, and organic coating. [00:00:53.220] Then another method of corrosion control is to use corrosion inhibitors. Again, we have two classes here: anodic inhibitors and catodic inhibitors. [00:01:02.500] The other method would be cathodic protection, and in that we are going to study about two methods in there, that is sacrificial anode method and the impressed cathodic current method. Finally, we will study about anodic protection, which is also known as impressed anodic current method. [00:01:21.508] Let us start with protective coatings. The first type of coating we are going to look at is metallic coating. So what would that be as the name suggests? Corrosion of the base metal can be prevented by coating it with a layer of another metal over it. [00:01:40.420] Now, since we are coating one metal over another metal, this coating metal could be either anodic to the base metal or it could be cathodic to the base metal. Therefore, we will have two types of coatings which are possible. [00:01:58.800] One is called anodic metal coating where we are going to coat a layer of metal which is anodic with respect to the base metal. A good example would be if you have an iron article and you are covering it with zinc metal, this would be [00:02:16.096] anodic metal coating and then there is another method where we which we call it as the cathodic metal coating where we are going to coat the metal with um another metal [00:02:28.624] Which is cathodic with respect to the base metal. And an example for that would be an iron article which is coated with tin. [00:02:38.788] Let us talk about anonic metal coating. As I said just now, the coating metal is more active with respect to the base metal. So even if the base metal [00:02:51.840] Is not completely covered, it is not going to undergo pitting corrosion. This is a very big advantage of anodic metal coating. So suppose we have a metal like this, say one metal M1, and we coat it. [00:03:10.468] With another metal, say this is M2. Now if M2 happens to be anodic with respect [00:03:19.556] To the base metal. So the coating is anodic with respect to the base metal. Therefore, even if there is a gap here, we will find that if the coating is not completely covering the surface, there is a gap on the surface. You have studied that. [00:03:35.936] This will not give rise to pitting corrosion because as long as the anode is in contact with the [00:03:44.864] Cathode, it is the anode which undergoes corrosion. So there cannot be any intense corrosion or pitting corrosion. That is the biggest advantage of having anodic metal coating. And a simple example would be coating of zinc or magnesium on iron. [00:04:02.084] Let us take the case of galvanization. Now, what is galvanizing? When we cover the metal with zinc on the surface, like zinc is acting as the coating, then this [00:04:18.096] process is known as galvanization. It's usually on iron. So we say what is galvanization? Coating a layer of zinc on iron would be referred to as galvanization. What would be the various steps involved in this process? [00:04:31.744] Now we know that basically we have to dip this iron article into zinc, but before doing that, we need to [00:04:42.000] Make sure that the surface is absolutely clean because we know any coating when it is made on a surface, the smoother the surface, the better the coating is going to adhere to the surface. So the first and foremost treatment before [00:04:57.424] Uh coating any surface is to make it absolutely clean. So in that the first step would be

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.212] The iron sheet is passed to an organic solvent or a caustic solution to remove any oil or grease which may be present on the surface. Next, it has to be washed with dilute sulfuric acid. This process is known as pickling. [00:05:26.628] The iron sheet is passed through dilute sulfuric acid so that [00:05:31.204] Any kind of rust, steel, scale, or dust can be removed from the surface. And now the next treatment is again

very important. It is going to pass through an aqueous solution of zinc chloride and ammonium chloride. Now this is a [00:05:48.224] a flux. What happens when we use this flux onto the surface is that it will prevent the oxidation. So this is very important using zinc chloride and ammonium chloride as flux. [00:06:03.600] Once that is done, it is dried. And finally, we come to the main tank, that is the galvanizing tank, where we are going to dip the article into a zinc into molten zinc. [00:06:18.144] Of course, if you want a coating of zinc onto the surface, it should be molten zinc, and that melting point of zinc is around 425, 430. So you have to maintain the temperature at that point so that the zinc is in the molten form and it gets coated onto the [00:06:33.712] uh surface of the iron sheet. And finally, excess zinc which is present on the iron sheet has to be removed by passing it through hot rollers. So this would [00:06:47.536] be the surfa the method of galvanization, the whole process of galvanization. [00:06:54.596] Let us see this picture and understand. So as I told you, first we need to prepare the surface before it is galvanized. So this is where the actual galvanization is taking place. So you can see very clear here, as I mentioned, degreasing is done and then it is washed and again pickling is done. [00:07:12.368] in with dilute estrus support again there is rinsing and then finally the flux and then it is dried and you have then it gets into the zinc bar. [00:07:23.616] Now the this is where molten zinc is there and iron sheet or any article it is going to get coated with this and then it excess zinc is removed from the surface and we uh it is then cooled and sent for inspection. [00:07:39.940] Now, this is called dry dry method. There's also a wet method where the flux need not be separate. What is done is the zinc bath itself is covered with this flux. So you need not dry in that case. So it's a wet method. That is also done in some cases. [00:07:58.724] Now, what are the applications for this galvanized iron? Galvanized iron, you know, we have heard of GI pipes and GI uh sheets. They are called galvanized iron sheets and pipes, which are which is used in [00:08:13.604] uh building construction material. So galvanization of iron is carried out to produce roofing sheets, fencing wire, buckets, bolts, nuts, pipes, etc. [00:08:25.412] So, as I mentioned just in the beginning of the class, the advantage of this galvanization is that even if the zinc coat peels off or there are gaps on the surface of the [00:08:41.456] uh uh uh gaps in the coating we find that the base metal iron will never get corroded because as long as zinc is present in contact with the iron it is zinc which gets corroded and not the uh iron article. [00:08:59.152] Which is underneath. So this is a very good advantage of galvanization. [00:09:04.960] But there is also a serious disadvantage. What is the disadvantage? That galvanized articles are not used for preparing and storing food because the zinc which is present on the surface is going to dissolve in dilute acids, and that is going to produce. [00:09:22.608] toxic compounds. So therefore we find that the uh galvanized article the galvanized sheets cannot be used for uh storing food [00:09:34.404] Uh making those storing uh food storing storing containers. So otherwise galvanization is very good because as long as zinc is there, we don't have any problem. So that means we need to look for another type of coating, and that is where we talk about the [00:09:52.176] Metal coating. Now what is done here? The coating metal is less active with respect to the base metal. Means it is more noble as compared to the base metal. Now because it is more noble, as long as it covers the entire surface, corrosion will be much less.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.744] But there is a serious problem here, which we have talked about many times. If the coating [00:10:17.936] is not uniform. So again we have this metal m1 and this metal m2 and if there are gaps on the surface this metal m2 is coating onto the m1 [00:10:30.692] Metal, if there are gaps then now what has happened

because the coating is cathodic with respect to the base metal, this gap is going to expose the [00:10:41.616] metal M1, which is now acting as the anode. So you have a case of small anode and large cathode, just like we understood in during pitting correct studying pitting corrosion, when we were studying pitting corrosion, and therefore this is going to cause severe [00:10:56.736] fitting corrosion and the metal actually underneath is going to get more damaged than than if it was left uncovered. So an example for this cathodic cathodic metal coating would be coating of tin [00:11:12.960] or nickel metal over iron. So let us look at the tinning process. Now what would be tinning? How would we define tinning? Coating a layer of tin over iron will be called the tinning process. So what are the steps? Again we understand the article [00:11:32.352] Has to be dipped. Now this method is known as the hot dipping method. In fact, even the galvanization which we did just now is known as the hot dipping method. It's understood why because we are dipping the article in the [00:11:49.392] molten zinc which is hot. Similarly, here we are going to do tinning. So therefore we need to dip the article in molten tin, but before that again we need to [00:12:00.484] Take care, I mean we need to do the surface preparation, and here specifically, because we know if there are gaps left on the surface, it is going to result in intense corrosion or pitting corrosion of the article which we want to protect. Therefore, we need to take special precautions. [00:12:18.256] So what can we do? First the steps are same. We are going to pass it through organic solvent or caustic solution to remove the oil or grease present on the surface, and then it is washed with dilute H2O4. This process is called pickling to remove any rust, scale, dust, etc. [00:12:35.824] And then next step is very important where it is going to be treated with zinc chloride flux. Now, what is the role of the zinc chloride flux? It is going to make sure that the tin coating adheres to the surface. [00:12:53.956] So zinc chloride flux is very important here and we have to take care that the coating sticks to the surface and that is done with the help of zinc chloride flux. And then it is dipped in the uh in molten tin. Now tin melts at around 230 degrees centigrade, so it is maintained at 230 degrees. [00:13:11.728] Centigrade, so that it's in the molten form, and then uh your article is dipped in it. Obviously, it is going to get a tin coating. Excess tin now is removed. [00:13:22.820] By passing through hot rollers, and now it is then immersed in palm oil. Now, what is the role of palm oil here? It prevents the surface oxidation of the freshly coated tin. So these are the [00:13:39.904] various precautions which we take when we are carrying out the tinning process. So these are the various now we know how tinning is done. [00:13:49.824] Let us look at this picture and understand. So we have a decrease tin sheet. Here we have not shown the organic solvent or the uh caustic solution treatment. Now the decrease tin sheet is put into the sulfuric acid. This is where the uh pickling is taking place. After pickling, you can see that the sheet. [00:14:07.504] It is passing through a bath of uh zinc chloride flux and then it comes to the molten tin area, and finally it is going to the palm oil, which is going to cover the surface and prevent oxidation of the freshly coated tin. Then it goes to the rollers and uh [00:14:25.072] All the excess tennis report and this is the process of tinning. What are the applications? It's understood that tin coated steel. [00:14:35.972] Why did we uh talk about this process in the first place? Because we know galvanized sheets cannot be used for making um containers which store food. So here, of course, tin coated steel is used for manufacturing containers used for storing food stuff such as jam, instant food. [00:14:53.744] Make fruits, pickles, etc. We know that because it's highly uh they will be slightly acidic in nature, and they could.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.804] That would provide the right environment for corrosion. And also, if it was zinc, it would have dissolved in this, but here it may we have used tin coating, so it is safe. And obviously, that would be the biggest advantage of tinning over galvanization. So tin coating [00:15:18.576] Tin coated steel can be used for making containers to store food, but of course, we know there is a serious disadvantage that there will be intense and accelerated corrosion if [00:15:31.556] Even a small crack is formed, or while coating any gaps left on the surface, this would result in pitting corrosion. [00:15:40.608] due to the formation of small anode and large cathode area. So this is how metallic coating is done anodic coating and cathodic coating. So in the next class we are going to talk about [00:15:55.812] Another method of protective coating and that is inorganic coating. Thank you.

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Full transcript: 2392 normalized words, 67 timestamped segments, 962.816 seconds duration. Low-confidence segments flagged: 0.

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14 Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II

This lesson is indexed from the source video titled Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000234 · EV0000235

Graphs and diagrams

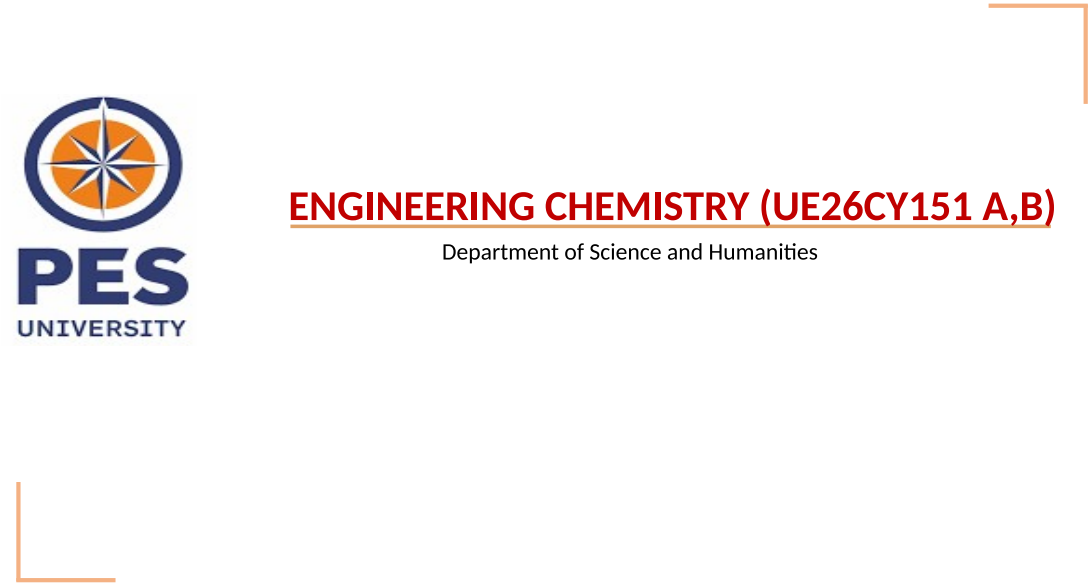

Figure G000122. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 1.

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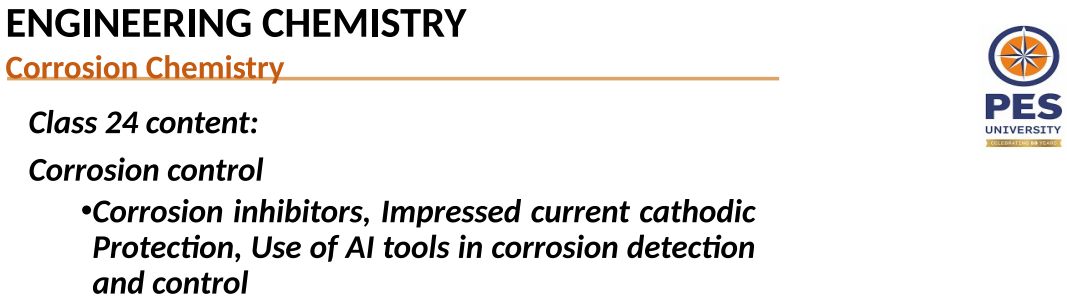

Figure G000123. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 2.

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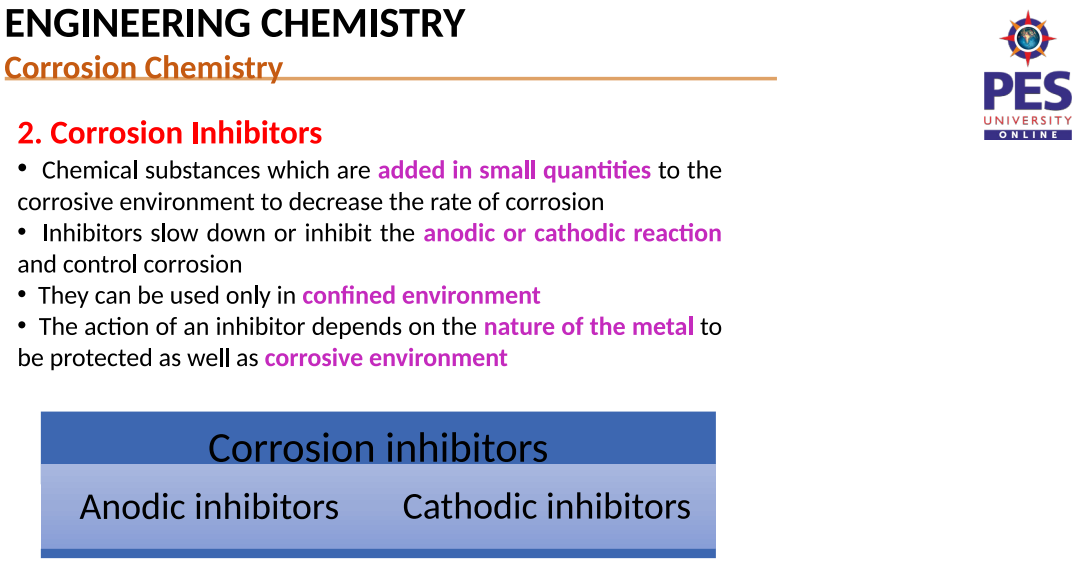

Figure G000124. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 3.

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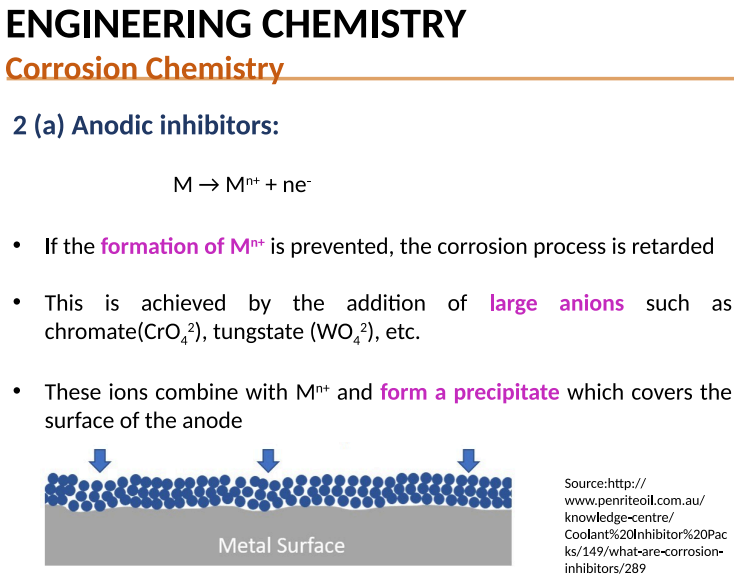

Figure G000125. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 4.

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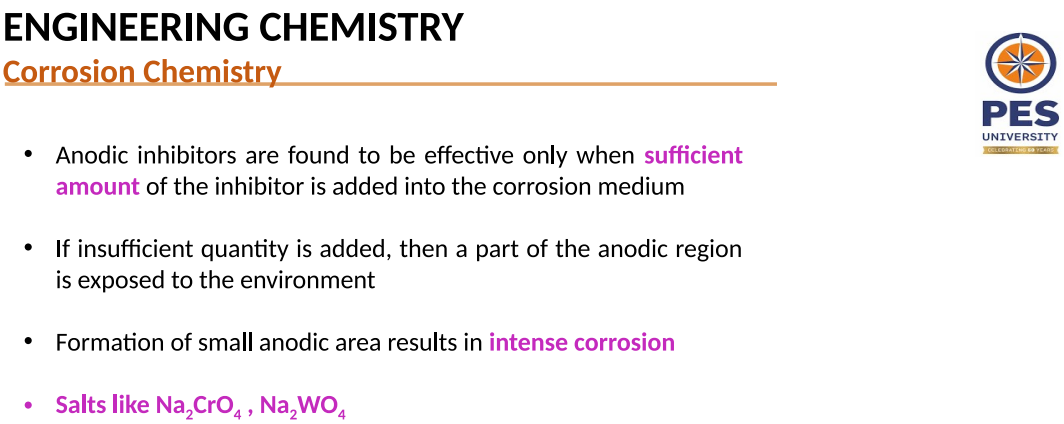

Figure G000126. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 5.

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Figure G000127. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 6.

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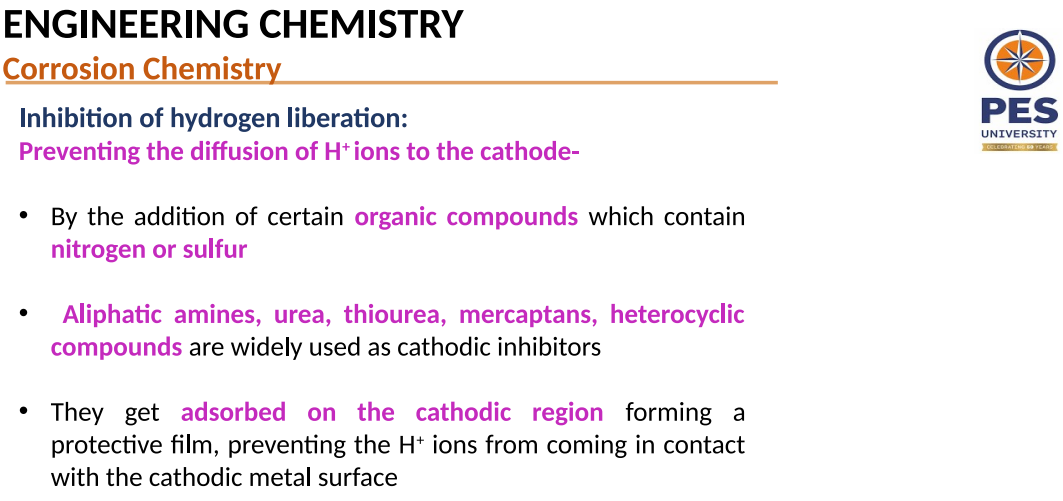

Figure G000128. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 7.

EV0000648

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Figure G000129. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 8.

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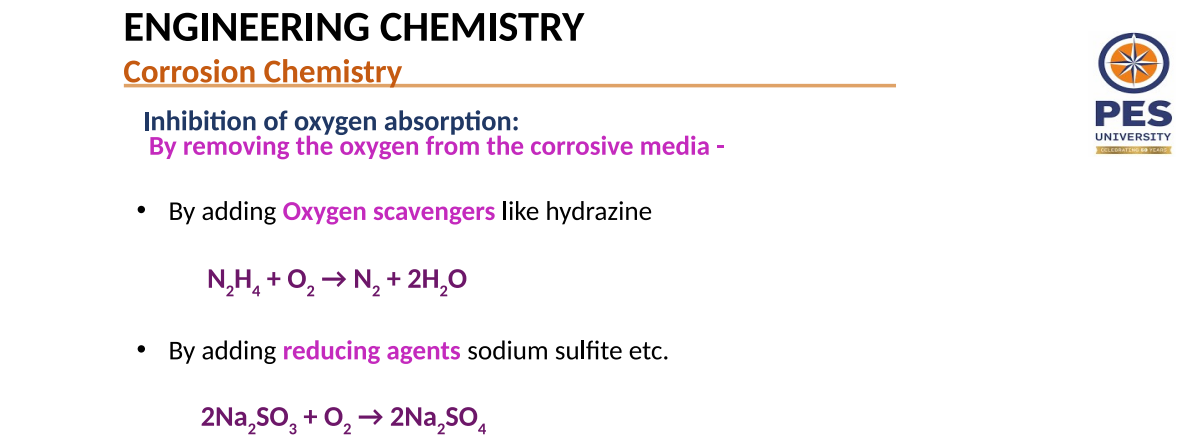

Figure G000130. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 9.

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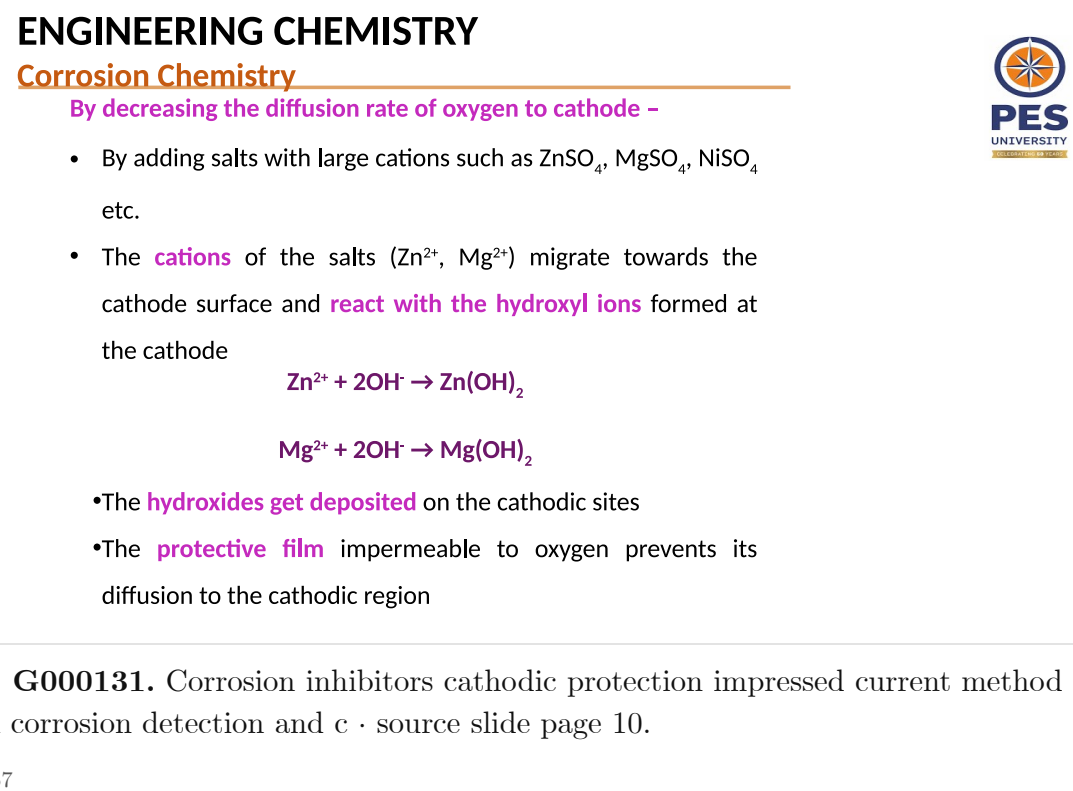

Figure G000131. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 10.

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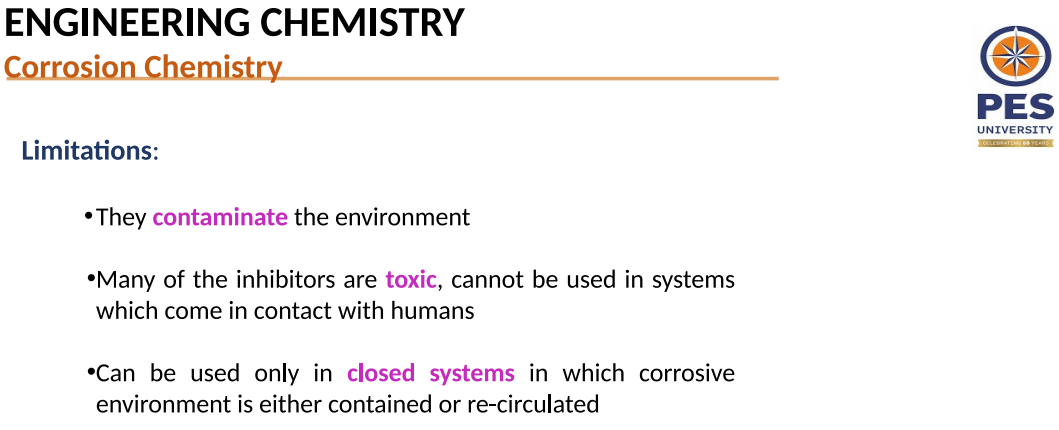

Figure G000132. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 11.

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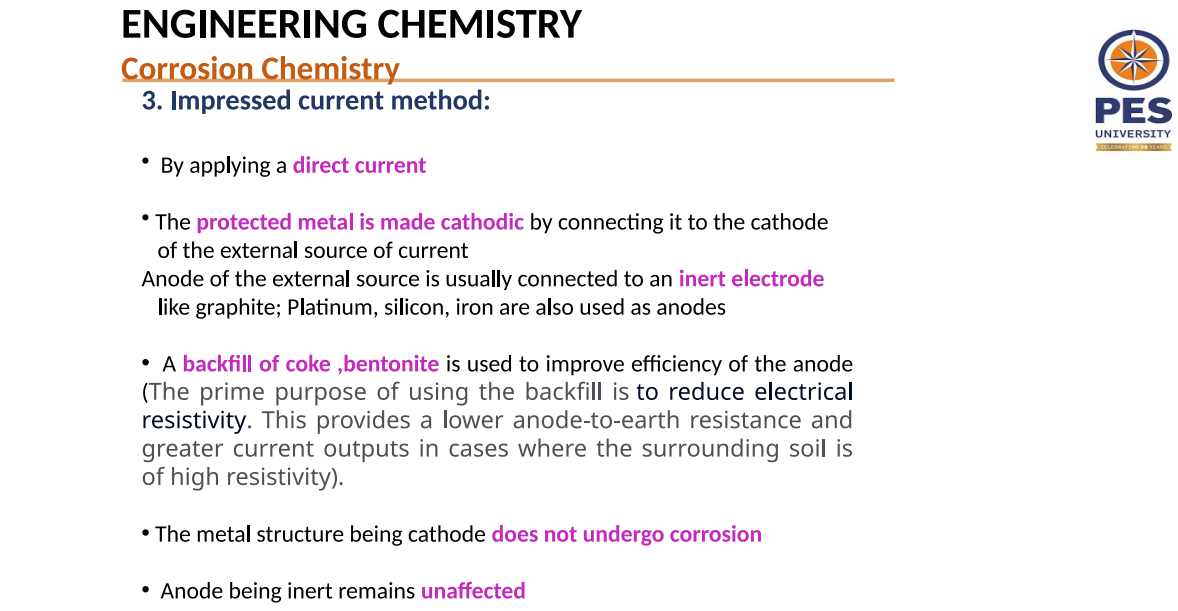

Figure G000133. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 12.

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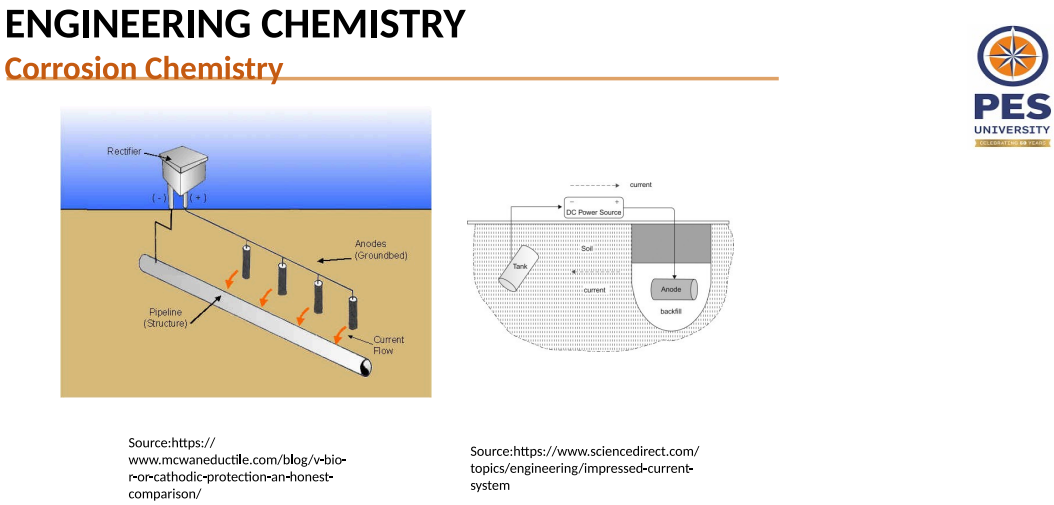

Figure G000134. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 13.

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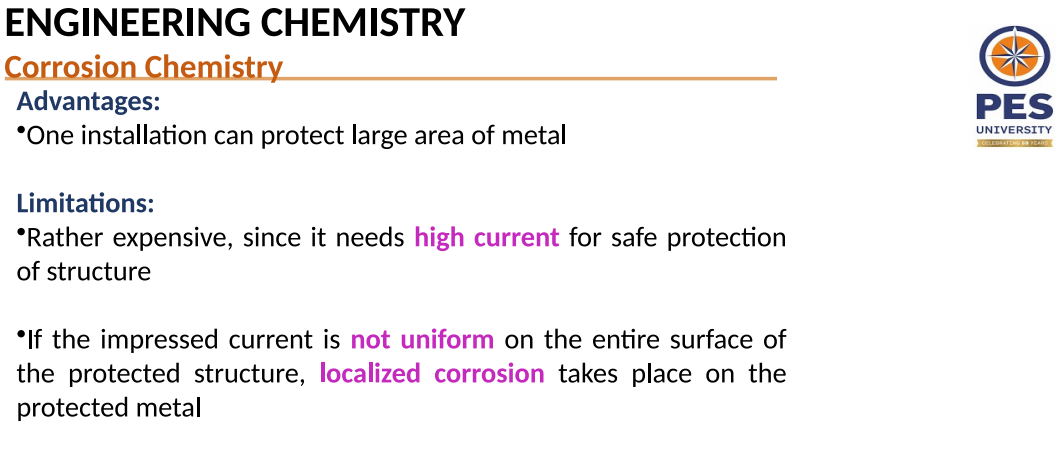

Figure G000135. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 14.

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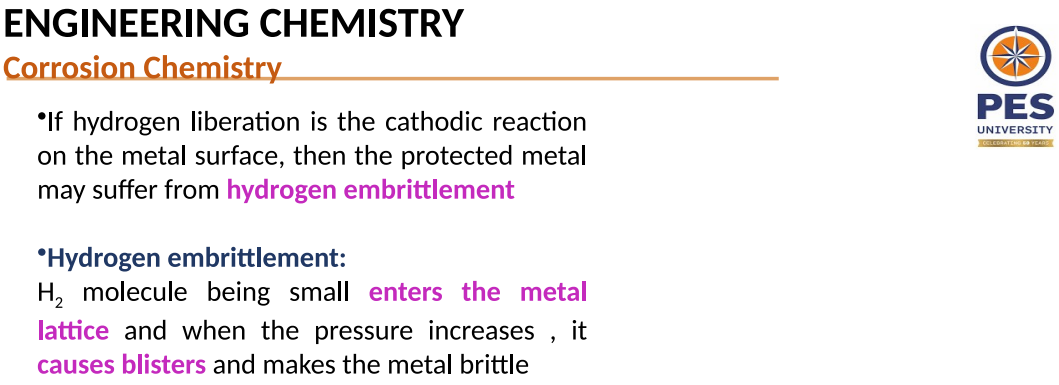

Figure G000136. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 15.

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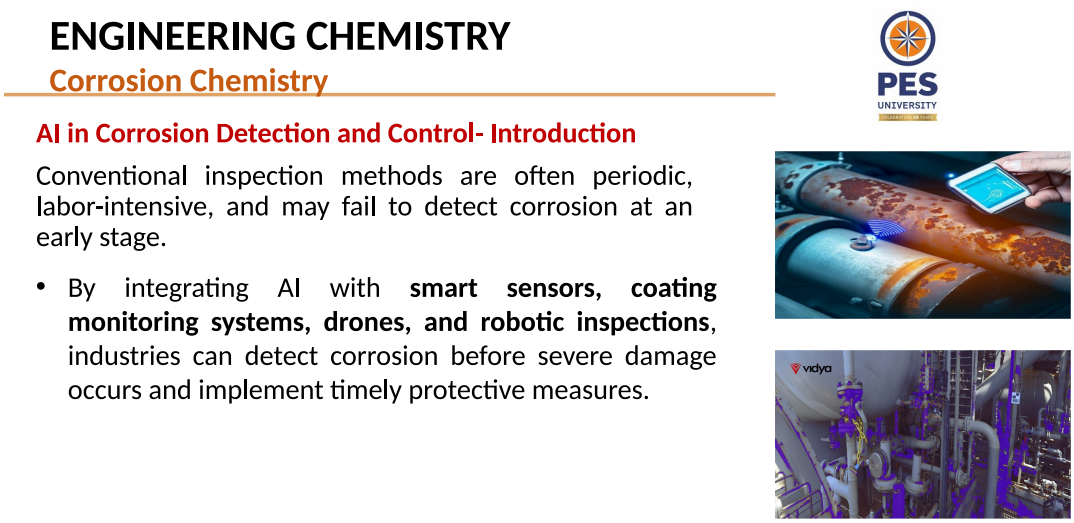

Figure G000137. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 16.

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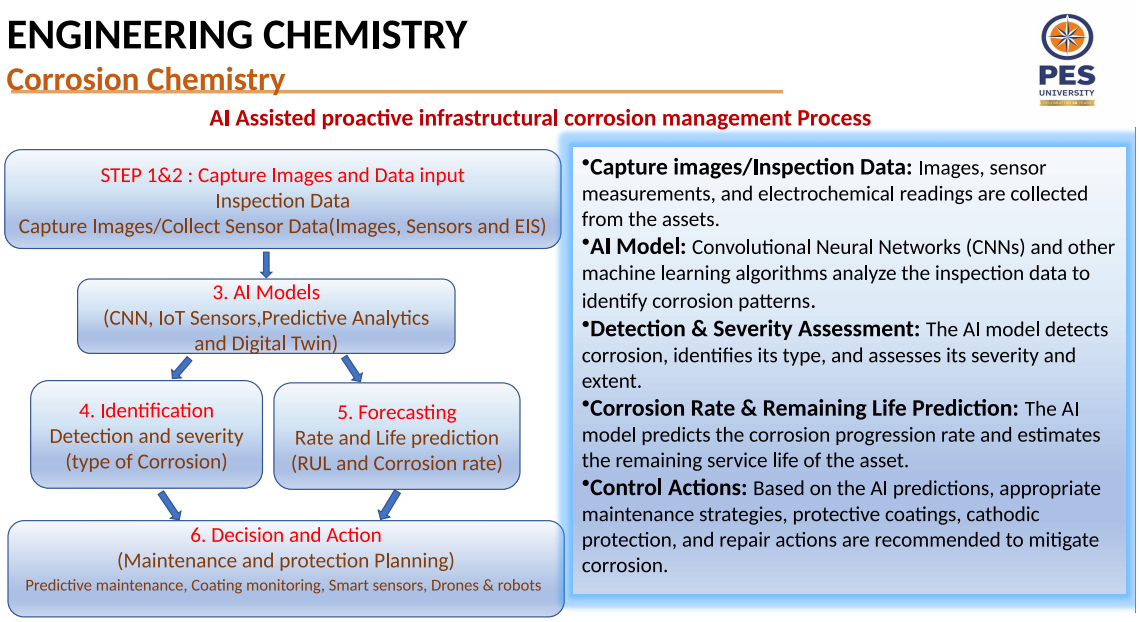

Figure G000138. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 17.

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Figure G000139. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 18.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000140. Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c · source slide page 19.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000629 · EV0000630 · EV0000631
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Class 24 content: Corrosion control Corrosion inhibitors, Impressed current cathodic Protection, Use of AI tools in corrosion detection and control EV0000632 · EV0000633 · EV0000634
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 2. Corrosion Inhibitors Chemical substances which are added in small quantities to the corrosive environment to decrease the rate of corrosion Inhibitors slow down or inhibit the anodic or cathodic reaction and control corrosion They can be used only in confined environment The action of an inhibitor depends on the nature of the metal to be protected as well as corrosive environment EV0000635 · EV0000636 · EV0000637
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 2 (a) Anodic inhibitors: M → Mn+ + ne- If the formation of Mn+ is prevented, the corrosion process is retarded This is achieved by the addition of large anions such as chromate(CrO42), tungstate (WO42), etc. These ions combine with Mn+ and form a precipitate which covers the surface of the anode [Image text, OCR] Inhibitors coat the surface and prevent corrosion Metal Surface Source:http://www.penriteoil.com.au/knowledge-centre/Coolant%20Inhibitor%20Packs/149/what-are-corrosion-inhibitors/289 EV0000638 · EV0000639 · EV0000640

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Anodic inhibitors are found to be effective only when sufficient amount of the inhibitor is added into the corrosion medium If insufficient quantity is added, then a part of the anodic region is exposed to the environment Formation of small anodic area results in intense corrosion Salts like Na2CrO4 , Na2WO4 EV0000641 · EV0000642 · EV0000643
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 2 (b) Cathodic inhibitors: liberation of hydrogen absorption of oxygen Inhibiting any of these reactions will slow down the cathodic reaction which in turn slows down the anodic reaction; corrosion is inhibited EV0000644 · EV0000645 · EV0000646
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Inhibition of hydrogen liberation: Preventing the diffusion of H+ ions to the cathode- By the addition of certain organic compounds which contain nitrogen or sulfur Aliphatic amines, urea, thiourea, mercaptans, heterocyclic compounds are widely used as cathodic inhibitors They get adsorbed on the cathodic region forming a protective film, preventing the H+ ions from coming in contact with the cathodic metal surface EV0000647 · EV0000648 · EV0000649
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry By increasing the hydrogen overvoltage on the metal surface – When Hydrogen overvoltage is high , liberation of H2 will be difficult By the addition of oxides of arsenic, antimony(As2O3, Sb2O3) or salts like sodium meta-arsenite(NaAsO2) They deposit on the cathode region The hydrogen overvoltage on these metals is very high; liberation of H2 is reduced EV0000650 · EV0000651 · EV0000652
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Inhibition of oxygen absorption: By removing the oxygen from the corrosive media - By adding Oxygen scavengers like hydrazine N2H4 + O2 → N2 + 2H2O By adding reducing agents sodium sulfite etc. 2Na2SO3 + O2 → 2Na2SO4 EV0000653 · EV0000654 · EV0000655
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry By decreasing the diffusion rate of oxygen to cathode – By adding salts with large cations such as ZnSO4, MgSO4, NiSO4 etc. The cations of the salts (Zn2+, Mg2+) migrate towards the cathode surface and react with the hydroxyl ions formed at the cathode Zn2+ + 2OH- → Zn(OH)2 Mg2+ + 2OH- → Mg(OH)2 The hydroxides get deposited on the cathodic sites The protective film impermeable to oxygen prevents its diffusion to the cathodic region EV0000656 · EV0000657 · EV0000658
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Limitations: They contaminate the environment Many of the inhibitors are toxic, cannot be used in systems which come in contact with humans Can be used only in closed systems in which corrosive environment is either contained or re-circulated EV0000659 · EV0000660 · EV0000661

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry 3. Impressed current method: By applying a direct current The protected metal is made cathodic by connecting it to the cathode of the external source of current Anode of the external source is usually connected to an inert electrode like graphite; Platinum, silicon, iron are also used as anodes A backfill of coke ,bentonite is used to improve efficiency of the anode (The prime purpose of using the backfill is to reduce electrical resistivity. This provides a lower anode-to-earth resistance and greater current outputs in cases where the surrounding soil is of high resistivity). The metal structure being cathode does not undergo corrosion Anode being inert remains unaffected EV0000662 · EV0000663 · EV0000664
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry [Image text, OCR] Rectifier- Anodes (Groundbed) Pipeline (Structure) Current Flow Source:https://www.mcwaneductile.com/blog/v-bio-r-or-cathodic-protection-an-honest-comparison/ Source:https://www.sciencedirect.com/topics/engineering/impressed-current-system EV0000665 · EV0000666 · EV0000667
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Advantages: One installation can protect large area of metal Limitations: Rather expensive, since it needs high current for safe protection of structure If the impressed current is not uniform on the entire surface of the protected structure, localized corrosion takes place on the protected metal EV0000668 · EV0000669 · EV0000670
15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry If hydrogen liberation is the cathodic reaction on the metal surface, then the protected metal may suffer from hydrogen embrittlement Hydrogen embrittlement: H2 molecule being small enters the metal lattice and when the pressure increases , it causes blisters and makes the metal brittle EV0000671 · EV0000672 · EV0000673
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry AI in Corrosion Detection and Control- Introduction Conventional inspection methods are often periodic, labor-intensive, and may fail to detect corrosion at an early stage. By integrating AI with smart sensors, coating monitoring systems, drones, and robotic inspections, industries can detect corrosion before severe damage occurs and implement timely protective measures. [Image text, OCR] vidya 38645 EV0000674 · EV0000675 · EV0000676

17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry AI Assisted proactive infrastructural corrosion management Process Capture images/Inspection Data: Images, sensor measurements, and electrochemical readings are collected from the assets. AI Model: Convolutional Neural Networks (CNNs) and other machine learning algorithms analyze the inspection data to identify corrosion patterns. Detection & Severity Assessment: The AI model detects corrosion, identifies its type, and assesses its severity and extent. Corrosion Rate & Remaining Life Prediction: The AI model predicts the corrosion progression rate and estimates the remaining service life of the asset. Control Actions: Based on the AI predictions, appropriate maintenance strategies, protective coatings, cathodic protection, and repair actions are recommended to mitigate corrosion. STEP 1&2 : Capture Images and Data input Inspection Data Capture Images/Collect Sensor Data(Images, Sensors and EIS) 3. AI Models (CNN, IoT Sensors,Predictive Analytics and Digital Twin) 4. Identification Detection and severity (type of Corrosion) 5. Forecasting Rate and Life prediction (RUL and Corrosion rate) 6. Decision and Action (Maintenance and protection Planning) Predictive maintenance, Coating monitoring, Smart sensors, Drones & robots EV0000677 · EV0000678 · EV0000679
18. Source slide page 18 (extracted text): ENGINEERING CHEMISTRY Corrosion Chemistry Artificial Intelligence (AI) is thus transforming corrosion management by enabling continuous monitoring, early defect detection, predictive maintenance, and intelligent decision-making. EV0000680 · EV0000681 · EV0000682
19. Source slide page 19 (extracted text): THANK YOU Department of Science and Humanities EV0000683 · EV0000684 · EV0000685

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.124] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module six on corrosion chemistry and today's class we are going to talk about yet another method of corrosion control and that is cathodic protection. [00:00:17.280] In that there are two ways, one is known as the sacrificial anon method, the other is known as impressed cathodic current method. So what do we mean by cathodic protection? So as the name suggests we are going to convert the structure [00:00:36.420] To be protected into cathode. So what happens? This is coming from the electrochemical theory of corrosion, where we understand that it's only the anode which undergoes corrosion, whereas the cathode remains unaffected. So that concept we are going to use, and the structure to be protected is being [00:00:56.240] converted into cathode. So what are we doing here? The structure [00:01:01.712] to be protected is completely converted into cathode and no part of it is allowed to act as anode. And we know that since cathode does not undergo corrosion, so the structure is completely protected from corrosion. [00:01:16.964] Now there are two ways by which we can do this. One is known as the sacrificial annode method, and the other is the impressed current method. Let us understand what we do in sacrificial annode method. Now in this method [00:01:32.740] Remember that we have to convert the metal into cathode. One good way would be if it is connected to a more active metal. So that that active metal is going to behave as the anode, and once we know that there is an anodic metal [00:01:51.552] In uh contact with the cathodic metal, the cathodic metal gets protected completely as long as the anodic metal is present in contact with the cathodic metal, the it is the anode which undergoes corrosion. So that [00:02:05.968] Concept is used here. So the protected metal structure is completely converted into

cathode by connecting it to a more active metal. So which metals are usually used? They are zinc and magnesium. [00:02:21.984] most of the times and these are referred to as auxiliary anode. Now since these metals are more active they act as anode and undergo preferential corrosion and thereby they protect the metallic structure which is now acting as the cathode. [00:02:40.228] Now why do we call it as the sacrificial anode method? This is referred to as the sacrificial anode method for the simple reason that in order to protect the structure, metallic structure, we are sacrificing the metal which is acting as the anode. [00:02:58.000] Hence, this is known as the sacrificial anode method. The only problem with this method is that as the anodes get exhausted, they have to be replaced periodically because we have to understand that the cathode structure now is protected only. [00:03:15.568] as long as it is in contact with an anode metal. So let us take up some good examples here. Suppose we have [00:03:26.752] Magnesium and zinc bars which are fixed to sides of ocean going ships. Now these bars which are attached are going to be more active than the [00:03:40.768] uh body of the ship now that is made of steel it's a steel vessel as we say and these magnesium and zinc bars are attached to the body the hull of the ship so that as long as these magnesium bars are there the entire structure the big [00:03:58.416] Ship is getting protected just by attaching a magnesium or zinc bar onto the surface, of course. [00:04:06.816] It is not that we put a small zinc bar and a whole big ship is protected. Yes, it is done by calculation and at regular intervals these magnesium you can see here also so many of them are placed here. Regular intervals these magnesium or zinc bars are pr uh are uh attached uh to the [00:04:24.496] uh body of the ship and that is going to protect the big huge ship from corrosion. So this would be a very good example for sacrificial anode method. [00:04:37.872] Another example would be when we have a very pipeline like these here, then we uh magnesium or zinc blocks are connected to it. So as long as these magnesium or zinc block is connected to the [00:04:53.248] Uh very pipeline, we find that the pipeline is protected from corrosion. Again, it has to be kept at regular intervals. Now sometimes instead of using magnesium and zinc blocks, there is a zinc wire which is running parallell with the

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.816] uh a pipeline which has to be protected and therefore we ensure that it is always in contact with the uh zinc wire so that the zinc undergoes corrosion and the pipeline metal is protected from corrosion. Usually it will be steel. [00:05:30.560] Now another way would be uh if you have uh oil storage tanks then there also magnesium block is connected. So we see that zinc and magnesium are usually used and they will act as auxiliary anodes and protect the uh structure from corrosion. [00:05:47.940] What are the advantages of this method? It's a very very simple method and the insulation cost is very low and it does not require any power supply. The magnesium bar or the zinc bar has to be just in contact with the structure to be protected. [00:06:05.712] would be the disadvantage. It involves recurring expenditure because we need to replace the consumed anode. So it's a very simple method by which [00:06:16.576] we can protect the structure from corrosion and we are not just talking about uh protecting it, it is completely unaffected by corrosion, it's only the anode which is undergoing corrosion here and the structure to be protected is absolutely unaffected by this by corrosion. [00:06:36.484] Let us look at the next method that is the impressed current method. Now here we are going to do something different. Here we are going to use direct current. Now again another way of making the protected structure into cathode would be [00:06:53.472] if we can connect the external connect an external store uh source the cathode terminal of the external source to the

protected structure so if you are forcing it to behave like the [00:07:08.512] cathode because it's connected to the cathode terminal of the external source and the anode will be uh connected to another uh metal over it. First let us concentrate on the cathode. So what are we doing here? We have to provide current of course but this now has a [00:07:25.216] The cathode terminal will be connected to the cathode to the protected structure so that you when you pass current you are forcing your metal to behave as cathode and that is what we talk about in cathodic protection. [00:07:37.860] So the protected metal is made cathode by connecting it to the cathode of the external source of current. Now we also have to connect something to the [00:07:47.840] uh anode of the external source and that is usually connected to an inert electrode like graphite and but sometimes platinum and silicon iron etc are also used. Of course you have to remember if [00:08:01.360] The anodes are not inert, in that case we need to keep replacing them. But if we use a most common use is that of the uh graphite where it is [00:08:12.032] It does not uh it is an inert anode. Uh therefore now your protected uh metal, the metal which you want to protect that is acting as the cathode, so it does not undergo corrosion and also your anode being inert remains unaffected. [00:08:29.860] Here we have to have a backfill of coke or bentonite which is used to improve the efficiency of the anode. So around the anode, usually what is done a deep trench is dug and there we are going to uh [00:08:47.748] Place the uh have a backfill around it. So this backfill is basically to improve the efficiency of the anode. Now in this case, so neither your cathode underwent corrosion and also the anode is unaffected. So if when we are obviously when we are saying that we are [00:09:05.520] digging a deep trench then and placing the anode over there that anode has to be an inert anode otherwise um if it's any other anode which is reactive then you need to keep replacing it and of course we can understand from this statement that this is most commonly used [00:09:23.088] for uh stationary uh uh objects stationary metal structures. [00:09:31.524] Now let us see an example here. So we have this pipeline and we have these anode bed over here. So you have these anode and there is the rectifier here which is going to supply the power. So your metal pipe the pipeline is now connected to the cathode and here we have [00:09:49.296] the anode which is connected to the anode terminal of the DC supply. So this is how it is done and then [00:09:58.384] Uh as I said to our stationary uh uh structures like bridges etc this is very commonly done then offshore structures uh where we have these uh oil wigs etc there also this kind of protection is used. Now here I talked about backfills.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:16.080] So you can see here that there is an anode is here, it uh a deep trench is there. And around this anode we have the backfill which is usually uh coke or uh coke and bentonite. [00:10:30.148] It's a type of clay. So that is being used to provide and to make it more uh to improve the efficiency, to improve the conductivity of this another because this is all in soil and then you have the tank over here or a pipe, whatever you have to protect is used and here we have [00:10:47.920] this DC supply and this is connected to the cathode this is connected to the anode terminal of the DC supply and that is how this protection is uh happening of the metal structure [00:11:02.244] Now what are the advantages here? That one installation can protect a large area of the metal, whereas in sacrificial anode uh we said that it had the these anode have to be placed at regular intervals. But here one installation can protect a large area of the metal. [00:11:19.600] Now there are some serious limitations which we need to pay attention. Of course, one is that it is rather expensive because we need a very high current for the safe protection of the structure. That is the first thing we

understand. Now this second point is very important. [00:11:37.216] If the impressed current is not uniform on the entire surface, then [00:11:45.476] What will happen? There will be local anode and cathode form because the current is not uniform over the entire surface. There will be local anode and cathode and that will give rise to localized corrosion of the protected metal. [00:12:00.880] So that has to be avoided, hence you have to have a high current and it should be uniform over the entire surface. Now there's a third point which is very very important and that is called hydrogen embrittlement is a possibility when we do cathodic protection. Now why? [00:12:20.068] If hydrogen liberation is the cathodic reaction. Now we are what we have done is we have made the entire structure into cathode. Now we are saying that it is totally unaffected by corrosion. Yes, that does happen, but we know that though the metal itself is not involved in the reaction, the environment has to take part in the [00:12:40.064] Cathodic reaction. Now suppose we have a highly acidic environment, what would be the reaction? H plus taking up electron and giving you half H two. Now hydrogen gas is getting produced. If this is the main reaction, or even in a neutral atmosphere, it is possible [00:12:57.232] that hydrogen liberation can take place. We have studied the reaction. Now if that happens [00:13:03.136] Hydrogen evolution if it is taking place either in an acidic environment like the reaction I've written written here or it could also happen in the uh neutral environment if that happens then a lot of hydrogen is getting uh produced and that will result in hydrogen embrittlement. So what is [00:13:20.848] this hydrogen M littlewend. Let us have a uh little understanding of this. So this hydrogen which is getting evolved is going to uh enter the metal lattice. Now [00:13:34.720] This being very small, hydrogen molecule is a very small molecule, so it enters the interstitial space of the metal lattice, and when the pressure starts building up, more and more hydrogen is getting involved. So the pressure starts building up and then uh if it becomes too much then it tries to [00:13:52.432] to come out and that will result in blisters on the metal surface and also it will make the metal brittle. So this phenomenon is referred to as hydrogen embrittlement and in cathodic protection we have made the entire structure into cathode. [00:14:10.000] So there is a possibility that if hydrogen evolution is the main reaction which is happening. So even if you have say water, it takes up electron, it is giving you in a neutral or an alkaline medium water taking up electron, and it is giving you OH mic. [00:14:27.568] And hydrogen liberation. So both these reactions are possible. Whether it is an acidic medium or it is in neutral or alkaline medium, we know that hydrogen liberation could be an important reaction which will take place at the cathode, and in that case, what is going to happen? It will result in hydrogen. [00:14:45.136] So these are some serious limitations of the cat uh cathodic protection and especially when we are using the [00:14:54.432] uh impressed anode uh cathodic current method. Of course this hydrogen embrittlement can take place even when we are using any kind of cathodic protection like even in the sacrificial anode method but uh yes there is a possibility that hydrogen embrittlement will take place when

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:12.080] Whenever we use cathodic protection. So [00:15:15.568] These uh this is about the uh how we are going to protect the metal structure using cathodic protection. The entire structure has been made into cathode and we have an anode which is uh undergoing uh corrosion if it is active, if it is an inert and [00:15:33.232] anode then it is also unaffected by corrosion. So this is what happens in cathodic protection. So now in the next class we are talking going to talk about another method of protection and that is anodic protection. Thank you.

EV0000246 · EV0000247 · EV0000248 · EV0000249

Full transcript: 2413 normalized words, 62 timestamped segments, 951.850667 seconds duration. Low-confidence segments flagged: 0.

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15 Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I

This lesson is indexed from the source video titled Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000250 · EV0000251

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.964] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module six on corrosion chemistry and in today's class we are going to study about corrosion control. [00:00:14.308] We will look at another method of corrosion control and that is using corrosion inhibitors. There are two types of corrosion inhibitors, anodic corrosion inhibitors and cathodic corrosion inhibitors. So we are going to study about them. [00:00:29.828] Let us understand what exactly are corrosion inhibitors. Now as the name suggests, they are substances which are going to inhibit the corrosion reaction. [00:00:42.640] So, how would we exactly define them? They are chemical substances which are added in small quantities to the corrosive environment in order to decrease the rate of corrosion. So here, wherever we see this corrosive environment, we are aware of what is the corrosive. [00:01:00.304] Environment and based on that, and also the nature of the metal, which is which needs to be protected, we are going to choose a corrosion inhibitor and use it, add it to the corrosive environment, and this should be added only in small quantities. [00:01:19.748] Now because we know there are two different reactions which happen at dianode and cathode, so there would be different types of corrosion inhibitors. So what exactly are these inhibitors doing? Basically, they are going to inhibit [00:01:35.056] either the anodic reaction or they'll inhibit the cathodic reaction and therefore they will help us to control corrosion. But the only [00:01:45.124] Problem with this corrosion inhibitor is that it can be used only in a confined environment. That can be understood because the concept is we are adding some substance into the corrosive environment. So obviously we cannot use corrosion inhibitors in an open environment. It has to be an enclosed [00:02:02.896] environment so we say it is used only in confined environment. Now the [00:02:09.424] Action of the corrosion inhibitor obviously will depend on the article which has to be protected as well as the corrosive environment. So you know it there is no general corrosion inhibitor which can be used in all the cases. We need to study the specific case. [00:02:27.088] what

is the nature of the metal, what is the corrosive environment, and then decide what corrosion inhibitor would be suitable to protect that article from corrosion. [00:02:37.472] Now since we have two sets of reactions, one happening at the anode and the other is happening at the cathode, we will have two types of corrosion inhibitors, anodic inhibitors and cathodic inhibitors. [00:02:50.468] Let us understand what is an anodic inhibitor. Now at the anode, what is the reaction? We know metal is getting oxidized forming mn plus plus n electron n electrons. [00:03:03.108] So we need to control this reaction. So that basically what we need to do is stop the formation of Mn. If we achieve that, then we are going to retard the corrosion process. Now, how can it be done? This is done with the help of [00:03:22.980] addition of large anions. So the corrosion inhibitor has to have a large anion and examples would be chromate and tung state. Now these [00:03:34.704] anions these large anions what do they do they are going to react with the Mn plus and form a precipitate which will now cover the surface of the anode. So you can see the metal surface here in this picture here and this precipitate which has got form [00:03:52.432] is going to cover the surface of the anode and therefore the [00:03:56.720] is it is like creating a barrier again it is just creating a barrier between the metal surface and the environment and therefore since the metal is now not exposed to the environment we find that this is going to inhibit the uh anodic reaction and hence control corrosion. [00:04:14.980] So this precipitate which is getting formed requires two uh large anions. It requires large anions because large anions like chromatic usually will form precipitate with these metal ions. So that is why we specifically say large anions. [00:04:35.300] So anodic inhibitors [00:04:37.760] are found to be effective only when sufficient amount of the inhibitor is added to the corrosion medium. What is the reason for that? We know that the metal surface is in the is acting as the anode. So when we add this corrosion inhibitor [00:04:55.504] They are going to form a precipitate as I said just now, and they are going to cover the surface of the anode. Now, if this covering is not uniform, if there are gaps, then your anode is getting exposed, and therefore this will

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.832] small amount of the anode area is getting exposed because the precipitate is not completely covering the [00:05:20.420] Surface and therefore this will lead to intense corrosion. So, therefore, whenever anodic inhibitors are used, we have to make sure that we add sufficient amount of the inhibitor. So, what is the corrosion? [00:05:38.192] Inhibitor we are using here so sodium chromate sodium tungstate. So these will be uh soluble in the environment and the chromate ions now react with the metal ions which are getting formed from a precipitate and create a barrier on the surface. So this is how anodic inhibitors work. Now let us come to [00:05:57.584] Cathodic inhibitors. Now when we come to the cathode, we know from the electrochemical theory of corrosion that there can be different types of reactions which will take place because the metal itself is not undergoing uh reduction. It cannot undergo reduction. And what were the the two types of reactions? [00:06:15.248] we had classified into they were cathodic reactions were either it could be liberation of hydrogen or it could be absorption of high oxygen. [00:06:24.768] So that means if we want to control corrosion, then we need to look at ways by which either we can stop the liberation of hydrogen or we can stop the absorption of oxygen. [00:06:40.608] So that is how we will be able to control corrosion. Now inhibiting any of these reactions is going to slow down the cathodic reaction. [00:06:53.604] So as a result what will happen? Now because the cathodic reaction has become slow, it is going to in turn slow down the anodic reaction. And once anodic reaction is not taking place, we know that the

[00:07:07.428] corrosion is under control or you have inhibit corrosion to a good extent. So let us see how we are going to either stop the liberation of hydrogen or absorption of oxygen. [00:07:23.492] First, let us take up hydrogen liberation. How do we inhibit the liberation of hydrogen? It can be done in two different ways. Let us study the first one where we are going to prevent the diffusion of H plus ions to the cathode. [00:07:40.656] This this reaction is happening at the cathode and one of the reasons for the reaction would be that H plus ions are [00:07:50.692] being reduced at the cathode. So if we can prevent the H plus ions uh coming close to the cathode and taking up electrons then the liberation of hydrogen is not going to uh take place. So it could be done by addition of certain organic compounds which contain nitrogen [00:08:08.464] and sulfur. So what would be the corrosion inhibitor in this case? Organic compounds containing nitrogen and sulfur. How are they doing it? First let us look at some examples. So like aliphatic amines, urea, thio urea, mercaptins, heterocyclic compounds. [00:08:26.832] All these are widely used as corrosion inhibitors and what is their role? They are going to prevent the diffusion of H plus ions to the cathode. [00:08:37.728] So urea if you remember NH2CO NH2 is urea and thyor urea would be NH2 C S NH2. So as I mentioned here nitrogen and sulfur containing organic compounds. So these can be [00:08:54.000] uh used here as corrosion uh cathodic inhibitor. So how what exactly do they do? They get absorbed on the [00:09:02.608] cathodic region they get absorbed on the cathodic surface and form a protective film. Now once there is this protective film on the surface the H plus ions cannot come in contact with the cathodic uh part of the metal [00:09:19.076] and hence we would have inhibited the reaction of liberation of hydrogen. [00:09:28.196] Another way of doing it is also liberation of hydrogen can be prevented by increasing the hydrogen overvoltage on the metal surface. Now when we studied factors affecting corrosion rate, one important [00:09:43.920] uh factor which we studied was hydrogen over voltage and what did we understand there that if the hydrogen over voltage is high then the liberation of hydrogen is going to be less and therefore we can control we can uh reduce the rate of reaction or the rate of uh

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.648] Corrosion is going to be less. Now that concept is being used here. What we have to do is we have to increase the hydrogen over voltage over the cathodic surface. Now once we achieve that, we know that the liberation of hydrogen becomes [00:10:18.320] uh difficult and then that would mean since hydrogen liberation is a cathodic reaction that would reduce the [00:10:25.392] uh extent of corrosion. So how is it done? It is done with the help of oxides of arsenic and antimony. Now these are added into the corrosive environment. So oxides of arsenic and antimony or salts like sodium metta arsenide. Now [00:10:41.956] Arsenic and antimony oxide, say for example, these will be the corrosion inhibitors. What happens is they are going to deposit arsenic and antimony onto the cathodic region. Arsenic and antimony have very high [00:11:00.036] Hydrogen over voltage. Now since you have raised the hydrogen over voltage of that particular surface, the liberation of hydrogen becomes difficult and therefore uh corrosion is going to [00:11:12.624] the rate of corrosion is going to fall down. So we are adding arsenic and antimony, but they will come to the cathode get reduced to basically arsenic and antimony which have high hydrogen. [00:11:26.560] over voltage. So this is how we are going to uh limit the liberation of hydrogen. [00:11:35.556] Let us talk of oxygen absorption. How do we inhibit oxygen absorption? So the easiest thing to do would be to remove the oxygen from the corrosive media. [00:11:48.900] And that should not be very difficult. All you have to do is oxygen has to be removed. Now that can be done with the help of oxygen scavengers like hydrazine. Now

what does hydrazine do? It is going to react with the oxygen and it will form nitrogen and H2O. Therefore [00:12:07.152] You would have removed the oxygen from the uh atmosphere. What's a scavenger? It is going to eat up the oxygen. So oxygen scavenger is being used here, it's going to take away all the oxygen from the atmosphere, and therefore, when there is no oxygen, obviously oxygen absorption will not. [00:12:24.816] happen meaning oxygen is not available to undergo the reduction reaction and hence you would have brought down the extent of cathodic reaction. Similarly to remove oxygen we could also use some reducing agents like sodium sulfide etc. [00:12:41.264] Now sodium sulphide when it reacts with oxygen it is forming sodium sulphate. So you can see that the oxidation number of sulfur has changed. [00:12:50.688] and therefore we can understand that there is a loss of electrons and therefore now we know that the reducing agents when we add it is going to take up the oxygen and that will [00:13:04.708] help us to remove oxygen from the atmosphere and therefore it will reduce the cathodic reaction. [00:13:11.556] Another way of reducing or removing the oxygen is or reducing oxygen absorption is by decreasing the diffusion rate of oxygen to the cathode. So basically we are not allowing the [00:13:28.688] oxygen to come in contact with the cathood. Now this can be done [00:13:33.904] by adding salts with large cations. So if you remember in anodic inhibitors we use large anions, the same concept is used here. We are using large cations like zinc sulfate, zinc and magnesium. So the salts [00:13:51.600] used are zinc sulphate and magnesium sulphate nickel sulphate etc why these salt because their cations are large zinc and mg2 plus zinc 2 plus mg2 plus now these are going to [00:14:05.760] migrate towards the cathodic surface and react with the hydroxyl ions. [00:14:11.760] Now these hydroxyons which are being produced at the cathode. So zinc reacts with OH minus to form zinc hydroxide. Again this is a precipitate. Magnesium also reacts with hydroxide and it forms zinc. [00:14:26.468] Magnesium hydroxide which is a precipitate. Now these hydroxides again they are getting deposited over on the cathodic sites and when that happens a protective film is formed over the cathodic sites and therefore oxygen cannot come in contact with the [00:14:44.240] the cathode and take up electrons and this is how you are preventing the absorption of oxygen. So these are the two ways by which you can reduce the absorption of oxygen at the cathode. [00:14:59.684] Now this is about the corrosion inhibitors. The anodic inhibitors and the cathodic inhibitors, how they act. So anodic inhibitors we learned that large anions are used like chromate and tungstate.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:15.424] In the case of cathodic inhibitors, depending on which reaction we are going to stop, if it is liberation of hydrogen, then we use urea, thiourea. [00:15:24.944] And uh another was to increase the hydrogen over voltage that was also done with the help of arsenic oxide and antimony oxide. So these are examples of corrosion inhibitors, specifically cathodic inhibitors. And there was another set of cathodic inhibitors where you could use oxygen scavengers like [00:15:42.608] hydrazine or a reducing agent like sodium sulfide and also if you want to prevent oxygen from coming in contact with the [00:15:52.164] cathode surface we could use uh zinc sulfate, uh magnesium sulphate etc. So this this is the whole set of corrosion inhibitors. Now let us look at some limitations of this corrosion inhibitors. [00:16:05.808] Now this is important, they will contaminate the environment because we know that corrosion inhibitors are substances we are adding directly into the environment. So of course they are going to contaminate the environment. And many of these [00:16:20.448] uh inhibitors are toxic, like for example arsenic oxide. Now these are toxic and therefore they cannot

be used in systems which come in contact with uh humans. [00:16:32.484] And the last fall, which we have already uh understood, is that they can be used only in closed system. We cannot use them uh in open uh system, but only in closed system in which the corrosion corrosive environment is either [00:16:48.976] Contained or it is recirculated. So these are the limitations of corrosion inhibitor inhibitors. Nevertheless, we have to study the nature of the metal, the and corrosive environment, and then decide what type of corrosion inhibitor we are going to use for that particular system. [00:17:08.896] So that is about corrosion inhibitors and in the next class we are going to talk about another method of corrosion control and that is cathodic protection. Thank you.

EV0000262 · EV0000263 · EV0000264 · EV0000265

Full transcript: 2477 normalized words, 74 timestamped segments, 1041.045333 seconds duration. Low-confidence segments flagged: 0.

EV0000250 · EV0000251 · EV0000252 · EV0000253

16 Source slide deck: Nernst equation and its derivation

This entry preserves the complete slide deck titled Nernst equation and its derivation as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_02, to avoid an unsupported match. EV0000299 · EV0000300

Graphs and diagrams

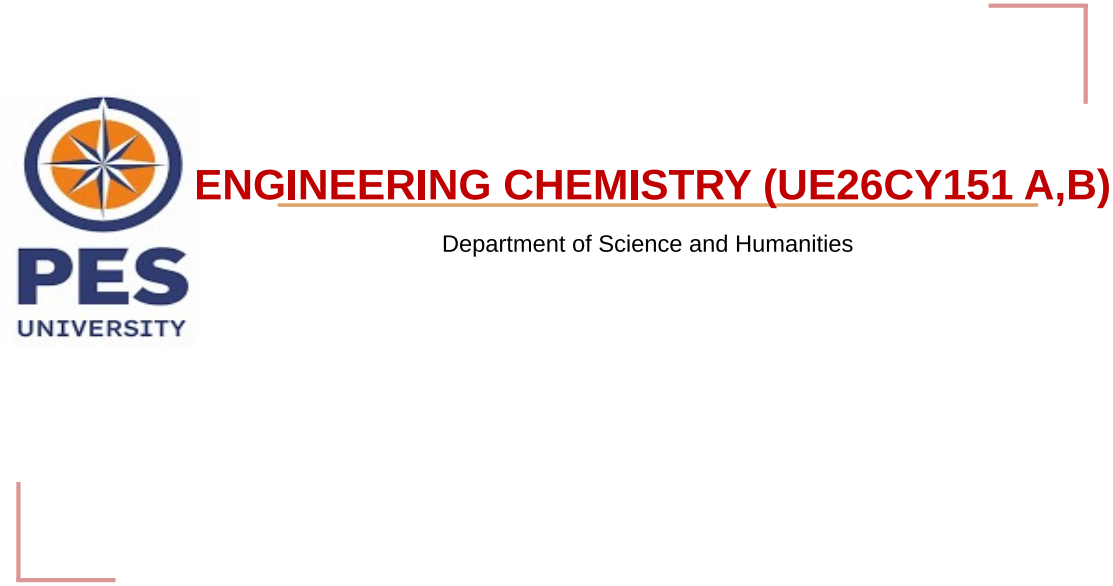

Figure G000012. Nernst equation and its derivation · source slide page 1.

EV0000300

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

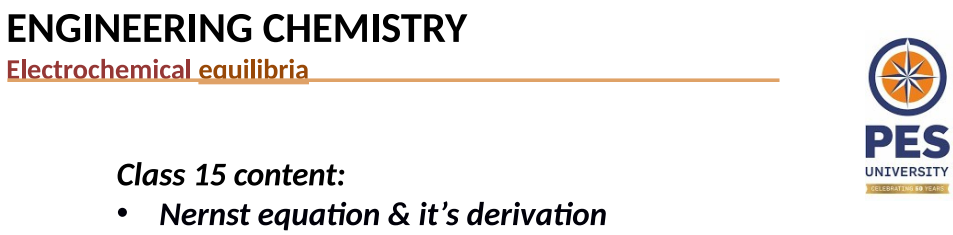

Figure G000013. Nernst equation and its derivation · source slide page 2.

EV0000303

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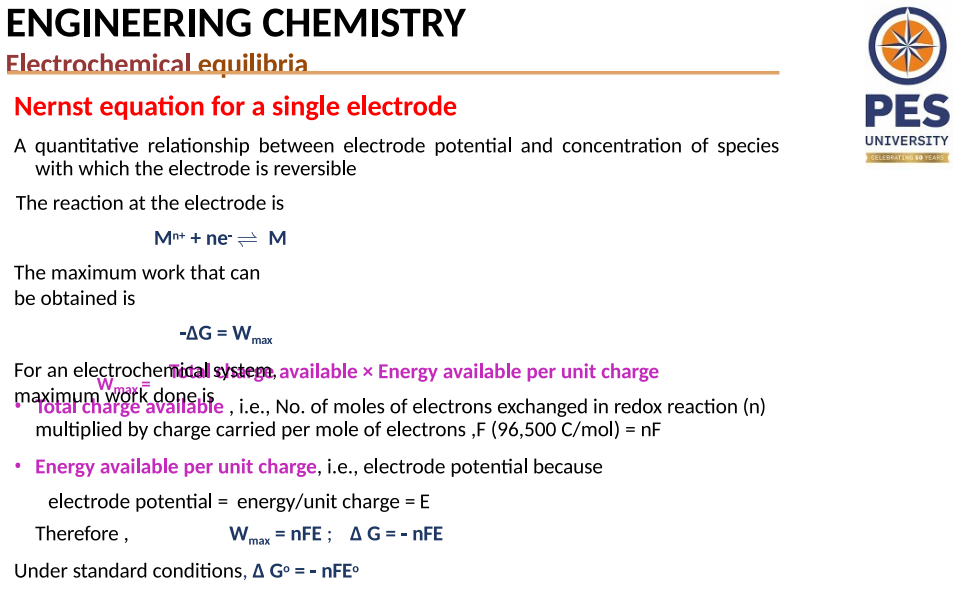

Figure G000014. Nernst equation and its derivation · source slide page 3.

EV0000306

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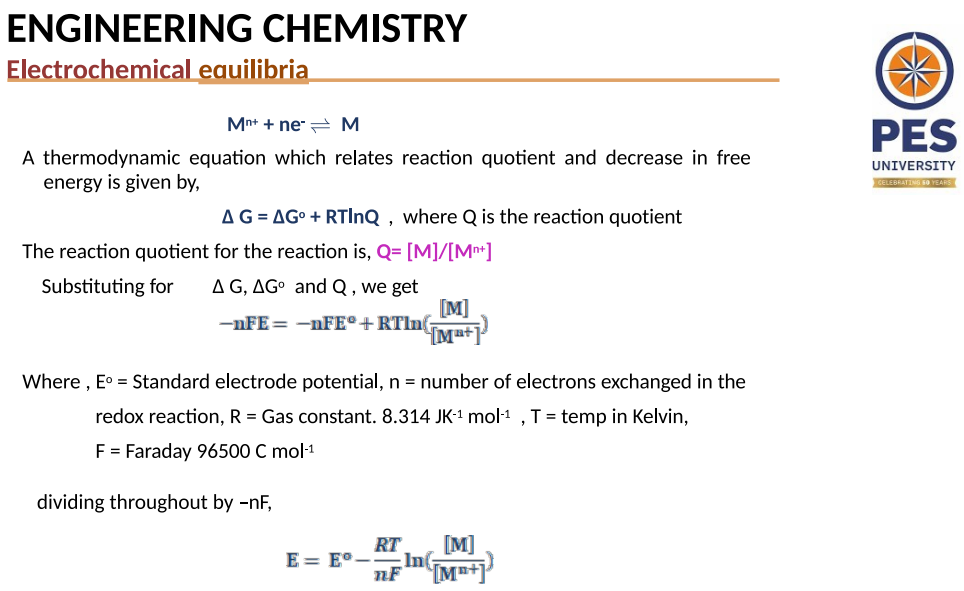

Figure G000015. Nernst equation and its derivation · source slide page 4.

EV0000309

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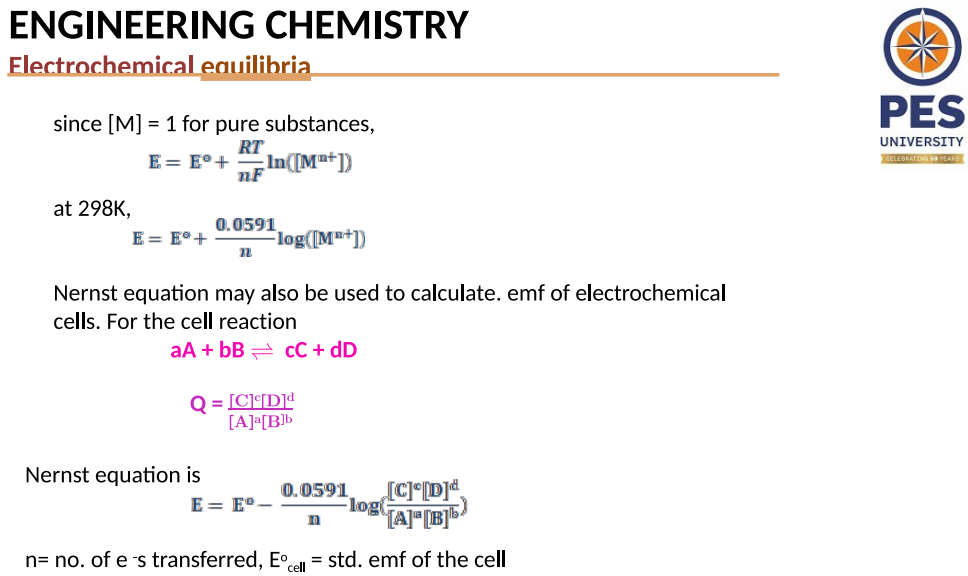

Figure G000016. Nernst equation and its derivation · source slide page 5.

EV0000312

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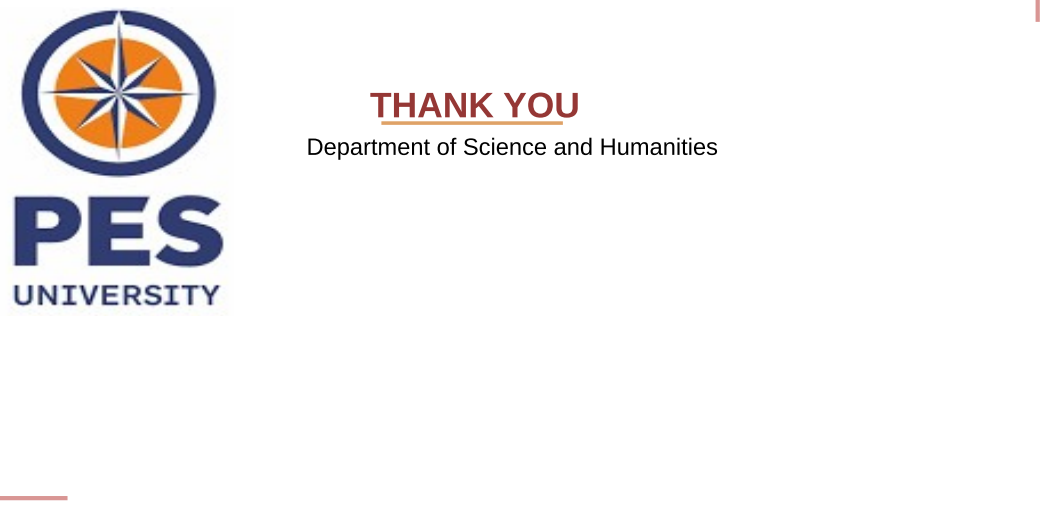

Figure G000017. Nernst equation and its derivation · source slide page 6.

EV0000315

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000299 · EV0000300 · EV0000301
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Class 15 content: Nernst equation & it's derivation EV0000302 · EV0000303 · EV0000304
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Nernst equation for a single electrode A quantitative relationship between electrode potential and concentration of species with which the electrode is reversible The reaction at the electrode is Mn+ + ne- ⇌ M The maximum work that can be obtained is -ΔG = Wmax For an electrochemical system, maximum work done is Total charge available × Energy available per unit charge Wmax = Total charge available , i.e., No. of moles of electrons exchanged in redox reaction (n) multiplied by charge carried per mole of electrons ,F (96,500 C/mol) = nF Energy available per unit charge, i.e., electrode potential because electrode potential = energy/unit charge = E Therefore , Wmax = nFE ; Δ G = - nFE Under standard conditions, Δ Go = - nFEo EV0000305 · EV0000306 · EV0000307
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria Mn+ + ne- ⇌ M A thermodynamic equation which relates reaction quotient and decrease in free energy is given by, Δ G = ΔGo + RTlnQ , where Q is the reaction quotient The reaction quotient for the reaction is, Q= [M]/[Mn+] Substituting for Δ G, ΔGo and Q , we get Where , Eo = Standard electrode potential, n = number of electrons exchanged in the redox reaction, R = Gas constant. 8.314 JK-1 mol-1 , T = temp in Kelvin, F = Faraday 96500 C mol-1 dividing throughout by –nF, EV0000308 · EV0000309 · EV0000310

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Electrochemical equilibria since [M] = 1 for pure substances, at 298K, Nernst equation may also be used to calculate. emf of electrochemical cells. For the cell reaction aA + bB ⇌ cC + dD **[C]c[D]d** Q = **[A]a[B]b** Nernst equation is n= no. of e -s transferred, Eocell = std. emf of the cell EV0000311 · EV0000312 · EV0000313
6. Source slide page 6 (extracted text): THANK YOU Department of Science and Humanities EV0000314 · EV0000315 · EV0000316

SHORT-NOTE RECAP

Source deck preserved: Nernst equation and its derivation. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000299

17 Source slide deck: Experiential learning Based on corrosion chemistry

This entry preserves the complete slide deck titled Experiential learning Based on corrosion chemistry as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_02, to avoid an unsupported match. EV0000686 · EV0000687

Graphs and diagrams

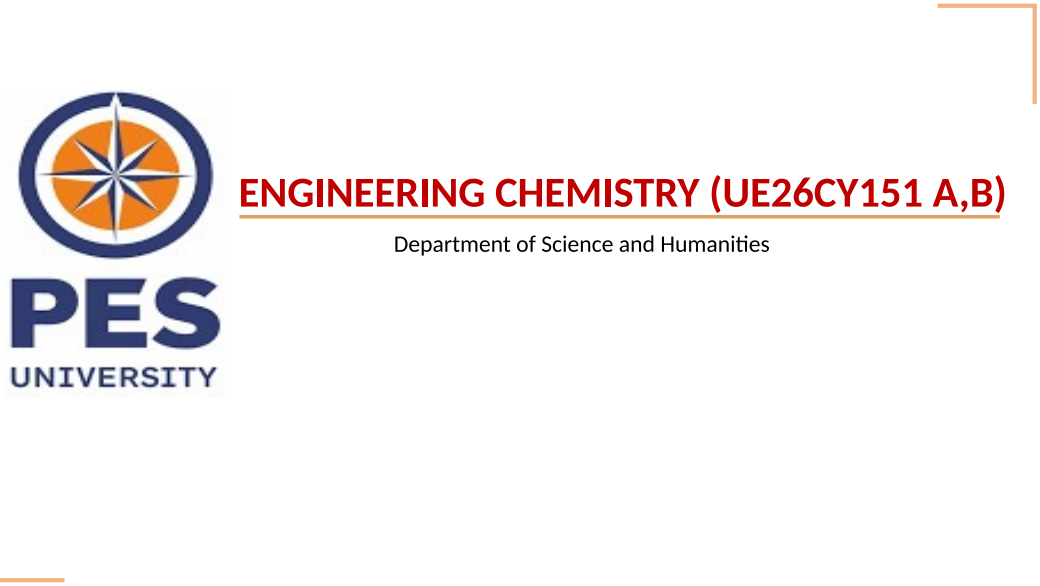

Figure G000141. Experiential learning Based on corrosion chemistry · source slide page 1.

EV0000687

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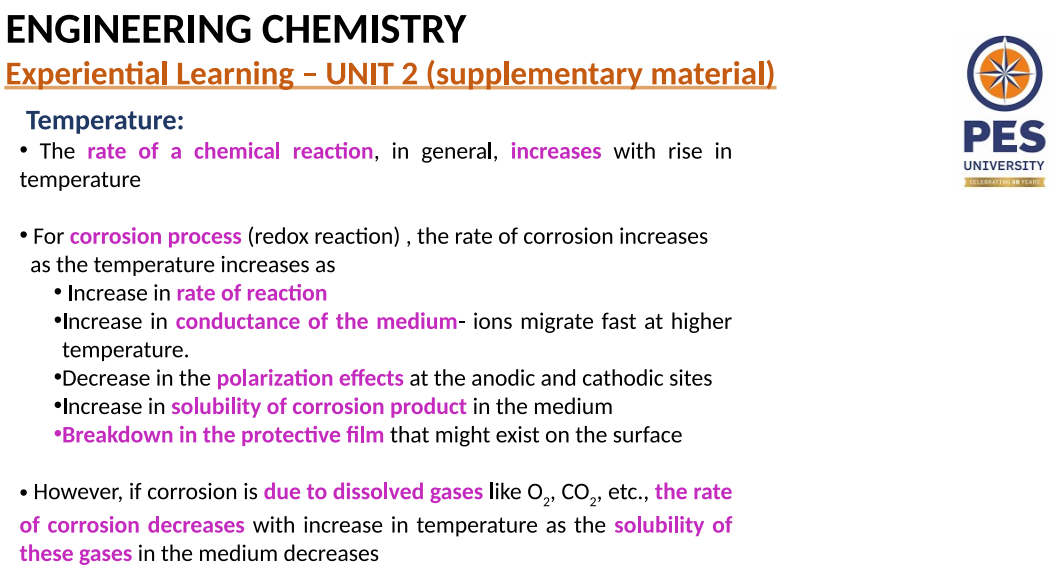

Figure G000142. Experiential learning Based on corrosion chemistry · source slide page 2.

EV0000690

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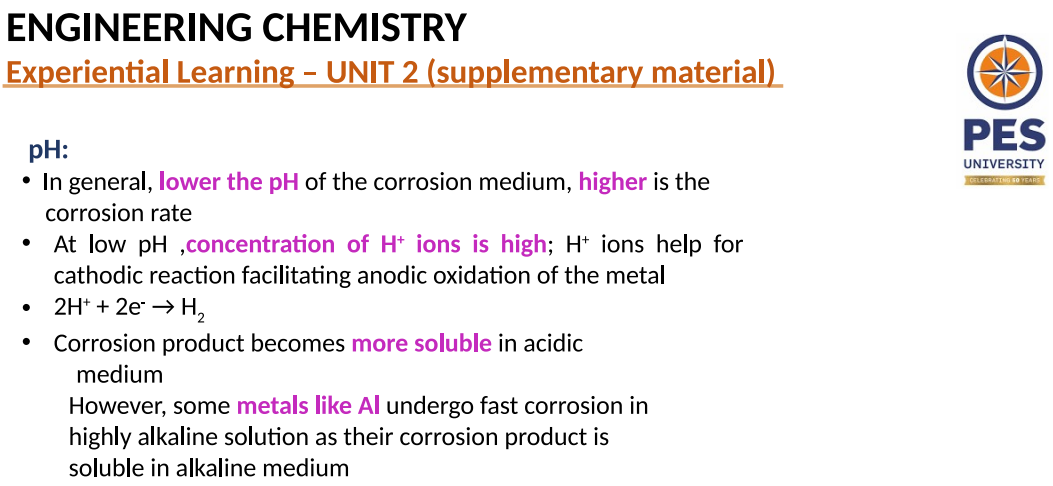

Figure G000143. Experiential learning Based on corrosion chemistry · source slide page 3.

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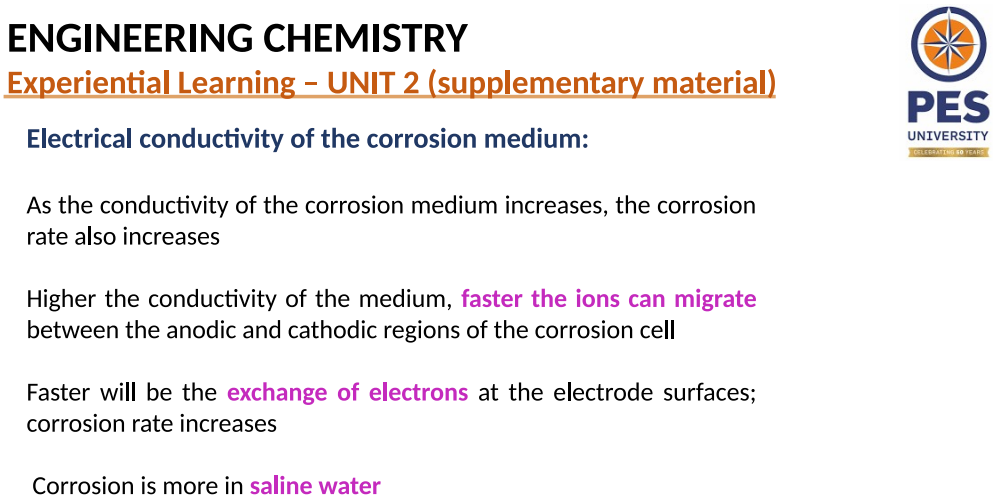

Figure G000144. Experiential learning Based on corrosion chemistry · source slide page 4.

EV0000696

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

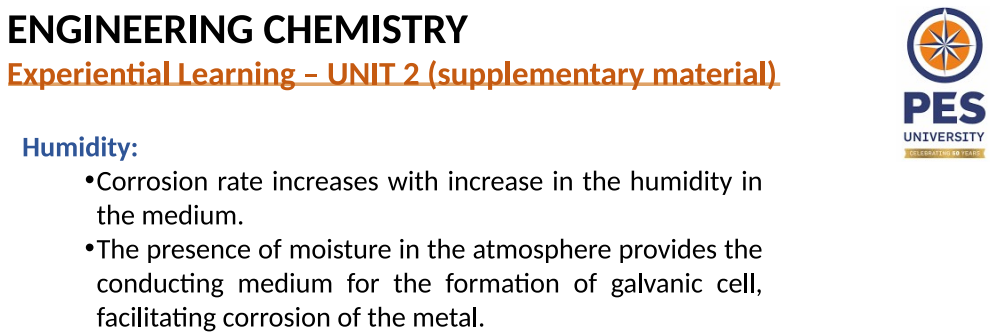

Figure G000145. Experiential learning Based on corrosion chemistry · source slide page 5.

EV0000699

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000146. Experiential learning Based on corrosion chemistry · source slide page 6.

EV0000702

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000686 · EV0000687 · EV0000688
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Experiential Learning – UNIT 2 (supplementary material) Temperature: The rate of a chemical reaction, in general, increases with rise in temperature For corrosion process (redox reaction) , the rate of corrosion increases as the temperature increases as Increase in rate of reaction Increase in conductance of the medium- ions migrate fast at higher temperature. Decrease in the polarization effects at the anodic and cathodic sites Increase in solubility of corrosion product in the medium Breakdown in the protective film that might exist on the surface However, if corrosion is due to dissolved gases like O2, CO2, etc., the rate of corrosion decreases with increase in temperature as the solubility of these gases in the medium decreases EV0000689 · EV0000690 · EV0000691
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Experiential Learning – UNIT 2 (supplementary material) pH: In general, lower the pH of the corrosion medium, higher is the corrosion rate At low pH ,concentration of H+ ions is high; H+ ions help for cathodic reaction facilitating anodic oxidation of the metal 2H+ + 2e- → H2 Corrosion product becomes more soluble in acidic medium However, some metals like Al undergo fast corrosion in highly alkaline solution as their corrosion product is soluble in alkaline medium EV0000692 · EV0000693 · EV0000694

4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Experiential Learning – UNIT 2 (supplementary material) Electrical conductivity of the corrosion medium: As the conductivity of the corrosion medium increases, the corrosion rate also increases Higher the conductivity of the medium, faster the ions can migrate between the anodic and cathodic regions of the corrosion cell Faster will be the exchange of electrons at the electrode surfaces; corrosion rate increases Corrosion is more in saline water EV0000695 · EV0000696 · EV0000697
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Experiential Learning – UNIT 2 (supplementary material) Humidity: Corrosion rate increases with increase in the humidity in the medium. The presence of moisture in the atmosphere provides the conducting medium for the formation of galvanic cell, facilitating corrosion of the metal. EV0000698 · EV0000699 · EV0000700
6. Source slide page 6 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU Department of Science and Humanities EV0000701 · EV0000702 · EV0000703

SHORT-NOTE RECAP

Source deck preserved: Experiential learning Based on corrosion chemistry. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review. EV0000686

18 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Engineering Chemistry, Unit_02 (Electrochemistry and corrosion) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000704 · EV0000705

Important points and why

1. Archive page 3 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.5; source text/OCR may be partial. (4.0 Marks) (5.0 Marks) (4.0 Marks) (2.0 Marks) 2.b. For a Pb-Ag system 1. Why is condensed phase rule applicable to this system? 2. Explain how silver can be extracted from Argentiferrous lead? 3.What is the Eutectic composition and Eutectic temperature? 2.c. A silver electrode containing 0.15M silver nitrate and copper electrode containing 0.19M copper sulphate were coupled using salt bridge. Write the cell representation, half cell reactions, calculate E , E at 25 C (Give E = 0.8V, E = 0.34V) 2.d. A Satuarated calomel electrode and glass electrode assembly was used to determine unknown pH. In the above cell following potentials are recorded a. 0.2836V in contact with buffer solution of pH = 4. b. 0.1949V in contact with solution of unknown pH. Write cell representation and calculate unknown pH (Given Esce = 0.24V) 2.e. i. Mention the factors affecting the conductance of a solution. 1M ii. In conductometric titration experiment, 25mL of acid mixture is titrated against 1M sodium hydroxide (NaOH) and the graph reveles V1 and V2 are 3.2 mL and 7.3 mL of NaOH respectively, Calculate the normality of acetic acid. 1M ocell cell o oAg /Ag + oCu /Cu 2+ EV0000704
2. Archive page 5 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.4167; source text/OCR may be partial. (2.0 Marks) (5.0 Marks) (4.0 Marks) 3.e. For a weak acid, pH = pKa, Substantiate using Henderson-Hasselbalch equation. 4.a. i. "Electro chemical Theory has provided satisfactory explaination for corrosion of metals" Justify the statement by taking iron as a demonstration sample metal, give the reactions of anode and cathode at different corroding environments. -4M How does iron corrods in presence of excess of oxygen and limited sypply of oxygen? - 1M 4.b. i. Identify the type of corrosion of the metal pipe in the picture ii. Briefly reason out the corroding environment and write electro chemical reactions at anode and cathode in the given picture - 3M EV0000705
3. Archive page 6 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (4.0 Marks) (4.0 Marks) (3.0 Marks) (4.0 Marks) 4.c. i. What are corrosion ihhibitors? Describe in a sentence or two how they control the corrosion rate - 2M ii. Anodizing and Phosphating are two techniques to protect a metal from corroding by coating the metal with an inorganic coating. Write the two compounds and the relevant chemical reactions for the anodization of aluminium and phosphating of iron. - 2M 4.d. i. Galvanization and tinning are two popular methods of protecting a metal by coating with another metal. If the reduction potentials of iron, Zinc and tin are -0.44V, -0.76V and -0.14V respectively, then comment on the situation; corrosion of iron would be observed in case a section of the protecting layer wears off. Briefly discuss the reason. ii. Write the corrosion cell that represents caustic embrittlement. What could go wrong with a boiler if caustic embrittlement is not taken care of? 4.e. i. In naphthalene–biphenyl system mention the number of components and phases at eutectic point. -2M

   ii. Comment on the following statement with respect to viscosity: "In an Ostwald viscometer, two liquids, A and B, take 60 seconds and 90 seconds respectively to flow between two marks of two identical capillaries" 5.a. What are biodegradable polymers? Give the two step process of degradation of polymers. Mention the factors influencing the degradation of polymers EV0000706

4. Archive page 4 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.5833; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 3.d. What happens when ammonia solution is added to cupric ions during 'Colorimetric estimation of copper in copper sulphate' experiment. Why is the analysis done at 620nm wavelength? 4.a. (i) Under what conditions can hydrazine be used as a corrosion inhibitor? How does it bring about corrosion inhibition, give the relevant reaction. (2 marks) (ii) What would be the effect of using a depolariser like cupric ion on the rate of corrosion reaction in a system that is cathodically polarised? (2 marks) (iii) What would be the cathodic reaction as per 'Electrochemical Theory of Wet Corrosion' if the corrosive medium is neutral and devoid of oxygen. (2 marks) 4.b. (i) What kind of corrosion would take place if steel bolts and nuts are present on a huge copper sheet? How would be the rate/severity of corrosion be affected by the ratio of anodic to cathodic area in this case. (3 marks) (ii) How would addition of insufficient amount of Na CrO (sodium chromate) to a confined corrosive environment affect the rate of corrosion. (3 marks) 4.c. (i) Why should the stainless steel tank being protected by the anodic protection technique be always maintained at a voltage corresponding to passive region? What would happen if the voltage strays into the trans passive region? (3 marks) (ii) A sample of brass of area 12.5 cm when exposed to a given electrolyte gave a corrosion current of 0.25 Amperes during Tafel studies, Calculate the rate of corrosion in mpy if the density and the equivalent weight of the brass sample is 8.73 g/cm and 32.068 respectively. (3 marks) 2 4 2 3 EV0000707
5. Archive page 3 from 1__1rBrYYqw3s.pdf; unit match candidate score 0.75; source text/OCR may be partial. (5.0 Marks) (6.0 Marks) (2.0 Marks) (6.0 Marks) 2.c. i) Explain the effects of following factors on corrosion of metal a) Nature of corrosion product b) Hydrogen overvoltage ii) Identify the type of corrosion taking place, when steel pipe connected with copper plumbing. 2.d. i) Explain the process of anodizing with the help of a diagram and relevant reaction. ii) How is sealing done for anodized articles ? 2.e. (i) Mention the role of sulphuric acid in the estimation of ferrous iron by potentiometric titration method. ii) Why pH is equals to pKa at half equivalence point in the pH metric titration of weak acid vs strong base ? 3.a. i) what are reserve batteries ? Give an example. ii) Calculate the capacity (Ah), energy density (Wh/kg) and power density (W/kg) of a battery if 2.4 g of lithium is stored in the battery and the battery lasts for 120 mins. The battery weighs 115 g and gives steady voltage of 3.4 V. (Given: F = 96500 C/mol, atomic mass of lithium = 7) EV0000708
6. Archive page 3 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 3/7 (4.0 Marks) (5.0 Marks) (8.0 Marks) 2.b. A decinormal calomel electrode is used to determine the potential of the following redox electrode: Pt/Cu (0.58M),Cu (0.08M). (i) write the cell representation (ii) write the half-cell reactions at both the electrodes (iii) calculate E and E at 298K (Given: E = 0.281 V, E = 0.153V) 2.c. (i) A glass electrode is coupled with saturated calomel electrode to measure unknown pH. The cell potentials measured are 0.215 V and 0.385 V in contact with a solution of pH7 and with solution of unknown pH respectively. Calculate the pH of unknown solution. Given: E = 0.244 V (2M) (ii) Discuss the construction of glass electrode with a labelled diagram (3M) 2.d. (i) Briefly describe the process of galvanization of iron (3M) (ii) Describe the cathodic protection by sacrificial anode method with its advantages and disadvantages (3M + 2M) 2+ + 0cell cell 0Hg/Hg2Cl2/Cl- 0Cu2+/Cu+ SCE EV0000709

7. Archive page 2 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 2/7 (6.0 Marks) (3.0 Marks) (6.0 Marks) 1.d. (i) Draw a neat labeled phase diagram of Lead-Silver system indicating the different areas and lines. Give the temperature and composition values corresponding to eutectic point. (ii) For Iron-Carbon system, mention the phases that are equilibrium with each other, corresponding temperature and weight % of carbon at the peritectic point. 1.e. i) Mention the factors affecting the conductance of a solution. ii) In conductometric titration experiment, 25 cm acid mixture is titrated against 1N NaOH and the graph revels V and V values at 2.8 cm and 6.5 cm of NaOH respectively. Calculate the normality of hydrochloric acid and acetic acid. 2.a. For the cell Pt/Hg/Hg Cl /Cl (0.1N)//Fe (0.05M), Fe (0.5M),/Pt i) Write the anodic, cathodic and overall cell reaction. (ii) Calculate E and E at 25 C (Given: E = 0.281V, E = 0.77V) . 3 1 2 3 3 2 2 - 3+ 2+ oCell Cell o o calomel oFe3+/Fe2+ EV0000710
8. Archive page 3 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 3/7 (6.0 Marks) (5.0 Marks) (6.0 Marks) 2.b. (i) What is ion selective electrode ? Write the cell representation when a glass electrode is coupled with a calomel electrode. (ii) When a buffer of pH = 4.0 is used an emf of 0.786 V is recorded at 298 K. At the same temperature when a solution of unknown pH is used an emf of 0.825 V is recorded. Calculate the pH of the unknown solution. (Given: E = 0.2422 V) 2.c. i) Explain the effects of following factors on corrosion of metal a) Nature of corrosion product b) Hydrogen overvoltage ii) Identify the type of corrosion taking place, when steel pipe connected with copper plumbing. 2.d. i) Explain the process of anodizing with the help of a diagram and relevant reaction. ii) How is sealing done for anodized articles ? SCE EV0000711
9. Archive page 3 from 5__1pIeTzocnU.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 3/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 2.b. i) What is the asymmetric electrode potential for a glass electrode? ii) A glass electrode dipped in a solution of pH of 4 offers an EMF of 0.2066 V with SCE at 298 K. Dipped in a solution of unknown pH, at the same temperature, in contact with SCE, recorded an EMF of 0.1076 V. Calculate the pH of the solution if ESCE is 0.2412 V 2.c. (i) Consider two situations: one where Iron (Fe) is in contact with copper (Cu) and the other where iron is in contact with silver (Ag). Given the standard reduction potentials of Fe, Cu, and Ag as -0.44 V, 0.34 V, and 0.8 V respectively, determine which scenario would lead to a faster corrosion of Fe and explain the reason. (ii) If the metal ions liberated at the anode accumulate in the anodic region, what impact would this have on the corrosion rate? Explain the observed effect. (iii) Describe the conditions under which oxides of antimony and arsenic can be employed for corrosion control. Explain how these oxides contribute to a reduction in the corrosion rate. 2.d. 1. What is anodizing of aluminum? How can one produce anodized aluminum articles? 2. How is sealing done during anodization, Give the relevant reaction. 3. How can colored anodized articles be produced? EV0000712

10. Archive page 2 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 2/7 (6.0 Marks) (3.0 Marks) (6.0 Marks) 1.d. (i) Draw a neat labeled phase diagram of Lead-Silver system indicating the different areas and lines. Give the temperature and composition values corresponding to eutectic point. (ii) For Iron-Carbon system, mention the phases that are equilibrium with each other, corresponding temperature and weight % of carbon at the peritectic point. 1.e. i) Mention the factors affecting the conductance of a solution. ii) In conductometric titration experiment, 25 cm acid mixture is titrated against 1N NaOH and the graph revels V and V values at 2.8 cm and 6.5 cm of NaOH respectively. Calculate the normality of hydrochloric acid and acetic acid. 2.a. For the cell Pt/Hg/Hg Cl /Cl (0.1N)//Fe (0.05M), Fe (0.5M),/Pt i) Write the anodic, cathodic and overall cell reaction. (ii) Calculate E and E at 25 C (Given: E = 0.281V, E = 0.77V) . 3 1 2 3 3 2 2 - 3+ 2+ oCell Cell o o calomel oFe3+/Fe2+ EV0000713
11. Archive page 3 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 3/7 (6.0 Marks) (5.0 Marks) (6.0 Marks) 2.b. (i) What is ion selective electrode ? Write the cell representation when a glass electrode is coupled with a calomel electrode. (ii) When a buffer of pH = 4.0 is used an emf of 0.786 V is recorded at 298 K. At the same temperature when a solution of unknown pH is used an emf of 0.825 V is recorded. Calculate the pH of the unknown solution. (Given: E = 0.2422 V) 2.c. i) Explain the effects of following factors on corrosion of metal a) Nature of corrosion product b) Hydrogen overvoltage ii) Identify the type of corrosion taking place, when steel pipe connected with copper plumbing. 2.d. i) Explain the process of anodizing with the help of a diagram and relevant reaction. ii) How is sealing done for anodized articles ? SCE EV0000714
12. Archive page 4 from 7__16pEjcSTVO.pdf; unit match candidate score 1.0; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 3.d. What is Beer- Lambert's law? 4.a. What is differential aeration corrosion? Draw a suitable diagram to illustrate the same, write the corrosion reaction (both anodic and cathodic) assuming the medium to be neutral and oxygen to be present.. 4.b. How do the below mentioned factors affect the rate of corrosion- (i) Rate of anodic to cathodic area (ii) pH of the medium (iii) Nature of metal 4.c. (i) Explain sacrificial anodic protection method of corrosion control with the help of a diagram. (ii) Explain the steps involved in galvanization technique of corrosion control. EV0000715
13. Archive page 5 from 8__1015pA2nzY.pdf; unit match candidate score 0.4167; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 5/7 (6.0 Marks) (5.0 Marks) (4.0 Marks) 4.a. What is anodizing? Explain the process of anodizing and give its application. 4.b. i) Why do the metals undergo corrosion? ii) Two metals such as iron and tin are in contact with each other and exposed to the corrosive environment, name the type of corrosion and explain with the relevant reactions. 4.c. Discuss how the following factors affect corrosion: i) Nature of the metal ii) pH EV0000716

14. Archive page 6 from 9__14fBQb1-hv.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 11/8/24, 1:02 PM ESA - DEC 2023 - UE22CY151B/UE22CY151A (set- 1) about:blank 6/9 (5.0 Marks) (4.0 Marks) (4.0 Marks) 4.a. i. Explain Electrochemical throry of Corrosion, with suitable example. (4M) ii. Comment on yellow rust and black rust of iron (1M) 4.b. i. "Corrosion of Iron (Fe) is faster when it is in contact with copper (Cu)" reason out the fact. (2M) ii. How does nature of corrosion product influence the rate of corroiosn? (2M) 4.c. i. what are cathodic inhibitors? (1M) ii. Mention any one chemical compound used an cathodic inhibitor which inhibit the hydrogen liberation at cathode. (1M) iii. Mention any two limitations of Cathodic inhibitors (2M) EV0000717

15. Archive page 2 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.75; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 2/6 (7.0 Marks) (6.0 Marks) (6.0 Marks) 1.d. (i) What is computational chemistry? Give any three applications of computational chemistry. (4 marks) (ii) During the conductometric titration of a mixture of strong and weak acid (HCl and CH COOH) against a strong base (NaOH), account for the nature of the conductance vs Volume of NaOH graph obtained. (3 marks) 2.a. For the below mentioned cell- Ag/AgCl/Cl−1(0.1) //Fe2+(0.02), Fe3+(0.1)/Pt Write the anodic and cathodic reaction, E and E values. Given : The standard reduction potential values of E0 Fe 3+ /Fe 2+ =0.77V and E0 Ag/AgCl/Cl −1 = 0.28V, R=8.314 J/K/mol and F=96500 C/mol 2.b. (i) When a metal M is in contact with solution containing its ions M and if ionization is faster than deposition, explain the formation of the electrical double layer at the interface of metal and solution. (3 Marks) (ii) Derive an expression for the 'Boundary Potential' established when a glass electrode is immersed in an external solution of H ions. Assume that the concentration of H ions in the external solution is c1 and the concentration of H ions in the internal reference solution is c2. (3 Marks) 3 0cell cell n+ + + + EV0000718

16. Archive page 3 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 3/6 (6.0 Marks) (7.0 Marks) (6.0 Marks) 2.c. (i) If overvoltage of hydrogen gas is high on a metal surface, how does this affect the rate of corrosion of metal. (Assume that the cathodic reaction is hydrogen evolution) (2 marks) (ii) Ship sailing in the sea or docked in the yard for a long time suffers from which type of corrosion. Give the anodic and cathodic regions formed and give the corresponding cathodic reactions. (4 marks) 2.d. (i) Give examples of - (4 marks) A corrosion product that offers protection An anodic corrosion inhibitor A cathodic protection technique for preventing corrosion The flux used in galvanization (ii) In the potentiometric titration experiment for estimation of ferrous iron, name the indicator and reference electrode. Justify. (3marks) 3.a. (i) A battery using 15 g Zinc as anode and discharges a constant current for 2 hours.The voltage of the battery is 1.9 V and the total weight of battery is 50g. Calculate the energy density of the battery in Wh/Kg. (Given : molar mass of Zn = 65 g, F = 96500 C/mol and no. of electrons involved in the redox reaction is 2.) (4 marks) (ii) Give the anodic and cathodic reactions taking place in a zinc-air battery. (2 marks) EV0000719

17. Archive page 2 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 2/7 (4.0 Marks) (2.0 Marks) (6.0 Marks) 1.d. Derive phase rule for a system of C components and P phases which are in equilibrium with each other. (4M) 1.e. i) Mention any two factors affecting the conductance. (1M) ii) Write any two advantages of potentiometric titration. (1M) 2.a. Derive Nernst equation for the given cell and calculate the emf of the cell. 2Al+3Cu2+(0.6M) →2Al3+(0.35M)+ Cu (Given:E =2.0V) 0cell EV0000720
18. Archive page 3 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.75; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 3/7 (5.0 Marks) (5.0 Marks) (6.0 Marks) 2.b. i) What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a saturated calomel electrode. (2M) (ii) When a glass electrode was coupled with a saturated calomel electrode to measure the pH of an unknown solution, an emf of 0.786V was recorded at 298K when pH = 4.2. At the same temperature when a solution of unknown pH was used, an emf of 0.825V was recorded. Calculate the pH of the unknown solution. (Given: E =0.2412V) (3M) 2.c. Two metals such as iron and tin are in contact with each other and exposed to the corrosive environment, name the type of corrosion and explain with the relevant reactions. 2.d. Account for the following: i) Influence of hydrogen over voltage on the rate of corrosion (2M) ii) Anodic corrosion inhibitors (2M) iii) Sacrificial anode method (2M) SCE EV0000721

SHORT-NOTE RECAP

18 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000704

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Origin of electrode potential, cell potential transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Origin of electrode potential, cell potential transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Origin of electrode potential, cell potential transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Origin of electrode potential, cell potential transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Reference electrodes, calomel electrode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Concentration cells, types of concentration cells transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Numerical problems based on electrode and cell potential determination transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9971; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9963; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9963; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9964; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Slide page text: Origin of electrode potential cell potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Origin of electrode potential cell potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Origin of electrode potential cell potential (converted PDF; original hash 2ec83ffe0ef9ffdc3cf9456eb8d23916168a1766212c76e6ea1d250785e1a9fe)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Nernst equation and its derivation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Nernst equation and its derivation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Nernst equation and its derivation (converted PDF; original hash 901ec34e6c7457fc05d8553aac54951d1d3cf815bd88ea702ba9d98dc08376c9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Types of electrodes	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Types of electrodes	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Types of electrodes (converted PDF; original hash 48a406f54e08258accb8bb124fb41fa06ec3d46d5d2d0e5677ae46877f62d136)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Reference electrodes calomel electrode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for Reference electrodes calomel electrode (converted PDF; original hash 523e549f45a848fead9f881858bff0e91e9842ecbf27ad439dd04e5317432d81)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Concentration cells types of concentration cells	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
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Page-render source for Concentration cells types of concentration cells (converted PDF; original hash 0c89dcbfb44000fe7dfaffb7f867bf3434272b52c2c683a1e33e6cea31cd00d9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Electrochemical theory of corrosion	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Electrochemical theory of corrosion	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Electrochemical theory of corrosion (converted PDF; original hash 06d6b3e6d9112aed3508b8efd82d97e4c6f9335e7bf49c38b421f7d3d89a5332)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Numerical problems based on electrode and cell potential determination	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
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Page-render source for Numerical problems based on electrode and cell potential determination (converted PDF; original hash d16e81ca0c88a84a96aa5c6872c372c7a562ff637112bbaa4e3dec1768a2a669)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio (converted PDF; original hash 3642374cacec4d437aba5c53e58b25c1ef51a91fbcbcfa18224d57dea5962bb3)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage (converted PDF; original hash 8a08dc91bae80c3389f9bbd219ecdaa28ea7542d2a73e4d9e864b959e10fd01c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Corrosion control metal coating galvanizing inorganic coating anodizing	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Corrosion control metal coating galvanizing inorganic coating anodizing	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Corrosion control metal coating galvanizing inorganic coating anodizing (converted PDF; original hash f9b7aec8a4e9b902b072e17d3e33cbd18a4b8199c8d22c74c4e48ef58ba6ac57)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c (converted PDF; original hash 012f8192c8f8ae555e88e96053fe4b311c1e5b1f31b00ef4aa6f67828573e804)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Based on corrosion chemistry	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Experiential learning Based on corrosion chemistry	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Based on corrosion chemistry (converted PDF; original hash 1f96475bab9ff13a23696c1d32bac7a405682eb92e45eec1a69787638ac47977)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
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Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
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Source	Path	Use
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Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
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Extracted historical pages: Chem-EC[2025]__1ZtWpoG9Dv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-EC[2025]__1ZtWpoG9Dv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV0000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0085. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0107. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0107. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0107. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0107. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000022	SRC00002/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0108 to s0111. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000023	SRC00003/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0108 to s0111. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00004/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0108 to s0111. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00005/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0108 to s0111. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000029	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000036	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000037	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000038	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000039	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000040	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000043	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000049	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000050	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000051	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000057	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000058	SRC00010/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00011/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00012/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00013/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000064	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000065	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000066	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000067	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000068	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000069	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000070	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000071	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000072	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000073	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000074	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000075	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000076	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000077	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000078	SRC00014/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000079	SRC00015/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000080	SRC00016/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000081	SRC00017/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00014/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0123. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00015/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0123. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00016/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0123. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000085	SRC00017/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0123. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000086	SRC00014/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0124 to s0139. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00015/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0124 to s0139. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000088	SRC00016/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0124 to s0139. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00017/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0124 to s0139. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000091	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000092	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000093	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000099	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000100	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000101	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000102	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000103	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000104	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000105	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000106	SRC00018/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000107	SRC00019/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00020/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00021/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000112	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000113	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000114	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000115	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000116	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000117	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000118	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000119	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000120	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000121	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000126	SRC00022/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0103. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000127	SRC00023/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0103. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000128	SRC00024/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0103. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000129	SRC00025/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0103. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000130	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000131	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000132	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000133	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000134	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000135	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000140	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 0.9971; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000141	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000142	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000146	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000147	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000148	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000149	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000150	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000151	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000152	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000153	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000154	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000155	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000156	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0063. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000162	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000163	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000168	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000169	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000170	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000171	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000172	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000174	SRC00034/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000175	SRC00035/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000176	SRC00036/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0101. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000177	SRC00037/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00034/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0102 to s0115. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00035/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0102 to s0115. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00036/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0102 to s0115. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000181	SRC00037/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0102 to s0115. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000182	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000183	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000184	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000185	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000186	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000187	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000188	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000189	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000190	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000191	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000192	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000193	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000194	SRC00038/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000195	SRC00039/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000196	SRC00040/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 97	SRC00041/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 98	SRC00038/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 99	SRC00039/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 00	SRC00040/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 01	SRC00041/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0092. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 02	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 03	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000204	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000205	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000206	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000207	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000208	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000209	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000210	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000211	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000212	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000213	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000214	SRC00042/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000215	SRC00043/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000216	SRC00044/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0076. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000217	SRC00045/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000218	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000219	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000220	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000221	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000222	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000223	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000224	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000225	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000226	SRC00046/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000227	SRC00047/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000228	SRC00048/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000229	SRC00049/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000230	SRC00046/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000231	SRC00047/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000232	SRC00048/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0067. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000233	SRC00049/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000234	SRC00050/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000235	SRC00051/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000236	SRC00052/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000237	SRC00053/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000238	SRC00050/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000239	SRC00051/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000240	SRC00052/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000241	SRC00053/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000242	SRC00050/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000243	SRC00051/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000244	SRC00052/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000245	SRC00053/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002 46	SRC00050/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 47	SRC00051/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 48	SRC00052/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 0.9963; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 49	SRC00053/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 50	SRC00054/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 51	SRC00055/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 52	SRC00056/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9964; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000253	SRC00057/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000254	SRC00054/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000255	SRC00055/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000256	SRC00056/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 0.9964; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000257	SRC00057/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000258	SRC00054/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000259	SRC00055/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000260	SRC00056/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare at similarity 0.9964; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000261	SRC00057/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000262	SRC00054/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000263	SRC00055/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000264	SRC00056/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0074. Sidecar variants compare at similarity 0.9964; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000265	SRC00057/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000266	SRC00058/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000267	SRC00060/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000268	SRC00059/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000269	SRC00058/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000270	SRC00060/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000271	SRC00059/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000272	SRC00058/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000273	SRC00060/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000274	SRC00059/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000275	SRC00058/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000276	SRC00060/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000277	SRC00059/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000278	SRC00058/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002279	SRC00060/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002280	SRC00059/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002281	SRC00058/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002282	SRC00060/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002283	SRC00059/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00002284	SRC00058/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000285	SRC00060/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000286	SRC00059/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000287	SRC00058/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000288	SRC00060/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000289	SRC00059/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000290	SRC00058/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000291	SRC00060/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000292	SRC00059/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000293	SRC00058/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000294	SRC00060/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000295	SRC00059/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000296	SRC00058/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 97	SRC00060/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 98	SRC00059/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 99	SRC00061/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 00	SRC00063/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 01	SRC00062/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 02	SRC00061/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 03	SRC00063/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000304	SRC00062/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000305	SRC00061/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000306	SRC00063/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000307	SRC00062/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000308	SRC00061/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000309	SRC00063/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000310	SRC00062/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 11	SRC00061/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 12	SRC00063/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 13	SRC00062/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 14	SRC00061/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 15	SRC00063/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 16	SRC00062/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 17	SRC00064/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000318	SRC00066/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000319	SRC00065/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000320	SRC00064/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000321	SRC00066/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000322	SRC00065/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000323	SRC00064/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000324	SRC00066/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000325	SRC00065/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000326	SRC00064/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000327	SRC00066/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000328	SRC00065/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000329	SRC00064/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000330	SRC00066/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000331	SRC00065/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000332	SRC00064/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000333	SRC00066/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000334	SRC00065/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000335	SRC00064/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000336	SRC00066/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000337	SRC00065/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000338	SRC00064/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000339	SRC00066/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000340	SRC00065/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000341	SRC00064/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000342	SRC00066/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000343	SRC00065/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000344	SRC00064/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000345	SRC00066/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000346	SRC00065/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000347	SRC00067/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000348	SRC00069/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000349	SRC00068/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000350	SRC00067/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000351	SRC00069/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000352	SRC00068/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000353	SRC00067/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000354	SRC00069/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000355	SRC00068/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000356	SRC00067/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000357	SRC00069/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000358	SRC00068/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000359	SRC00067/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000360	SRC00069/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000361	SRC00068/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000362	SRC00067/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000363	SRC00069/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000364	SRC00068/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000365	SRC00067/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000366	SRC00069/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000367	SRC00068/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000368	SRC00067/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000369	SRC00069/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000370	SRC00068/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000371	SRC00067/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000372	SRC00069/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000373	SRC00068/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000374	SRC00070/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000375	SRC00072/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000376	SRC00071/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000377	SRC00070/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000378	SRC00072/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000379	SRC00071/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000380	SRC00070/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000381	SRC00072/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000382	SRC00071/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000383	SRC00070/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000384	SRC00072/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000385	SRC00071/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000386	SRC00070/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000387	SRC00072/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000388	SRC00071/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000389	SRC00070/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000390	SRC00072/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000391	SRC00071/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000392	SRC00070/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000393	SRC00072/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000394	SRC00071/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000395	SRC00070/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000396	SRC00072/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000397	SRC00071/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 98	SRC00070/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 99	SRC00072/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 00	SRC00071/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 01	SRC00070/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 02	SRC00072/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 03	SRC00071/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000404	SRC00073/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000405	SRC00075/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000406	SRC00074/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000407	SRC00073/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000408	SRC00075/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000409	SRC00074/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000410	SRC00073/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000411	SRC00075/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000412	SRC00074/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000413	SRC00073/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000414	SRC00075/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000415	SRC00074/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000416	SRC00073/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000417	SRC00075/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000418	SRC00074/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000419	SRC00073/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000420	SRC00075/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000421	SRC00074/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000422	SRC00073/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000423	SRC00075/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000424	SRC00074/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004025	SRC00073/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004026	SRC00075/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004027	SRC00074/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00004028	SRC00073/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004029	SRC00075/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004030	SRC00074/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00004031	SRC00073/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000432	SRC00075/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000433	SRC00074/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000434	SRC00073/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000435	SRC00075/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000436	SRC00074/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000437	SRC00073/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000438	SRC00075/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000439	SRC00074/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000440	SRC00073/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000441	SRC00075/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000442	SRC00074/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000443	SRC00076/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000444	SRC00078/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000445	SRC00077/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000446	SRC00076/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000447	SRC00078/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000448	SRC00077/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000449	SRC00076/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000450	SRC00078/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000451	SRC00077/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000452	SRC00076/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000453	SRC00078/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000454	SRC00077/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000455	SRC00076/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000456	SRC00078/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000457	SRC00077/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000458	SRC00076/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000459	SRC00078/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000460	SRC00077/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000461	SRC00076/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000462	SRC00078/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000463	SRC00077/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000464	SRC00076/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000465	SRC00078/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000466	SRC00077/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000467	SRC00076/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000468	SRC00078/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000469	SRC00077/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000470	SRC00076/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000471	SRC00078/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000472	SRC00077/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000473	SRC00079/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000474	SRC00081/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000475	SRC00080/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 76	SRC00079/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 77	SRC00081/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 78	SRC00080/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 79	SRC00079/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 80	SRC00081/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 81	SRC00080/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000482	SRC00079/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000483	SRC00081/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000484	SRC00080/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000485	SRC00079/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000486	SRC00081/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000487	SRC00080/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000488	SRC00079/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000489	SRC00081/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000490	SRC00080/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000491	SRC00079/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000492	SRC00081/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000493	SRC00080/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000494	SRC00079/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000495	SRC00081/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000496	SRC00080/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000497	SRC00079/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000498	SRC00081/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000499	SRC00080/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000500	SRC00079/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000501	SRC00081/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000502	SRC00080/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000503	SRC00079/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000504	SRC00081/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000505	SRC00080/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000506	SRC00079/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000507	SRC00081/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000508	SRC00080/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000509	SRC00079/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000510	SRC00081/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000511	SRC00080/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000512	SRC00079/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000513	SRC00081/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000514	SRC00080/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000515	SRC00079/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000516	SRC00081/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000517	SRC00080/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000518	SRC00079/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000519	SRC00081/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000520	SRC00080/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000521	SRC00079/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000522	SRC00081/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000523	SRC00080/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000524	SRC00079/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000525	SRC00081/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000526	SRC00080/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000527	SRC00079/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000528	SRC00081/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000529	SRC00080/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000530	SRC00079/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000531	SRC00081/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000532	SRC00080/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000533	SRC00079/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000534	SRC00081/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000535	SRC00080/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000536	SRC00079/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000537	SRC00081/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000538	SRC00080/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000539	SRC00079/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000540	SRC00081/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000541	SRC00080/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000542	SRC00082/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000543	SRC00084/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000544	SRC00083/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000545	SRC00082/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000546	SRC00084/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000547	SRC00083/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000548	SRC00082/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000549	SRC00084/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000550	SRC00083/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000551	SRC00082/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000552	SRC00084/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000553	SRC00083/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000554	SRC00082/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000555	SRC00084/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000556	SRC00083/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000557	SRC00082/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000558	SRC00084/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000559	SRC00083/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000560	SRC00082/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000561	SRC00084/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000562	SRC00083/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000563	SRC00082/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000564	SRC00084/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000565	SRC00083/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000566	SRC00082/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000567	SRC00084/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000568	SRC00083/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000569	SRC00082/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000570	SRC00084/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000571	SRC00083/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000572	SRC00082/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000573	SRC00084/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000574	SRC00083/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000575	SRC00082/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000576	SRC00084/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000577	SRC00083/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000578	SRC00082/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000579	SRC00084/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000580	SRC00083/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000581	SRC00082/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000582	SRC00084/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000583	SRC00083/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000584	SRC00085/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000585	SRC00087/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000586	SRC00086/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000587	SRC00085/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000588	SRC00087/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000589	SRC00086/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000590	SRC00085/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000591	SRC00087/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000592	SRC00086/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000593	SRC00085/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000594	SRC00087/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000595	SRC00086/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 96	SRC00085/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 97	SRC00087/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 98	SRC00086/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 99	SRC00085/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 00	SRC00087/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006 01	SRC00086/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000602	SRC00085/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000603	SRC00087/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000604	SRC00086/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000605	SRC00085/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000606	SRC00087/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000607	SRC00086/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000608	SRC00085/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000609	SRC00087/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000610	SRC00086/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000611	SRC00085/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000612	SRC00087/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000613	SRC00086/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000614	SRC00085/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000615	SRC00087/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000616	SRC00086/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000617	SRC00085/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000618	SRC00087/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000619	SRC00086/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000620	SRC00085/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000621	SRC00087/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000622	SRC00086/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000623	SRC00085/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000624	SRC00087/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000625	SRC00086/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000626	SRC00085/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000627	SRC00087/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000628	SRC00086/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000629	SRC00088/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000630	SRC00090/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000631	SRC00089/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000632	SRC00088/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000633	SRC00090/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000634	SRC00089/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000635	SRC00088/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000636	SRC00090/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000637	SRC00089/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000638	SRC00088/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000639	SRC00090/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000640	SRC00089/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000641	SRC00088/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000642	SRC00090/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000643	SRC00089/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000644	SRC00088/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000645	SRC00090/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000646	SRC00089/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000647	SRC00088/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000648	SRC00090/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000649	SRC00089/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000650	SRC00088/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000651	SRC00090/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000652	SRC00089/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000653	SRC00088/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000654	SRC00090/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000655	SRC00089/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000656	SRC00088/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000657	SRC00090/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000658	SRC00089/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000659	SRC00088/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000660	SRC00090/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000661	SRC00089/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000662	SRC00088/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000663	SRC00090/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000664	SRC00089/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000665	SRC00088/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000666	SRC00090/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000667	SRC00089/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000668	SRC00088/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000669	SRC00090/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000670	SRC00089/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000671	SRC00088/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000672	SRC00090/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000673	SRC00089/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000674	SRC00088/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000675	SRC00090/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000676	SRC00089/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000677	SRC00088/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000678	SRC00090/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000679	SRC00089/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000680	SRC00088/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000681	SRC00090/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000682	SRC00089/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000683	SRC00088/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000684	SRC00090/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000685	SRC00089/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000686	SRC00091/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000687	SRC00093/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000688	SRC00092/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000689	SRC00091/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000690	SRC00093/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000691	SRC00092/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000692	SRC00091/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000693	SRC00093/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000694	SRC00092/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000695	SRC00091/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000696	SRC00093/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000697	SRC00092/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00006 98	SRC00091/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 99	SRC00093/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 00	SRC00092/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 01	SRC00091/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 02	SRC00093/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 03	SRC00092/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 04	SRC00094/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV0000705	SRC00094/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000706	SRC00094/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000707	SRC00095/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV0000708	SRC00096/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV0000709	SRC00097/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV0000710	SRC00098/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000711	SRC00098/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV0000712	SRC00099/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV0000713	SRC00100/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000714	SRC00100/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV0000715	SRC00101/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV0000716	SRC00102/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV0000717	SRC00103/q6	covered	Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
EV0000718	SRC00104/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000719	SRC00104/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV0000720	SRC00105/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV00007 21	SRC00105/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.

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